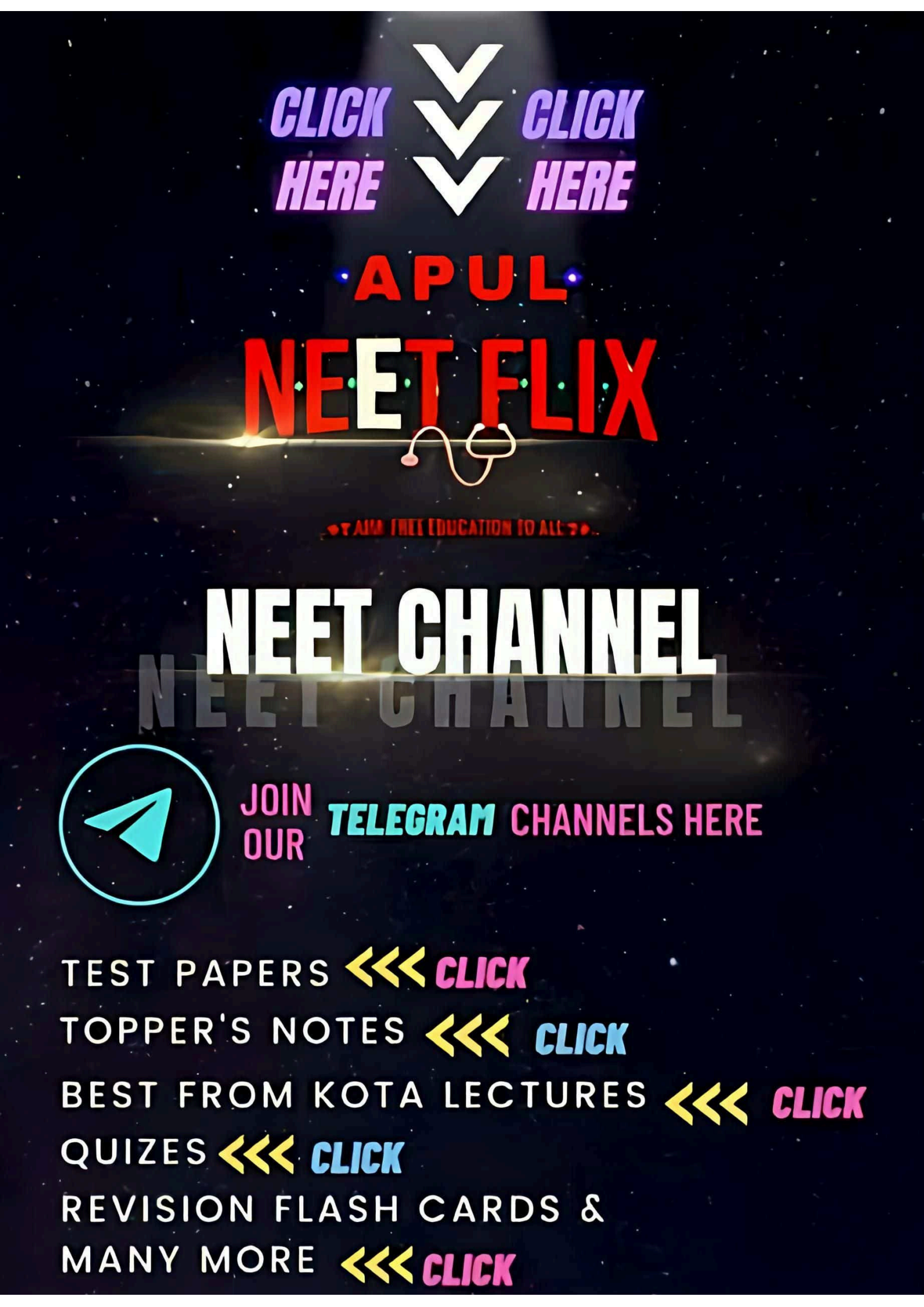

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Exercise - I (Conceptual Questions) Build Up Your Understanding

QUESTIONS BASED ON MOLES

1. The number of atoms present in 16 g of oxygen is
(1) $6.02 \times 10^{11.5}$ (2) 3.01×10^{23}
(3) $3.01 \times 10^{11.5}$ (4) 6.02×10^{23}
CMC001

2. The number of atoms in 4.25 g of NH_3 is approx:-
(1) 1×10^{23} (2) 1.5×10^{23}
(3) 2×10^{23} (4) 6×10^{23}
CMC002

3. Which of the following contains maximum number of oxygen atoms ?
(1) 1 g of O
(2) 1 g of O_2
(3) 1 g of O_3
(4) all have the same number of atoms
CMC003

4. The number of atoms present in 0.5 g atom of nitrogen is same as the atoms in –
(1) 12 g of C (2) 32 g of S
(3) 8 g of oxygen (4) 24 g of Mg
CMC004

5. Which of the following contains maximum number of atoms ?
(1) 4 g of H_2 (2) 16 g of O_2
(3) 28 g of N_2 (4) 18 g of H_2O
CMC005

6. Number of neutrons present in 1.7 g of ammonia is -
(1) N_A (2) $N_A/10 \times 4$
(3) $(N_A/10) \times 7$ (4) $N_A \times 10 \times 7$
CMC006

7. 5.6 L of oxygen at STP contains -
(1) 6.02×10^{23} atoms
(2) 3.01×10^{23} atoms
(3) 1.505×10^{23} atoms
(4) 0.7525×10^{23} atoms
CMC007

8. Number of oxygen atoms in 8 g of ozone is -
(1) 6.02×10^{23} (2) $\frac{6.02 \times 10^{23}}{2}$
(3) $\frac{6.02 \times 10^{23}}{3}$ (4) $\frac{6.02 \times 10^{23}}{6}$
CMC008

9. Sum of number of protons, electrons and neutrons in 12g of $^{12}_{6}C$ is :-
(1) 1.8 (2) 12.044×10^{23}
(3) 1.084×10^{25} (4) 10.84×10^{23}
CMC009

10. The weight of one atom of Uranium is 238 amu. Its actual weight is .... g.
(1) 1.43×10^{26} (2) 3.94×10^{-22}
(3) 6.99×10^{-23} (4) 1.53×10^{-22}
CMC010

11. The actual weight of a molecule of water is -
(1) 18 g
(2) 2.99×10^{-23} g
(3) both (1) & (2) are correct
(4) 1.66×10^{-24} g
CMC011

12. What is the mass of a molecule of CH_4 :–
(1) 16 g (2) 26.6×10^{22} g
(3) 2.66×10^{-23} g (4) 16 N_A g
CMC012

13. Which of the following has the highest mass ?
(1) 1 g atom of C
(2) 1/2 mole of CH_4
(3) 10 mL of water
(4) 3.011×10^{23} atoms of oxygen
CMC013

14. Which of the following contains the least number of molecules ?
(1) 4.4 g CO_2 (2) 3.4 g NH_3
(3) 1.6 g CH_4 (4) 3.2 g SO_2
CMC014

15. The number of molecule in 4.25 g of NH_3 is -
(1) 1.505×10^{23} (2) 3.01×10^{23}
(3) 6.02×10^{23} (4) None of these
CMC015

16. Elements A and B form two compounds B_2A_3 and B_2A. 0.05 moles of B_2A_3 weight 9.0 g and 0.10 mole of B_2A weight 10 g. Calculate the atomic weight of A and B :-
(1) 20 and 30 (2) 30 and 40
(3) 40 and 30 (4) 30 and 20
CMC016

17. 5.6 L of oxygen at NTP is equivalent to –
(1) 1 mol (2) 1/2 mol
(3) 1/4 mol (4) 1/8 mol
CMC017

18. 4.4 g of an unknown gas occupies 2.24 L of volume at STP. The gas may be :-
(1) N_2O (2) CO
(3) CO_2 (4) 1 & 3 both
CMC018

19. Which contains least number of molecules :–
(1) 1 g CO_2 (2) 1 g N_2
(3) 1 g O_2 (4) 1 g H_2
CMC019

20. If V mL of the vapours of substance at NTP weight W g. Then molecular weight of substance is:-
(1) (W/V) × 22400 (2) $\frac{V}{W} \times 22.4$
(3) (W - V) × 22400 (4) $\frac{W \times 1}{V \times 22400}$
CMC020

21. If 3.01×10^{20} molecules are removed from 98 mg of H_2SO_4, then the number of moles of H_2SO_4 left are :–
(1) 0.1×10^{-3} (2) 0.5×10^{-3}
(3) 1.66×10^{-3} (4) 9.95×10^{-2}
CMC021

22. A gas is found to have the formula $(CO)_x$. It's VD is 70. The value of x must be:-
(1) 7 (2) 4 (3) 5 (4) 6
CMC022

23. Vapour density of gas is 11.2. Volume occupied by 2.4 g of this at STP will be -
(1) 11.2 L (2) 2.24 L (3) 22.4 L (4) 2.4 L
CMC023

24. The volume of a gas in discharge tube is 1.12×10^{-7} mL at STP. Then the number of molecule of gas in the tube is -
(1) 3.01×10^{4} (2) 3.01×10^{15}
(3) 3.01×10^{12} (4) 3.01×10^{16}
CMC024

25. The number of electron in 3.1 mg NO_3^- is ($N_A = 6 \times 10^{23}$)
(1) 32 (2) 1.6×10^{-3}
(3) 9.6×10^{20} (4) 9.6×10^{23}
CMC025

26. Given that one mole of N_2 at NTP occupies 22.4 L the density of N_2 is -
(1) 1.25 g L^{-1} (2) 0.80 g L^{-1}
(3) 2.5 g L^{-1} (4) 1.60 g L^{-1}
CMC026

27. The number of carbon atoms present in a signature, if a signature written by carbon pencil, weighing 1.2×10^{-3} g is
(1) 12.04×10^{20} (2) 6.02×10^{19}
(3) 3.01×10^{19} (4) 6.02×10^{20}
CMC027

QUESTIONS BASED ON PERCENTAGE, EMPIRICAL FORMULA & MOLECULAR FORMULA

28. A compound of X and Y has equal mass of them. If their atomic weights are 30 and 20 respectively. Molecular formula of the compound is :-
(1) X_2Y_2 (2) X_3Y_3 (3) X_2Y_3 (4) X_3Y_2
CMC028

29. An oxide of sulphur contains 50% of sulphur in it. Its empirical formula is -
(1) SO_2 (2) SO_3 (3) SO (4) S_2O
CMC029

30. A hydrocarbon contains 80% of carbon, then the hydrocarbon is -
(1) CH_4 (2) C_2H_4 (3) C_2H_6 (4) C_2H_2
CMC030

31. Empirical formula of glucose is -
(1) $C_6H_{12}O_6$ (2) $C_3H_6O_3$ (3) $C_2H_4O_2$ (4) CH_2O
CMC031

32. An oxide of metal M has 40% by mass of oxygen. Metal M has atomic mass of 24. The empirical formula of the oxide is :-
(1) M_2O (2) M_2O_3 (3) MO (4) M_3O_4
CMC032

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33. A compound contains 38.8% C, 16.0% H and 45.2% N. The formula of the compound would be

(1) CH_3NH_2 (2) CH_3CN

(3) C_2H_5CN (4) $CH_2(NH)_2$

CMC033

34. The simplest formula of a compound containing 50%(w/w) of element X(at wt. = 10) and 50% of element Y(at wt. = 20) is:-

(1) XY (2) X_2Y (3) XY_2 (4) X_3Y

CMC034

35. Which of the following compound has same empirical formula as that of glucose:-

(1) CH_3CHO (2) CH_3COOH

(3) CH_3OH (4) C_2H_6

CMC035

36. 2.2 g of a compound of phosphorous and sulphur has 1.24 g of 'P' in it. Its empirical formula is -

(1) P_2S_3 (2) P_3S_2 (3) P_3S_4 (4) P_4S_3

CMC036

37. On analysis, a certain compound was found to contain iodine and oxygen in the mass ratio of 254:80. The formula of the compound is :

(At mass I = 127, O = 16)

(1) IO (2) I_2O (3) I_5O_2 (4) I_2O_5

CMC037

38. The number of atoms of Cr and O are 4.8×10^{10} and 9.6×10^{10} respectively. Its empirical formula is –

(1) Cr_2O_3 (2) CrO_2 (3) Cr_2O_4 (4) CrO_5

CMC038

39. Insulin contains 3.4% sulphur by mass. The minimum molecular weight of insulin is :

(1) 941.176 (2) 944 (3) 945.27 (4) None

CMC039

40. Caffine has a molecular weight of 194. It contains 28.9% by mass of nitrogen Number of atoms of nitrogen in one molecule of it is :-

(1) 2 (2) 3 (3) 4 (4) 5

CMC040

QUESTIONS BASED ON STOICHIOMETRY

41. An organic compound contains carbon, hydrogen and oxygen. Its elemental analysis gives 38.71% of C and 9.67% of H. The empirical formula of the compound would be :-

(1) CHO (2) CH_4O (3) CH_3O (4) CH_2O

CMC041

42. The amount of water (g) produced by combustion of 286 g of propane is

(1) 168 g (2) 200 g (3) 468 g (4) 693 g

CMC042

43. In a gaseous reaction of the type

$aA + bB \longrightarrow cC + dD$,

which statement is wrong ?

(1) A litre of A combines with b litre of B to give C and D

(2) A mole of A combines with b moles of B to give C and D

(3) A g of A combines with b g of B to give C and D

(4) A molecules of A combines with b molecules of B to give C and D

CMC043

44. Assuming that petrol is octane (C_8H_{18}) and has density 0.8 g mL^{-1}. 1.425 L of petrol on complete combustion will consume.

(1) 50 mole of O_2 (2) 100 mole of O_2

(3) 125 mole of O_2 (4) 200 mole of O_2

CMC044

45. In a given reaction, 9 g of Al will react with

$2Al + \frac{3}{2} O_2 \rightarrow Al_2O_3$

(1) 6 g O_2 (2) 8 g O_2 (3) 9 g O_2 (4) 4 g O_2

CMC045

46. The equation :

$2Al_{(s)} + \frac{3}{2} O_2(g) \rightarrow Al_2O_{3(s)}$ shows that :-

(1) 2 mol of Al reacts with $\frac{3}{2}$ mol of O_2 to produce $\frac{7}{2}$ mol of Al_2O_3

(2) 2 g of Al reacts with $\frac{3}{2}$ g of O_2 to produce one mol of Al_2O_3

(3) 2 g of Al reacts with $\frac{3}{2}$ L of O_2 to produce 1 mol of Al_2O_3

(4) 2 mol of Al reacts with $\frac{3}{2}$ mol of O_2 to produce 1 mol of Al_2O_3

CMC046

47. 1 L of CO_2 is passed over hot coke. When the volume of reaction mixture becomes 1.4 L, the composition of reaction mixture is–

(1) 0.6 L CO

(2) 0.8 L CO_2

(3) 0.6 L CO_2 and 0.8 L CO

(4) None

CMC047

48. 26 cc of CO_2 are passed over red hot coke. The volume of CO evolved is :–

(1) 15 cc (2) 10 cc (3) 32 cc (4) 52 cc

CMC048

49. If 1/2 mol of oxygen combine with Aluminium to form Al_2O_3 then weight of Aluminium metal used in the reaction is (Al=27) –

(1) 27 g (2) 18 g (3) 54 g (4) 40.5 g

CMC049

50. If 0.5 mol of $BaCl_2$ is mixed with 0.2 mol of Na_3PO_4, the maximum number of moles of $Ba_3(PO_4)_2$ that can be formed is –

$3BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$

(1) 0.7 (2) 0.5 (3) 0.3 (4) 0.1

CMC050

51. If 8 mL of uncombined O_2 remain after exploding O_2 with 4 mL of hydrogen, the number of mL of O_2 originally were -

(1) 12 (2) 2 (3) 10 (4) 4

CMC051

52. 4 g of hydrogen are ignited with 4 g of oxygen. The weight of water formed is -

(1) 0.5 g (2) 3.5 g (3) 4.5 g (4) 2.5 g

CMC052

53. For the reaction $A + 2B \longrightarrow C$,

5 mol of A and 8 mol of B will produce

(1) 5 mole of C (2) 4 mole of C

(3) 8 mole of C (4) 13 mole of C

CMC053

54. If 1.6 g of SO_2 and 1.5×10^{22} molecules of H_2S are mixed and allowed to remain in contact in a closed vessel until the reaction

$2H_2S + SO_2 \longrightarrow 3S + 2H_2O$,

proceeds to completion. Which of the following statement is true ?

(1) Only 'S' and 'H_2O' remain in the reaction vessel.

(2) 'H_2S' will remain in excess

(3) 'SO_2' will remain in excess

(4) None

CMC054

55. 12 L of H_2 and 11.2 L of Cl_2 are mixed and exploded. The composition by volume of mixture is–

(1) 24 L of HCl (g)

(2) 0.8 L Cl_2 and 20.8 L HCl (g)

(3) 0.8 L H_2 and 22.4 L HCl (g)

(4) 22.4 L HCl (g)

CMC055

56. 10 mL of gaseous hydrocarbon on combustion give 40 mL of $CO_2(g)$ and 50 mL of H_2O (vap.). The hydrocarbon is -

(1) C_4H_5 (2) C_8H_{10} (3) C_4H_8 (4) C_4H_{10}

CMC056

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QUESTIONS BASED ON LAWS OF CHEMICAL COMBINATION

57. The law of multiple proportion was proposed by :

(1) Lavoisier (2) Dalton

(3) Proust (4) Gaylussac

CMC101

58. Which one of the following pairs of compound illustrate the law of multiple proportions ?

(1) H_2O, Na_2O

(2) MgO, Na_2O

(3) Na_2O, BaO

(4) $SnCl_2$, $SnCl_4$

CMC102

59. In the reaction

$N_2 + 3H_2 \longrightarrow 2\,NH_3$, ratio by volume of N_2, H_2 and NH_3 is 1 : 3 : 2. This illustrates law of -

(1) Definite proportion

(2) Multiple proportion

(3) Law of conservation of mass

(4) Gaseous volumes

CMC103

60. Different proportions of oxygen in the various oxides of nitrogen prove the law of -

(1) Equivalent proportion

(2) Multiple proportion

(3) Constant proportion

(4) Conservation of matter

CMC104

61. The law of conservation of mass holds good for all of the following except -

(1) All chemical reactions

(2) Nuclear reactions

(3) Endothermic reactions

(4) Exothermic reactions

CMC106

62. Number of molecules in 100 mL of each of O_2, NH_3 and CO_2 at STP are –

(1) In the order $CO_2 < O_2 < NH_3$

(2) In the order $NH_3 < O_2 < CO_2$

(3) The same

(4) $NH_3 = CO_2 < O_2$

CMC107

63. The empirical formula of an organic compound containing carbon and hydrogen is CH_2. The mass of one litre of this organic gas is exactly equal to that of one litre of N_2 at same temperature and pressure. Therefore, the molecular formula of the organic gas is –

(1) C_2H_4 (2) C_3H_6 (3) C_6H_{12} (4) C_4H_8

CMC108

64. Four one litre flasks are separately filled with the gases hydrogen, helium, oxygen and ozone at same room temperature and pressure. The ratio of total number of atoms of these gases present in the different flasks would be -

(1) 1 : 1 : 1 : 1 (2) 1 : 2 : 2 : 3

(3) 2 : 1 : 2 : 3 (4) 2 : 1 : 3 : 2

CMC109

65. A container of volume V, contains 0.28 g of N_2 gas. If same volume of an unknown gas under similar condition of temperature and pressure weighs 0.44 g, the molecular mass of the gas is

(1) 22 (2) 44 (3) 66 (4) 88

CMC110

66. When 100 g of ethylene polymerizes to polyethylene according to equation $nCH_2 = CH_2 \rightarrow -(-CH_2 - CH_2 -)_n-$. The weight of polyethylene produced will be:-

(1) $\frac{n}{2}$ g (2) 100 g

(3) $\frac{100}{n}$ g (4) 100 ng

CMC111

67. A chemical equation is balanced according to the law of –

(1) Multiple proportion

(2) Constant composition

(3) Gaseous volume

(4) Conservation of mass

CMC112

68. Two flasks A & B of equal capacity of volume contain NH_3 and SO_2 gas respectively under similar conditions. Which flask has more number of moles:-

(1) A

(2) B

(3) Both have same moles

(4) None

CMC113

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	4	3	1	3	2	2	3	2	2	3	1	4	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	4	1	1	2	3	4	3	3	1	2	3	1	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	3	1	2	2	4	4	2	1	3	3	3	3	3	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	3	4	2	4	3	3	2	3	3	4	2	4	4	2
Question	61	62	63	64	65	66	67	68							
Answer	2	3	1	3	2	2	4	3							

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Equal masses of H_2,O_2 and methane have been taken in a container of volume V at temperature 27°C at identical conditions. The ratio of the volumes of gases $H_2 : O_2 : CH_4$ would be :
(1) 8 : 16 : 1 (2) 16 : 8 : 1
(3) 16 : 1 : 2 (4) 8 : 1 : 2
CMC118

2. When 22.4 L of $H_2(g)$ is mixed with 11.2 L of $Cl_2(g)$ at S.T.P., the moles of HCl(g) formed is equal to:-
(1) 1 mol of HCl (g) (2) 2 mol of HCl (g)
(3) 0.5 mol of HCl (g) (4) 1.5 mol of HCl (g)
CMC119

3. 1.0 g of magnesium is burnt with 0.56 g O_2 in a closed vessel. Which reactant is left in excess and by how much ?
(At. wt. Mg = 24 ; O = 16)
(1) Mg, 0.16 g (2) O_2, 0.16 g
(3) Mg, 0.44 g (4) O_2, 0.28 g
CMC120

AIPMT 2015

4. A mixture of gases contains H_2 and O_2 gases in the ratio of 1 : 4 (w/w). What is the molar ratio of the two gases in the mixture ?
(1) 4 : 1 (2) 16 : 1 (3) 2 : 1 (4) 1 : 4
CMC121

Re-AIPMT 2015

5. The number of water molecules is maximum in-
(1) 18 g of water
(2) 18 mol of water
(3) 18 molecules of water
(4) 1.8 g of water
CMC122

6. If avogadro number N_A, is changed from 6.022×10^{23} mol^{-1} to 6.022×10^{20} mol^{-1}, this would change :
(1) The ratio of chemical species to each other in a balanced equation
(2) The ratio of elements to each other in a compound
(3) The definition of mass in units of grams
(4) The mass of one mole of carbon
CMC123

7. 20.0 g of a magnesium carbonate sample decomposes on heating to give carbon dioxide and 8.0 g magnesium oxide. What will be the percentage purity of magnesium carbonate in the sample ? (Atomic weight of Mg = 24)
(1) 60 (2) 84 (3) 75 (4) 96
CMC124

NEET-II 2016

8. Suppose the elements X and Y combine to form two compounds XY_2 and X_3Y_2. When 0.1 mole of XY_2 weighs 10 g and 0.05 mole of X_3Y_2 weighs 9 g, the atomic weights of X and Y are
(1) 20, 30 (2) 30, 20 (3) 40, 30 (4) 60, 40
CMC125

NEET(UG) 2018

9. A mixture of 2.3 g formic acid and 4.5 g oxalic acid is treated with conc. H_2SO_4. The evolved gaseous mixture is passed through KOH pellets. Weight (in g) of the remaining product at STP will be

$$HCOOH(\ell) \xrightarrow{H_2SO_4} H_2O(\ell) + CO(g)$$

$$H_2C_2O_4(\ell) \xrightarrow{H_2SO_4} H_2O(\ell) + CO(g) + CO_2(g)$$

(1) 1.4 (2) 3.0 (3) 2.8 (4) 4.4
CMC126

10. In which case is the number of molecules of water maximum?
(1) 18 mL of water
(2) 0.18 g of water
(3) 0.00224L of water vapours at 1 atm and 273K
(4) 10^{-3} mol of water
CMC127

NEET (UG) 2019

11. The number of moles of hydrogen molecules required to produce 20 moles of ammonia through Haber's process is :-
(1) 10 (2) 20 (3) 30 (4) 40
CMC128

NEET (UG) (Odisha) 2019

12. The volume occupied by 1.8 g of water vapour at 374 °C and 1 bar pressure will be :-
[Use R = 0.083 bar L $K^{-1}mol^{-1}$]
(1) 96.66 L (2) 55.87 L (3) 3.10 L (4) 5.37 L
CMC129

NEET (UG) 2020

13. Which one of the following has maximum number of atoms?
(1) 1g of Li(s) [Atomic mass of Li = 7]
(2) 1g of Ag(s) [Atomic mass of Ag = 108]
(3) 1g of Mg(s) [Atomic mass of Mg = 24]
(4) 1g of O_2(g) [Atomic mass of O = 16]

CMC130

NEET (UG) 2020 (Covid-19)

14. One mole of carbon atom weighs 12 g, the number of atoms in it is equal to,
(Mass of carbon – 12 is 1.9926×10^{-23} g)
(1) 1.2×10^{23} (2) 6.022×10^{22}
(3) 12×10^{22} (4) 6.022×10^{23}

CMC131

NEET (UG) 2021

15. An organic compound contains 78% (by wt.) carbon and remaining percentage of hydrogen. The right option for the empirical formula of this compound is [Atomic wt. of C is 12, H is 1]
(1) CH (2) CH_2 (3) CH_3 (4) CH_4

CMC132

NEET (UG) 2022

16. What mass of 95% pure $CaCO_3$ will be required to neutralise 50 mL of 0.5 M HCl solution according to the following reaction ?
$CaCO_{3(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + CO_{2(g)} + H_2O_{(l)}$
[Calculate upto second place of decimal point]
(1) 1.32 g (2) 3.65 g (3) 9.50 g (4) 1.25 g

CMC133

17. The density of the solution is 2.15 g mL^{-1} , then mass of 2.5 mL solution in correct significant figures is
(1) 5375×10^{-3} g (2) 5.4 g
(3) 5.38 g (4) 53.75 g

CMC134

18. What fraction of Fe exists as Fe(III) in $Fe_{0.96}O$?
(Consider $Fe_{0.96}$ to be made up of Fe(II) and Fe(III) only)
(1) $\frac{1}{12}$ (2) 0.08 (3) $\frac{1}{16}$ (4) $\frac{1}{20}$

CMC135

NEET (UG) 2023

19. The **right** option for the mass of CO_2 produced by heating 20 g of 20% pure limestone is
(Atomic mass of Ca = 40)
[$CaCO_3 \xrightarrow{1200\,K} CaO + CO_2$]
(1) 1.76 g (2) 2.64 g (3) 1.32 g (4) 1.12 g

CMC151

NEET (UG) 2024

20. The highest number of helium atoms is in :
(1) 4 mol of helium
(2) 4 u of helium
(3) 4 g of helium
(4) 2.271098 L of helium at STP

CMC169

21. 1 gram of sodium hydroxide was treated with 25 mL of 0.75 M HCL solution, the mass of sodium hydroxide left unreacted is equal to
(1) 750mg (2) 250mg (3) Zero mg (4)200mg

CMC170

22. A compound X contains 32% of A, 20% of B and remaining percentage of C. Then, the empirical formula of X is :
(Given atomic masses of A = 64; B = 40; C = 32 u)
(1) A_2BC_2 (2) ABC_3 (3) AB_2C_2 (4) ABC_4

CMC171

NEET (UG) 2024 (Re-examination)

23. 1.0 g of H_2 has same number of molecules as in
(1) 14 g of N_2 (2) 18 g of H_2O
(3) 16 g of CO (4) 28 g of N_2

CMC172

24. On complete combustion, 0.3 g of an organic compound gave 0.2 g of CO_2 and 0.1 g of H_2O. The percentage composition of carbon and hydrogen in the compound, respectively is :
(1) 4.07 % and 15.02% (2) 18.18% and 3.70%
(3) 15.02% and 4.07% (4) 3.70% and 18.18%

CMC173

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	1	1	2	4	2	3	3	1	3	4	1	4	3
Question	16	17	18	19	20	21	22	23	24						
Answer	1	2	1	1	1	2	2	1	2						

Exercise – III (Analytical Questions) Master Your Understanding

1. Given below are two statements:

Statement-I : A molecule of butane, C_4H_{10} has a mass of 58 amu.

Statement-II : One mole of butane contains 6.022×10^{23} molecules and has a mass of 58g.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement-I is correct while II is incorrect.

(2) Statement-I and II both are correct.

(3) Both Statement-I and II are incorrect.

(4) Statement-I is incorrect and II is correct.

CMC137

2. Given below are two statements:

Statement-I : Both 12g of carbon and 27g of Aluminium will have 6.02×10^{23} atoms.

Statement-II : Gram atomic mass of an element contains Avogadro's no. of atoms.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement-I and II both are correct.

(2) Statement-I is correct while II is incorrect.

(3) Both Statement-I and II are incorrect.

(4) Statement-I is incorrect and II is correct.

CMC138

3. Given below are two statements:

Statement-I : $2A + 3B \rightarrow C$

$\frac{4}{3}$ moles of 'C' are produced when 3 moles of 'A' and 4 moles of 'B' are added.

Statement-II : 'A' is limiting reagent for the given data.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement-I and II both are correct.

(2) Statement-I is correct while II is incorrect.

(3) Both Statement-I and II are incorrect.

(4) Statement-I is incorrect and II is correct.

CMC139

4. Given below are two statements:

Statement-I : In the reaction

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$

One mole of $CH_4(g)$ reacts with two moles of $O_2(g)$ to give one mole of $CO_2(g)$ and two moles of $H_2O(g)$.

Statement-II : 1g of $CH_4(g)$ reacts with 2g of $O_2(g)$ to give 1g of $CO_2(g)$ and 2g of $H_2O(g)$.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement-I and II both are correct.

(2) Statement-I is correct while II is incorrect.

(3) Both Statement-I and II are incorrect.

(4) Statement-I is incorrect and II is correct.

CMC140

5. Given below are two statements:

Statement-I : For a reaction between 10L $H_2(g)$ and 10L $O_2(g)$, only 10L of liquid water is obtained, under the same conditions of P & T.

Statement-II : When gases combine or are produced in a chemical reaction they do so in a simple ratio by volume as their stoichiometric coefficients, provided that all gases are present under the same conditions of pressure & temperature.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement-I and II both are correct.

(2) Statement-I is correct while II is incorrect.

(3) Both Statement-I and II are incorrect.

(4) Statement-I is incorrect and II is correct.

CMC141

6. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.

Assertion (A) : Hydrogen combines with oxygen to form two compounds namely, water and hydrogen peroxide.

Reason (R) : The above example is in accordance with law of multiple proportion.

In the light of the assertion reason statements, choose the **most appropriate** answer from the options given below :

(1) If (A) & (R) both are correct and (R) is the correct explanation of (A).

(2) If (A) & (R) both are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is incorrect.

(4) (A) & (R) both are incorrect.

CMC142

7. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion (A) : The average mass of one C atom is 12.011 amu, which is not the actual mass of one C atom.

Reason (R) : Three isotopes ^{12}C, ^{13}C and ^{14}C, of C are found in nature.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) If (A) & (R) both are correct and (R) is the correct explanation of (A).

(2) If (A) & (R) both are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is incorrect.

(4) (A) & (R) both are incorrect.

CMC143

8. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.

Assertion (A) : Equal volume of all gases contains equal no. of atoms, under same conditions of P and T.

Reason (R) : Atom is the fundamental entity which takes part in chemical reactions.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) If (A) & (R) both are correct and (R) is the correct explanation of (A).

(2) If (A) & (R) both are correct but (R) is not the correct explanation of (A).

(3) (A) is incorrect and (R) is correct.

(4) (A) & (R) both are incorrect.

CMC144

9. Select the correct option for $(NH_4)_3PO_4$.

(A) Ratio of number of oxygen atom to number of hydrogen atom is 1 : 3.

(B) Ratio of no. of cation to no. of anion is 3 : 1.

(C) Ratio of no. of g-atom of nitrogen to g-atom of oxygen is 3 : 2.

(D) Total no. of atoms in one mole of $(NH_4)_3PO_4$ is 20.

(1) A, B both are correct.

(2) A, B & D are correct.

(3) C & D both are correct.

(4) B & C both are correct.

CMC145

10. Select the incorrect statement -

(A) 1g molecule of any substance contains equal no. of atoms.

(B) If 2g of $H_2(g)$ completely reacts with 71g of $Cl_2(g)$ then 36.5 g of HCl is produced.

(C) Reactant with minimum no. of moles acts as limiting reagent.

(D) During a chemical reaction, no. of total moles may increase, decrease or remain constant

(1) A, B, C (2) B, C, D

(3) A, B, C, D (4) A, C, D

CMC147

11. Match the column.

	Column I		Column II
(A)	Amount in gram in 20 mol H_2SO_4	(P)	20 N_A
(B)	No. of mol of H_2O in 540 g	(Q)	6.02×10^{13}
(C)	No. of atoms in 10^{-10} g H_2	(R)	30
(D)	No. of amu in 20 g	(S)	1960

(1) (A) → (S), (B) → (R), (C) → (Q), (D) → (P)

(2) (A) → (P), (B) → (R), (C) → (S), (D) → (Q)

(3) (A) → (R), (B) → (S), (C) → (P), (D) → (Q)

(4) (A) → (P), (B) → (R), (C) → (Q), (D) → (S)

CMC148

12. The sample containing same no. of 'Na' atoms as there are 'Na' atoms in 5.3 g of Na_2CO_3 is -

(A) 4g of NaOH

(B) 5.85 g of NaCl

(C) 0.25 mol of Na_2SO_4

(D) 5.6 g of Na_3PO_4

(1) A & C both are correct.

(2) B & D both are correct.

(3) A & B are correct.

(4) C & D both are correct.

CMC149

13. Match the column.

	Column I		Column II
(A)	8g-atoms of Hydrogen	(P)	2 moles C_3H_4
(B)	6g-atoms of Carbon	(Q)	3 moles CH_3COOH
(C)	6g-atoms of oxygen	(R)	4 moles N_2H_2
(D)	4g-molecules of Nitrogen	(S)	2 moles N_2O_3

(1) (A) → (P,Q) (B) → (R,S) (C) → (P,S) (D) → (P)

(2) (A) → (P,R) (B) → (P,Q) (C) → (Q,S) (D) → (R)

(3) (A) → (P,R) (B) → (P,R) (C) → (R,S) (D) → (S)

(4) (A) → (R,S) (B) → (P,S) (C) → (Q,R) (D) → (Q)

CMC150

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer	2	1	2	2	4	1	1	3	1	1	1	3	2

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Some Basic Concepts of Chemistry-Mole Concept

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (4)

16 g oxygen (O_2)

$n_{O_2} = \frac{16}{32} = \frac{1}{2}$

Number of O-atoms

$= n_{O_2} \times N_A \times$ atomicity

$= \frac{1}{2} \times 6.022 \times 10^{23} \times 2$

$= 6.022 \times 10^{23}$

2. Ans. (4)

$n_{NH_3} = \frac{4.25}{17} = \frac{1}{4}$

Number of atoms

$= n_{NH_3} \times N_A \times$ atomicity

$= \frac{1}{4} \times N_A \times 4$

$= N_A = 6 \times 10^{23}$

3. Ans. (4)

Mass	Mole	Number of atoms [n × N_A × atomicity]
1 g O	1/16	$\frac{1}{16} \times N_A \times 1 = \frac{N_A}{16}$
1 g O_2	1/32	$\frac{1}{32} \times N_A \times 2 = \frac{N_A}{16}$
1 g O_3	1/48	$\frac{1}{48} \times N_A \times 3 = \frac{N_A}{16}$

All have the same number of atoms.

4. Ans. (3)

Number of atom in 0.5g atom of nitrogen

$= 0.5 \times N_A \times 2 = N_A$

Number of atom in 8g oxygen

$= \frac{8}{16} \times N_A \times 2 = N_A$

5. Ans. (1)

Mass	Moles	Number of atoms
4g H_2	$\frac{4}{2} = 2$	$2 \times N_A \times 2 = 4N_A$
16g O_2	$\frac{16}{32} = \frac{1}{2}$	$\frac{1}{2} \times N_A \times 2 = N_A$
28g N_2	$\frac{28}{28} = 1$	$1 \times N_A \times 2 = 2N_A$
18g H_2O	$\frac{18}{18} = 1$	$1 \times N_A \times 3 = 3N_A$

6. Ans. (3)

Number of neutrons

$= n \times N_A \times$ (number of neutrons)

$= \frac{1.7}{17} \times N_A \times 7$

$= \frac{N_A}{10} \times 7$

7. Ans. (2)

Number of atoms of oxygen

$= n \times N_A \times$ number of atoms

$= \frac{5.6}{22.4} \times 6.022 \times 10^{23} \times 2$

$= 3.01 \times 10^{23}$ atoms.

8. Ans. (2)

Number of atoms in [O_3]

$= n \times N_A \times$ number of atoms

$= \frac{8}{48} \times N_A \times 3$

$= \frac{N_A}{2} = \frac{6.022 \times 10^{23}}{2}$

9. Ans. (3)

Sum of protons, electrons and neutrons in 12g of $^{12}_{6}C$ $= n \times N_A \times$ [sum of e, p & n]

$= \frac{12}{12} \times N_A \times 18$

$= 18 \times 6.022 \times 10^{23}$

$= 1.08 \times 10^{25}$

10. Ans. (2)

1 "U" atom = 238 amu

1 amu = 1.66×10^{-24}g

1 "U" atom = $238 \times (1.66 \times 10^{-24}\text{g})$

$\simeq 3.94 \times 10^{-22}$g

11. Ans. (2)

1 H_2O molecule = 18 amu

Actual mass of 1 H_2O molecule

$= 18 \times (1.66 \times 10^{-24}\text{g})$

$\simeq 2.99 \times 10^{-23}$

12. Ans. (3)

1 molecule CH_4 = 16 amu

1 amu = $(1.66 \times 10^{-24}\text{g})$

mass of 1 molecule of $CH_4 = 16 \times (1.66 \times 10^{-24}\text{g})$

$= 2.66 \times 10^{-23}$g

13. Ans. (1)

1g atom of C = 1 mole of C

= 12g

14. Ans. (4)

Mass	Moles	Number of atoms
4.4g CO_2	$\frac{4.4}{44} = 0.1$	0.1 N_A
3.4g NH_3	$\frac{3.4}{17} = 0.2$	0.2 N_A
1.6g CH_4	$\frac{1.6}{16} = 0.1$	0.1 N_A
3.2g SO_2	$\frac{3.2}{64} = 0.05$	0.05 N_A

15. Ans. (1)

$n_{NH_3} = \frac{4.25}{17} = \frac{1}{4}$

Number of NH_3 molecules

$= \frac{1}{4} \times 6.022 \times 10^{23}$

$= 1.505 \times 10^{23}$

16. Ans. (3)

Let atomic wt. of A = a

Atomic wt. of B = b

Molar mass of

$B_2A_3 = (2 \times b) + (3 \times a)$

= mass of 1 mole substance

0.05 mol $B_2A_3 \rightarrow 9$g

1 mol of $B_2A_3 = \frac{9}{0.05} = 180$g

$2b + 3a = 180$ ...(1)

molar mass of $B_2A = (2b + a)$

0.1 mol of $B_2A \rightarrow 10$g

1 mol of $B_2A \rightarrow \frac{10}{0.1} = 100$

$\Rightarrow$ $2b + a = 100$ ...(2)

On solving $\boxed{a = 40, b = 30}$

17. Ans. (3)

$n_{O_2} = \frac{5.6}{22.4} = \frac{1}{4}$

18. Ans. (4)

$n_{gas} = \frac{2.24}{22.4} = 0.1$

Molar mass = $\frac{w_{gas}}{n_{gas}}$

$= \frac{4.4}{0.1} = 44$

CO_2 & N_2O

19. Ans. (1)

Mass	Moles	Number of atoms
1g CO_2	$\frac{1}{44}$	$\frac{1}{44} N_A$
1g N_2	$\frac{1}{28}$	$\frac{1}{28} N_A$
1g O_2	$\frac{1}{32}$	$\frac{1}{32} N_A$
1g H_2	$\frac{1}{2}$	$\frac{1}{2} N_A$

20. Ans. (1)

mass of gas = $w_{(g)}$

Vol. of gas = V mL (at STP)

moles of gas = $\frac{\text{mass}}{\text{mol.wt.}} = \frac{w}{\text{mol.wt.}}$

moles of gas = $\frac{V_{STP}(ml)}{22400}$

$\frac{w}{\text{mol.wt.}} = \frac{V_{STP}(ml)}{22400}$

Mol. wt. = $\frac{w}{V_{(ml)}} \times 22400$

21. Ans. (2)

Initial mass of H_2SO_4 = 98mg = 98 × 10^{-3}g

Initial moles of H_2SO_4 = $\frac{98\times10^{-3}}{98}=10^{-3}$mol

Number of molecules removed = 3.01 × 10^{20}

Number of moles removed

= $\frac{3.01\times10^{20}}{6.02\times10^{23}}=0.5\times10^{-3}$mol

Moles of H_2SO_4 left = 10^{-3} – [0.5 × 10^{-3}]

= 0.5 × 10^{-3}

22. Ans. (3)

Formula of gas = $[CO]_x$

Mol. wt. of gas = $[12 + 16]_x$

= 28x

28x = 2 × V.D.

28x = 2 × 70

28x = 140

x = $\frac{140}{28}=5$

23. Ans. (4)

Mol. wt. = 2 × 11.2 = 22.4g

moles of gas = $\frac{2.4}{22.4}$

Vol. of gas (at STP) = moles × 22.42

= $\frac{2.4}{22.4}\times22.4=2.42$

24. Ans. (3)

$n_{gas}=\frac{V_{STP}(ml)}{22400}$

= $\frac{1.12\times10^{-7}}{22400}=\frac{1}{2}\times10^{-11}$

molecules of gas = moles × N_A

= $\frac{1}{2}\times10^{-11}\times6.02\times10^{23}$

= 3.01 × 10^{12}

25. Ans. (3)

Number of electrons in

$NO_3^-=n_{NO_3^-}\times N_A$ (number of electrons)

= $\frac{3.1\times10^{-3}}{62}\times6.022\times10^{23}\times32$

= 9.6 × 10^{20}

26. Ans. (1)

Vol. of 1 mole N_2 [at N.T.P.] = 22.4 L

mass of 1 mole N_2 = 28g

Density of N_2 = $\frac{\text{mass}}{\text{volume}}=\frac{28g}{22.4L}=1.25g\ L^{-1}$

27. Ans. (2)

number of carbon atoms

= $n_c \times N_A \times$ atomicity

= $\frac{1.2\times10^{-3}}{12}\times6.022\times10^{23}\times1$

= 6.022 × 10^{19}

28. Ans. (3)

	X	Y
mass	w	w
moles	$\frac{w}{30}$	$\frac{w}{20}$
simple ratio × 60	$\frac{1}{30}\times60$	$\frac{w}{20}\times60$
	2	3

Molecular formula = X_2Y_3

29. Ans. (1)

	mass	moles	simples ratio
S	50	$\frac{50}{32}=1.5$	$\frac{1.5}{1.5}=1$
O	50	$\frac{50}{16}=3.125$	$\frac{3.125}{1.5}\simeq2$

Empirical formula = SO_2

30. Ans. (3)

	Mass%	atomic ratio	simples ratio
C	80	$\frac{80}{12}=6.66$	$\frac{6.66}{6.66}=1$
H	20	$\frac{20}{1}=20$	$\frac{20}{6.66}\simeq3$

C_1 H_3

1 : 3

C_2 H_6

31. Ans. (4)

Molecular formula of glucose = $C_6H_{12}O_6$

C H O

6 : 12 : 6

1 : 2 : 1

E. F. of glucose = CH_2O

32. Ans. (3)

	Mass%	atomic ratio	simplest ratio
M	60	$\frac{60}{24}=2.5$	$\frac{2.5}{2.5}=1$
O	40	$\frac{40}{16}=2.5$	$\frac{2.5}{2.5}=1$

$\boxed{\text{E. F} = \text{MO}}$

33. Ans. (1)

	Mass%	atomic ratio	simples ratio
C	38.8	$\frac{38.8}{12}=3.23$	$\frac{3.23}{3.23}=1$
H	16	$\frac{16}{1}=16$	$\frac{16}{3.23}\simeq 5$
N	45.2	$\frac{45.2}{14}\simeq 3.23$	$\frac{3.23}{3.23}=1$

$\boxed{\text{E.F.} = CH_5N \text{ or } CH_3NH_3}$

34. Ans. (2)

	Mass%	atomic ratio	simplest ratio
X	50	$\frac{50}{10}=5$	$\frac{5}{2.5}=2$
Y	50	$\frac{50}{20}=2.5$	$\frac{2.5}{2.5}=1$

Simplest formula = X_2Y

35. Ans. (2)

E. F. of glucose = $C_6H_{12}O_6$

$\Downarrow \div 6$

E. F. = CH_2O

$CH_3COOH \Rightarrow C_2H_4O_2$

$\Downarrow \div 2$

E. F. CH_2O

36. Ans. (4)

	Mass%	atomic ratio	simple ratio
P	1.24g	$\frac{1.24}{31}=0.04$	4
S	(2.2–1.24) = 0.96	$\frac{0.96}{32}=0.03$	3

$\boxed{\text{E.F.} = P_4S_3}$

37. Ans. (4)

	Mass%	Atomic ratio	simple ratio
I	254	$\frac{254}{127}=2$	2
O	80	$\frac{80}{10}=5$	5

$\boxed{\text{Formula} = I_2O_5}$

38. Ans. (2)

E. f. $\Rightarrow$ obtain from simples ratio of atom

Gives Cr = 4.8×10^{10} , 0 = 9.6×10^{10}

Simplest ratio 1 : 2

Empirical formula = CrO_2

39. Ans. (1)

$$\% \text{ of S} \Rightarrow 3.4 = \frac{1\times 32}{\text{Min.mol.wt.}}\times 100$$

$$\text{Minimum molecular wt.} = \frac{32\times 100}{3.4}\simeq 941.176$$

40. Ans. (3)

$$\% \text{ of N} = \frac{\text{mass of N}\times\text{no.of atoms}}{\text{molecolar wt. of Insulin}}\times 100$$

$$28.9 = \frac{14\times n}{194}\times 100 \Rightarrow n = \frac{28.9\times 194}{14\times 100}\simeq 4$$

41. Ans. (3)

	Mass	atomic ratio	simples ratio
C	38.71	$\frac{38.71}{12}=3.22$	$\frac{3.22}{3.22}=1$
H	9.67	$\frac{9.67}{1}=9.67$	$\frac{9.67}{3.21}=3$
O	51.62	$\frac{51.62}{16}=3.22$	$\frac{3.22}{3.22}=1$

$\boxed{\text{E.f.} = CH_3O}$

42. Ans. (3)

$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

$n = \frac{286}{44} = 6.5$

1 mol $C_3H_8 \rightarrow 4$ mol of H_2O

6.5 mol $C_3H_8 \rightarrow 4 \times 6.5$ mol of H_2O

$n_{H_2O} = 26$

$w_{H_2O} = 26 \times 18 = 468g$

43. Ans. (3)

$aA + bB \rightarrow cC + dD$

mass ratio is not same as coffi. Ratio so statement is wrong statement for this reaction.

44. Ans. (3)

$C_8H_{18} + \frac{25}{2} O_2 \rightarrow 8CO_2 + 9H_2O$

Mass of $C_3H_{18} = 0.8 \times 1.425$

$= 1140g$

$n_{C_3H_8} = \frac{1140}{114} = 10$

1 mol $C_8H_{18} \rightarrow \frac{25}{2}$ mol O_2

10 mol $C_8H_{18} \rightarrow \frac{25}{2} \times$ mol O_2

$n_{O_2} = \frac{250}{2} = 125$

45. Ans. (2)

$2Al + \frac{3}{2} O_2 \rightarrow Al_2O_3$

$n = \frac{9}{27} = \frac{1}{3}$

2 mol Al $\rightarrow \frac{3}{2}$ mol O_2

1 mol Al $\rightarrow \frac{3}{4}$ mol O_2

$\frac{1}{3}$ mol Al $\rightarrow \frac{3}{4} \times \frac{1}{3}$ mol O_2

$n_{O_2} = \frac{1}{4}$

wt. of $O_2 = \frac{1}{4} \times 32 = 8g$

46. Ans. (4)

$2 Al_{(s)} + \frac{3}{2} O_{2(g)} \rightarrow Al_2O_{3(s)}$

2 mol of Al reacts with $\frac{3}{2}$ mol of O_2 to produce 1 mol of Al_2O_3

47. Ans. (3)

	$C_{(s)} + CO_{2(g)} \rightarrow$	$2CO_{(g)}$
t = 0	1	–
t = t	1 – x	2x

Vol. after = 1 – x + 2x = 1.4

reaction $\boxed{x = 0.4}$

V_{CO_2} left = 1 – 0.4 = 0.6 L

V_{CO} prepared = 2 × 0.4 = 0.8 L

48. Ans. (4)

$C + CO_2 \rightarrow 2CO$

26 CC

1 Vol. of $CO_2 \rightarrow$ 2Vol. of CO

26 CC of $CO_2 \rightarrow 2 \times 26$ CC of CO

$\boxed{V_{CO} = 52CC}$

49. Ans. (2)

$4Al + 3O_2 \rightarrow 2Al_2O_3$

3 mol $O_2 \rightarrow$ 4 mol of Al

1 mol $O_2 \rightarrow \frac{4}{3}$ mol of Al

$\frac{1}{2}$ mol $O_2 \rightarrow \frac{4}{3} \times \frac{1}{2}$ mol of Al

$n_{Al} = \frac{4}{6}$

$w_{Al} = \frac{\cancel{4}^{2}}{\cancel{6}_{3}} \times \cancel{27}^{9} = 18g$

50. Ans. (4)

$3 BaCl_2 + 2 Na_3 PO_4 \rightarrow Ba_3(PO_4)_2 + 6 NaCl$

0.5 0.2

L.R. $\frac{0.5}{3} = 0.16$ $\frac{0.2}{2} = 0.1$ L.R.

2 mol $Na_3PO_4 \rightarrow 1$ mol $Ba_3(PO_4)_2$

1 mol $Na_3PO_4 \rightarrow \frac{1}{2}$ mol $Ba_3(PO_4)_2$

0.2 mol $Na_3PO_4 \rightarrow \frac{1}{2} \times 0.2$ mol $Ba_3(PO_4)_2$

$\boxed{n_{Ba_3(PO_4)_2} = 0.1}$

51. Ans. (3)

$2H_2 + O_2 \rightarrow 2H_2O$

2vol. H_2 react with 1 vol. of O_2

4 ml H_2 will react with 2 ml of O_2

Total vol. of $O_2 = VO_{2(reacted)} + VO_{2(unreacted)}$

$VO_{2(total)}$ = 2 ml + 8 ml = 10 ml

52. Ans. (3)

$$2H_2 + O_2 \rightarrow 2H_2O$$

moles $\quad \frac{4}{2}=2 \quad \frac{4}{32}=\frac{1}{8}$

L. R. $\quad \frac{2}{2}=1 \quad \underset{\text{L.R.}}{\frac{\frac{1}{8\times1}=\frac{1}{8}}}$

$$n_{H_2O}=\frac{1}{8}\times2=\frac{1}{4}$$

$$w_{H_2O}=\frac{1}{4}\times18=4.5$$

53. Ans. (2)

$$A + 2B \rightarrow C$$

moles $\quad 5 \quad 8$

L.R. $\quad \frac{5}{1} \quad \frac{8}{2}=4$

$n_C = \quad 4 \times 1 = 4$

54. Ans. (3)

$$2H_2S + SO_2 \rightarrow 3S + 2H_2O$$

Moles $\quad \frac{1.6}{34}=0.04; \quad \frac{1.5\times10^{22}}{6\times10^{23}}=0.025$

L. R. $\quad \underset{\text{L.R.}}{\frac{\frac{0.04}{2}=0.02}{}} \quad \underset{\text{E.R.}}{\frac{\frac{0.025}{1}=0.025}{}}$

SO_2 will remain in excess

55. Ans. (3)

$$H_2 + Cl_2 \rightarrow 2HCl$$

Vol. $\quad 12\text{ L} \quad 11.2\text{ L}$

L.R. $\quad \underset{\text{E.R.}}{\frac{\frac{12}{1}}{}} \quad \underset{\text{L.R.}}{\frac{\frac{11.2}{1}}{}}$

11.2 L of Cl_2 will react with 11.2 L H_2 to product 22.4 L HCl.

Remaining H_2 = 12 – 11.2 = 0.8 L

56. Ans. (4)

$$C_xH_y + \left(x+\frac{y}{4}\right)O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O$$

10ml $\quad$ - $\quad$ 40ml $\quad$ 50ml

From stoichiometry, $x = 4, \frac{y}{2} = 5 \Rightarrow y = 10$

Which means C_4H_{10}

57. Ans. (2)

Dalton

58. Ans. (4)

$SnCl_2$, $SnCl_4$ $\Rightarrow$ show law of multiple proportions.

59. Ans. (4)

Gaseous volumes

60. Ans. (2)

Multiple proportions.

61. Ans. (2)

Nuclear reaction

62. Ans. (3)

Number of molecule $= n \times N_A$

n is same for all O_2, NH_3 & CO_2.

63. Ans. (1)

volume $\propto$ moles

$$n = \frac{\text{molecular mass}}{\text{Empirical mass}}=\frac{28}{14}=2$$

M. F. = 2 × $CH_2 = C_2H_4$

64. Ans. (3)

$$H_2 : He : O_2 : O_3$$

moles $\quad \frac{1}{22.4}:\frac{1}{22.4}:\frac{1}{22.4}:\frac{1}{22.4}$

moleculs $\quad \frac{1}{22.4}N_A:\frac{1}{22.4}N_A:\frac{1}{22.4}N_A:\frac{1}{22.4}N_A$

atoms $\quad \frac{2\,N_A}{22.4}:\frac{N_A}{22.4}:\frac{2\,N_A}{22.4}:\frac{3\,N_A}{22.4}$

65. Ans. (2)

$$\frac{M_{gas}}{M_{N_2}}=\frac{0.44}{0.28}$$

$$M_{gas}=\frac{0.44}{0.28}\times28$$

= 44

66. Ans. (2)

Entire mass of ethylene is converted to polyethylene so according to LOMA 100 g

67. Ans. (4)

Conservation of mass

68. Ans. (3)

	A	B
	NH_3	SO_2
Moles	$\frac{x}{22.4}$	$\frac{x}{22.4}$

Same moles.

Exercise-II (Previous Year Questions)

1. Ans. (3)

	H_2		O_2		CH_4
moles	$\frac{x}{2}$		$\frac{x}{32}$		$\frac{x}{16}$
Vol.	$\frac{x}{2}\times22.4$		$\frac{x}{32}\times22.4$		$\frac{x}{16}\times22.4$
	$\frac{1}{2}$		$\frac{1}{32}$		$\frac{1}{16}$
	16	:	1	:	2

2. Ans. (1)

	H_2	+	Cl_2	$\rightarrow$	2HCl
moles	$\frac{22.4}{22.4}=1$		$\frac{11.2}{22.4}=\frac{1}{2}$		

L.R. $x_1=\frac{1}{1}$ $\quad x_2=\frac{1}{\frac{2\times1}{\text{L.R.}}}$

$n_{HCl}=\frac{1}{2}\times2=1$

3. Ans. (1)

	2Mg	+	O_2	$\rightarrow$	2MgO
moles	$\frac{1}{24}$		$\frac{0.56}{32}$		
	$=0.041$		$=0.0175$		

L.R. $x_1=\frac{0.041}{2}$ $\quad x_2=\frac{\frac{0.0175}{1}}{\text{L.R.}}$

$=0.02$

$n_{HCl}=\frac{1}{2}\times2=1$

Mg left in excess = 0.0416 – (0.0175) × 2

wt. = 0.0066 × 24 = 0.16g

4. Ans. (1)

	H_2	+	O_2
moles	$\frac{100}{2}=50$		$\frac{400}{32}=12.5$
molar ratio =	50	:	12.5
	4	:	1

5. Ans. (2)

number of molecules = n × N_A

(1) $\frac{18}{18}\times N_A=N_A$

(2) $18N_A$

(3) 18

(4) $\frac{1.8}{1.8}\times N_A=\frac{N_A}{10}$

6. Ans. (4)

If N_A change from 6.022×10^{23} to 6.027×10^{20} then the mass of 1 mole of carbon will also change.

7. Ans. (2)

	$MgCO_3$	$\xrightarrow{\Delta}$	MgO	+	CO_2
moles	$\frac{20}{84}$		$\frac{8}{40}=\frac{1}{5}$		

1 mol $MgCO_3$ $\longrightarrow$ 1 mol MgO

$n_{MgO}=n_{MgCO_3}=\frac{1}{5}$

Mass of $MgCO_3=\frac{1}{5}\times84=\frac{84}{5}=16.8$

% Purity $=\frac{16.8}{20}\times100=84\%$

8. Ans. (3)

$X + 2Y \rightarrow XY_2$

$0.1\rightarrow$ 10g mass of 1 mol XY_2 = 100g

$3X + 2Y \rightarrow X_3Y_2$

$0.05\rightarrow$ 9g mass of 1 mol X_3Y_2 = 180g

$X + 2Y = 100$

$3X + 2Y = 180$

$-2X = -80 \Rightarrow X = 40$

$40 + 2Y = 100$

$2Y = 60 \Rightarrow Y = 30$

9. Ans. (3)

$HCOOH_{(\ell)} \xrightarrow{H_2SO_4} H_2O_{(\ell)} + CO_{(g)}$

$n=\frac{2.3}{46}=\frac{1}{20}$

$H_2C_2O_4 \xrightarrow{H_2SO_4} H_2O_{(\ell)} + CO(g) + CO_2(g)$

$n=\frac{4.5}{90}=\frac{1}{20}$

$\frac{1}{20}$ mole HCOOH produce $\frac{1}{20}$ mole $CO_{(g)}$

$\frac{1}{20}$ mole $H_2C_2O_4$ produce $\frac{1}{20}$ mole CO & $\frac{1}{20}$ mole CO_2.

KOH consume CO_2, So gas left are

$\left(\frac{1}{20}+\frac{1}{20}\right)CO$

$n_{CO} = \frac{1}{10}$

mass of CO = $\frac{1}{10} \times 28 = 2.8g$

10. Ans. (1)

Number of molecules = n × N_A

(1) $\frac{18}{18}\times N_A = N_A$

(2) $\frac{0.18}{18}\times N_A = \frac{N_A}{100}$

(3) $\frac{0.00224}{22.4}\times N_A = \frac{N_A}{10000}$

(4) $10^{-3}\ N_A$

11. Ans. (3)

$N_2 + 3H_2 \rightarrow 2NH_3$

2 mole NH_3 produce by 3 mole H_2

so 20 mole NH_3 will produce by $\frac{60}{2}$ mole of H_2.

$n_{H_2} = 30$

12. Ans. (4)

$P_V = nRT$

$v = \frac{1.8\times0.083\times647}{18\times 1} = 5.372$

13. Ans. (1)

Number of atoms = n × N_A × atomicity

(1) $\frac{1}{7}\times N_A \times 1 = \frac{N_A}{7}$

(2) $\frac{1}{108}\times N_A \times 1 = \frac{N_A}{108}$

(3) $\frac{1}{24}\times N_A \times 1 = \frac{N_A}{24}$

(4) $\frac{1}{32}\times N_A \times 2 = \frac{N_A}{16}$

14. Ans. (4)

Number of C–atom = $\frac{12}{12}$ × N_A × 1

= $N_A = 6.022 \times 10^{23}$

15. Ans. (3)

C	78	$\frac{78}{12}=6.5$	$\frac{6.5}{6.5}=1$
H	22	$\frac{22}{1}=22$	$\frac{22}{6.5}=3.3\simeq3$

CH_3

16. Ans. (1)

$CaCO_{3(s)} + 2HCl_{(aq.)} \rightarrow CaCl_{2(aq.)} + CO_{2(g)} + H_2O_{(\ell)}$

$\frac{50\times0.5}{1000}=0.025$

1 mol $CaCO_3$ react with 2 mole HCl

0.025 mol HCl react with $\frac{1}{2}$ × 0.025 mol $CaCO_3$

n_{CaCO_3} = 0.0125

mass of $CaCO_3$ = 0.0125 × 100

= 1.25 g

$95 = \frac{1.25}{w}\times100$

$w = \frac{125}{95}=1.32g$

17. Ans. (2)

mass = Vol. × density

mass = 2.5 × 2.15

= 5.375

= 5.4g

18. Ans. (1)

$Fe_{0.96}O$

$Fe_x^{+3}\ Fe_{(0.96-x)}^{+2}O$

3x × 2(0.96 – x) – 2 = 0

3x + 1.92 – 2x –2 = 0

x = 0.08

fraction of Fe = $\frac{0.08}{0.96}=0.083=\frac{1}{12}$

19. Ans. (1)

Weight of impure limestone = 20 g

Weight of pure limestone ($CaCO_3$) = 20% of 20 g

= $\frac{20}{100}\times20$

= 4g

$n_{CaCO_3} = \frac{4}{100}$ = 0.04

$CaCO_3 \rightarrow CaO + CO_2$

n=0.04 n=0.04

n_{CO_2} = 0.04

W_{CO_2} = 0.04 × 44

= 1.76 g

20. Ans. (1)

(1) 4 mol of He ; Number of He atom = $4 \times N_A$

(2) 4u of He ; Number of He atom = 1

(3) 4 g of He ; Number of He atom = N_A

(4) 0.1 mol of He ; Number of He atom = $0.1\ N_A$

21. Ans. (2)

$NaOH + HCl \rightarrow NaCl + H_2O$

Given $\Rightarrow$ 1g (0.75 M, 25 ml)

moles of NaOH = $\frac{1}{40}$ mol = 0.025 mol

moles of HCl = $\frac{0.75 \times 25}{1000} = 0.01875$ mol

now find the L.R. $\Rightarrow$ Value = $\frac{\text{given amount}}{\text{S.C}}$

NaOH ; HCl

$\frac{0.025}{1} > \frac{0.01875}{1}$

So, L.R will be HCl

Unreacted moles of NaOH = 0.025 – 0.01875

= 0.00625 mol

Unreacted mass = 0.00625 × 40 g

= 0.25 g

= 250 mg

22. Ans. (2)

Element	%	Moles	Simple Ratio
A	32	$\frac{32}{64} = \frac{1}{2}$	1
B	20	$\frac{20}{40} = \frac{1}{2}$	1
C	48	$\frac{48}{32} = \frac{3}{2}$	3

Empirical formula = ABC_3

23. Ans. (1)

1g of H_2 = $\frac{1}{2}$ mol H_2 = $\frac{N_A}{2}$ molecules

similarity, 14 g N_2 = $\frac{14}{28}$ mol = $\frac{N_A}{2}$ molecules

24. Ans. (2)

$C\% = \frac{12}{44} \times \frac{0.2}{0.3} \times 100 = 18.18\%$

$H\% = \frac{2}{18} \times \frac{0.1}{0.3} \times 100 = 3.70\%$

Exercise-III (Analytical Questions)

1. Ans. (2)

Mass of 1 butane (C_4H_{10}) molecule is 58 amu.

Mass of 1 mole butane = 58 amu × N_A

= N_A × amu = 1 g = 58g

1 mole has = 6.022×10^{23} molecules.

2. Ans. (1)

→ $n_C = \frac{12}{12} = 1$ & $n_{Al} = \frac{27}{27} = 1$

Moles are same so number of atoms also same number of atoms

$n \times N_A \times 1$

$1 \times N_A \times 1 = N_A$ atoms.

→ Gram atomic mass represents mass of 1 mole of atoms & 1 mol has 6.022×10^{23} atoms.

3. Ans. (2)

→ $2A + 3B \rightarrow C$

n = 3 4

L.R. $\frac{3}{2} = 1.5$ $\frac{4}{3} = 1.33$

L.R.

$n_C = \frac{4}{3} \times 1 = \frac{4}{3}$

B is L.R. & A. is E. R.

Ist correct, IInd incorrect.

4. Ans. (2)

$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(g)}$

→ 1mole CH_4 reacts with 2mole O_2 to produce 1 mole CO_2 & 2 mole H_2O.

→ Coefficients ratio will not be same as mass ratio.

Ist correct, IInd incorrect.

5. Ans. (4)

$2H_2 + O_2 \rightarrow 2H_2O$

10L 10L

L.R. $\frac{10}{2} = 5$ $\frac{10}{1} = 1$

V_{H_2O} $\frac{10}{2} \times 2 = 10L$

O_2 will also left as excess reagent.

Stoichiometric coefficients ratio is same as volume ratio.

Ist incorrect, IInd correct.

6. Ans. (1)

$2H_2 + O_2 \rightarrow 2H_2O$

$H_2 + O_2 \rightarrow H_2O_2$

Both reaction follows law of multiple proportion A & R both are correct & R is correct explanation of A.

7. Ans. (1)

→ Average mass of C atom is 12.011 amu & actual mass of C is $12.011 \times 1.67 \times 10^{-24}$g.

→ Average mass can be calculate by the help of taking Average of Isotopes mass. ^{12}C, ^{13}C, ^{14}C.

A & R both are correct also R is correct explanation of A.

8. Ans. (3)

→ Equal volume of gas.

$n = \frac{v}{22.4(L)}$ moles are same for all gases.

But number of atoms can be different for different gases.

→ atom is fundamental quantity which take in chemical reaction.

9. Ans. (1)

$[NH_4]_3 PO_4$.

(a) $\frac{\text{number of O}-\text{atom}}{\text{number of H atom}} = \frac{4}{12} = \frac{1}{3}$

(b) $\frac{\text{number of cation}}{\text{number of anion}} = \frac{3}{1}$

(c) $\frac{\text{g}-\text{atom of N}}{\text{g}-\text{atom of O}} = \frac{3}{4}$

(d) total number of atom is 1 mol $(NH_4)_3PO_4$

$(5 \times 3) + 1 + 4 = 20\ N_A$

A, B both are correct.

10. Ans. (1)

(A) number of atoms = $n \times N_A \times$ atomicity

Atomicity for substance can be different.

(B) $H_2 + Cl_2 \longrightarrow 2HCl$

$n = \frac{2}{2} = 1$ $\quad \frac{71}{71} = 1$

1 mol H_2 react with 1 mol Cl_2 to produce 2 mol HCl $n_{HCl} = 2$

Mass of HCl = $2 \times 36.5 = 71$ g

(C) Reactant which completely consume in reaction will act as L.R.

(D) [A, B, C]

11. Ans. (1)

(A) $n_{H_2SO_4} = 20$

Mass = $20 \times 98 = 1960$ (S)

(B) $n_{H_2O} = \frac{540}{18} = 30$ (R)

(C) Number of atoms

$= \frac{10^{-10}}{2} \times 6.022 \times 10^{23} \times 2$

$= 6.022 \times 10^{13}$ (Q)

(D) Number of amu in 20g

$1\text{ g} = \frac{1}{1.67 \times 10^{-24}\text{g}}$

$20\text{ g} = 20 \times \frac{1}{1.67 \times 10^{-24}\text{g}}$

$\frac{1}{\text{amu}} = N_A$

$= 20\ N_A$ (P)

12. Ans. (3)

Number of N_A atoms = $\frac{5.3}{106} \times N_A \times 2 = \frac{N_A}{10}$

(a) Number of N_A atoms = $\frac{4}{40} \times N_A \times 1 = \frac{N_A}{10}$

(b) Number of N_A atoms = $\frac{5.85}{58.5} \times N_A \times 1 = \frac{N_A}{10}$

(c) Number of N_A atoms = $0.25 \times N_A \times 2 = \frac{N_A}{2}$

(d) Number of N_A atoms = $\frac{5.6}{164} \times N_A \times 3 = 0.1N_A$

A & B are correct.

13. Ans. (2)

(A) Number of moles of Hydrogen = 8

P → 2 moles of C_3H_4 has 8 mol H atoms.

R → 4 moles of N_2H_2 has 8 mol H atoms.

(B) Number of moles of carbon = 6

P → 2 moles of C_3H_4 has 6 mol C atoms.

Q → 3 moles of CH_3COOH has 6 mol C atoms.

(C) Number of moles of oxygen = 6

Q → 3 moles CH_3COOH has = 6 mole O atoms.

S → 2 moles N_2O_3 has = 6 mole O atoms.

(D) Number of moles of N_2 molecule = 4

R → number of molecule of N in N_2H_4 = 4.

Exercise - I (Conceptual Questions) Build Up Your Understanding

INTRODUCTION

1. Rutherford's α-particle scattering experiment proved that atom has :-
(1) Electrons (2) Neutrons
(3) Nucleus (4) Orbitals
CAS001

2. A and B are two elements which have same atomic weight and are having atomic number 27 and 30 respectively. If the atomic weight of A is 57 then number of neutron in B is :-
(1) 27 (2) 33
(3) 30 (4) 40
CAS002

3. Find out the atoms which are isoneutronic :-
(1) $^{14}_{6}C$, $^{15}_{7}N$, $^{17}_{9}F$ (2) $^{12}_{6}C$, $^{14}_{7}N$, $^{19}_{9}F$
(3) $^{14}_{6}C$, $^{14}_{7}N$, $^{17}_{9}F$ (4) $^{14}_{6}C$, $^{14}_{7}N$, $^{19}_{9}F$
CAS003

4. Species which are isoelectronic to one another are
(a) CN^- (b) OH^- (c) CH_3^+ (d) N_2
(e) CO
Correct answer is
(1) a, b, c (2) a, c, d
(3) a, d, e (4) b, c, d
CAS004

5. For any anion X^{-3}, the mass number is 14. If anion has 10 electrons, then number of neutrons in X_2 nucleus :–
(1) 10 (2) 14 (3) 7 (4) 5
CAS005

6. Which of the following pairs is correctly matched :
(1) Isotopes $^{40}_{20}Ca$, $^{40}_{19}K$
(2) Isotones $^{30}_{14}Si$, $^{31}_{15}P$, $^{32}_{16}S$
(3) Isobars $^{16}_{8}O$, $^{17}_{8}O$, $^{18}_{8}O$
(4) Isoelectronic N^{-3}, O^{-2}, Cr^{+3}
CAS006

7. (i) $^{54}_{26}Fe$, $^{56}_{26}Fe$, $^{57}_{26}Fe$, $^{58}_{26}Fe$ (a) Isotopes
(ii) $^{3}_{1}H$, $^{3}_{2}He$ (b) Isotones
(iii) $^{76}_{32}Ge$, $^{77}_{33}As$ (c) Isodiaphers
(iv) $^{235}_{92}U$, $^{231}_{90}Th$ (d) Isobars
(v) $^{1}_{1}H$, $^{2}_{1}D$, $^{3}_{1}T$
Match the above correct terms:-
(1) [(i), - a], [(ii) - d], [(iii) - b], [(iv) - c], [(v) - a]
(2) [(i) - a] [(ii) - d], [(iii) - d] [(iv) - c] [v - a]
(3) [v -a] [(iv) - c]. [(iii) - d] [(ii) - b] [(i) - a]
(4) None of them
CAS007

8. Choose the false statement about deuterium:-
(1) It is an isotope of hydrogen
(2) It contains [(1e) + (1p) + (1n)]
(3) It contains only [(1p) + (1n)]
(4) D_2O is called as heavy water
CAS008

9. The relative abundance of two rubidium isotopes of atomic weights 85 and 87 are 75% and 25% respectively. The average atomic weight of rubidium is:–
(1) 75.5 (2) 85.5 (3) 86.5 (4) 87.5
CAS009

10. The ratio of specific charge of a proton and an α–particle is :–
(1) 2 : 1 (2) 1 : 2
(3) 1 : 4 (4) 1 : 1
CAS010

11. In an atom $^{27}_{13}Al$, number of proton is (a), electron is (b) and neutron is (c). Hence ratio will be [in order c : b : a] :-
(1) 13 : 14 : 13 (2) 13 : 13 : 14
(3) 14 : 13 : 13 (4) 14 : 13 : 14
CAS011

12. Atomic weight of Ne is 20.2. Ne is mixture of ^{20}Ne and ^{22}Ne, relative abundance of heavier isotope is :-
(1) 90 (2) 20 (3) 40 (4) 10
CAS012

13. Number of protons, neutrons & electrons in the element $^{231}_{89}\gamma$ is :-

(1) 89, 231, 89 (2) 89, 89, 242
(3) 89, 142, 89 (4) 89, 71, 89

CAS013

14. Atoms $^{13}_{6}C$ and $^{17}_{8}O$ are related to each other as:-

(1) Isotones (2) Isoelectronic
(3) Isodiaphers (4) Isosters

CAS014

15. The e/m ratio is maximum for :-

(1) D^+ (2) He^+
(3) H^+ (4) He^{2+}

CAS015

16. An isotone of $^{76}_{32}Ge$ is :-

(i) $^{77}_{32}Ge$ (ii) $^{77}_{33}As$ (iii) $^{77}_{34}Se$ (iv) $^{78}_{34}Se$

(1) (ii) & (iii) (2) (i) & (ii)
(3) (ii) & (iv) (4) (ii) & (iii) & (iv)

CAS016

17. In $^{14}_{7}N$ if mass attributed to electrons were doubled & the mass attributed to protons were halved, the atomic mass would become approximately :-

(1) Halved (2) Doubled
(3) Reduced by 25% (4) Remain same

CAS017

18. The value of planck's constant is 6.63×10^{-34} Js. The velocity of light is 3.0×10^8 m s^{-1}. Which value is closest to the wavelength in metres of a quantum of light with frequency of 8×10^{15} s^{-1} ?

(1) 3×10^7 (2) 2×10^{-25}
(3) 5×10^{-18} (4) 3.75×10^{-8}

CAS018

19. The energy of one mole photons of radiation having frequency 5.01×10^{14} Hz is :

(1) 160 KJ mol^{-1}
(2) 180 KJ mol^{-1}
(3) 200 KJ mol^{-1}
(4) 220 KJ mol^{-1}

CAS019

BOHR'S ATOMIC MODEL

20. Angular momentum in second Bohr orbit of H-atom is x. Then find out angular momentum in I^{st} excited state of Li^{+2} ion :

(1) 3x (2) 9x (3) $\frac{x}{2}$ (4) x

CAS020

21. Multiplication of electron velocity and radius for a orbit in an atom is :-

(1) Proportional to mass of electron
(2) Proportional to square of mass of electron
(3) Inversely proportional to mass of electron
(4) Does not depend upon mass of electron

CAS021

22. The radius of a shell for H-atom is 4.761Å. The value of n is :-

(1) 3 (2) 9 (3) 5 (4) 4

CAS022

23. In Bohr's atomic model radius of I^{st} orbit of Hydrogen is 0.053 nm then radius of 3rd orbit of Li^{+2} is :

(1) 0.159 nm (2) 0.053 nm
(3) 0.023 nm (4) 0.026 nm

CAS023

24. The radius ratio of first three Bohr orbits is :-

(1) 1 : 0·5 : 0·5 (2) 1 : 2 : 3
(3) 1 : 4 : 9 (4) 1 : 8 : 27

CAS024

25. For Li^{+2} ion, $r_2 : r_5$ will be :-

(1) 9 : 25 (2) 4 : 25 (3) 25 : 4 (4) 25 : 9

CAS025

26. The ratio of the radii of two Bohr orbits of H-atom is 4 : 1, what would be their nomenclature ?

(1) K & L (2) L & K
(3) N & L (4) 2 & 3 both

CAS026

27. The velocity of electron in third excited state of Be^{3+} ion will be :–

(1) $\frac{3}{4}(2.188 \times 10^8)$ ms^{-1}

(2) $\frac{3}{4}(2.188 \times 10^6)$ ms^{-1}

(3) (2.188×10^6) Kms^{-1}
(4) (2.188×10^3) Kms^{-1}

CAS027

28. The Bohr orbit radius for the hydrogen atom (n = 1) is approximately 0.530 Å. The radius for the first excited state (n = 2) will be :-
(1) 0.13 Å (2) 1.06 Å
(3) 4.77 Å (4) 2.12 Å

CAS028

29. The ratio of radius of first orbit in hydrogen to the radius of first orbit in deuterium will be :-
(1) 1 : 1 (2) 1 : 2 (3) 2 :1 (4) 4 : 1

CAS029

30. For any H like system, the ratio of velocities of electron in I, II & III orbit i.e., $V_1 : V_2 : V_3$ will be
(1) 1 : 2 : 3 (2) 1 : 1/2 : 1/3
(3) 3 : 2 : 1 (4) 1 : 1 : 1

CAS030

31. The energy of H–atom in n^{th} orbit is E_n, then energy in n^{th} orbit of singly ionised helium ion will be:-
(1) $4E_n$ (2) $E_n/4$ (3) $2E_n$ (4) $E_n/2$

CAS031

32. The energy of second Bohr orbit of the hydrogen atom is –328 kJ/mol. Hence the energy of fourth Bohr orbit should be :
(1) –41 kJ/mol (2) –1312 kJ/mol
(3) –164 kJ/mol (4) –82 kJ/mol

CAS032

33. In a hydrogen atom, if energy of an electron in ground state is –13.6 eV, then energy in the 2nd excited state is :-
(1) –1.51 eV (2) –3.4 eV
(3) –6.04 eV (4) –13.6 eV

CAS033

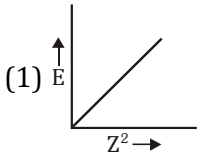
34. The ratio between kinetic energy and the total energy of the electron of hydrogen atom according to Bohr's model is :-
(1) 2 : 1 (2) 1 : 1 (3) 1 : –1 (4) 1 : 2

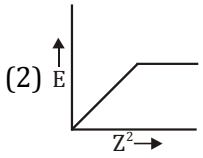
CAS034

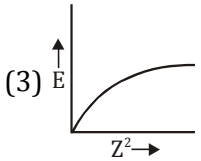
35. Potential energy is – 27.2 eV in second orbit of He^+, then calculate double of total energy in first excited state of hydrogen atom :–
(1) – 13.6 eV (2) – 54.4 eV
(3) – 6.8 eV (4) – 27.2 eV

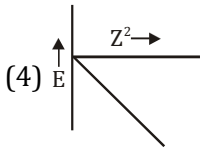
CAS035

36. The graphical representation of energy of electron and atomic number is :-

(1) 

(2) 

(3) 

(4) 

CAS036

37. Going from K–shell to N–shell in case of H–atom :-
(1) Kinetic energy decreases
(2) Total energy decreases
(3) Potential energy decreases
(4) None of these

CAS037

38. Maximum frequency of emission is obtained for the transition :–
(1) n = 2 to n = 1 (2) n = 6 to n = 2
(3) n = 1 to n = 2 (4) n = 2 to n = 6

CAS038

39. If the ionization energy of hydrogen is 313.8 kcal per mol, then the energy of the electron in 2nd excited state will be :-
(1) –113.2 kcal/mol (2) –78.45 kcal/mol
(3) –313.8 kcal/mol (4) –35 kcal/mol

CAS039

40. Which of the following electron transition will require the largest amount of energy in a hydrogen atom :-
(1) From n = 1 to n = 2
(2) From n = 2 to n = 3
(3) From n = ∞ to n = 1
(4) From n = 3 to n = 5

CAS040

41. If the potential energy (PE) of electron of hydrogen atom is –3.02 eV then in which of the following excited level is electron present :-
(1) 1st (2) 2nd
(3) 3rd (4) 4th

CAS041

42. The radiation of low frequency will be emitted in which transition of hydrogen atom :–
(1) n = 1 to n = 4 (2) n = 2 to n = 5
(3) n = 3 to n = 1 (4) n = 5 to n = 2
CAS042

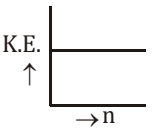
43. The ratio of energies of hydrogen atom for first and second excited state is :-
(1) 4/1 (2) 1/4 (3) 4/9 (4) 9/4
CAS043

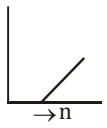
44. $E_n = -313.6/n^2$. If the value of $E_n = -34.84$ then to which of the following values does 'n' correspond :-
(1) 1 (2) 2 (3) 3 (4) 4
CAS044

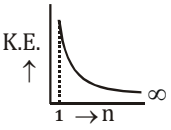
45. The ratio of potential energy and total energy of an electron in a Bohr orbit of hydrogen like species is :-
(1) 2 (2) –2 (3) 1 (4) –1
CAS045

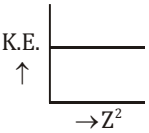
46. Which is not a correct order of energy for 1st, 2nd & 3rd orbit :-
(1) $E_1 > E_2 > E_3$
(2) $(PE)_1 < (PE)_2 < (PE)_3$
(3) $(KE)_1 > (KE)_2 > (KE)_3$
(4) '1' & '3' both
CAS046

47. Which is correct for any H like species :-
(1) $(E_2 - E_1) > (E_3 - E_2) > (E_4 - E_3)$
(2) $(E_2 - E_1) < (E_3 - E_2) < (E_4 - E_3)$
(3) $(E_2 - E_1) = (E_3 - E_2) = (E_4 - E_3)$
(4) $(E_2 - E_1) = 1/4\ (E_3 - E_2) = 1/9\ (E_4 - E_3)$
CAS047

48. Which of the following is a correct graph :-
(1) 

(2) 

(3) 

(4) 

CAS048

49. First excitation potential of H atom is :
(1) 10.2 eV (2) 3.4 eV
(3) 1.51 eV (4) – 3.4 eV
CAS049

50. Energy required to remove an e^- from M shell of H-atom is 1.51 eV, then energy of first excited state will be :-
(1) –1.51 eV (2) +1.51 eV
(3) –3.4 eV (4) –13.6 eV
CAS050

51. The ionisation potential of the hydrogen atom is 13.6 eV. The energy needed to ionise a hydrogen atom which is in its second excited state is about:-
(1) 13.6 eV (2) 10.2 eV
(3) 3.4 eV (4) 1.51 eV
CAS051

52. The ionisation energy for excited hydrogen atom in eV will be :–
(1) 13.6 (2) Less than 13.6
(3) Greater than 13.6 (4) 3.4 or less
CAS052

53. The energy required to excite an electron of H-atom from first orbit to second orbit is :-
(1) $\frac{3}{4}$ of its ionisation energy
(2) $\frac{1}{2}$ of its ionisation energy
(3) $\frac{1}{4}$ of its ionisation energy
(4) None of these
CAS053

54. The ionisation potential of a singly ionised helium ion is equivalent to :-
(1) Kinetic energy of first orbit
(2) Energy of last orbit
(3) Average energy in orbits
(4) Maximum energy in orbits
CAS054

SPECTRUM AND SPECTRAL LINES

55. The spectrum of He is expected to be similar to that of :-
(1) H (2) Na (3) He^+ (4) Li^+
CAS055

56. Third line of Balmer series is produced by which transition in spectrum of H–atom
(1) n = 5 to n = 2 (2) n = 5 to n = 1
(3) n = 4 to n = 2 (4) n = 4 to n = 1
CAS056

57. Which one of the following electronic transition between energy levels produces the line of shortest wavelength in hydrogen spectrum ?
(1) $n_2 \rightarrow n_1$ (2) $n_3 \rightarrow n_1$
(3) $n_4 \rightarrow n_1$ (4) $n_4 \rightarrow n_3$
CAS057

58. Which series have highest energy in hydrogen spectrum :-
(1) Balmer (2) Brackett
(3) Pfund (4) Lyman
CAS058

59. The ratio of minimum frequency of Lyman & Balmer series will be :–
(1) 1.25 (2) 0.25 (3) 5.4 (4) 10
CAS059

60. Which transition emits photon of maximum frequency :–
(1) Second spectral line of Balmer series
(2) Second spectral line of Paschen series
(3) Fifth spectral line of Humphery series
(4) First spectral line of Lyman series
CAS060

61. Which one of the following species will give a series of spectral lines similar to that of Mg^{2+} :-
(1) Al^{3+} (2) Na (3) Mg^{+} (4) F
CAS061

62. The ratio of minimum wavelengths of Lyman & Balmer series will be :–
(1) 1.25 (2) 0.25 (3) 5 (4) 10
CAS062

63. The wavelength of photon obtained by electron transition between two levels in H–atom and singly ionised He are λ_1 and λ_2 respectively, then :-
(1) $\lambda_2 = \lambda_1$ (2) $\lambda_2 = 2\lambda_1$
(3) $\lambda_2 = \lambda_1/2$ (4) $\lambda_2 = \lambda_1/4$
CAS063

64. Find out ratio of following for photon
$(\nu_{max.})_{Lyman} : (\nu_{max})_{Brackett}$
(1) 1 : 16 (2) 16 : 1
(3) 4 : 1 (4) 1 : 4
CAS064

65. The ratio of wavelengths of first line of Lyman series in Li^{+2} and first line of Lyman series in deuterium $\left({}_1^2H\right)$ is :-
(1) 1 : 9 (2) 9 : 1 (3) 1 : 4 (4) 4 : 1
CAS065

66. In an electronic transition, atom cannot emit :-
(1) Visible light (2) γ - rays
(3) Infra red light (4) Ultra violet light
CAS066

67. The first Lyman transition in the hydrogen spectrum has $\Delta E = 10.2$ eV. The same energy change is observed in the second Balmer transition of :-
(1) Li^{2+} (2) Li^{+} (3) He^{+} (4) Be^{3+}
CAS067

68. The limiting line in Balmer series will have a frequency of :-
(1) $3.65 \times 10^{14} sec^{-1}$ (2) $3.29 \times 10^{15} sec^{-1}$
(3) $8.22 \times 10^{14}\ sec^{-1}$ (4) $-8.22 \times 10^{14}\ sec^{-1}$
CAS068

69. The first emission line in the H–atom spectrum in the Balmer series will have wave number :-
(1) $\frac{5R}{36}\ cm^{-1}$ (2) $\frac{3R}{4}\ cm^{-1}$
(3) $\frac{7R}{144}\ cm^{-1}$ (4) $\frac{9R}{400}\ cm^{-1}$
CAS069

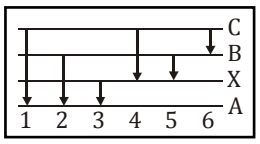
70. What transition in He^{+} will have the same λ as the 1st line in Lyman series of H–atom :-
(1) $n = 5 \rightarrow n = 3$ (2) $n = 3 \rightarrow n = 2$
(3) $n = 6 \rightarrow n = 4$ (4) $n = 4 \rightarrow n = 2$
CAS070

71. In H–atom, electron transits from 6th orbit to 2nd orbit in multi step. Then total spectral lines (without Balmer series) will be :–
(1) 6 (2) 10 (3) 4 (4) 0
CAS071

72. An atom has x energy level, then total number of lines in its spectrum are:-
(1) 1 + 2 + 3 .....(x + 1) (2) 1 + 2 + 3 ..... (x^2)
(3) 1 + 2 + 3 ..... (x - 1) (4) (x + 1)(x + 2)(x + 4)
CAS072

73. The figure indicates the energy level diagram for the origin of six spectral lines in emission spectrum (e.g. line no. 5 arises from the transition from level B to X) which of the following spectral lines will not occur in the absorption spectrum :–

(1) 1, 2, 3
(2) 3, 2
(3) 4, 5, 6
(4) 3, 2, 1



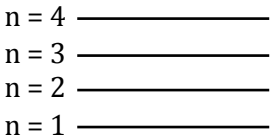
CAS073

74. A certain electronic transition from an excited state to ground state of the H atom in one or more step gives rise to three lines in the ultra violet region of the spectrum. How many lines does this transition produce in the infrared region of the spectrum :–

(1) 1 (2) 2 (3) 3 (4) 4

CAS074

75. Four lowest energy levels of H–atom are shown in the figure. The number of emission lines could be :-



(1) 3
(2) 4
(3) 5
(4) 6

CAS075

76. In the above problem, the number of absorption lines could be :-

(1) 3 (2) 4 (3) 5 (4) 6

CAS076

77. If 9.9 eV energy is supplied to H atom, the no. of spectral lines emitted is equal to :-

(1) 0 (2) 1 (3) 2 (4) 3

CAS077

78. Frequency of photons emitted in Paschen series is given by $\nu = 3.29 \times 10^{15}$ (Hz) $\left[\frac{1}{3^2}-\frac{1}{n^2}\right]$. Value of n for photon having wavelength 1285 nm will be :

(1) n = 3 (2) n = 4
(3) n = 5 (4) n = 6

CAS078

de-BROGLIE CONCEPT AND HEISENBERG'S UNCERTAINTY PRINCIPLE

79. An electron has a kinetic energy of 2.8×10^{-23} J. de-Broglie wavelength will be nearly :- ($m_e = 9.1 \times 10^{-31}$ kg)

(1) 9.28×10^{-24} m (2) 9.28×10^{-7} m
(3) 9.28×10^{-8} m (4) 9.28×10^{-10} m

CAS079

80. What is the de-Broglie wavelength associated with the hydrogen electron in its third orbit :-

(1) 9.96×10^{-10} cm (2) 9.96×10^{-8} cm
(3) 9.96×10^{4} cm (4) 9.96×10^{8} cm

CAS080

81. If the de-Broglie wavelength of the fourth Bohr orbit of hydrogen atom is 4Å, the circumference of the orbit will be :-

(1) 4 Å (2) 4 nm
(3) 16 Å (4) 16 nm

CAS081

82. Number of waves in fourth orbit is :–

(1) 4 (2) 5 (3) 0 (4) 1

CAS082

83. What is the ratio of the de-Broglie wavelengths for electrons accelerated through 200 volts and 50 volts :-

(1) 1 : 2 (2) 2 : 1 (3) 3 : 10 (4) 10 : 3

CAS083

84. For a valid Bohr orbit, its circumference should be

(1) = n λ (2) = (n – 1)λ
(3) > n λ (4) < n λ

CAS084

85. The number of waves made by a Bohr electron in an orbit of maximum magnetic quantum number +2:-

(1) 3 (2) 4 (3) 2 (4) 1

CAS085

86. The uncertainty in position of an electron & helium atom are same. If the uncertainty in momentum for the electron is 32×10^5 g cm s^{-1}, then the uncertainty in momentum of helium atom will be

(1) 32×10^5 g cm s^{-1} (2) 16×10^5 g cm s^{-1}
(3) 8×10^5 g cm s^{-1} (4) None of these

CAS086

87. Calculate the uncertainty in the position of an electron (mass 9.1×10^{-28}g) moving with a velocity of 3×10^4 cm sec^{-1}, if the uncertainty in velocity is 0.011% ?

(1) 1.92 cm (2) 7.68 cm
(3) 0.175 cm (4) 3.84 cm

CAS087

88. Heisenberg Uncertainty principle is not valid for

(1) Moving electron (2) Motor car
(3) Stationary particles (4) 2 & 3 both

CAS088

89. What should be the momentum (in gram centimetre per second) of a particle if its de-Broglie wavelength is 1Å and the value of h is 6.6252×10^{-27} erg second ?

(1) 6.6252×10^{-19} gcm/s
(2) 6.6252×10^{-21} gcm/s
(3) 6.6252×10^{-24} gcm/s
(4) $6.6252 \times 10^{--27}$ gcm/s

CAS089

90. Which of the following has least de-Broglie wavelength, moving with same speed ?

(1) e^- (2) p (3) CO_2 (4) SO_2

CAS090

QUANTUM NUMBERS

91. The following quantum no. are possible for how many orbitals n = 3, ℓ = 2, m = +2 ?

(1) 1 (2) 2 (3) 3 (4) 4

CAS091

92. Number of possible orbitals (all types) in n = 3 energy level is :-

(1) 1 (2) 3 (3) 4 (4) 9

CAS092

93. Which sub–shell is not permissible :–

(1) 2d (2) 4f (3) 6p (4) 3s

CAS093

94. Nodal plane is found in which orbital :-

(1) n = 2, ℓ = 0 (2) n = 3, ℓ = 0
(3) n = 2, ℓ = 1 (4) n = 1, ℓ = 0

CAS094

95. No. of nodal surface in 2s orbital :-

(1) 0 (2) 1 (3) 2 (4) 3

CAS095

96. Number of orbitals in h sub-shell is

(1) 11 (2) 15 (3) 17 (4) 19

CAS096

97. How many quantum numbers are required to specify the position of electron :-

(1) 1 (2) 2 (3) 3 (4) 4

CAS097

98. Which of the following is correct for a 4d–electron

(1) n = 4, ℓ = 2, s = + $\frac{1}{2}$
(2) n = 4, ℓ = 2, s = 0
(3) n = 4, ℓ = 3, s = 0
(4) n = 4, ℓ = 3, s = + $\frac{1}{2}$

CAS098

99. If n = 3, then which value of 'ℓ' is correct :–

(1) 0 (2) 1
(3) 2 (4) All of them

CAS099

100. Energy of atomic orbitals in a particular shell is in order:-

(1) s < p < d < f (2) s > p > d > f
(3) p < d < f < s (4) f > d > s > p

CAS100

101. Which statement is not correct for n = 5, m = 2 :-

(1) ℓ = 4
(2) ℓ = 0, 1,2,3 ; s = + 1/2
(3) ℓ= 3
(4) ℓ= 2, 3, 4

CAS101

102. Spin angular momentum for electron is :-

(1) $\sqrt{s(s+1)}\ \frac{h}{2\pi}$ (2) $\sqrt{2s(s+1)}\ \frac{h}{2\pi}$
(3) $\sqrt{s(s+2)}\ \frac{h}{2\pi}$ (4) None

CAS102

103. The maximum number of electrons in a p–orbital with n = 6 and m = 0 can be :-
(1) 14 (2) 6 (3) 2 (4) 10
CAS103

104. The total number of value of m for the electrons in n = 4 is -
(1) 4 (2) 8 (3) 16 (4) 32
CAS104

105. In an atom, for how many electrons, the quantum numbers will be n = 3, ℓ = 2, m = + 2, s = + $\frac{1}{2}$:-
(1) 18 (2) 6 (3) 24 (4) 1
CAS105

106. Which orbital is represented by the complete wave function ψ_{420} :-
(1) $4d_{z^2}$ (2) $3d_{z^2}$
(3) $4p_z$ (4) 4s
CAS106

107. An electron is in one of 4d orbital. Which of the following quantum number value is not possible :-
(1) n = 4 (2) ℓ = 1
(3) m = 1 (4) m = 2
CAS107

108. A neutral atom of an element has 2K, 8L, 11 M and 2N electrons. The number of s-electron in the atom are
(1) 2 (2) 8
(3) 10 (4) 6
CAS108

109. If ℓ = 3 then type and number of orbital is :–
(1) 3p, 3 (2) 4f, 14
(3) 5f, 7 (4) 3d, 5
CAS109

110. Any nf–orbital can accomodate upto :–
(1) 14 electron
(2) Six electrons
(3) Two electrons with parallel spin
(4) Two electrons with opposite spin
CAS110

111. n, ℓ and m values of an electron in $3p_y$ orbital are :-
(1) n = 3 ; ℓ = 1 and m = 1
(2) n = 3 ; ℓ = 1 and m = –1
(3) Both 1 and 2 are correct
(4) None of these
CAS111

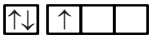
112. The maximum probability of finding an electron in the d_{xy} orbital is :-
(1) Along the x-axis
(2) Along the y-axis
(3) At an angle of 45° from the x and y axis
(4) At an angle of 90° from the x and y axis
CAS112

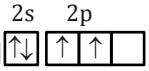
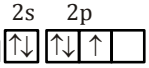
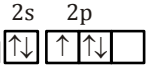
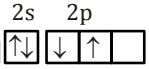
113. Which orbital has two angular nodal planes :-
(1) s (2) p (3) d (4) f
CAS113

114. An orbital with ℓ = 0 is symmetrical about the:-
(1) x-axis only (2) y-axis only
(3) z-axis only (4) nucleus
CAS114

115. If n & ℓ are principal and azimuthal quantum no. respectively then the expression for calculating the total no. of electron in any energy level is :-
(1) $\sum_{\ell=0}^{\ell=n} 2(2\ell+1)$ (2) $\sum_{\ell=1}^{\ell=n-1} 2(2\ell+1)$
(3) $\sum_{\ell=0}^{\ell=n+1} 2(2\ell+1)$ (4) $\sum_{\ell=0}^{\ell=n-1} 2(2\ell+1)$
CAS115

RULES FOR FILLING OF ORBITALS

116. Which configuration does not obey Pauli's exclusion principle :-
(1)  (2) 
(3)  (4) 
CAS116

117. Which of the following configuration follows the Hund's rule :–
(1) [He]  (2) [He] 
(3) [He]  (4) [He] 
CAS117

118. The basis of three unpaired electrons present in the configuration of nitrogen is :–
(1) Aufbau principle
(2) Pauli's principle
(3) Hund's principle
(4) Uncertainty principle

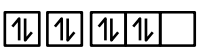
CAS118

119. The orbital with maximum energy is :-
(1) 3d (2) 5p (3) 4s (4) 6d

CAS119

120. No. of all subshells having $n + \ell = 7$ is:-
(1) 4 (2) 5 (3) 6 (4) 7

CAS120

121. Electronic configuration 
has violated :–
(1) Hund's rule (2) Pauli's principle
(3) Aufbau principle (4) $(n + \ell)$ rule

CAS121

122. The total spin resulting from a d^9 configuration is:-
(1) $\frac{1}{2}$ (2) 2 (3) 1 (4) $\frac{3}{2}$

CAS122

123. Which of the following transition neither shows absorption nor emission of energy in case of hydrogen atom :–
(1) $3p_x \rightarrow 3s$ (2) $3d_{xy} \rightarrow 3d_{yz}$
(3) $3s \rightarrow 3d_{xy}$ (4) All the above

CAS123

124. In ground state of $_{24}Cr$, number of orbitals with paired and unpaired electron :-
(1) 10 (2) 12 (3) 15 (4) 18

CAS124

125. For Na (Z = 11) set of quantum numbers for last electron is:–

(1) $n = 3, \ell = 1, m = 1, s = +\frac{1}{2}$

(2) $n = 3, \ell = 0, m = 0, s = +\frac{1}{2}$

(3) $n = 3, \ell = 0, m = 1, s = +\frac{1}{2}$

(4) $n = 3, \ell = 1, m = 1, s = -\frac{1}{2}$

CAS125

126. Which of the following set of quantum numbers is correct for the 19th electron of Chromium :-

	n	**ℓ**	**m**	**s**
(1)	3	0	0	1/2
(2)	3	2	–2	1/2
(3)	4	0	0	1/2
(4)	4	1	–1	1/2

CAS126

127. Which set of quantum number is correct for an electron in 3p orbital :-

(1) $n = 3, \ell = 2, m = 0, s = +\frac{1}{2}$

(2) $n = 3, \ell = 0, m = +1, S = +\frac{1}{2}$

(3) $n = 3, \ell = -2\ m = -1, s = +\frac{1}{2}$

(4) $n = 3\ \ell = 1\ m = 0, s = +\frac{1}{2}$

CAS127

128. An atom of Cr [Z = 24] loses 2 electrons. How many unpaired electrons shall be there in Cr^{+2}:
(1) 4 (2) 3 (3) 2 (4) 1

CAS128

129. The atomic number of an element is 17, the number of orbitals containing electron pairs in the valence shell is:-
(1) 8 (2) 2 (3) 3 (4) 6

CAS129

130. A transition metal 'X' has a configuration [Ar] $3d^5$ in its + 3 oxidation state. Its atomic number is:–
(1) 22 (2) 26 (3) 28 (4) 19

CAS130

131. $4s^2$ is the configuration of the outermost orbit of an element. Its atomic number would be :–
(1) 29 (2) 24 (3) 30 (4) 19

CAS131

132. Sum of the paired electrons present in the orbital with $\ell = 2$ in all the species Fe^{2+}, Co^{2+} and Ni^{+2} are:–
(1) 9 (2) 12 (3) 6 (4) 15

CAS132

133. The quantum number of 20th electron of Fe(Z = 26) would be :–

(1) 3, 2, – 2, – ½ (2) 3, 2, 0, ½

(3) 4, 0, 0, + ½ (4) 4, 1, – 1, + ½

CAS133

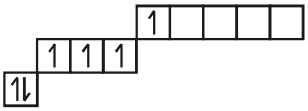
134. The atomic number of the element having maximum number of unpaired 3p electrons is (in ground state):-

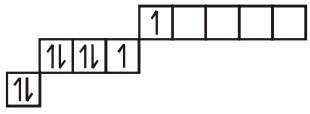
(1) 15 (2) 10

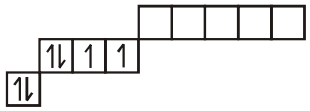
(3) 12 (4) 8

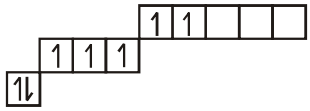
CAS134

135. Which one represent ground state configuration :-

(1) 

(2) 

(3) 

(4) 

CAS135

136. In an atom having 2K, 8L, 8M and 2N electrons, the number of electrons with m = 0; $s = +\frac{1}{2}$ are :-

(1) 6 (2) 2 (3) 8 (4) 16

CAS136

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	1	3	2	2	1	3	2	1	3	4	3	3	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	4	3	4	3	1	1	3	2	4	4	4	1	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	4	1	3	3	4	1	1	4	1	2	4	4	3	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	1	3	1	3	4	4	1	1	4	1	3	4	3	4
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	2	4	2	1	2	3	3	1	4	1	3	3	1	4
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	1	1	3	3	2	3	1	1	1	1	1	3	4	1	4
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	1	4	1	3	2	1	3	1	4	1	2	1	3	3	4
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	1	2	2	3	4	3	3	3	4	4	2	1	3	4	1
Question	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Answer	1	1	4	3	2	3	4	1	3	2	3	2	3	1	3
Question	136														
Answer	1														

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. What is the maximum number of orbitals that can be identified with the following quantum numbers. n = 3, ℓ = 1, m_ℓ = 0 ?

(1) 1 (2) 2 (3) 3 (4) 4

CAS150

2. Calculate the energy in joule corresponding to light of wavelength 45 nm : (Planck's constant h = 6.63×10^{-34} Js; speed of light c = 3×10^{8} ms^{-1})

(1) 6.67×10^{15} (2) 6.67×10^{11}

(3) 4.42×10^{-15} (4) 4.42×10^{-18}

CAS151

3. Magnetic moment 2.83 BM is given by which of the following ions ?

(At. no. Ti = 22, Cr = 24, Mn = 25, Ni = 28):-

(1) Ti^{3+} (2) Ni^{2+} (3) Cr^{3+} (4) Mn^{2+}

CAS152

AIPMT 2015

4. Which of the following pairs of ions are isoelectronic and isostructural ?

(1) ClO_3^-, CO_3^{2-} (2) SO_3^{2-}, NO_3^-

(3) ClO_3^-, SO_3^{2-} (4) CO_3^{2-}, SO_3^{2-}

CAS153

5. The number of d-electrons in Fe^{2+} (Z = 26) is not equal to the number of electrons in which one of the following?

(1) p-electrons in Cl (Z = 17)

(2) d-electrons in Fe (Z = 26)

(3) p-electrons in Ne (Z = 10)

(4) s-electrons in Mg (Z = 12)

CAS154

6. Magnetic moment 2.84 B.M. is given by :-

(At. no.), Ni = 28, Ti = 22, Cr = 24, Co = 27)

(1) Ti^{3+} (2) Cr^{2+} (3) Co^{2+} (4) Ni^{2+}

CAS155

7. The angular momentum of electron in 'd' orbital is equal to :-

(1) $\sqrt{2}\hbar$ (2) $2\sqrt{3}\hbar$ (3) $0\,\hbar$ (4) $\sqrt{6}\hbar$

CAS156

RE-AIPMT 2015

8. Which is the correct order of increasing energy of the listed orbitals in the atom of titanium ? (At. no. Z = 22)

(1) 3s 3p 3d 4s (2) 3s 3p 4s 3d

(3) 3s 4s 3p 3d (4) 4s 3s 3p 3d

CAS157

NEET-I 2016

9. Two electrons occupying the same orbital are distinguished by :-

(1) Principal quantum number

(2) Magnetic quantum number

(3) Azimuthal quantum number

(4) Spin quantum number

CAS158

NEET-II 2016

10. Which of the following pairs of d-orbitals will have electron density along the axis ?

(1) $d_{z^2}, d_{x^2-y^2}$ (2) $d_{xy}, d_{x^2-y^2}$

(3) d_{z^2}, d_{xz} (4) d_{xz}, d_{yz}

CAS159

11. How many electrons can fit in the orbital for which n = 3 and ℓ = 1 ?

(1) 10 (2) 14 (3) 2 (4) 6

CAS160

NEET(UG) 2017

12. Which one is the wrong statement ?

(1) The uncertainty principle is $\Delta E \times \Delta t \geq h/4\pi$

(2) Half filled and fully filled orbitals have greater stability due to greater exchange energy, greater symmetry and more balanced arrangement.

(3) The energy of 2s orbital is less than the energy of 2p orbital in case of Hydrogen like atoms

(4) de-Broglie's wavelength is given by $\lambda = \frac{h}{mv}$, where m = mass of the particle, v = velocity of the particle

CAS161

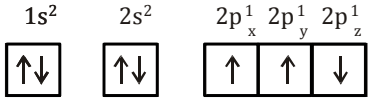
NEET(UG) 2018

13. Which one is a **wrong** statement ?

(1) Total orbital angular momentum of electron in 's' orbital is equal to zero

(2) An orbital is designated by three quantum numbers while an electron in an atom is designated by four quantum numbers.

(3) The electronic configuration of N atom is



(4) The value of m for d_{z^2} is zero

CAS162

NEET(UG) 2019

14. Which of the following series of transitions in the spectrum of hydrogen atom falls in visible region?
(1) Lyman series (2) Balmer series
(3) Paschen series (4) Brackett series
CAS163

Odisha NEET(UG) 2019

15. Orbital having 3 angular nodes and 3 total nodes is :-
(1) 5 p (2) 3 d (3) 4 f (4) 6 d
CAS164

16. In hydrogen atom, the de Broglie wavelength of an electron in the second Bohr orbit is :-
[Given that Bohr radius, $a_0 = 52.9$ pm]
(1) 211.6 pm (2) 211.6 π pm
(3) 52.9 π pm (4) 105.8 pm
CAS165

NEET (UG) 2020

17. The number of protons, neutrons and electrons in $^{175}_{71}Lu$, respectively, are :
(1) 175, 104 and 71 (2) 71, 104 and 71
(3) 104, 71 and 71 (4) 71, 71 and 104
CAS166

NEET (UG) 2020 (Covid-19)

18. The number of angular nodes and radial nodes in 3s orbital are
(1) 0 and 2, respectively
(2) 1 and 0, respectively
(3) 3 and 0, respectively
(4) 0 and 1, respectively
CAS167

NEET (UG) 2021

19. A particular station of All India Radio, New Delhi, broadcasts on a frequency of 1,368 kHz (kilohertz). The wavelength of the electromagnetic radiation emitted by the transmitter is: [speed of light $c = 3.0 \times 10^8$ ms^{-1}]
(1) 219.3 m (2) 219.2 m
(3) 2192 m (4) 21.92 cm
CAS168

NEET (UG) 2022

20. Identify the **incorrect** statement from the following.
(1) All the five 4d orbitals have shapes similar to the respective 3d orbitals.
(2) In an atom, all the five 3d orbitals are equal in energy in free state.
(3) The shapes of d_{xy}, d_{yz}, and d_{zx} orbitals are similar to each other; and $d_{x^2-y^2}$ and d_{z^2} are similar to each other.
(4) All the five 5d orbitals are different in size when compared to the respective 4d orbitals
CAS169

21. If radius of second Bohr orbit of the He^+ ion is 105.8 pm, what is the radius of third Bohr orbit of Li^{2+} ion?
(1) 15.87 pm (2) 1.587 pm
(3) 158.7 Å (4) 158.7 pm
CAS170

22. Match **List-I** with **List-II** :

	List-I (quantum number)		**List-II (Orbital)**
(a)	n = 2, ℓ = 1	(i)	2 s
(b)	n = 3, ℓ = 2	(ii)	3 s
(c)	n = 3, ℓ = 0	(iii)	2 p
(d)	n = 2, ℓ = 0	(iv)	3 d

Choose the **correct answer** from the options given below :
(1) (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii)
(2) (a) – (iv), (b) – (iii), (c) – (i), (d) – (ii)
(3) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)
(4) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)
CAS171

23. When electromagnetic radiation of wavelength 300 nm falls on the surface of a metal, electrons are emitted with the kinetic energy of 1.68×10^5 J mol^{-1}. What is the minimum energy needed to remove an electron from the metal ?
($h = 6.626 \times 10^{-34}$ Js, $c = 3 \times 10^8$ ms^{-1}, $N_A = 6.022 \times 10^{23}$ mol^{-1})
(1) 2.31×10^6 J mol^{-1} (2) 3.84×10^4 J mol^{-1}
(3) 3.84×10^{-19} J mol^{-1} (4) 2.31×10^5 J mol^{-1}
CAS172

NEET (UG) 2023

24. Select the **correct** Statements from the following:
(A) Atoms of all elements are composed of two fundamental particles.
(B) The mass of the electron is 9.10939×10^{-31} kg.
(C) All the isotopes of a given elements show same chemical properties.
(D) Protons and electrons are collectively known as nucleons.
(E) Dalton's atomic theory, regarded the atom as an ultimate particle of matter.
Choose the **correct** answer from the options given below.
(1) C, D and E only (2) A and E only
(3) B,C and E only (4) A,B and C only
CAS193

25. The relation between n_m, (n_m = the number of permissible values of magnetic quantum number (m)) for a given value of azimuthal quantum number (l), is

(1) $l = 2n_m + 1$ (2) $n_m = 2l^2 + 1$

(3) $n_m = l + 2$ (4) $l = \frac{n_m - 1}{2}$

CAS194

NEET (UG) 2024

26. Match List I with List II.

List I Quantum Number	List II Information provided
A. m_ℓ	I. shape of orbital
B. m_s	II. size of orbital
C. ℓ	III. orientation of orbital
D. n	IV. orientation of spin of electron

Choose the correct answer from the options given below :

(1) A-I, B-III, C-II, D-IV (2) A-III, B-IV, C-I, D-II

(3) A-III, B-IV, C-II, D-I (4) A-II, B-I, C-IV, D-III

CAS227

27. The energy of an electron in the ground state (n = 1) for He^+ ion is –xJ, then that for an electron in n = 2 state for Be^{3+} ion in J is :

(1) – x (2) $-\frac{x}{9}$ (3) –4x (4) $-\frac{4}{9}x$

CAS228

NEET (UG) 2024 (Re-examination)

28. The quantum numbers of four electrons are given below :

I. n = 4, ℓ = 2, m_ℓ = –2, s = $-\frac{1}{2}$

II. n = 3, ℓ = 2, m_ℓ = 1, s = $+\frac{1}{2}$

III. n = 4, ℓ = 1, m_ℓ = 0, s = $+\frac{1}{2}$

IV. n = 3, ℓ = 1, m_ℓ = –1, s = $+\frac{1}{2}$

The **correct** decreasing order of energy of these electrons is :

(1) IV > II > III > I (2) I > III > II > IV

(3) III > I > II > IV (4) I > II > III > IV

CAS229

29. Given below are two statements :

Statement-I : The Balmer spectral line for H atom with lowest energy is located at $\frac{5}{36}R_H cm^{-1}$

(R_H = Rydberg constant)

Statement-II : When the temperature of blackbody increases, the maxima of the curve (intensity vs wavelength) shifts to shorter wavelength.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

CAS230

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	2	3	1	4	4	2	4	1	3	3	3	2	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
Answer	2	2	1	1	3	4	4	4	3	4	2	1	2	3	

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Exercise - III (Analytical Questions) Master Your Understanding

1. Given below are two statements:
Statement I : Azimuthal quantum number 'ℓ' identifies the subshell and determines the shape of orbital.
Statement II : There are $(2\ell+1)$ orbitals of each type in a subshell.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement I and statement II are correct.
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct.
(4) Both statements I and statements II are incorrect.

CAS173

2. Given below are two statements:
Statement I : Two electrons can have the same value of three quantum numbers n, l and m in same atom.
Statement II : No two electrons in an atom can have, the same set of four quantum numbers.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement I and statement II are correct.
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct.
(4) Both statements I and statements II are incorrect.

CAS174

3. Given below are two statements:
Statement I : The energy of 3d orbital is greater than 4p orbital.
Statement II : If two orbital have the same value of $(n+\ell)$, the orbital with lower value of n will have the lower energy.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement I and statement II are correct.
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct.
(4) Both statements I and statements II are incorrect.

CAS177

4. Given below are two statements:
Statement I : Energy of stationary state of an electron is $E_n = R_H\left(\frac{-1}{n^2}\right)$.
Statement II : When the electron present in n^{th} orbit is attracted by nucleus then its energy is less compare to when it is present at infinite distance.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement I and statement II are correct.
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct.
(4) Both statements I and statements II are incorrect.

CAS178

5. True statements for Cathode Ray are
(a) Travel in straight line in absence of electric or magnetic field.
(b) Characteristics depend upon the material of electrodes of tube.
(c) Characteristics does not depend upon the nature of gas present in cathode ray tube.
(1) a, c (2) a, b (3) b, c (4) a, b, c

CAS181

6. According to Thomson model of atom :
(a) Negative charge is uniformly distributed in an atom.
(b) Atom possesses spherical shape.
(c) Mass of the atom is assumed to be uniformly distributed over the atom.
(d) Overall neutrality of atom cannot be explained.
(1) b, c (2) b, c, d (3) a, b, c (4) a, c, d

CAS182

7. For an electrically neutral atom, choose correct statement -
(a) Number of electron is equal to number of proton.
(b) Number of electron is equal to number of neutron.
(c) Positive charge of the nucleus is due to protons.
(d) Mass of nucleus is due to proton, neutrons and electrons.
(1) a and c (2) a, c, d (3) b, d (4) a, b, c

CAS183

8. Match the Column – I and Column - II

Column - I		Column - II	
(a)	all s orbitals have	(p)	The minimum value of principal quantum number is 3
(b)	for $\ell = 2$	(q)	different from other d-orbitals
(c)	Shape of d_{z^2} is	(r)	extra stability is associated
(d)	with completely and half filled subshells	(s)	Probability of finding the electron at a given distance is equal in all directions.

(1) (a) – (s), (b) – (p), (c) – (q), (d) – (r)
(2) (a) – (p), (b) – (r), (c) – (s), (d) – (q)
(3) (a) – (q), (b) – (p), (c) – (s), (d) – (r)
(4) (a) – (p), (b) – (q), (c) – (r), (d) – (s)

CAS188

9. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .
Assertion (A) : In hydrogen atom, energy of 4s is more than 3d.
Reason (R) : An orbital with lower value of $(n + \ell)$ has smaller energy than the orbital with higher value of $(n + \ell)$.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**
(2) **(A)** is correct but **(R)** is not correct.
(3) **(A)** is not correct but **(R)** is correct.
(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

CAS189

10. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .
Assertion (A) : p orbital has dumb-bell shape.
Reason (R) : Electrons present in p-orbital can have one of three value for 'm'.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**
(2) **(A)** is correct but **(R)** is not correct.
(3) **(A)** is not correct but **(R)** is correct.
(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

CAS190

11. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .
Assertion (A) : No two electrons in an atom can have the same value of four quantum numbers
Reason (R) : No two electrons in an atom can be simultaneously in the same shell, same subshell same orbitals and have same spin.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**
(2) **(A)** is correct but **(R)** is not correct.
(3) **(A)** is not correct but **(R)** is correct.
(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

CAS192

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	1	1	3	1	1	1	1	1	1	1	4

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Atomic Structure

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (3)

Rutherford's α scattering experiment proves that atom has Nucleus.

2. Ans. (1)

$_{27}A^{57}$ $_{30}B^{57}$

$n_B = 57-30 = 27$

3. Ans. (1)

$_6C^{14}$ $_7N^{15}$ $_9F^{17}$

n = 14 – 6 = 8 n = 15 – 7 = 8 n = 17 – 9 = 8

4. Ans. (3)

$CN^- \rightarrow 6 + 8 = 14e^-$ ✓

$OH^- \rightarrow 8 + 2 = 10e^-$

$CH_3^+ \rightarrow 6 + 2 = 8e^-$

$N_2 \rightarrow 7 + 7 = 14e^-$ ✓

$CO \rightarrow 6 + 8 = 14e^-$ ✓

5. Ans. (2)

$_Z^AX^{3-}$

no. of e^- = Z + 3 = 10

Z = 7

$_7^{14}X^{-3} \rightarrow$ no. of neutrons = 14 – 7 = 7

no. of neutrons in X_2 = 7 + 7 = 14

6. Ans. (2)

Isotones → n same

$_{14}^{30}Si \Rightarrow 30 - 14 = 16$

$_{15}^{31}P \Rightarrow 31 - 15 = 16$

$_{16}^{32}S \Rightarrow 32 - 16 = 16$

7. Ans. (1)

(i) same Z → Isotopes

(ii) same A (p+n) → Isobars

(iii) same n → Isotones

(iv) same n – p → Isodiaphers

(v) same Z → Isotopes.

8. Ans. (3)

Deuterium $_1H^2$ p → 1, e → 1, n → 1

9. Ans. (2)

Average Atomic weight

$$= \frac{85\times75 + 87\times25}{75+25}$$

$$= \frac{6375 + 2175}{100} = \frac{8550}{100} = 85.5$$

10. Ans. (1)

$$\text{Specific charge} = \frac{\text{Charge (e)}}{\text{mass (m)}}$$

$$\frac{S.C._p}{S.C._\alpha} = \frac{e_p/m_p}{e_\alpha/m_\alpha} = \frac{1/1}{2/4} = \frac{2}{1}$$

11. Ans. (3)

$_{13}Al^{27} \rightarrow$

a p b e c n

c : b : a

14 : 13 : 13

12. Ans. (4)

Let Percentage of ^{20}Ne = x%

^{22}Ne = 100 – x %

Average atomic weight ⇒ 20.2

$$= \frac{(20\times x) + 22(100-x)}{100}$$

2x = 180

x = 90

^{20}Ne = 90%

^{22}Ne = 10% → heavier Isotope.

13. Ans. (3)

$_{89}^{231}\gamma \rightarrow$ p 89, e 89, n 142

p, n, e

89 142 89

14. Ans. (3)

$_6^{13}C \rightarrow 7 - 6 = 1$ $_8^{17}O \rightarrow 9 - 8 = 1$

Same (n – p) Isodiaphers

15. Ans. (3)

D^+	He^+	H^+	He^{2+}
$[_1H^2]^{+1}$	$[_2He^4]^{+1}$	$[_1H^1]^{+1}$	$[_2He^4]^{2+}$
$\frac{e}{m}$ $\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{1}$	$\frac{2}{4} = \frac{1}{2}$

Maximum e/m ration is of H^+.

16. Ans. (3)

Same number of (n–p) $\Rightarrow$ Isotones

$^{77}_{32}Ge \Rightarrow 76 - 32 = 44$ = no. of n

no. of n of $^{77}_{33}As \Rightarrow 77 - 33 = 44$

no. of n of $^{78}_{34}Se \Rightarrow 78 - 34 = 44$

17. Ans. (3)

$^{14}_{7}N \rightarrow P$	n	A	
7	7	14	14– 10.5 = 3.5
$\frac{7}{2}$	7	10.5	

$$\% \text{ change} = \frac{3.5}{14} \times 100 = 25\% \text{ Reduces.}$$

18. Ans. (4)

$$\nu = \frac{c}{\lambda} \Rightarrow \lambda = \frac{c}{\nu} = \frac{3\times10^{8}}{8\times10^{15}}$$

$$= 0.375 \times 10^{-7}\text{m}$$

$$= 3.75 \times 10^{-8}\text{m}$$

19. Ans. (3)

$E = h\nu$

$E = 6.626 \times 10^{-34} \times 5.01 \times 10^{14}$

Thus the energy of 1 mole

$E = 33.19 \times 10^{-20} \times 6.023 \times 10^{23} = 200$ KJ/mole

20. Ans. (4)

$$mvr = \frac{2h}{2\pi} = x \text{ (given) [for H–atom]}$$

$$mvr = \frac{2h}{2\pi} = \text{[for Ist exited state of } Li^{+2} \text{ Ion]}$$

$$\boxed{mvr = x}$$

21. Ans. (3)

$$mvr = \frac{nh}{2\pi}$$

$$vr = \frac{nh}{2\pi m}$$

$$vr \propto \frac{1}{m}$$

22. Ans. (1)

$$r = 0.529 \times \frac{n^2}{Z} \text{ A°}$$

$$n^2 = \frac{4.761\times1}{0.529} = 9$$

$n = 3$

23. Ans. (1)

$$r_H = 0.053 \text{ nm} = 0.053 \times \frac{(1)^2}{1}$$

$$r_{Li^{+2}} = 0.053 \times \frac{(3)^2}{3}$$

$r_{Li^{+2}} = 0.0159$ nm

24. Ans. (3)

Ist orbit : IInd orbit : IIIrd orbit

$$0.529 \times \frac{(1)^2}{Z} : 0.529 \times \frac{(2)^2}{Z} : 0.529 \times \frac{(3)^2}{Z}$$

1 : 4 : 9

25. Ans. (2)

r_2 : r_5

$$0.529 \times \frac{(2)^2}{3} : 0.529 \times \frac{(5)^2}{3}$$

4 : 25

26. Ans. (4)

L : K	N : L
$(2)^2 : (1)^2$	$(4)^2 : (2)^2$
4 : 1	4 : 1

27. Ans. (4)

$$v = 2.188 \times 10^{6} \times \frac{z}{n} \text{ m/s}$$

$$= 2.188 \times 10^{6} \times \frac{4}{4} \text{ ms}^{-1}$$

$$= 2.188 \times 10^{3} \times \text{Kms}^{-1}$$

28. Ans. (4)

$r_{H_{(1)}} = 0.53$ A°

$$r_{H_{(2)}} = 0.53 \times \frac{(2)^2}{1}$$

$$= 0.53 \times \frac{4}{1} = 2.12 \text{ Å}$$

29. Ans. (1)

H :	D
n = 1 :	n = 1
z = 1 :	z = 1
$0.529 \times \frac{1}{1}$:	$0.529 \times \frac{1}{1}$

1 : 1

30. Ans. (2)

$V_1 : V_2 : V_3$

$2.188\times10^6\times\frac{1}{1} : 2.188\times10^6\times\frac{1}{2} : 2.188\times10^6\times\frac{1}{3}$

$1 : \frac{1}{2} : \frac{1}{3}$

31. Ans. (1)

For H–atom

$\Rightarrow E_n = -13.6\times\frac{1}{n^2}$

For He^+ (Z=2)

$\Rightarrow E_n = -13.6\times\frac{4}{n^2} = 4E_n$

32. Ans. (4)

$E_2 = -13.6\times\frac{1}{(2)^2} = -328 kJ/mol$

$E_4 = -13.6\times\frac{1}{(4)^2}$

$E_4 = -328\times\frac{1}{4} = -82 kJ/mol$

33. Ans. (1)

$\frac{E_1}{E_3}\Rightarrow\frac{-13.6\times\left(\frac{1}{12}\right)}{-13.6\times\left(\frac{1}{3^2}\right)} = \frac{-13.6}{E_3}$

$E_3 = -13.6\times\frac{1}{9}$

$E_3 = -1.51eV$

34. Ans. (3)

As we know T.E. = –K.E.

T.E. : – K.E.

1 : – 1

35. Ans. (3)

P.E. = 2E

P.E. = 4E [Double of Total Energy]

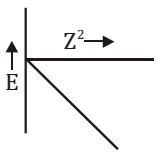
$E = \frac{P.E.}{4} = \frac{-27.2ev}{4} = -6.8eV$

36. Ans. (4)

$E = -13.6\times\frac{z^2}{n^2}$

$\Downarrow \quad \Downarrow$

$y = -mx$



37. Ans. (1)

As we know K.E. $\propto \frac{1}{r}$

So going from K–shell to N–shell radius increases so we can say K.E. decreases

38. Ans. (1)

e^- jumps from higher to lower energy realises & lower to higher energy absorbed.

$\frac{1}{\lambda} = R_H z^2\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$ --

$\frac{c}{\lambda} = R_H c z^2\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$ --

$\nu = R_H c z^2\left[\frac{1}{1}-\frac{1}{4}\right] = \frac{3R_H c z^2}{4}$ [minimum]

39. Ans. (4)

I.E. = 313.8 $[n_1 = 1, n_2 = \infty]$

Energy in IInd exited state

$E = -313.8\times\frac{1}{3^2} = -34.86 \simeq -35$ kcal/mol

40. Ans. (1)

In transition, closer to nucleus will require largest amount of energy, i.e. n = 1 to n = 2.

41. Ans. (2)

$E = \frac{P.E.}{2} = \frac{-3.02}{2} = -1.51$

$-1.51 = -13.6\times\frac{(1)^2}{(n)^2}$

$n^2 = \frac{13.6}{1.51} \simeq 9$

n = 3 [IInd exited state]

42. Ans. (4)

In transition, away from, lowest frequency radiation will be emitted.

43. Ans. (4)

Ist exited state (n=2)

IInd exited state (n=3)

$\frac{E_2}{E_3} = \frac{-13.6\times\frac{(1)^2}{(2)^2}}{-13.6\times\frac{(1)^2}{(3)^2}} = \frac{\frac{1}{4}}{\frac{1}{9}} = \frac{9}{4}$

44. Ans. (3)

$E_n = -313.6/n^2 = -34.84$

$n^2 = \frac{313.6}{34.84} \simeq 9$

$n = 3$

45. Ans. (1)

$\frac{P.E.}{2} = T.E$

$P.E. = 2\ T.E.$

$\frac{P.E.}{T.E} = \frac{2}{1} = 2$

46. Ans. (1)

$E = -13.6 \times \frac{z^2}{n^2}$

$E \propto \frac{-1}{n^2}$

$n \uparrow \quad E \uparrow$

Energy order $E, > E_2 > E_3$ is not true real order will be $E_3 > E_2 > E_1$

47. Ans. (1)

$E_2 - E_1 = 10.2ev$

$E_3 - E_2 = 1.89ev$

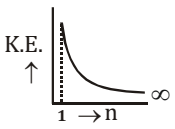
$E_4 - E_3 = 0.66ev$

$[E_2 - E_1] > [E_3 - E_2] > [E_4 - E_3]$

48. Ans. (3)

$K.E. = \frac{kZe^2}{2r}$

$K.E. \propto \frac{1}{n}$



K. E. ↑ then n ↓

49. Ans. (1)

First excitation potential of H

$E_2 - E_1 = 10.2eV$

50. Ans. (3)

M shell [n = 4]

$E_2 = -13.6 \times \frac{(1)^2}{(2)^2}$ First exited state (n = 2)

$E_2 = -13.6 \times \frac{1}{4}$

$E_2 = -3.4eV$

51. Ans. (4)

Second exited state [n = 3]

$I.E. = 13.6 \times \frac{1}{n^2} = 13.6 \times \frac{1}{9} = 1.51eV$

52. Ans. (4)

$E_2 = -13.6 \times \frac{(1)^2}{(2)^2}$

$= -3.4eV$

3.4 or less

53. Ans. (1)

$E_2 - E_1 = -13.6 \times (1)^2 \left[\frac{1}{4} - \frac{1}{1}\right]$

$E_2 - E_1 = 13.6 \left[\frac{3}{4}\right]$

54. Ans. (1)

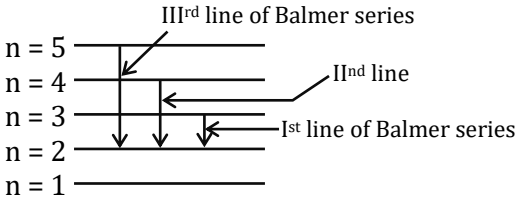
I. P. of He^+ Ion is equivalent to K.E. of first orbit.

Singly Ionized He^+ Contain only 1 electron & also e^- present in I^{st} orbit hence this energy is equal to K.E. I^{st} orbit.

55. Ans. (4)

He^+ is expected to be similar to Li^{+2} because number of electrons are similar.

56. Ans. (1)



N = 5 to n = 2

57. Ans. (3)

Energy difference maximum then wavelength will be lower.

$n_4 \longrightarrow n_1$

58. Ans. (4)

Lyman (n = 1) have highest energy in hydrogen spectrum.

$E \propto \frac{1}{n^2}$

$n \uparrow . E \downarrow$

59. Ans. (3)

$$\frac{\nu_L}{\nu_B}=\frac{RZ^2\left[\frac{1}{1}-\frac{1}{4}\right]}{RZ^2\left[\frac{1}{4}-\frac{1}{4}\right]}$$

$$\frac{\nu_L}{\nu_B}=\frac{\frac{3}{4}}{\frac{5}{36}}$$

$$\frac{\nu_L}{\nu_B}=\frac{3\times36}{5\times4}=\frac{27}{5}=5.4$$

60. Ans. (4)

$n_1 = 1, \quad n_2 = 2$

$$E_{2\to1}=h\nu_{2\to1}=13.6\times\left[\frac{1}{4}-\frac{1}{1}\right]$$

$$=13.6\times\frac{3}{4}=10.2\text{ev}$$

$$\Delta E\propto\nu\propto\frac{1}{\lambda}$$

61. Ans. (1)

$Mg^{+2}\to 10e^-$

$Al^{+3}\to 10e^-$

Same number of electrons.

62. Ans. (2)

$$\frac{\frac{1}{\lambda_L}}{\frac{1}{\lambda_B}}=\frac{\left(RZ^2\right)\left[\frac{1}{1^2}-\frac{1}{\infty^2}\right]}{\left(RZ^2\right)\left[\frac{1}{2^2}-\frac{1}{\infty^2}\right]}$$

$$=\frac{\lambda_L}{\lambda_B}=\frac{1}{4}=0.25$$

63. Ans. (4)

H atom $\frac{1}{\lambda_1}=R(1)^2\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$

He^+ atom $\frac{1}{\lambda_2}=R(2)^2\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$

$$\frac{\lambda_1}{\lambda_2}=\frac{R\times4\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]}{R\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]}$$

$$\frac{\lambda_1}{\lambda_2}=\frac{4}{1}$$

$$\lambda_2=\frac{\lambda_1}{4}$$

64. Ans. (2)

$$\nu_{maxL}=cRZ^2\left[\frac{1}{1^2}-\frac{1}{\infty^2}\right]=cRZ^2$$

$$\nu_{maxB}=cRZ^2\left[\frac{1}{4^2}-\frac{1}{\infty^2}\right]=\frac{cRZ^2}{16}$$

$$\frac{\nu_{maxL}}{\nu_{maxB}}=\frac{cRZ^2}{\frac{cRZ^2}{16}}=\frac{16}{1}$$

65. Ans. (1)

$$\lambda_{Li^{+2}}=\frac{1}{R(3)^2\left[\frac{1}{1^2}-\frac{1}{2^2}\right]}=\frac{1}{9R\left(\frac{3}{4}\right)}$$

$$\lambda_D=\frac{1}{R(1)^2\left[\frac{1}{1^2}-\frac{1}{2^2}\right]}=\frac{1}{\left(\frac{3}{4}\right)R}$$

$$\frac{\lambda_{Li^{+2}}}{\lambda_D}=\frac{R\left(\frac{3}{4}\right)}{9R\left(\frac{3}{4}\right)}=\frac{1}{9}$$

66. Ans. (2)

Visible $\to$ Balmer series

Infra red $\to$ Pashend, bracket, Pfund, Humphrey

U.V. light $\to$ Lyman series.

Atom can not emit γ–rays.

67. Ans. (3)

Lyman [2 to 1] $\to$ H $\to$ 10.2ev

Balmer [4 to 2] $\to$ 10.2ev

$$\Delta E=-13.6\times Z^2\left[\frac{1}{n_2^2}-\frac{1}{n_1^2}\right]$$

$$10.2=-13.6\times2^2\left[\frac{1}{4^2}-\frac{1}{2^2}\right]$$

$Z^2 = 4 \Rightarrow Z = 2\ (He^+)$

68. Ans. (3)

$$\nu=cRZ^2\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$$

$$\nu=3\times10^8\times1.1\times10^7\times(1)^2\left[\frac{1}{2^2}-\frac{1}{\infty^2}\right]$$

$\nu = 8.22\times10^{14}\ sec^{-1}$

69. Ans. (1)

$\bar{\nu} = Rz^2\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$

$\bar{\nu} = R(1)^2\left[\frac{1}{4}-\frac{1}{9}\right]$

$\bar{\nu} = \frac{5R}{36}cm^{-1}$

70. Ans. (4)

Ist Lyman series [2 to 1] $\rightarrow \lambda = 10.2ev\ [E_2 - E_1]$

$He^+ (n_2 \rightarrow n_1) \rightarrow \lambda = \Delta E = 10.2ev$

$(E_{n_2} - E_{n_1})_Z = (E_{n_2} - E_{n_1})_H \times Z^2$

$(10.2)He^+ = (E_{n_2} - E_{n_1})_H \times (2)^2$

$\Rightarrow (E_{n_2} - E_{n_1})_H = \frac{10.2}{4} = 2.55ev$

So $n_2 = 4$ & $n_1 = 2$

71. Ans. (1)

$n_2 = 6$ to $n_1 = 3$

Total spectral lines = $\frac{(n_2-n_1)(n_2-n_1+1)}{2}$

$= \frac{(6-3)(6-3+1)}{2} = 6$

72. Ans. (3)

Let n = 7

Total spectral lines = $\frac{(n_2-n_1)(n_2-n_1+1)}{2}$

$= \frac{(7-1)(7-1+1)}{2} = \frac{6\times7}{2} = 21$

73. Ans. (3)

1, 2, 3 $\rightarrow$ Coming to energy level A but 4, 5, 6 $\rightarrow$ are not coming to energy level A So they will not appear in the absorption spectrum.

74. Ans. (1)

U.V. region $\rightarrow$ Lyman series $\rightarrow$ 3

Ist Line $\rightarrow 2 \rightarrow 1$

IInd Line $\rightarrow 3 \rightarrow 1$

IIIrd Line $\rightarrow 4 \rightarrow 1$

$n_2 = 4$

$4 \rightarrow 3 \rightarrow$ Total lines = 1 (in paschen series)

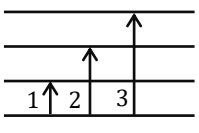
75. Ans. (4)

Total spectral Lines = $\frac{(n_2-n_1)(n_2-n_1+1)}{2}$

$= n_2 = 4, n_1 = 1$

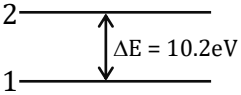
T. S. Lines = $\frac{(4-1)(4-1+1)}{2} = 6$

76. Ans. (1)



Absorption always take place from ground state. So Total 3 absorption lines.

77. Ans. (1)



$E_2 - E_1 = 10.2eV$

e^- will reach in 2nd orbit it we supply min. 10.2eV energy. So here no spectral line will be formed as energy supplied according to question is 9.9eV

78. Ans. (3)

$\nu = \frac{C}{\lambda} = \frac{3\times10^8}{1285\times10^9}$

Also $\nu = 3.39\times10^{15}\left[\frac{1}{3^2}-\frac{1}{n^2}\right]$

$\frac{1}{n^2} = \frac{1}{9} - \frac{3\times10^8}{1285\times10^{-9}} \times \frac{1}{3.29\times10^{15}}$

$n^2 = 25 \Rightarrow n = 5$

79. Ans. (3)

K. E. = 2.8×10^{-23}J

$m_e = 9.1 \times 10^{-31}$ kg

$\lambda = \frac{h}{\sqrt{2m_e\ KE}}$

$= \frac{6.6\times10^{-34}}{\sqrt{2\times9.1\times10^{-31}\times2.8\times10^{-23}}}$

$\lambda = 9.28 \times 10^{-8}$m

80. Ans. (2)

$2\pi r_n = n\lambda$

$2\pi\ 0.529\frac{n^2}{Z} = n\lambda$

$\lambda = 2\pi\ 0.529\times\frac{n}{Z}$

$Z = 1, n = 3$

$\lambda = 9.96 \times 10^{-8}$cm

81. Ans. (3)

n = 4, λ = 4 A°

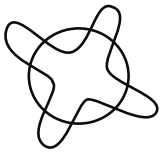
Circumference = 2πr = nλ

Circumference = nλ

Circumference = 4 × 4 A°

Circumference = 16 A°

82. Ans. (1)



Orbit is 4 so number of wave is also 4.

83. Ans. (1)

$$\lambda = \sqrt{\frac{150}{\nu}}$$

$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{\nu_2}{\nu_1}} = \sqrt{\frac{50}{200}} = \frac{1}{2}$$

84. Ans. (1)

2πr = nλ

Circumference = 2πr

= nλ

85. Ans. (1)

m = +2

ℓ = +2

n = +3

number of wave made by Bohr electron is 3.

86. Ans. (1)

$\Delta Pe = 32 \times 10^5$

$\Delta X_e = \Delta X_{He}$

ΔP_{He} = ?

$$\frac{\Delta Pe}{P_{He}} = \frac{\Delta X_{He}}{\Delta X_e} \Rightarrow \frac{32\times10^5}{\Delta P_{He}} = \frac{1}{1}$$

$\Delta P_{He} = 32 \times 10^5$

87. Ans. (3)

$m_e = 9.1 \times 10^{-28}$

$h = 6.6 \times 10^{-27} gcm^2/sec.$

Δv = 0.011% of 3×10^4

= 3.3cm/sec.

$$\Delta x = \frac{h}{4\pi m \Delta v} = \frac{6.6\times10^{-27}}{4\pi\times9.1\times10^{-28}\times3.3}$$

= 0.1748cm

88. Ans. (4)

Motor car and stationary Particles (2 and 3 both)

As we know it is not valid for macroscopic particles & stationary particles.

89. Ans. (1)

$$\lambda = \frac{h}{P} \Rightarrow P = \frac{h}{\lambda}$$

λ = 1 Å = 10^{-8}cm

$$P = \frac{6.6252\times10^{-27}\,erg/s}{1\times10^{-8}\,cm}$$

$P = 6.6252 \times 10^{-19} gcm/sec.$

90. Ans. (4)

$$\lambda = \frac{h}{mv}$$

$$\lambda \propto \frac{1}{m}$$

SO_2 has maximum mass so it will have least de-Broglie wavelength.

91. Ans. (1)

n = 3, l = 2, m = 2

i.e. $3d_{x^2-y^2}$ so 1 orbital is possible.

92. Ans. (4)

Number of possible orbitals = n^2

n = 3

So number of orbitals = $3^2 = 9$

93. Ans. (1)

2d ℓ should be less than n

n = 2

ℓ = 2

94. Ans. (3)

(1) 2s

(2) 3s

(3) 2p

(4) 1s

s orbital has 0 nodal plans so [3] is ans.

95. Ans. (2)

2s → n = 2

ℓ = 0

n – ℓ – 1 = 2 – 0 – 1 = 1

96. Ans. (1)

Number of orbitals $= 2\ell + 1$

$= 2\,[5] + 1$

$= 10 + 1 = 11$

97. Ans. (3)

n, ℓ, m $\Rightarrow$ 3 are required to specify the position of electron.

98. Ans. (1)

4d $\rightarrow$ n = 4

$\ell = 2$

m = –2, –1, 0, +1, +2

$s = \frac{+1}{2}, \frac{-1}{2}$

99. Ans. (4)

n = 3

ℓ = 0 to n –1

ℓ = 0 to 2

ℓ = 0, 1, 2

100. Ans. (1)

s < p < d < f

101. Ans. (2)

n = 5, m = 2

$\ell = 0 \rightarrow$ m = 0

$\ell = 1 \rightarrow$ m = –1, 0 +1

ℓ = 2, 3, 4 $\rightarrow$ m = +2 possible

102. Ans. (1)

Spin angular momentum = $\sqrt{s(s+1)}\ \frac{h}{2\pi}$

103. Ans. (3)

n = 6, m = 0, $\ell = 1$

6 $P_z \Rightarrow 2e^-$(maximum)

104. Ans. (3)

n = 4, total no. of orbitals will be $n^2 = 4^2 = 16$

105. Ans. (4)

n = 3, ℓ =2, m = +2, $s = +\frac{1}{2}$

1 electron

106. Ans. (1)

$\Psi_{n\ell m} = \Psi_{420}$

$4d_{z^2}$

107. Ans. (2)

4d $\rightarrow$ n = 4

$\ell = 2$

m = –2, –1, 0, +1, +2

$s = \frac{+1}{2}, \frac{-1}{2}$

ℓ = 1 is not possible----

108. Ans. (2)

2K, 8L, 11M, 2N $23e^-$

Z = 23

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^3$

$8e^-$s

109. Ans. (3)

$\ell = 3 \rightarrow$ f

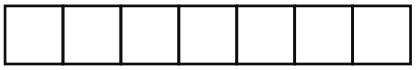
(1) 3p, 3

(2) 4f, 14

(3) 5f, 7 number of orbital

(4) 3d, 5

110. Ans. (4)



nf–orbital has $2e^-$ with Opposite spin.

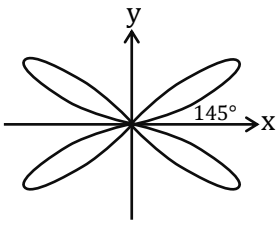
111. Ans. (3)

$3P_y$ n = 3

$\ell = 1$

m = –1 or +1----

112. Ans. (3)



At an angle of 45° from x & y axis

113. Ans. (3)

Angular nodal planer $= \ell$

$= 2$

d–orbital.

114. Ans. (4)

$\ell = 0 \Rightarrow$ s–orbital

s-orbital shape is symmetrical around the nucleus.

115. Ans. (4)

$\sum_{\ell=0}^{\ell=n-1} 2(2\ell+1)$

or

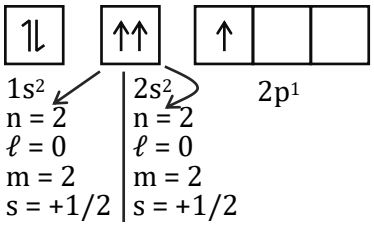
maximum $e^- = 2n^2$

let $n = 1 \Rightarrow$ maximum $2e^-$

Put $n = 1$ in all option we get $2e^-$

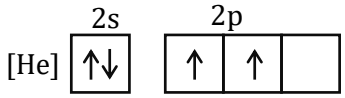
116. Ans. (2)

Acc. to pauli's exclusion Principle all 4 quantum number should be different for different electrons.

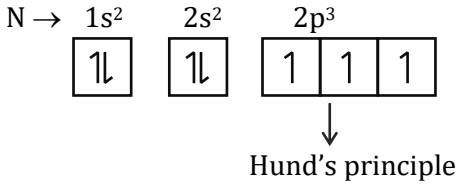


117. Ans. (1)

Acc. to hund's rule paring of electrons in orbitals belonging to same subshell does not takes place until each orbital has got on electron each with same spin.



118. Ans. (3)



119. Ans. (4)

3d $n + \ell = 3 + 2 = 5$

5p $n + \ell = 5 + 1 = 6$

4s $n + \ell = 4 + 0 = 4$

6d $n + \ell = 6 + 2 = 8$

$n + \ell \uparrow$ the maximum energy.

120. Ans. (1)

$n + \ell = 7$

$7 + 0 = 7 \Rightarrow 7s$ $\quad n > \ell$

$6 + 1 = 7 \Rightarrow 6p$ $\quad n > \ell$

$5 + 2 = 7 \Rightarrow 5d$ $\quad n > \ell \Rightarrow$ 4 Subshell

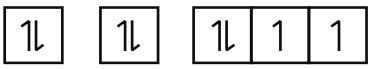
$4 + 3 = 7 \Rightarrow 4f$ $\quad n > \ell$

$3 + 4 = 7$ $\quad n > \ell$

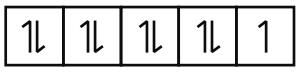
$2 + 5 = 7$ $\quad n > \ell$

$1 + 6 = 7$ $\quad n > \ell$

121. Ans. (1)



122. Ans. (1)

$d^9 \Rightarrow$ 

Total spin = $\frac{1}{2}$ × (number of unpaired electrons)

$= \frac{1}{2} \times 1 = \frac{1}{2}$

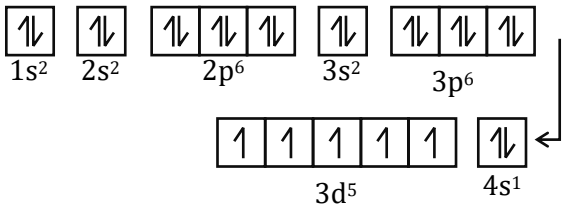
123. Ans. (4)

In case of H-atom n = 3

all options have value n = 3 so neither absorption nor emission takes place.

124. Ans. (3)

$_{24}Cr \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$



Total orbital with paired e^- = 15

125. Ans. (2)

$_{11}Na \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^1$

$3s^1 \Rightarrow$ $n = 3, \ell = 0, m = 0$

$s = +\frac{1}{2}, -\frac{1}{2}$

126. Ans. (3)

$_{24}Cr \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$

$4s^1 \Rightarrow$ $n = 4, \ell = 0, m = 0$

$s = +\frac{1}{2}, -\frac{1}{2}$

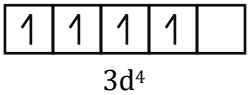
127. Ans. (4)

$3p \Rightarrow$ $n = 3, \ell = 1, m = -1, 0, +1$

$s = +\frac{1}{2}, -\frac{1}{2}$

128. Ans. (1)

$Cr^{+2} \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^4\ 4s^\circ$

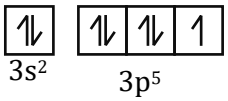


Total 4 unpaired electrons.

129. Ans. (3)

Z = 17

$_{17}Cl \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$



Total 3 orbitals.

130. Ans. (2)

[Ar] $3d^5$

Actual number of e^-s = 23 + 3 = 26

131. Ans. (3)

$_{29}Cu \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^{10}$

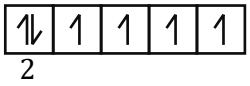
$_{24}Cr \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$

$_{30}Zn \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}$

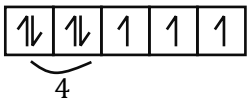
$_{19}K \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$

132. Ans. (2)

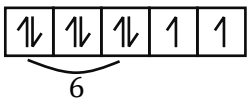
$_{26}Fe^{+2} \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^6$



$_{27}Co^{+2} \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^7$



$_{28}Ni^{+2} \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^8$



Total = 2 + 4 + 6 = $12e^-$

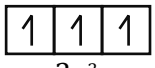
133. Ans. (3)

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^6$

n = 4, ℓ = 0, m = 0

$s = +\frac{1}{2}, -\frac{1}{2}$

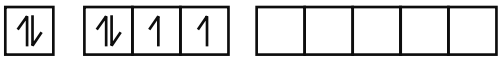
134. Ans. (1)



maximum 3 unpaired e^-

$15e^- \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$

135. Ans. (3)



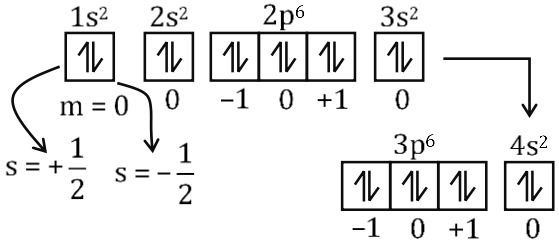
It shows ground state configuration

136. Ans. (1)

2K, 8L, 8M, 2N

Number of e^- is 20

Z = 20



6 electrons $\Rightarrow$ m = 0

$S = +\frac{1}{2}$

Exercise-II (Previous Year Questions)

1. Ans. (1)

n = 3, ℓ = 1, m_ℓ = 0 $\Rightarrow$ 1 orbital

2. Ans. (4)

$\lambda = 45nm = 45 \times 10^{-9}m$

$h = 6.63 \times 10^{-34}$ J.s

$c = 3 \times 10^8$ m/s

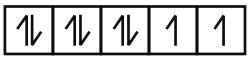
$E = \frac{hc}{\lambda} = \frac{6.63\times10^{-34}\times3\times10^8}{45\times10^{-9}}$

$E = 0.442 \times 10^{-17}$ J

$E = 4.42 \times 10^{-18}$ J

3. Ans. (2)

$_{28}Ni^{2+}\ 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^8$



$3d^8 \rightarrow$ 2 unpaired electrons

Magnetic moment $\Rightarrow \sqrt{n(n+2)}$

$= \sqrt{2(2+2)} = \sqrt{8}$ = 2.83 B.M.

4. Ans. (3)

ClO_3^- and SO_3^{-2} are isoelectronic and isostructural.

5. Ans. (1)

$_{26}Fe^{+2} \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^6$

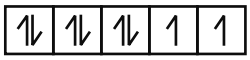
$\Rightarrow$ 6 d- electrons

$_{17}Cl \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5 \Rightarrow 6 + 5 = 11$ electrons.

p-electrons in Cl

6. Ans. (4)

$_{28}Ni^{+2} \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^8$



Number of unpaired e^- s = 2

Magnetic moment $\Rightarrow \sqrt{n(n+2)}$

$= \sqrt{2(2+2)} = \sqrt{8}$ = 2.84 BM.

7. Ans. (4)

Orbital angular momentum = $\sqrt{\ell(\ell+1)}\,\frac{h}{2\pi}$

for d-orbital $\ell = 2$, $\frac{h}{2\pi} = \hbar$

The Orbital angular momentum

$= \sqrt{2(2+1)}\,\hbar$

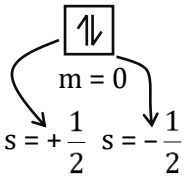
$= \sqrt{6}\,\hbar$

8. Ans. (2)

$_{22}Ti \Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^2$

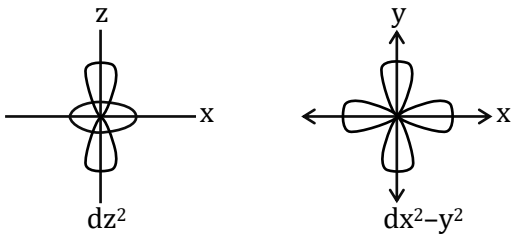
3s 3p 4s 3d

9. Ans. (4)



Spin quantum number (5)

10. Ans. (1)



Both have e^- density along the axis

11. Ans. (3)

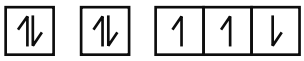
n = 3, $\ell = 1$

3p maximum 2 electrons are possible in 1 orbital

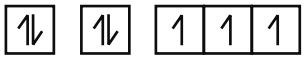
12. Ans. (3)

1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f < ---

13. Ans. (3)

N → 

This is wrong statement



14. Ans. (2)

In Hydrogen atom spectrum Balmer series Transitions falls in visible region.

15. Ans. (3)

4 f

Orbital angular node (e) = 3

Total node = n – ~~ℓ~~ – 1 + ~~ℓ~~

= n = 1

= 4 – 1 = 3

16. Ans. (2)

$mvr_n = \frac{nh}{2\pi}$

$P = \frac{nh}{2\pi r_n}$ ...(1)

$\lambda = \frac{h}{p} \Rightarrow \lambda = \frac{n}{2\pi r_n}$

we know $r_n = 52.9n^2$

$r_2 = 52.9(2)^2 = 211.6$ pm

so $\lambda = \frac{2\pi(211.6)}{2}$ = 211.6 π pm

17. Ans. (2)

$^{175}_{71}Lu \Rightarrow p \to 71$

$e \to 71$

$n = 175 - 71 = 104$

p	n	e
71	104	71

18. Ans. (1)

3s angular nodes $\ell = 0$

Radial nodes n – ℓ – 1

= 3 – 0 – 1 = 2

19. Ans. (1)

$E = \frac{hc}{\lambda} = h\nu$

So $\frac{c}{\lambda} = \nu \Rightarrow \lambda = \frac{c}{\nu}$

$\lambda = \frac{3\times10^8}{1368\times10^3} = 219.3\text{m}$

20. Ans. (3)

d_{xy}, d_{y^2}, d_{zx} & $d_{x^2-y^2}$ these 4 orbitals has similar shape but d_{z^2} has different shape.

21. Ans. (4)

$$r_n \propto \frac{n^2}{z}$$

$$\frac{r_3(Li^{+2})}{r_2(He^{+})} = \frac{(3)^2}{3} \times \frac{2}{(2)^2}$$

$$\frac{r_3(Li^{+2})}{105.8} = \frac{3}{2}$$

$r_3(Li^{+2}) = \frac{3}{2} \times 105.8 = 158.7$ pm

22. Ans. (4)

(a) $n = 2, \ell = 1 \Rightarrow 2p \Rightarrow$ (iii)
(b) $n = 3, \ell = 2 \Rightarrow 3d \Rightarrow$ (iv)
(c) $n = 3, \ell = 0 \Rightarrow 3s \Rightarrow$ (ii)
(a) $n = 2, \ell = 0 \Rightarrow 2s \Rightarrow$ (i)

23. Ans. (4)

$\lambda = 300$ nm $= 300 \times 10^{-9}$m

$$E = \frac{hc}{\lambda} = \frac{(6.626\times10^{-34})(3\times10^{8})}{300\times10^{-9}}$$

$= 6.626 \times 10^{-9}$ J

for 1 mole $\Rightarrow 6.626 \times 10^{-9} \times 6.023 \times 10^{23}$

$= 3.99 \times 10^{5}$ J/mole

K.E. = 1.68×10^{5} J/mole

$h\nu_0 = h\nu$ – K.E.

$h\nu_0 = 3.99 \times 10^{5} - 1.68\ 10^{5}$

$= 2.31 \times 10^{5}$ J/mole

minimum energy = 2.31×10^{5} J/mole^{-1}

24. Ans. (3)

It is statement based question.
Statements B, C & E are correct.
(B) Mass of the electron is 9.10939×10^{-31} kg
(C) All the isotopes of given elements show same chemical properties.
(E) Dalton's atomic theory, regarded the atom as an ultimate particle of matter.

25. Ans. (4)

Number of permissible values of magnetic quantum number for a given value of azimuthal quantum (ℓ) $\Rightarrow n_m = 2\ell + 1$

$$\Rightarrow \ell = \frac{n_m - 1}{2}$$

26. Ans. (2)

$m_\ell \rightarrow$ orientation of orbital

$m_s \rightarrow$ orientation of spin of e$^-$

$\ell \rightarrow$ shape of orbital

$n \rightarrow$ size of orbital

27. Ans. (1)

$$E \propto \frac{Z^2}{n^2}$$

$$\frac{E_1}{E_2} = \frac{Z_1^2}{n_1^2} \times \frac{n_2^2}{Z_2^2}$$

$He^{+} \rightarrow n_1 = 1$ & $Z_1 = 2$

$Be^{+3} \rightarrow n_2 = 2$ & $Z_2 = 4$

$$\frac{E_1}{E_2} = \frac{2^2}{1^2} \times \frac{2^2}{4^2} = 1$$

$E_1 = E_2$

28. Ans. (2)

$\uparrow(n+\ell)$ $\uparrow$Energy so correct one is

I > III > II > IV

29. Ans. (3)

Both statement I and statement II are true.

Exercise-III (Analytical Questions)

1. Ans. (1)

Azimuthal quantum number (ℓ) $\Rightarrow$ represents subshell also determines the shape of orbital.
Number of orbitals in subshell = $(2\ell + 1)$

2. Ans. (1)

n, ℓ, m can be same for two electrons. But s value should be different.

eg. $\underset{2s}{\boxed{\uparrow\downarrow}}$ $n = 2, \ell = 0, m = 0, s = +\frac{1}{2}$ & $-\frac{1}{2}$

3. Ans. (3)

$\rightarrow$ Energy order.

$1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f$ ----

(I) Incorrect.

$(n + \ell) \Rightarrow$ same then less n will have less energy. (II) Correct

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4. Ans. (1)

→ Energy of stationary state $e^- \Rightarrow E_n = R_H \left[\frac{-1}{n^2}\right]$ (I)correct

→ If electron is present in n^{th} orbit then it feel more attraction compare to electron present at infinity distance. So energy is less for n^{th} orbit of electron.

5. Ans. (1)

Cathode ray can travel in straight line in absence of electric or magnetic field. Also Characteristics does not depend upon the nature of gas because Characteristics particles are electrons.

6. Ans. (1)

As per the Thomson model of atom. b, c are correct.

7. Ans. (1)

For neutral atom

→ number of electrons = number of protons.

→ positive charge of nucleus is due to protons.

8. Ans. (1)

9. Ans. (1)

10. Ans. (1)

11. Ans. (4)

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Exercise - I (Conceptual Questions) Build Up Your Understanding

OXIDATION NUMBER

1. In $[Ni(CO)_4]$, the oxidation state of Ni is :
(1) 4 (2) 0 (3) 2 (4) 8
CRR001

2. The oxidation number of nitrogen in NH_2OH is:
(1) 0 (2) +1 (3) –1 (4) –2
CRR002

3. Of the following elements, which one has the same oxidation state in all of its compounds ?
(1) Hydrogen (2) Fluorine
(3) Carbon (4) Oxygen
CRR003

4. Oxidation number of fluorine in OF_2 is :
(1) +1 (2) +2 (3) –1 (4) –2
CRR004

5. The oxidation number of C in CH_4, CH_3Cl, CH_2Cl_2, $CHCl_3$ and CCl_4 are respectively :
(1) +4, +2, 0, –2, –4
(2) +2, +4, 0, –4, –2
(3) –4, –2, 0, +2, +4
(4) –2, –4, 0, +4,. +2
CRR005

6. Oxidation state of oxygen in hydrogen peroxide is
(1) –1 (2) +1 (3) 0 (4) –2
CRR007

7. Which one of the following statements is not correct?
(1) Oxidation state of S in $(NH_4)_2S_2O_8$ is +6
(2) Oxidation number of Os in OsO_4 is +8
(3) Oxidation state of S in H_2SO_5 is +8
(4) Oxidation number of O in KO_2 is $-\frac{1}{2}$
CRR008

8. Which of the following shows highest oxidation number in combined state :
(1) Os (2) Ru
(3) Both (1) and (2) (4) None
CRR009

9. Oxidation number of sodium in sodium amalgam(Na-Hg) is :
(1) +2 (2) +1 (3) –3 (4) Zero
CRR010

10. Oxidation number of C in HNC is :
(1) +2 (2) –3 (3) +3 (4) Zero
CRR011

11. Oxidation number of Fe in $Fe_{0.94}O$ is :
(1) 200 (2) 200/94
(3) 94/200 (4) None
CRR012

12. Oxidation number of carbon in carbon suboxide (C_3O_2) is :
(1) $\frac{+2}{3}$ (2) $\frac{+4}{3}$ (3) +4 (4) $\frac{-4}{3}$
CRR013

13. Oxidation number of sulphur in $Na_2S_2O_3$ would be :-
(1) +2 (2) +4 (3) –2 (4) 0
CRR014

14. Two oxidation states for chlorine are found in the compound :
(1) $CaOCl_2$ (2) KCl (3) $KClO_3$ (4) Cl_2O_7
CRR015

15. –1/3 oxidation state of nitrogen will be obtained in case of :
(1) Ammonia (NH_3)
(2) Hydrazoic acid (N_3H)
(3) Nitric oxide (NO)
(4) Nitrous oxide (N_2O)
CRR016

16. Compound $YBa_2Cu_3O_7$ is a super conductor. The O.N. of the copper in the compound will be: [O. No. of Y=+3]
(1) +7/3 (2) zero (3) +2 (4) +1
CRR017

17. The oxidation state of iodine in $H_4IO_6^-$ is :-
(1) +7 (2) –1 (3) +5 (4) +1
CRR018

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18. Amongst the following, identify the species with an atom in + 6 oxidation state:-
(1) MnO_4^- (2) $Cr(CN)_6^{3-}$
(3) NiF_6^{2-} (4) CrO_2Cl_2
CRR019

19. In which of the following compounds iron has lowest oxidation state:-
(1) $FeSO_4(NH_4)_2SO_4.6H_2O$
(2) $K_4[Fe(CN)_6]$
(3) $[Fe(CO)_5]$
(4) $Fe_{0.94}O$
CRR021

20. Select the compound in which the oxidation number of oxygen is –1:-
(1) H_2O (2) O_2F_2 (3) Na_2O (4) BaO_2
CRR022

21. Match List - I (compound) with list - II (Oxidation state of N) and select the correct answer using the codes given below the list:-

List - I	List-II
(A) KNO_3	(a) – 1/3
(B) HNO_2	(b) – 3
(C) NH_4Cl	(c) 0
(D) NaN_3	(d) + 3
	(e) + 5

Codes are:-

	A	B	C	D
(1)	e	d	b	a
(2)	e	b	d	a
(3)	d	e	a	c
(4)	b	c	d	e

CRR023

22. In which of the following pair, oxidation number of Fe is same :-
(1) $K_3[Fe(CN)_6]$, Fe_2O_3
(2) $Fe(CO)_5$, Fe_2O_3
(3) Fe_2O_3, FeO
(4) $Fe_2(SO_4)_3$, $K_4[Fe(CN)_6]$
CRR024

23. In the conversion of Br_2 to BrO_3^-, the oxidation state of bromine changes from :-
(1) 0 to 5 (2) 1 to 5
(3) 0 to –3 (4) 2 to 5
CRR025

24. The sum of oxidation states of sulphur in $H_2S_2O_8$ is :-
(1) +2 (2) +6 (3) +7 (4) +12
CRR026

25. In which of the following compounds of Cr, the oxidation number of Cr is not +6 :-
(1) CrO_3 (2) CrO_2Cl_2
(3) Cr_2O_3 (4) $K_2Cr_2O_7$
CRR027

26. Oxidation state of cobalt in $[Co(NH_3)_4(H_2O)Cl]SO_4$ is
(1) 0 (2) +4 (3) –2 (4) +3
CRR028

27. Oxidation number of carbon in graphite is :-
(1) Zero (2) +1 (3) +4 (4) +2
CRR029

28. Oxidation number of 'N' in N_3H (hydrazoic acid) is :-
(1) $-\frac{1}{3}$ (2) –3
(3) +3 (4) $+\frac{2}{3}$
CRR030

29. Which of the following doesn't have +5 oxidation state of phosphorus ?
(1) Phosphorus acid (H_3PO_3)
(2) Orthophosphoric acid (H_3PO_4)
(3) Meta phosphoric acid (HPO_3)
(4) Pyro phosphoric acid ($H_4P_2O_7$)
CRR031

30. The oxidation number of arsenic atom in H_3AsO_4 is :-
(1) –1 (2) –3 (3) +3 (4) +5
CRR032

31. In substance $Mg(HXO_3)$, the oxidation number of X is :-
(1) 0 (2) +2 (3) +3 (4) +4
CRR033

32. The oxidation number of phosphorus in PH_4^+, PO_2^{3-}, PO_4^{3-} and PO_3^{3-} are respectively :–
(1) –3, +1, +3, +5 (2) –3, +3, +5, +1
(3) +3, –3, +5, +1 (4) –3, +1, +5, +3
CRR034

33. Which of the following compounds are arranged in increasing oxidation number of S :–
(1) H_2SO_3, H_2S, H_2SO_4, $H_2S_2O_3$
(2) $H_2S_2O_3$, H_2SO_3, H_2S, H_2SO_4
(3) H_2S, H_2SO_3, H_2SO_4, $H_2S_2O_3$
(4) H_2S, $H_2S_2O_3$, H_2SO_3, H_2SO_4

CRR035

34. Iodine shows the highest oxidation state in the compound :–
(1) KI (2) KI_3 (3) IF_5 (4) KIO_4

CRR036

35. The sum of the oxidation states of all the carbon atoms present in the compound C_6H_5CHO is :
(1) –4 (2) 3 (3) + 5 (4) – 4/7

CRR037

APPLICATIONS OF REDOX REACTIONS

36. A reducing agent is a substance which can :
(1) Accept electrons
(2) Donate electrons
(3) Accept protons
(4) Donate protons

CRR038

37. The reaction $H_2S + H_2O_2 \rightarrow S + 2H_2O$ manifests :
(1) Oxidising action of H_2O_2
(2) Reducing nature of H_2O_2
(3) Acidic nature of H_2O_2
(4) Alkaline nature of H_2O_2

CRR039

38. If an element is in its lowest oxidation state, under proper conditions it can act as :
(1) Reducing agent
(2) An oxidising agent
(3) Oxidising as well as reducing agent
(4) Neither oxidising nor reducing agent

CRR040

39. In a reaction of
H_2O (steam) + C (red hot) $\rightarrow$ CO + H_2
(1) H_2O is the reducing agent
(2) H_2O is the oxidising agent
(3) Carbon is the oxidising agent
(4) Oxidation-reduction does not occur

CRR041

40. The compound that can work both as an oxidising as well as reducing agent is :
(1) $KMnO_4$ (2) H_2O_2
(3) $Fe_2(SO_4)_3$ (4) $K_2Cr_2O_7$

CRR042

41. HNO_2 acts as an oxidant with which one of the following reagent:-
(1) $KMnO_4$ (2) H_2S
(3) $K_2Cr_2O_7$ (4) Br_2

CRR043

42. In which of the following reaction H_2O_2 acts as reducing agent :-
(1) $2FeCl_2 + 2HCl + H_2O_2 \rightarrow 2FeCl_3 + 2H_2O$
(2) $Cl_2 + H_2O_2 \rightarrow 2HCl + O_2$
(3) $2HI + H_2O_2 \rightarrow 2H_2O + I_2$
(4) $H_2SO_3 + H_2O_2 \rightarrow H_2SO_4 + H_2O$

CRR044

43. When H_2 reacts with Na, it acts as :-
(1) Oxidising agent
(2) Reducing agent
(3) Both
(4) Cannot be predicted

CRR045

44. In the course of a chemical reaction an oxidant –
(1) Loses electron
(2) Gains electron
(3) Both loses and gain electrons
(4) Electron change does not occur

CRR046

45. In the reaction:-
$C + 4HNO_3 \rightarrow CO_2 + 2H_2O + 4NO_2$
HNO_3 acts as :–
(1) An oxidising agent
(2) An acid
(3) A reducing agent
(4) A base

CRR047

46. A compound contains atoms A, B and C. The oxidation number of A is +2, of B is +5 and of C is –2. The possible formula of the compound is :
(1) ABC_2 (2) $B_2(AC_3)_2$
(3) $A_3(BC_4)_2$ (4) $A_3(B_4C)_2$

CRR048

47. Equivalent weight of N_2 in the change $N_2 \rightarrow NH_3$ is

(1) $\frac{28}{6}$ (2) 28 (3) $\frac{28}{2}$ (4) $\frac{28}{3}$

CRR049

48. Equivalent weight of NH_3 in the change $N_2 \rightarrow NH_3$ is :

(1) $\frac{17}{6}$ (2) 17 (3) $\frac{17}{2}$ (4) $\frac{17}{3}$

CRR050

49. In the reaction, $2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$, the eq. wt. of $Na_2S_2O_3$ is equal to its :

(1) Mol. wt. (2) Mol. wt./2
(3) 2 x Mol. wt. (4) Mol. wt./6

CRR051

50. In the reaction, $VO + Fe_2O_3 \rightarrow FeO + V_2O_5$, the eq. wt. of V_2O_5 is equal to its :

(1) Mol. wt. (2) Mol. wt./8
(3) Mol .wt./6 (4) Mol. wt./2

CRR052

51. Molecular weight of $KBrO_3$ is M. What is its equivalent weight, if the reaction is :

$BrO_3^- \rightarrow Br^-$ (acidic medium)

(1) M (2) M/4 (3) M/6 (4) 6M

CRR053

52. In the reaction : $A^{-n_2} + xe^- \rightarrow A^{-n_1}$, here x will be

(1) $n_1 + n_2$ (2) $n_2 - n_1$
(3) $n_1 - n_2$ (4) $n_1 . n_2$

CRR054

53. What would be the equivalent weight of the reductant in the reaction :

$[Fe(CN)_6]^{-3} + H_2O_2 + 2OH^- \rightarrow 2[Fe(CN)_6]^{4-} + 2H_2O + O_2$

[Given : Fe = 56, C = 12, N = 14, O = 16, H = 1]

(1) 17 (2) 212 (3) 34 (4) 32

CRR055

54. Equivalent weight of FeC_2O_4 in the change : $FeC_2O_4 \rightarrow Fe^{3+} + CO_2$ is :

(1) M/3 (2) M/6 (3) M/2 (4) M/1

CRR056

55. The number of mole of oxalate ions oxidised by one mole of MnO_4^- is :

(1) 1/5 (2) 2/5 (3) 5/2 (4) 5

CRR057

56. In a reaction 4 mole of electrons are transferred to one mole of HNO_3 when it acts as an oxidant. The possible reduction product is:

(1) (1/2) mole N_2 (2) (1/2) mole N_2O
(3) 1 mole of NO_2 (4) 1 mole NH_3

CRR058

57. The equivalent weight of $MnSO_4$ is half of its molecular weight when it is converted to :-

(1) Mn_2O_3 (2) MnO_2
(3) MnO_4^- (4) MnO_4^{-2}

CRR059

58. In the following change,

$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

If the atomic weight of iron is 56, then its equivalent weight will be :-

(1) 42 (2) 21 (3) 63 (4) 84

CRR060

59. $Cr_2O_7^{-2} + I^- + H^+ \rightarrow Cr^{+3} + I_2 + H_2O$

The equivalent weight of the reductant in the above equation is :– (At. wt. of Cr=52, I=127)

(1) 26 (2) 127 (3) 63.5 (4) 10.4

CRR061

60. How many moles of $KMnO_4$ are reduced by 1 mole of ferrous oxalate in acidic medium:-

(1) $\frac{1}{5}$ (2) $\frac{5}{3}$ (3) $\frac{1}{3}$ (4) $\frac{3}{5}$

CRR062

REDOX REACTIONS

61. Which one of the following is a redox reaction?

(1) $H_2 + Br_2 \rightarrow 2HBr$
(2) $2NaCl + H_2SO_4 \rightarrow Na_2SO_4 + 2HCl$
(3) $HCl + AgNO_3 \rightarrow AgCl + HNO_3$
(4) $NaOH + HCl \rightarrow NaCl + H_2O$

CRR063

62. Which of the following is not a redox change ?

(1) $2H_2S + SO_2 \rightarrow 2H_2O + 3S$
(2) $2BaO + O_2 \rightarrow 2BaO_2$
(3) $BaO_2 + H_2SO_4 \rightarrow BaSO_4 + H_2O_2$
(4) $2KClO_3 \rightarrow 2KCl + 3O_2$

CRR064

63. In the reaction, $Cl_2 + OH^- \rightarrow Cl^- + ClO_4^- + H_2O$, chlorine is :
(1) Oxidised
(2) Reduced
(3) Oxidised as well as reduced
(4) Neither oxidised nor reduced

CRR065

64. Which is a redox reaction :
(1) $2CuI_2 \rightarrow CuI + I_2$
(2) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
(3) $NH_4Cl + NaOH \rightarrow NH_3 + NaCl + H_2O$
(4) $Cr_2(SO_4)_3 + 6KOH \rightarrow 2Cr(OH)_3 + 3K_2SO_4$

CRR066

65. Which of the following example does not represent disproportionation -
(1) $MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$
(2) $2H_2O_2 \rightarrow 2H_2O + O_2$
(3) $4KClO_3 \rightarrow 3KClO_4 + KCl$
(4) $3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$

CRR067

66. The decomposition of $KClO_3$ to KCl and O_2 on heating is an example of :
(1) Intermolecular redox change
(2) Intramolecular redox change
(3) Disproportionation or auto redox change
(4) Comproportionation

CRR068

67. Which of the following change represents a disproportionation reaction (s) :
(1) $Cl_2 + 2OH^- \rightarrow ClO^- + Cl^- + H_2O$
(2) $Cu_2O + 2H^+ \rightarrow Cu + Cu^{2+} + H_2O$
(3) $2HCuCl_2 \xrightarrow[\text{Water}]{\text{dilution with}} Cu + Cu^{2+} + 4Cl^- + 2H^+$
(4) All of the above

CRR069

68. How many electrons should X_2H_4 liberate so that in the new compound X shows oxidation number of -½ (E.N. X > H)
(1) 10 (2) 4 (3) 3 (4) 2

CRR070

69. Which one of the following is not a redox reaction :-
(1) $CaCO_3 \rightarrow CaO + CO_2$
(2) $2H_2 + O_2 \rightarrow 2H_2O$
(3) $Na + H_2O \rightarrow NaOH + \frac{1}{2}H_2$
(4) $MnCl_3 \rightarrow MnCl_2 + \frac{1}{2}Cl_2$

CRR071

70. In the reaction -
$MnO_4^- + SO_3^{2-} + H^+ \rightarrow SO_4^{-2} + Mn^{+2} + H_2O$
(1) MnO_4^- and H^+ both are reduced
(2) MnO_4^- is reduced and H^+ is oxidised
(3) MnO_4^- is reduced and SO_3^{2-} is oxidised
(4) MnO_4^- is oxidised and SO_3^{2-} is reduced

CRR072

71. $I_2 + KI \rightarrow KI_3$
In the above reaction:-
(1) Only oxidation taken place
(2) Only reduction takes place
(3) Both the above
(4) Neither oxidation nor reduction

CRR073

72. Which of the following reaction represents the oxidising behaviour of H_2SO_4 :-
(1) $2PCl_5 + H_2SO_4 \rightarrow 2POCl_3 + 2HCl + SO_2Cl_2$
(2) $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$
(3) $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$
(4) $2HI + H_2SO_4 \rightarrow I_2 + SO_2 + 2H_2O$

CRR074

73. Select the example of disproportionation reaction
(1) $BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + 2HCl$
(2) $NH_4NO_3 \rightarrow N_2O + 2H_2O$
(3) $4H_3PO_3 \rightarrow PH_3 + 3H_3PO_4$
(4) $AgCl + 2NH_3 \rightarrow Ag(NH_3)_2Cl$

CRR075

74. Which of the following reaction involves oxidation & reduction :-
(1) $NaBr + HCl \rightarrow NaCl + HBr$
(2) $HBr + AgNO_3 \rightarrow AgBr + HNO_3$
(3) $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$
(4) $H_2 + Br_2 \rightarrow 2HBr$

CRR076

75. The reaction
$2K_2MnO_4 + Cl_2 \rightarrow 2KMnO_4 + 2KCl$
is an example of
(1) Redox
(2) Reduction only
(3) Neutralization
(4) Disproportionation

CRR077

76. Which of the following reaction involves neither oxidation nor reduction :–
(1) $CrO_4^{2-} \rightarrow Cr_2O_7^{2-}$ (2) $Cr \rightarrow CrCl_3$
(3) $Na \rightarrow Na^+$ (4) $2S_2O_3^{2-} \rightarrow S_4O_6^{2-}$
CRR078

77. $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
Zn undergoes –
(1) Reduction
(2) Oxidation
(3) Both oxidation and reduction
(4) Neither oxidation nor reduction
CRR079

BALANCING OF REDOX REACTIONS

78. Balance the following given half reaction for the unbalanced whole reaction :
$CrO_4^{2-} \rightarrow CrO_2^- + OH^-$ is :
(1) $CrO_4^- + 2H_2O + 3e^- \rightarrow CrO_2^- + 4OH^-$
(2) $2CrO_4^{-2} + 8H_2O \rightarrow CrO_2^- + 4H_2O + 8OH^-$
(3) $CrO_4^{-2} + H_2O \rightarrow CrO_2^- + H_2O + OH^-$
(4) $3CrO_4^{-2} + 4H_2O + 6\,e^- \rightarrow 2CrO_2^{-1} + 8OH^-$
CRR080

79. Choose the set of coefficients that correctly balances the following equation :
$x\,Cr_2O_7^{2-} + yH^+ + z\,e^- \rightarrow a\,Cr^{+3} + bH_2O$

	x	y	z	a	b
(1)	2	14	6	2	7
(2)	1	14	6	2	7
(3)	2	7	6	2	7
(4)	2	7	6	1	7

CRR081

80. In the reaction:
$MnO_4^- + xH^+ + ne^- \rightarrow Mn^{2+} + yH_2O$
What is the value of n :
(1) 5 (2) 8 (3) 6 (4) 3
CRR082

81. The number of electrons required to balance charge in the following equation –
$MnO_4^- + 2H_2O \longrightarrow MnO_2 + 4OH^-$ is
(1) 5 (2) 4
(3) 3 (4) 2
CRR083

82. $2KMnO_4 + 5H_2S + 6H^+ \rightarrow 2Mn^{2+} + 2K^+ + 5S + 8H_2O$
In the above reaction, how many electrons would be involved in the oxidation of 1 mole of reductant?
(1) Two (2) Five (3) Ten (4) One
CRR084

83. The value of *n* in :
$MnO_4^- + 8\,H^+ + ne^- \rightarrow Mn^{2+} + 4\,H_2O$ is
(1) 5 (2) 4 (3) 3 (4) 2
CRR085

84. For the redox reaction
$Zn + NO_3^- \rightarrow Zn^{2+} + NH_4^+$
in basic medium, coefficients of Zn, NO_3^- and OH^- in the balanced equation respectively are :
(1) 4, 1, 7 (2) 7, 4, 1
(3) 4, 1, 10 (4) 1, 4, 10
CRR086

85. In the balanced equation-
$[Zn + H^+ + NO_3^- \rightarrow NH_4^+ + Zn^{+2} + H_2O]$
coefficient of NH_4^+ is:-
(1) 4 (2) 3 (3) 2 (4) 1
CRR087

86. In the balanced equation
$MnO_4^- + H^+ + C_2O_4^{2-} \rightarrow Mn^{2+} + CO_2 + H_2O$, the moles of CO_2 formed are :-
(1) 2 (2) 4 (3) 5 (4) 10
CRR088

87. In the following reaction the value of 'X' is
$H_2O + SO_3^{2-} \rightarrow SO_4^{2-} + 2H^+ + X$
(1) $4e^-$ (2) $3e^-$ (3) $2e^-$ (4) $1e^-$
CRR089

88. The number of electrons required to balance the following equation are:
$NO_3^- + 4H^+ \rightarrow 2H_2O + NO$
(1) 2 on right side
(2) 3 on left side
(3) 3 on right side
(4) 5 on left side
CRR090

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	2	3	3	1	3	3	4	1	2	2	1	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	1	4	3	4	1	1	1	4	3	4	1	1	1	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	4	4	4	1	2	1	1	2	2	2	2	1	2	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	1	4	1	3	3	3	1	1	3	2	2	2	2	4
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	3	3	1	1	2	4	3	1	3	3	4	3	4	1
Question	76	77	78	79	80	81	82	83	84	85	86	87	88		
Answer	1	2	1	2	1	3	1	1	3	4	4	3	2		

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. The reaction of aqueous $KMnO_4$ with H_2O_2 in acidic conditions gives :-
(1) Mn^{4+} and O_2
(2) Mn^{2+} and O_2
(3) Mn^{2+} and O_3
(4) Mn^{4+} and MnO_2

CRR094

RE-AIPMT 2015

2. Assuming complete ionization, same moles of which of the following compounds will require the least amount of acidified $KMnO_4$ for complete oxidation?
(1) FeC_2O_4
(2) $Fe(NO_2)_2$
(3) $FeSO_4$
(4) $FeSO_3$

CRR095

NEET-II 2016

3. Hot concentrated sulphuric acid is a moderately strong oxidizing agent. Which of the following reactions does not show oxidizing behaviour ?
(1) $C + 2H_2SO_4 \rightarrow CO_2 + 2SO_2 + 2H_2O$
(2) $CaF_2 + H_2SO_4 \rightarrow CaSO_4 + 2HF$
(3) $Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$
(4) $3S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$

CRR096

NEET(UG) 2018

4. For the redox reaction
$MnO_4^- + C_2O_4^{2-} + H^+ \longrightarrow Mn^{2+} + CO_2 + H_2O$
the correct coefficients of the reactants for the balanced equation are

	MnO_4^-	$C_2O_4^{2-}$	H^+
(1)	16	5	2
(2)	2	5	16
(3)	2	16	5
(4)	5	16	2

CRR097

NEET(UG) 2019

5. Which of the following reactions are disproportionation reaction ?
(a) $2Cu^+ \rightarrow Cu^{2+} + Cu^0$
(b) $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$
(c) $2KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$
(d) $2MnO_4^- + 3Mn^{2+} + 2H_2O \rightarrow 5MnO_2 + 4H^{\oplus}$
Select the **correct** option from the following :-
(1) (a) and (b) only (2) (a), (b) and (c)
(3) (a), (c) and (d) (4) (a) and (d) only

CRR098

NEET (UG) 2020

6. What is the change in oxidation number of carbon in the following reaction ?
$CH_4(g) + 4Cl_2(g) \rightarrow CCl_4(l) + 4HCl(g)$
(1) 0 to –4 (2) +4 to +4
(3) 0 to +4 (4) –4 to +4

CRR099

NEET (UG) 2021

7. The equivalent weight of the salt $KHC_2O_4.H_2C_2O_4.4H_2O$ used as reducing agent is
(1) $\frac{Mol.wt}{1}$
(2) $\frac{Mol.wt}{2}$
(3) $\frac{Mol.wt}{3}$
(4) $\frac{Mol.wt}{4}$

CRR100

8. Which of the following reactions is the metal displacement reaction ? Choose the right option.
(1) $2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2$
(2) $Cr_2O_3 + 2Al \xrightarrow{\Delta} Al_2O_3 + 2Cr$
(3) $Fe + 2HCl \rightarrow FeCl_2 + H_2\uparrow$
(4) $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2\uparrow$

CRR101

NEET (UG) 2022

9. Which of the following reactions is a decomposition redox reaction ?

(1) $2\ Pb(NO_3)_2(s) \rightarrow 2\ PbO(s) + 4\ NO_2(g) + O_2(g)$

(2) $N_2(g) + O_2(g) \rightarrow 2\ NO(g)$

(3) $Cl_2(g) + 2OH^-(aq) \rightarrow ClO^-(aq) + Cl^-(aq) + 4H_2O(\ell)$

(4) $P_4(s) + 3OH^-(aq) + 3H_2O(\ell) \rightarrow PH_3(g) + 3H_2PO^-_2(aq)$

CRR102

NEET (UG) 2023

10. On balancing the given redox reaction,

$$aCr_2O_7^{2-} + bSO_3^{2-}(aq) + cH^+(aq) \rightarrow$$

$$2aCr^{3+}(aq) + bSO_4^{2-}(aq) + \frac{c}{2}H_2O(\ell)$$

the coefficients a, b and c are found to be, respectively -

(1) 3, 8, 1 (2) 1, 8, 3

(3) 8, 1, 3 (4) 1, 3, 8

CRR113

NEET (UG) 2024

11. Which reaction is **NOT** a redox reaction ?

(1) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

(2) $2\ KClO_3 + I_2 \rightarrow 2\ KIO_3 + Cl_2$

(3) $H_2 + Cl_2 \rightarrow 2\ HCl$

(4) $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$

CRR136

NEET (UG) 2024 (Re-examination)

12. The oxidation states **not** shown by Mn in given reaction is :

$$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$$

A. +6 B. +2 C. +4

D. +7 E. +3

Choose the **most appropriate** answer from the options given below :

(1) D and E only (2) B and D only

(3) A and B only (4) B and E only

CRR137

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12
Answer	2	3	2	2	1	4	4	2	1	4	4	4

Exercise – III (Analytical Questions) Master Your Understanding

1. In the reaction given below :

(A) $H_2S_{(g)} + Cl_{2(g)} \longrightarrow 2HCl_{(g)} + S_{(s)}$

(B) $3Fe_3O_{4(s)} + 8Al_{(s)} \longrightarrow 9Fe_{(s)} + 4Al_2O_{3(s)}$

The related statements are :

Statement-I : In (A) H_2S is oxidised because a more electronegative element Cl_2 is added to hydrogen.

Statement-II : In (B) Aluminium is oxidised because oxygen is added to it.

Statement-III : In (B) Ferrous ferric oxide is reduced.

Statement-IV : Individual oxidation or reduction reactions never occur but redox reactions occur.

The correct one is :

(1) Statement-I, II, III and IV are correct

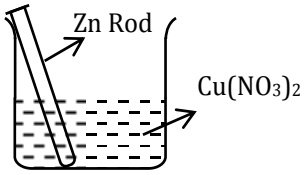
(2) Statement-I, II are correct but III and IV are incorrect

(3) Statement-I, II, III and IV are incorrect

(4) Statement-II, IV are incorrect but I and III are correct

CRR103

2.



(Zn rod in $Cu(NO_3)_2$ solution)

Statement-I: Zn looses electron to form Zn^{2+} so Zn is getting oxidised.

Statement-II: Cu acts as reducing agent.

The correct statement is :

(1) Statement-I is correct but statement-II is wrong

(2) Statement-I is incorrect but statement-II is correct

(3) Statement-I and statement-II both are correct

(4) Statement-I and statement-II both are incorrect

CRR104

3. Match the following column-I and column-II :

Column-I		Column-II	
(a)	H_2S	(p)	Oxidising agent
(b)	$KMnO_4$	(q)	Reducing agent
(c)	H_2O_2	(r)	Can act as both oxidising and reducing agent

(1) (a-r), (b-p), (c-q) (2) (a-q), (b-p), (c-r)

(3) (a-p), (b-q), (c-r) (4) (a-r), (b-q), (c-p)

CRR105

4. Given below are two statements:

Statement-I : Fractional oxidation number is unconvincing to us because electrons are never shared or transferred in fraction.

Statement-II : Fractional oxidation state is the average oxidation state of element.

The correct one is :

(1) Both statement-I and statement-II are correct

(2) Statement-I is correct but statement-II is wrong

(3) Statement-I is wrong but statement-II is correct

(4) Both statement-I and statement-II are wrong

CRR108

5. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion :- Oxidation state of Hydrogen is +1 in H_2O while –1 in CaH_2.

Reason :- CaH_2 is a metal hydride and for metal hydrides, hydrogen is assigned the oxidation number of +1.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**.

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is not correct but **(R)** is correct.

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.

CRR109

6. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion :- The average oxidation no. of sulphur in $Na_2S_4O_6$ is 2.5

Reason :- Two S-atoms are not directly linked with O-atoms.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is not correct but **(R)** is correct.

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

CRR110

7. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion :- The decomposition of H_2O_2 to form H_2O and O_2 is an example of disproportionation reaction.

Reason :- The oxygen of peroxide is in –1 oxidation state and it is converted to zero oxidation state in O_2 and –2 oxidation state in H_2O.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**.

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is not correct but **(R)** is correct.

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.

CRR111

8. From the following options :

(A) Oxidation state of Br in Br_3O_8 is +10/3

(B) Oxidation state of C in C_3O_2 is +1

(C) Oxidation state of S in $S_4O_6^{2-}$ is +2.5

(D) Oxidation state of H in metal hydrides is –1

(E) Oxidation state of P in P_4 is –1/4

The correct option is :

(1) A, B, C, D and E one correct

(2) A, C, D and E are correct

(3) B, C, D and E are correct

(4) C and D are correct

CRR112

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	1	1	2	1	2	1	4	4

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Redox reactions

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (2)

In metal carbonyl, O.S. of metal is always zero.

2. Ans. (3)

NH_2OH

$x + 2 + (-2) + 1 = 0$

$x = -1$

3. Ans. (2)

Fluorine always exhibits –1 O.S. in all compounds

4. Ans. (3)

O.S. of Fluorine is always –1

5. Ans. (3)

$\underline{C}H_4 = x + 4 = 0 \Rightarrow x = -4$

$\underline{C}H_3Cl = x + 3 - 1 = 0 \Rightarrow x = -2$

$CH_2Cl_2 \Rightarrow x + 2 - 2 = 0 \Rightarrow x = 0$

$CHCl_3 \Rightarrow x + 1 - 3 = 0 \Rightarrow x = 2$

$CCl_4 \Rightarrow x - 4 = 0 \Rightarrow x = +4$

6. Ans. (1)

In $H_2\underline{O}_2$; $+2 + 2x = 0$

$x = -1$,

O.S. of O is –1

7. Ans. (3)

O.S. of sulphur in H_2SO_5 is +6

(Max. O.S. of sulphur is +6)

8. Ans. (3)

Os & Ru shows highest O.S.

9. Ans. (4)

In always, O.S. of metal is always zero.

10. Ans. (1)

$\overset{+1}{H}—\overset{-2}{\underset{-1}{N}}\equiv\overset{+2}{C}$

O.S. of C is +2

11. Ans. (2)

$Fe_{0.94}O = 0.94x - 2 = 0$

$0.94x = 2$

$$x = \frac{2}{0.94} = \frac{200}{94}$$

12. Ans. (2)

C_3O_2

$3x + 2(-2) = 0$

$3x = 4$

$$x = \frac{4}{3}$$

13. Ans. (1)

$Na_2S_2O_3$

$+2 + 2x - 6 = 0$

$2x = 4$

$x = +2$

14. Ans. (1)

In $CaOCl_2$; O.S. of Cl is –1 & +1.

15. Ans. (2)

In $\underline{N}_3H$, O.S. of N & $\frac{-1}{3}$

$3x + 1 = 0$

$3x = -1$

$$x = \frac{-1}{3}$$

16. Ans. (1)

Y $Ba_2Cu_3O_7$

$+3 + 2(+2) + 3x - 14 = 0$

$3x = 7$

$$x = \frac{7}{3}$$

17. Ans. (1)

$H_4\underline{I}O_6^-$

$+4 + x - 12 = -1$

$x = +7$

18. Ans. (4)

$\underline{Cr}O_2Cl_2$, O.S. of Cr is +6

$x + 2(-2) + 2(-1) = 0$

$x = +6$

19. Ans. (3)

In $[Fe(CO)_5]$, iron has 'zero' Oxidation state.

20. Ans. (4)

$BaO_2 = 2 + 2x = 0$

$x = -1$

21. Ans. (1)

$K\underline{N}O_3 = 1 + x - 6 = 0 \Rightarrow x = +5$(e)

$H\underline{N}O_2 = 1 + x + 2(-2) = 0$

$\Rightarrow x = 3$(d)

$\underline{N}H_4Cl = x + 4 - 1 = 0$

$x = -3$ (b)

$Na\underline{N}_3 = 1 + 3x = 0$

$x = -\frac{1}{3}$ (a)

22. Ans. (1)

$K_3[Fe(CN)_6] = +3 + x - 6 = 0$

$x = +3$

$\underline{Fe}_2O_3 = 2x - 6 = 0$

$x = +3$

23. Ans. (1)

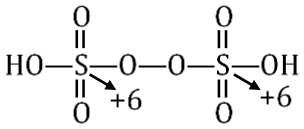
$\underline{Br}_2 \longrightarrow \underline{Br}O_3^-$

$x = 0 \quad x - 6 = -1$

$x = +5$

0 to +5

24. Ans. (4)



O.S. of sulphur in $H_2S_2O_8$ is +6

25. Ans. (3)

$\underline{Cr}O_2Cl_2 = x + 2(-2) + 2(-1)$

$= 0$

$x = +6$

26. Ans. (4)

$[\underline{Co}(NH_3)_4(H_2O)Cl]SO_4$

$x + 0 - 1 - 2 = 0$

$x = +3$

27. Ans. (1)

Oxidation no. of C in graphite (free state) is zero

28. Ans. (1)

$N_3H = 3x + 1 = 0$

$x = -\frac{1}{3}$

29. Ans. (1)

$H_3PO_3 = 3 + x + 3(-2) = 0$

$x = +3$

$H_3PO_4 = 3 + x + 4(-2) = 0$

$x = +5$

$HPO_3 = 1 + x + 3(-2) = 0$

$x = +5$

$H_4P_2O_7 = 4 + 2x + 7(-2) = 0$

$2x = 10$

$x = +5$

30. Ans. (4)

$H_3AsO_4 = +3 + x + 4(-2) = 0$

$x = +5$

31. Ans. (3)

$Mg(HXO_3) = 2 + 1 + x - 6 = 0$

$x = 3$

32. Ans. (4)

$PH_4^+ = x + 4 = +1 = x = -3$

$PO_2^{-3} = x - 4 = -3 \Rightarrow x = +1$

$PO_4^{-3} \Rightarrow x + 4(-2) = -3 \Rightarrow x = 5$

$PO_3^{-3} = x + (-6) = -3$

$x = +3$

33. Ans. (4)

$H_2\underline{S} = 2(+1) + x = 0$

$x = -2$

$H_2\underline{S}O_3 = 2(+1) + x + 3(-2) = 0$

$x = 4$

$H_2\underline{S}O_4 = 2(+1) + x + 4(-2) = 0$

$x = +6$

$H_2S_2O_3 = 2(+1) + 2x + 3(-2) = 0$

$x = 2$

$H_2S < H_2S_2O_3 < H_2SO_3 < H_2SO_4$

34. Ans. (4)

$K\underline{I} = +1 + x = 0 \Rightarrow x = -1$

$K\underline{I}_3 = 1 + 3x = 0 \Rightarrow x = -\frac{1}{3}$

$\underline{I}F_5 = x + 5(-1) = 0$

$x = +5$

$K\underline{I}O_4 = 1 + x + 4(-2) = 0$

$x = +7$

35. Ans. (1)

$C_6H_5\underline{C}HO = 7x + 6(1) + (-2) = 0$

$7x = -4$

Sum of oxidation state is –4

36. Ans. (2)

Reducing Agent :- Which oxidised itself & during oxidation, e^- are released/donate.

37. Ans. (1)

H_2O_2 oxidises sulphur from –2 to 0 hence it shows oxidising behaviour of H_2O_2.

38. Ans. (1)

It an element is in its lowest oxidation state it can Act as only reducing agent because it will undergo oxidation itself & reduces others.

39. Ans. (2)

C is oxidised to CO. Hence H_2O is the oxidising agent.

40. Ans. (2)

$H_2\underline{O}_2$:- Oxygen is in –1 oxidation state. So it can work both as an oxidising as well as reducing agent.

41. Ans. (2)

In H_2S, sulphur is in its lowest O.S. (–2). Hence HNO_2 can act only as an oxidant with H_2S.

42. Ans. (2)

In the reaction $Cl_2 + H_2O_2 \longrightarrow 2HCl + O_2$

Oxidation state of Cl changes from 0 to –1. Hence H_2O_2 act as reducing agent.

43. Ans. (1)

$\overset{0}{H}_2 + \overset{0}{Na} \longrightarrow 2\overset{+1}{Na}\overset{-1}{H}$

Oxidation of Na takes place which means H_2 act as O.A.

44. Ans. (2)

Oxidant mean oxidation of other & reduction of self. During reduction of itself it gains e^-.

45. Ans. (1)

$\overset{0}{C} + 4HNO_3 \longrightarrow \overset{+4}{CO}_2 + 2H_2O + 4NO_2$

Oxidation of C from 0 to +4 is carried with HNO_3 Hence HNO_3 will act as oxidising agent.

46. Ans. (3)

$A_3(BC_4)_2 = 3(+2) + 2[5 + 4(-2)] = 6 + (-6) = 0$

47. Ans. (1)

$\overset{0}{N}_2 \longrightarrow \overset{-3}{N}H_3$

n_f = no. of atoms x | change in O.S. |

$= 2 \times |0-(-3)| = 6$

$E_w = \frac{M_w}{n_f} = \frac{28}{6}$

48. Ans. (4)

$\overset{0}{N}_2 \longrightarrow \overset{-3}{N}H_3$

n_f = no. of atoms | change in O.S. |

$n_{NH_3} = 1 \times |-3-0| = 3$

$E_W = \frac{M_W}{n_f} = \frac{17}{3}$

49. Ans. (1)

$2S_2O_3^{-2} + I_2 \longrightarrow S_4O_6^{-2} + 2I^-$

$S_2O_3^{-2} = 2x + 3(-2) = -2$

$x = 2$

$S_4O_6^{-2} = 4x + 6(-2) = -2$

$x = 2.5$

n_f of $S_2O_3^{-2}$ = no. of atoms | change in O.S.

$= 2 \times |2-2.5| = 1$

E_w of $S_2O_3^{-2} = \frac{M_w}{n_f} = \frac{M_w}{1} = M_w$

50. Ans. (3)

$\overset{+2}{V}O + Fe_2O_3 \longrightarrow FeO + \overset{+5}{V}_2O_5$

n_f of $V_2O_5 = 2 \times |2-5| = 6$

$E_w = \frac{M_w}{n_f} = \frac{M}{6}$

51. Ans. (3)

$\overset{+5}{Br}O_3^- \longrightarrow \overset{-1}{B}r^-$

$n_{BrO_3^-}$ = no. of atoms | change in O.N.|

$= 1 \times |5-(-1)|$

$= 6$

$E_w = \frac{M}{6}$

52. Ans. (3)

$A^{-n_2} + xe^- \longrightarrow A^{-n_1}$

x = change in O.N.

$= -n_2-(-n_1)$

$= n_1-n_2$

53. Ans. (1)

$[Fe(CN)_6]^{-3} + H_2O_2 + 2OH^- \longrightarrow 2[Fe(CN)_6]^{4-} + 2H_2O + O_2$

Here O.S. of Fe changes from +3 to +2

Hence H_2O_2 is reducing agent.

E_w of H_2O_2 is 17 $\left\{\because E_w = \frac{M_w}{2} = \frac{34}{2} = 17\right\}$

54. Ans. (1)

$\overset{+2}{Fe}\overset{+3}{C_2}O_4 \longrightarrow Fe^{+3} + \overset{+4}{CO_2}$

$Fe^{+2} \longrightarrow Fe^{+3}$ $\qquad$ $\overset{+3}{C_2}O_4^{-2} \longrightarrow CO_2$

n_f = no. of atoms x | change in O.N.

= 1 | 2 – 3 | +2 | 3 – 4 | = 3

$E_w = \frac{M_w}{n_f} = \frac{M}{3}$

55. Ans. (3)

$(\text{no.of g eq})_{C_2O_4^{-2}} = (\text{no.of g eq})_{MnO_4^-}$

$(n_f \times \text{mole})_{C_2O_4^{-2}} = (n_f \times \text{mole})_{MnO_4^-}$

2 × mole = 5 × 1

Mole of oxalate ion = $\frac{5}{2}$

56. Ans. (2)

$\overset{5}{HNO_3} \longrightarrow \frac{1}{2}\overset{+1}{N_2}O$

$n_f = 1 \times |5 - 1| = 4$

4 mole of e^- are transferred when 1 mole HNO_3 changes to N_2O

57. Ans. (2)

$\overset{+2}{Mn}SO_4 \longrightarrow \overset{+4}{Mn}O_2$ $\quad (\because n_f = 2)$

$E_w = \frac{M_w}{n_f} = \frac{M}{2}$

58. Ans. (2)

$3\overset{0}{Fe} + 4H_2O \longrightarrow \overset{8/3}{Fe_3}O_4 + 4H_2$

n_f of Fe = $1 \times \left|0 - \frac{8}{3}\right| = \frac{8}{3}$

$E_w = \frac{\text{At wt}}{n_f} = \frac{56}{\left(\frac{8}{3}\right)} = 7 \times 3 = 21$

59. Ans. (2)

$H^+ + \overset{+6}{Cr_2}O_7^{2-} + I^- \longrightarrow Cr^{+3} + I_2 + H_2O$

Here I^- is reductant

$E_w = \frac{M_w}{n_f}$

n_f of I^- = 1|–1–0| = 1

$E_w = \frac{M_w}{1} = \frac{127}{1} = 127$

60. Ans. (4)

In Acidic medium

$MnO_4^- \longrightarrow Mn^{+2}$ $\qquad$ $(n_f = 5)$

$FeC_2O_4 \longrightarrow Fe^{+3} + 2CO_2$ $\qquad$ $(n_f = 3)$

$(n_f \times \text{mole})_{MnO_4^-} = (n_f \times \text{mole})_{FeC_2O_4}$

5 × n = 3 × 1

$n = \frac{3}{5}$

61. Ans. (1)

Option (2), (3), (4) are ion exchange reaction,

In option (1), O.N. changes

$\overset{0}{H_2} + \overset{0}{Br_2} \longrightarrow 2\overset{+1}{H}\overset{-1}{Br}$

H_2 is oxidised & Br_2 is reduced.

62. Ans. (3)

In option (3)

O.N. does not change. Hence it is not a redox reaction

$\overset{+2}{Ba}\overset{-1}{O_2} + \overset{+1}{H_2}SO_4 \longrightarrow \overset{+2}{Ba}SO_4 + H_2\overset{-1}{O_2}$

63. Ans. (3)

$\overset{0}{Cl_2} + OH^- \longrightarrow \overset{-1}{Cl^-} + \overset{+7}{ClO_4^-}$

O.N. of Cl_2 is decreased as well as increased. Hence Cl_2 is oxidised as well as reduced.

64. Ans. (1)

$2\overset{+2}{Cu}\overset{-1}{I_2} \longrightarrow \overset{+1}{Cu}\overset{-1}{I} + \overset{0}{I_2}$ is a redox

Reaction because both oxidation & reduction occurs in the same reaction.

65. Ans. (1)

$\overset{+4}{Mn}O_2 + 4H\overset{-1}{Cl} \longrightarrow \overset{+2}{Mn}Cl_2 + \overset{0}{Cl_2} + 2H_2O$

This is a redox reaction because MnO_2 is reduced & HCl is oxidised.

66. Ans. (2)

$K\overset{+5}{Cl}\overset{-2}{O_3} \longrightarrow K\overset{-1}{Cl} + \overset{0}{O_2}$

This is an example of Intramolecular redox reaction because in this, different elements same compound are being oxidised & reduced.

67. Ans. (4)

All reaction are examples of disproportionation reaction, because, disproportionation reaction same elements in same compound undergoes oxidation & reduction both.

68. Ans. (3)

$\overset{+2}{X_2}H_4 \longrightarrow \overset{-1/2}{X}$

$n_f = 2\left|-2-\left(-\frac{1}{2}\right)\right| = 3$

3 mole of e^- will be liberated in the following conversion.

69. Ans. (1)

$\overset{+2}{Ca}\overset{+4}{C}\overset{-2}{O_3} \longrightarrow \overset{+2}{Ca}\overset{-2}{O} + \overset{+4}{C}\overset{-2}{O_2}$

This is not a redox reaction, because there is no change in O.N.

70. Ans. (3)

$H^+ + \overset{+7}{Mn}O_4^- + \overset{+4}{SO_3^{2-}} \longrightarrow \overset{+6}{SO_4^{-2}} + \overset{+2}{Mn^{+2}} + H_2O$

Here MnO_4^- is reduced to Mn^{+2} & SO_3^{-2} is oxidised to SO_4^{-2}.

71. Ans. (3)

$\overset{0}{I_2} + K\overset{-1}{I} \longrightarrow K\overset{-1/3}{I_3}$

Here I_2 is reduced to KI_3 & KI is oxidised to KI_3.

72. Ans. (4)

$2H\overset{-1}{I} + H_2SO_4 \longrightarrow \overset{0}{I_2} + SO_2 + 2H_2O$

Here HI is oxidised to I_2 by H_2SO_4. Hence H_2SO_4 act as oxidising agent.

73. Ans. (3)

$4H_3\overset{+3}{P}O_3 \longrightarrow \overset{-3}{P}H_3 + 3H_3\overset{+5}{P}O_4$

Here O.N. of phosphorous is decreased as well as increased in the same compound. Hence it is an example of disproportionation reaction.

74. Ans. (4)

$\overset{0}{H_2} + \overset{0}{Br_2} \longrightarrow 2\overset{+1}{H}\overset{-1}{Br}$

Here H_2 is oxidised to HBr & Br_2 is reduced. Hence it involves oxidation & reduction.

75. Ans. (1)

$\overset{0}{Cl_2} + 2K_2\overset{+6}{Mn}O_4 \longrightarrow 2K\overset{+7}{Mn}O_4 + 2K\overset{-1}{Cl}$

Here Cl_2 is reduced & K_2MnO_4 is oxidised. So it is an example of redox reaction.

76. Ans. (1)

$\overset{+6}{Cr}O_4^{-2} \longrightarrow \overset{+6}{Cr_2}O_7^{2-}$

Here O.N. of Cr does not change.

77. Ans. (2)

$\overset{0}{Zn} + H_2SO_4 \longrightarrow \overset{+2}{Zn}SO_4 + H_2$

Here O.N. of Zn is increased. So Zn undergoes oxidation.

78. Ans. (1)

$CrO_4^{-2} \longrightarrow CrO_2^- + OH^-$

$3e^- + \underbrace{\overset{+6}{Cr}O_4^{-2} \longrightarrow \overset{+3}{Cr}O_2^-}_{n_f=3}$

Balance oxygen atom by adding H_2O to the deficient side

$CrO_4^{-2} + 3e^- \longrightarrow CrO_2^- + 2H_2O$

Now add H^+ to balance Hydrogen atom.

$4H^+ + CrO_4^{-2} + 3e^- \longrightarrow CrO_2^- + 2H_2O$

Add 4 OH^- to both sides.

$4H_2O + CrO_4^{-2} + 3e^- \rightarrow CrO_2^- + 2H_2O + 4OH^-$

net reaction

$2H_2O + CrO_4^{-2} + 3e^- \longrightarrow CrO_2^- + 4OH^-$

79. Ans. (2)

$\underbrace{\overset{+6}{Cr_2}O_7^{2-} \longrightarrow 2Cr^{+3}}_{n_f=2|6-3|=6}$

Balance oxygen atom

$6e^- + Cr_2O_7^{2-} \longrightarrow 2Cr^{+3} + 7H_2O$

Balance Hydrogen atom

$6e^- + 14H^+ + Cr_2O_7^{2-} \longrightarrow 2Cr^{+3} + 7H_2O$

80. Ans. (1)

$\underbrace{\overset{+7}{Mn}O_4^- \longrightarrow Mn^{+2}}_{n_f=|7-2|=5}$

Gain of $5e^-$

$MnO_4^- + 5e^- \longrightarrow Mn^{+2}$

n = 5

81. Ans. (3)

$\underbrace{\overset{+7}{Mn}O_4^- + 2H_2O \longrightarrow \overset{+4}{Mn}O_2 + 4OH^-}_{n_f=|7-4|=3}$

Gain of $3e^-$; $3e^-$ must be added to LHS to balance charge

$3e^- + MnO_4^- + 2H_2O \longrightarrow MnO_2 + 4OH^-$

82. Ans. (1)

$2KMnO_4 + 5H_2S + 6H^+ \rightarrow 2Mn^{+2} + 2K^+ + 5S + 8H_2O$

Here reductant is H_2S

$\overset{-2}{H_2S} \longrightarrow \overset{0}{S}$

$n_f = 1 \mid -2-0 \mid = 2$

83. Ans. (1)

$$\underset{n_f=|7-2|=5}{\overset{+7}{Mn}O_4^- + 8H^+ + ne^- \longrightarrow Mn^{+2} + 4H_2O}$$

n = 5

84. Ans. (3)

$$\underset{n_f=|0-2|=2\times4}{\overset{+5}{\overset{n_f=|5-(-3)|=8}{4Zn + NO_3^-}} \longrightarrow 4Zn^{+2} + \overset{-3}{NH_4^+}}$$

$4Zn + NO_3^- \longrightarrow 4Zn^{+2} + NH_4^+$

Balance oxygen atom by adding H_2O to deficient side.

$4Zn + NO_3^- \longrightarrow 4Zn^{+2} + NH_4^+ + 3H_2O$

Add H^+ & then OH^- to neutralise it

$4Zn + NO_3^- + 10H^+ + 10\,OH^- \longrightarrow 4Zn^{+2} + 4NH_4^+ + 3H_2O + 10OH^-$

net balance

$4Zn + NO_3^- + 7H_2O \rightarrow 4Zn^{+2} + NH_4^+ + 10(OH^-)$

85. Ans. (4)

$$\underset{n_f=|0-2|=2\times4}{\overset{+5}{\overset{n_f=|5-(-3)|=8}{4Zn + NO_3^-}} \longrightarrow 4Zn^{+2} + \overset{-3}{NH_4^+}}$$

$6H^+ + 4Zn + NO_3^- \longrightarrow 4Zn^{+2} + NH_4^+ + 3H_2O$

Coefficient of $NH_4^+ = 1$

86. Ans. (4)

$$\underset{n_f=|7-2| \atop =5\times2}{\overset{+7}{MnO_4^-}} + C_2O_4^{-2} \rightarrow \underset{n_f=2|3-4|=2\times5}{Mn^{+2}} + 2CO_2$$

$2MnO_4^- + 5C_2O_4^{-2} \longrightarrow 2Mn^{+2} + 10CO_2$

Balance oxygen atom by adding H_2O to deficient side.

$2MnO_4^- + 5C_2O_4^{-2} \longrightarrow 2Mn^{+2} + 10CO_2 + 8H_2O$

Now balance Hydrogen atom by adding H^+ to deficient side.

$2MnO_4^- + 5C_2O_4^{-2} \rightarrow 2Mn^{+2} + 10CO_2 + 8H_2O + 16H^+$

87. Ans. (3)

$$H_2O + \underset{n_f=|4-6|=2}{\overset{+4}{S}O_3^{-2} \longrightarrow \overset{+6}{S}O_4^{-2}} + 2H^+ + x$$

So x = $2e^-$

88. Ans. (2)

$$\underset{n_f=|5-2|=3}{\overset{+5}{NO_3^-} + 4H^+ \longrightarrow \overset{+2}{NO}} + 2H_2O$$

$3e^-$ on LHS

Exercise-II (Previous Year Questions)

1. Ans. (2)

Reduction

$MnO_4^- \longrightarrow Mn^{+2}$ (in Acidic medium)

$\overset{-1}{H_2O_2} \longrightarrow \overset{0}{O_2}$ (Oxidation)

2. Ans. (3)

Least amount of $KMnO_4$ will be required for which n_f is least.

$n_{f\,FeC_2O_4} = 3$ $\quad$ $n_{f\,FeSO_4} = 1$

$n_{f\,Fe(NO_2)_2} = 5$ $\quad$ $n_{f\,FeSO_3} = 3$

So, Ans will be $FeSO_4$.

3. Ans. (2)

In option (2), there is no change on O.N. so this does not show oxidising behaviour.

4. Ans. (2)

$$\underset{n_f=5\times2}{\overset{+7}{MnO_4^-}} + \overset{n_f=2|3-4|=2\times5}{C_2O_4^{-2}} \rightarrow Mn^{+2} + 2CO_2$$

$2MnO_4^- + 5C_2O_4^{-2} \longrightarrow 2Mn^{+2} + 10CO_2$

Add H_2O to balance oxygen atoms

$2MnO_4^- + 5C_2O_4^{-2} \longrightarrow 2Mn^{+2} + 10CO_2 + 8H_2O$

Add H^+ to balance Hydrogen atoms.

$16H^+ + 2MnO_4^- + 5C_2O_4^{-2} \rightarrow 2Mn^{+2} + 10CO_2 + 8H_2O$

Coefficient of $MnO_4^- = 2$

$C_2O_4^{-2} = 5$

$H^+ = 16$

5. Ans. (1)

$2Cu^+ \rightarrow Cu^{+2} + Cu^0$

(Disproportionation reaction)

$\overset{+6}{M}nO_4^{-2} \rightarrow \overset{+7}{Mn}O_4^- + \overset{+4}{Mn}O_4$

(Disproportionation reaction)

6. **Ans. (4)**

$CH_4 \longrightarrow \underline{C}Cl_4$

$\downarrow$

$x + 4(1) = 0$ $\quad$ $x + 4(-1) = 0$

$x = -4$ $\quad$ $x = 4$

-4 to $+4$

7. **Ans. (4)**

$KHC_2O_4.H_2C_2O_4.4H_2O$; when it act as reducing agent $C_2O_4^{-2}$ will convert to CO_2.

$2\overset{+3}{C}_2O_4^{-2} \rightarrow 4\overset{+4}{C}O_2$

$\Rightarrow 4|3-4| = 4$

$$\boxed{E_W = \frac{M_W}{4}}$$

8. **Ans. (2)**

$\underline{Cr}_2O_3 + 2Al \longrightarrow \underline{Al}_2O_3 + 2Cr$

Al, more reactive metal displace less reactive metal from its compound.

9. **Ans. (1)**

$2Pb(NO_3)_2 \longrightarrow 2PbO + 4NO_2 + O_2$

is a decomposition redox reaction.

10. **Ans. (4)**

Reaction has to be balanced in acidic medium 'O' atoms are balanced by adding H_2O and then H-atom is balanced by adding H^+ ions and charge is balanced by e^1.

Oxidation: $SO_3^{2-} + H_2O \rightarrow SO_4^{2-} + 2H^+ + 2e^{\ominus}]\times 3$

Reduction: $Cr_2O_7^{2-} + 14H^+ + 6e^{\ominus} \rightarrow 2Cr^{3+} + 7H_2O$

$Cr_2O_7^{2-} + 3SO_3^{2-} + 8H^{\oplus} \rightarrow 2Cr^{3+} + 3SO_4^{2-} + 4H_2O$

a = 1

b = 3

c = 8

11. **Ans. (4)**

$+2\ -1 \quad +1\ +6\ -2 \quad +2\ +6\ -2 \quad +1\ -1$

$BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$

As there is no change in oxidation state of any element therefore this is not a redox reaction.

12. **Ans. (4)**

$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

$\downarrow \quad\quad \downarrow \quad \downarrow$

$(+6) \quad\quad (+7) \quad (+4)$

incorrect oxidation state is +2 & +3

Exercise-III (Analytical Questions)

1. **Ans. (1)**

Addition of more EN elements and oxygen is oxidation while addition of hydrogen and electropositive element is known as reduction. Oxidation and reduction occures simultaneously.

2. **Ans. (1)**

$Zn \rightarrow Zn^{+2} + 2e^-$ (oxidation)

$Cu^{+2} + 2e^- \rightarrow Cu$ (reduction)

Here Zn is oxidised and Cu^{+2} is oxidising agent.

3. **Ans. (2)**

$H_2\underline{S}$:- S is in –2 oxidation state (lowest)

Hence it will act as Reducing agent.

$K\underline{Mn}O_4$:- Mn is in +7 oxidation state (maximum)

Hence it will act as oxidising agent.

$H_2\underline{O}_2$:- Oxygen is in -1 oxidation state (Intermediate)

Hence it can act as O.A. as well as R.A.

4. **Ans. (1)**

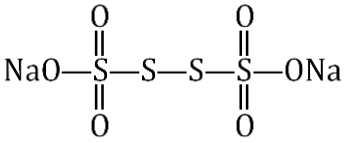
Fractional oxidation number is average of oxidation state.

5. **Ans. (2)**

In metal hydride, O.S. of hydrogen is –1.

6. **Ans. (1)**

Structure of $Na_2S_4O_6$:- Average oxidation number of sulphur is 2.5.



7. **Ans. (4)**

In disproportionation reaction, same element in same compound is oxidised and reduced.

8. **Ans. (4)**

$Br_3O_8 = 3x + 8(-2) = 0$

$$x = \frac{16}{3}$$

$C_3O_2 = 3x + 2(-2) = 0$

$3x = 4$

$$x = \frac{4}{3}$$

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$S_4O_6^{2-} = 4x + 6(-2) = -2$

$4x = 10$

$x = 2.5$

oxidation state of hydrogen in metal hydride is –1.

$P_4 \Rightarrow 4x = 0;$

$x = 0$

Oxidation state of P is P_4 is zero.

(Homoatomic molecule)

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Exercise - I (Conceptual Questions) Build Up Your Understanding

INTRODUCTION

1. Thermodynamics is concerned with :-
(1) Total energy of a system
(2) Energy changes in a system
(3) Rate of a chemical change
(4) Mass changes in nuclear reactions
CTD001

2. A well stoppered thermos flask contains some ice cubes. This is an example of :–
(1) Closed system
(2) Open system
(3) Isolated system
(4) Non-thermodynamic system
CTD002

3. Identify the intensive quantities from the following -
(1) Enthalpy and temperature
(2) Volume and temperature
(3) Enthalpy and volume
(4) Temperature and refractive index
CTD003

4. Which of the following is an extensive property
(1) Mass (2) Enthalpy
(3) Energy (4) All of these
CTD004

5. For an adiabatic process which of the following relations is correct
(1) $\Delta E = 0$ (2) $P\Delta V = 0$
(3) $q = 0$ (4) $q = + W$
CTD005

6. In which of the following process work is independent of path :
(1) Isothermal (2) Isochoric
(3) Adiabatic (4) Isobaric
CTD006

7. When a gas is compressed adiabatically and reversibly, the final temperature is-
(1) Higher than the initial temperature
(2) Lower than the initial temperature
(3) The same as initial temperature
(4) Dependent upon the rate of compression
CTD007

8. Which one is dependent on initial and final state?
(1) Heat supplied at constant pressure
(2) Heat supplied at constant volume
(3) Enthalpy
(4) All of the above
CTD008

9. Out of boiling point (I), entropy (II), pH (III) and emf of a cell (IV), intensive properties are :
(1) I, III, IV (2) I, II
(3) I, II, III (4) All of these
CTD009

10. The work done by a weightless piston in causing an expansion ΔV (at constant temperature), when the opposing pressure P is variable, is given by :
(1) $W = -\int PdV$ (2) $W = 0$
(3) $W = -P\Delta V$ (4) None
CTD010

11. The work done by 100 calorie of heat in isothermal expansion of ideal gas is :-
(1) 418.4 J (2) 4.184 J
(3) 41.84 J (4) None
CTD011

12. Temperature and volume are not :-
(1) Extensive properties
(2) Intensive properties
(3) Intensive and extensive properties respectively
(4) Extensive and intensive properties respectively
CTD012

13. $q = -w$ is not true for :-
(1) Isothermal process
(2) Adiabatic process
(3) Cyclic process
(4) 1 and 3 both
CTD013

14. The temperature of an ideal gas increase in an -
(1) Adiabatic compression
(2) Adiabatic expansion
(3) Isothermal expansion
(4) Isothermal compression

CTD014

15. Which statement is true for reversible process :–
(1) It takes place in single step
(2) Driving force is much greater than opposing force
(3) Work obtained is minimum
(4) None

CTD015

FIRST LAW OF THERMODYNAMICS ($\Delta E = q + W$)

16. Both q & w are_ _ _ _ _ function :-
(1) State (2) State, Path
(3) Path, State (4) Path

CTD016

17. If work done by the system is 300 joule when 100 cal. heat is supplied to it. The change in internal energy during the process is :-
(1) – 200 Joule (2) 400 Joule
(3) 720 Joule (4) 120 Joule

CTD017

18 A system has internal energy equal to E_1, 450 J of heat is taken out of it and 600 J of work is done on it. The final energy of the system will be -
(1) $(E_1 + 150)$ (2) $(E_1 + 1050)$
(3) $(E_1 - 150)$ (4) None of these

CTD018

19. The work done by a system is 8J when 40J heat is supplied to it. The change in internal energy of the system during the process :
(1) 32 J (2) 40 J (3) 48 J (4) –32 J

CTD019

ENTHALPY [$\Delta H = \Delta E + P\Delta V / \Delta H = \Delta E + \Delta n_g RT$]

20. Internal energy change during a reversible isothermal expansion of an ideal gas is :-
(1) Always negative
(2) Always positive
(3) Zero
(4) May be positive or negative

CTD020

21. Under which of the following conditions is the relation, $\Delta H = \Delta E + P\Delta V$ valid for a system :–
(1) Constant pressure
(2) Constant temperature
(3) Constant temperature and pressure
(4) Constant temperature, pressure and composition

CTD021

22. The difference between heats of reaction at constant pressure and constant volume for the reaction
$2C_6H_6(l) + 15O_2(g) \longrightarrow 12CO_2(g) + 6H_2O(l)$ at 25^0C in KJ is
(1) + 7.43 (2) +3.72
(3) – 7.43 (4) – 3.72

CTD022

23. For a gaseous reaction,
$A(g) + 3B(g) \longrightarrow 3C(g) + 3D(g)$
Δ E is 17 kCal at 27^0C assuming
R = 2 Cal K^{-1} mol^{-1}, the value of ΔH for the above reaction is:
(1) 15.8 Kcal (2) 18.2 Kcal
(3) 20.0 Kcal (4) 16.4 Kcal

CTD023

24. Which of the following statements is correct for the reaction ;
$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$ at constant temperature and pressure
(1) $\Delta H = \Delta E$ (2) $\Delta H < \Delta E$
(3) $\Delta H > \Delta E$ (4) None of the above

CTD024

25. For the reaction
$Ag_2O(s) \longrightarrow 2Ag(s) + ½\ O_2(g)$, which one of the following is true :
(1) $\Delta H = \Delta E$ (2) $\Delta H = ½\ \Delta E$
(3) $\Delta H < \Delta E$ (4) $\Delta H > \Delta E$

CTD025

26. A mixture of 2 moles of carbon monoxide and one mole of oxygen in a closed vessel is ignited to get carbon dioxide. If ΔH is the enthalpy change and ΔE is the change in internal energy, then :–
(1) $\Delta H > \Delta E$ (2) $\Delta H < \Delta E$
(3) $\Delta H = \Delta E$ (4) Not definite

CTD026

27. For the gaseous reaction involving the complete combustion of isobutane -
(1) $\Delta H = \Delta E$ (2) $\Delta H > \Delta E$
(3) $\Delta H = \Delta E = 0$ (4) $\Delta H < \Delta E$
CTD027

28. For the reversible isothermal expansion of one mole of an ideal gas at 300 K, from a volume of 10 dm^3 to 20 dm^3, ΔH is -
(1) 1.73 kJ (2) –1.73 kJ
(3) 3.46 kJ (4) Zero
CTD028

29. For $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ at 977°C, ΔH = 174 KJ/mol ; then ΔE is :-
(1) 160 kJ (2) 163.6 kJ
(3) 186.4 kJ (4) 180 kJ
CTD029

30. Heat of reaction for,
$CO(g) + ½ \quad O_2(g) \rightarrow CO_2(g)$ at constant V is –67.71 kcal at 17°C. The heat of reaction at constant P at 17°C is :-
(1) –68.0 kCal (2) + 68.0 kCal
(3) – 67.42 kCal (4) None
CTD030

31. The enthalpy of vaporisation of water at 100^0C is 40.63 kJ mol^{-1}. The value ΔE for this process would be:-
(1) 37.53 kJ mol^{-1} (2) 39.08 kJ mol^{-1}
(3) 42.19 kJ mol^{-1} (4) 43.73 kJ mol^{-1}
CTD031

32. For the system $S(s) + O_2(g) \rightarrow SO_2(g)$:–
(1) $\Delta H = \Delta E$ (2) $\Delta H > \Delta E$
(3) $\Delta E > \Delta H$ (4) $\Delta H = 0$
CTD032

33. For the reaction $CO(g) + \frac{1}{2} O_2(g) \rightarrow CO_2(g)$
Which one of these statement is correct at constant T and P ?
(1) $\Delta H = \Delta E$
(2) $\Delta H < \Delta E$
(3) $\Delta H > \Delta E$
(4) ΔH is Independent of physical state of reactants
CTD033

34. Which is true for the combustion of sucrose ($C_{12}H_{22}O_{11}$) at 25°C :-
(1) $\Delta H > \Delta E$ (2) $\Delta H < \Delta E$
(3) $\Delta H = \Delta E$ (4) None
CTD034

35. For which change $\Delta H \neq \Delta E$:-
(1) $H_2(g) + I_2(g) \rightarrow 2HI(g)$
(2) $HCl(\ell) + NaOH(\ell) \rightarrow NaCl(s) + H_2O(\ell)$
(3) $C(s) + O_2(g) \rightarrow CO_2(g)$
(4) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
CTD035

36. The heat of combustion of ethanol determined in a bomb calorimeter is – 670.48 kCal $mole^{-1}$ at 27°C. What is ΔH at 27°C for the reaction :–
(1) – 335.24 kCal (2) – 671.08 kCal
(3) – 670.48 kCal (4) + 670.48 kCal
CTD036

37. The difference in ΔH and ΔE for the combustion of methane at 25°C would be :–
(1) Zero (2) 2 × 298 × – 2 Cal
(3) 2 × 298 × – 3 Cal (4) 2 × 25 × – 3 Cal
CTD037

38. For which of the following reactions ΔH is less than ΔE :–
(1) $C_{12}H_{22}O_{11}(s) + 12O_2(g) \rightarrow 12CO_2(g) + 11H_2O(\ell)$
(2) $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$
(3) $N_2O_4(g) \rightarrow 2NO_2(g)$
(4) $N_2(g) + O_2(g) \rightarrow 2NO(g)$
CTD038

39. For a reaction $2X(s) + 2Y(s) \rightarrow 2C(\ell) + D(g)$
The q_p at 27°C is – 28 kCal mol^{-1}.
The q_V is _____ kCal mol^{-1} :–
(1) – 27.4 (2) + 27.4 (3) – 28.6 (4) 28.6
CTD039

WORK DONE IN DIFFERENT PROCESS

40. The work (in ergs) for a reversible expansion of one mole of an ideal gas from a volume of 10 litres to 20 litres at 25^0C is :
(1) $-2.303 \times 8.314 \times 10^7 \times 298 \log 2$
(2) $-2.303 \times 0.0821 \times 298 \log 2$
(3) $-2.303 \times 0.0821 \times 298 \log 0.5$
(4) $-2.303 \times 2 \times 298 \log 2$
CTD040

41. Two moles of an ideal gas expand spontaneously into vacuum. The work is :–
(1) Zero (2) 2 J (3) 4 J (4) 8 J
CTD041

42. One mole of a gas occupying $3dm^3$ expands against a constant external pressure of 1 atm to a volume of 13 L. Find work is :–
(1) – 10 atm dm^3 (2) – 20 atm dm^3
(3) – 39 atm dm^3 (4) – 48 atm dm^3
CTD042

ENTROPY/SECOND LAW OF THERMODYNAMICS

43. For which reaction from the following, ΔS will be maximum ?
(1) $Ca(s) + ½\ O_2(g) \longrightarrow CaO(s)$
(2) $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$
(3) $C(s) + O_2(g) \longrightarrow CO_2(g)$
(4) $N_2(g) + O_2(g) \longrightarrow 2NO(g)$
CTD043

44. An adiabatic reversible process is one in which :-
(1) Temperature of the system does not change
(2) The system is not closed to heat transfer
(3) There is no entropy change
(4) None of these
CTD044

45. Entropy means
(1) Disorderness (2) Randomness
(3) Orderness (4) Both 1 & 2
CTD045

46. ΔS for the reaction;
$MgCO_3(s) \longrightarrow MgO(s) + CO_2(g)$ will be :
(1) 0 (2) –ve
(3) +ve (4) ∞
CTD046

47. Change in entropy is negative for
(1) $Br_2(\ell) \longrightarrow Br_2(g)$
(2) $C(s) + H_2O(g) \longrightarrow CO(g) + H_2(g)$
(3) $N_2(g, 10\ atm) \longrightarrow N_2(g, 1\ atm)$
(4) Fe(at 400 K) $\longrightarrow$ Fe(at 300 K)
CTD047

48. In which reaction ΔS is positive :-
(1) $H_2O(\ell) \rightarrow H_2O(s)$
(2) $3O_2(g) \rightarrow 2O_3(g)$
(3) $H_2O(\ell) \rightarrow H_2O(g)$
(4) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
CTD048

49. When the egg is hard boiled, there is-
(1) Increase in disorder
(2) Decrease in disorder
(3) No change in disorder
(4) ΔG is negative
CTD049

50. If S^0 for H_2, Cl_2 and HCl are 0.13, 0.22 and 0.19 kJ K^{-1} mol^{-1} respectively. The total change in standard entropy for the reaction
$H_2 + Cl_2 \longrightarrow 2HCl$ is :
(1) 30 JK^{-1} mol^{-1} (2) 40 JK^{-1} mol^{-1}
(3) 60 JK^{-1} mol^{-1} (4) 20 JK^{-1} mol^{-1}
CTD050

51. Which has the least entropy :
(1) Graphite (2) Diamond
(3) $N_2(g)$ (4) $N_2O(g)$
CTD051

52. When two gases are mixed the entropy :–
(1) Remains constant (2) Decreases
(3) Increases (4) Becomes zero
CTD052

53. The enthalpy of vaporisation of per mole of ethanol (B.pt. = 79.5^0C and $\Delta S = 109.8\ JK^{-1}\ mol^{-1}$) is :–
(1) 27.35 kJ mol^{-1} (2) 32.19 kJ mol^{-1}
(3) 38.70 kJ mol^{-1} (4) 42.37 kJ mol^{-1}
CTD053

54. If 900J/g of heat is exchanged at boiling point of water, then what is increase in entropy?
(1) 43.4 $JK^{-1}mole^{-1}$ (2) 87.2 $JK^{-1}mole^{-1}$
(3) 900 $JK^{-1}mole^{-1}$ (4) Zero
CTD054

55. 5 mole of an ideal gas expand reversibly from a volume of 8 dm^3 to $80dm^3$ at a temperature of 27^0C. The change in entropy is :–
(1) 41.57 JK^{-1} (2) – 95.73 JK^{-1}
(3) 95.73 JK^{-1} (4) – 41.57 JK^{-1}
CTD055

56. In a spontaneous irreversible process the total entropy of the system and surroundings
(1) Remains constant (2) Increases
(3) Decreases (4) Zero
CTD056

57. The total entropy change for a system & its surroundings increases if the process is :
(1) Reversible (2) Spontaneous
(3) Exothermic (4) Endothermic
CTD057

58. Calculate the entropy of $Br_2(g)$ in the reaction $H_2(g) + Br_2(g) \rightarrow 2HBr(g)$, $\Delta S° = 20.1\ JK^{-1}$ given, entropy of H_2 and HBr is 130.6 and 198.5 J $mol^{-1}\ K^{-1}$:-
(1) 246.3 JK^{-1} (2) 123.15 JK^{-1}
(3) 24.63 JK^{-1} (4) 20 kJK^{-1}
CTD058

59. In which of the following case entropy decreases–
(1) Solid changing to liquid
(2) Expansion of a gas
(3) Crystals dissolve
(4) Polymerisation
CTD059

60. Which of the following quantity is not zero for element in standard state :–
(1) Enthalpy of formation
(2) Entropy
(3) Free energy of formation
(4) All of these
CTD060

61. Entropy of an adiabatic reversible process is:-
(1) Positive (2) Zero
(3) Negative (4) Constant
CTD061

GIBBS FREE ENERGY

62. A gas is allowed to expand under reversible adiabatic conditions then :-
(1) $\Delta U = 0$ (2) $\Delta T = 0$
(3) $\Delta S = 0$ (4) None of these
CTD062

63. For a reaction at 25^0C enthalpy change (ΔH) and entropy change (ΔS) are $-11.7 \times 10^3\ Jmol^{-1}$ and $-105\ J\ mol^{-1}\ K^{-1}$ respectively. The reaction is :
(1) Spontaneous
(2) Non spontaneous
(3) At equilibrium
(4) Can't say anything
CTD063

64. If $\Delta H > 0$ and $\Delta S > 0$, the reaction proceeds spontaneously when :-
(1) $\Delta H > 0$ (2) $\Delta H < T\Delta S$
(3) $\Delta H = T\Delta S$ (4) None
CTD064

65. The temperature at which the reaction
$Ag_2O(s) \longrightarrow 2Ag(s) + ½\ O_2(g)$
is at equilibrium is .....;
Given $\Delta H = 30.5\ kJ\ mol^{-1}$
and $\Delta S = 0.066\ kJK^{-1}\ mol^{-1}$:
(1) 462.12 K (2) 362.12 K
(3) 262.12 K (4) 562.12 K
CTD065

66. Which of the following is true for the reaction $H_2O(\ell) \rightleftharpoons H_2O(g)$ at 100^0C and 1 atmosphere
(1) $\Delta S = 0$ (2) $\Delta H = 0$
(3) $\Delta H = \Delta E$ (4) $\Delta H = T\Delta S$
CTD066

67. For the reaction $A(s) \longrightarrow B(s) + C(g)$ the value of $\Delta H = 30.56\ kJ\ mol^{-1}$ and $\Delta S = 66\ JK^{-1}\ mol^{-1}$. The temperature at which the free energy change for the reaction will be zero is :–
(1) 373 K (2) 413 K (3) 463 K (4) 493 K
CTD067

68. For hypothetical reversible reaction
$½\ A_2(g) + ⅗\ B_2(g) \longrightarrow AB_3(g)$; $\Delta H = -20$ kJ if standard entropies of A_2, B_2 and AB_3 are 60, 40 and 50 JK^{-1} $mole^{-1}$ respectively. The above reaction will be in equilibrium at :–
(1) 400 K (2) 500 K (3) 250 K (4) 200 K
CTD068

69. For the precipitation of AgCl by Ag^+ ions and HCl
(1) $\Delta H = 0$ (2) $\Delta G = 0$
(3) $\Delta G = -ve$ (4) $\Delta H = \Delta G$
CTD069

70. What is the sign of ΔG for the process of ice melting at 1 atm, 283 K is ?
(1) $\Delta G > 0$ (2) $\Delta G = 0$
(3) $\Delta G < 0$ (4) None of these
CTD070

71. A reaction A + B $\longrightarrow$ C + D + q is found to have a positive entropy change, the reaction will be -
(1) Possible at high temperature
(2) Possible only at low temperature
(3) Not possible at any temperature
(4) Possible at any temperature
CTD071

72. Equilibrium constant of a reaction is related to :
(1) Standard free energy change ΔG^0
(2) Free energy change ΔG
(3) Entropy change
(4) None
CTD072

73. The Vant Hoff equation is :
(1) $\Delta G° = RT \log_e K_P$ (2) $-\Delta G° = RT \log_e K_P$
(3) $\Delta G° = RT^2 \ell n K_P$ (4) None
CTD073

74. If $\Delta G^0 > 0$ for a reaction then :
(1) $K_P > 1$
(2) $K_P < 1$
(3) The products predominate in the equilibrium mixture
(4) None
CTD074

75. If the equilibrium constant for a reaction is 10, then the value of ΔG^0 will be
(R = 8JK^{-1} mol^{-1}, T = 300 K)
(1) + 5.527 kJ mol^{-1} (2) – 5.527 kJ mol^{-1}
(3) +55.27 kJ mol^{-1} (4) – 55.27 kJ mol^{-1}
CTD075

76. The process of evaporation of a liquid is accompanied by :
(1) Increase in enthalpy
(2) Decrease in free energy
(3) Increase in entropy
(4) All
CTD076

77. For the process, $CO_2(s) \longrightarrow CO_2(g)$:
(1) Both ΔH and ΔS are +ve
(2) ΔH is negative and ΔS is +ve
(3) ΔH is +ve and ΔS is –ve
(4) Both ΔH and ΔS are –ve
CTD077

78. Which of the following provide exceptions to third law of thermodynamics
(1) CO (2) ice
(3) CO_2 (4) All the above
CTD078

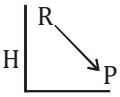
79. The Gibbs free energy change of a reaction at 27°C is –26 kCal and its entropy change is – 60 Cal K^{-1}. ΔH for the reaction is :–
(1) – 44 kCal (2) – 18 kCal
(3) 34 kCal (4) – 24 kCal
CTD079

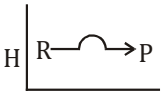
80. Which of the following reaction is expected never to be spontaneous :–
(1) $2O_3(g) \rightarrow 3O_2(g)$
ΔH = – Ve, ΔS = +Ve
(2) $Mg(s) + H_2(g) \rightarrow MgH_2$
ΔH = – Ve, ΔS = – Ve
(3) $Br_2(l) \rightarrow Br_2(g)$
ΔH = + Ve, ΔS = + Ve
(4) $2Ag(s) + 3N_2(g) \rightarrow 2AgN_3$
ΔH = +Ve, ΔS= – Ve
CTD080

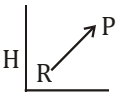
THERMOCHEMICAL REACTION

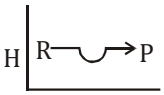
81. The formation of water from $H_2(g)$ and $O_2(g)$ is an exothermic process because :
(1) The chemical energy of $H_2(g)$ and $O_2(g)$ is more than that of water
(2) The chemical energy of $H_2(g)$ and $O_2(g)$ is less than that of water
(3) The temperature of $H_2(g)$ and $O_2(g)$ is higher than that of water
(4) The temperature of $H_2(g)$ and $O_2(g)$ is lower than that of water
CTD081

82. Which plot represents for an exothermic reaction:

(1) 

(2) 

(3) 

(4) 

CTD082

83. Which one of the following is not applicable for a thermochemical equation :

(1) It tells about physical state of reactants and products

(2) It tells whether the reaction is spontaneous

(3) It tells whether the reaction is exothermic or endothermic

(4) It tells about the allotropic form (if any) of the reactants

CTD083

84. The correct thermochemical equation is :

(1) $C + O_2 \longrightarrow CO_2$; $\Delta H = -94$ kCal

(2) $C + O_2 \longrightarrow CO_2$; $\Delta H = +94.0$ kCal

(3) $C(s) + O_2(g) \longrightarrow CO_2(g)$; $\Delta H = -94$ kCal

(4) $C(s) + O_2(g) \longrightarrow CO_2(g)$; $\Delta H = +94$ kCal

CTD084

85. The enthalpy changes of formation of the gaseous oxide of nitrogen (N_2O and NO) are positive because of :

(1) The high bond energy of the nitrogen molecule

(2) The high electron affinity of oxygen atoms

(3) The high electron affinity of nitrogen atoms

(4) The tendency of oxygen to form O^{2-}

CTD085

86. ΔH for transition of carbon from diamond form to graphite form is – 453.5 Cal. This suggests that :

(1) Graphite is chemically different from diamond

(2) Graphite is as stable as diamond

(3) Graphite is more stable than diamond

(4) Diamond is more stable than graphite

CTD086

87. Which of the following values of heat of formation indicates that the product is least stable

(1) – 94 kCal (2) – 231.6 kCal

(3) + 21.4 kCal (4) + 64.8 kCal

CTD087

88. Heat of formation, ΔH_f° of an explosive compound like NCl_3 is –

(1) Positive

(2) Negative

(3) Zero

(4) Positive or negative

CTD088

89. According to the following reaction

$C(s) + 1/2\ O_2(g) \rightarrow CO(g)$, $\Delta H = -26.4$ kCal

(1) CO is an endothermic compound

(2) CO is an exothermic compound

(3) The reaction is endothermic

(4) None of the above

CTD089

90. Which of the following represents an exothermic reaction:-

(1) $N_2(g) + O_2(g) \rightarrow 2NO(g)$, $\Delta H = 180.5$ kJ

(2) $H_2O(g) + C(s) \rightarrow CO(g) + H_2(g)$, $\Delta E = 131.2$ kJ

(3) $2HgO(s) + 180.4\ KJ \rightarrow 2Hg(\ell) + O_2(g)$

(4) $2Zn(s) + O_2(g) \rightarrow 2ZnO(s)$, $\Delta E = -693.8$ kJ

CTD090

91. Consider the reaction $3O_2 \rightarrow 2O_3$; $\Delta H = +$ Ve, from the reaction, we can say that :–

(1) Ozone is more stable then oxygen

(2) Ozone is less stable then oxygen and ozone decomposes forming oxygen readily

(3) Oxygen is less stable than ozone and oxygen decomposes forming ozone readily

(4) None of the above

CTD091

92. From the reaction P(White) $\rightarrow$ P(Red) ; $\Delta H = -18.4$ kJ, it follows that :-

(1) Red P is readily formed from white P

(2) White P is readily formed from red P

(3) White P can not be converted to red P

(4) White P can be converted into red P and red P is more stable

CTD092

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FACTORS AFFECTING HEAT OF REACTION

93. In Kirchoff's equation which factor affects the heat of reaction :
(1) Pressure (2) Temperature
(3) Volume (4) Atomicity
CTD093

94. The enthalpy of a reaction at 273 K is –3.57 kJ. what will be the enthalpy of reaction at 373 K if ΔC_p = zero :–
(1) – 3.57 kJ (2) Zero
(3) $-3.57 \times \frac{373}{273}$ kJ (4) – 375 kJ
CTD094

95. For the reactions,
(i) $H_2(g) + Cl_2(g) \longrightarrow 2HCl(g) + x$ kJ
(ii) $H_2(g) + Cl_2(g) \longrightarrow 2HCl(\ell) + y$ kJ
Which one of the following statement is correct :
(1) x > y (2) x < y
(3) x = y (4) More data required
CTD095

HEAT OF FORMATION

96. Since the enthalpy of formation of the elements in their standard states is taken to be zero. The heat of formation (ΔH_f) of compounds :
(1) Is always negative
(2) Is always positive
(3) Is zero
(4) May be positive or negative
CTD096

97. Reaction $H_2(g) + I_2(g) \longrightarrow 2HI$; ΔH = 12.40 kCal. According to this, heat of formation of HI will be –
(1) 12.40 kCal (2) – 12.40 kCal
(3) – 6.20 kCal (4) 6.20 kCal
CTD097

98. Which of the following equations respresents standard heat of formation of CH_4 ?
(1) $C_{(diamond)} + 2H_{2\,(g)} \longrightarrow CH_{4(g)}$
(2) $C_{(graphite)} + 2H_{2\,(g)} \longrightarrow CH_{4(g)}$
(3) $C_{(diamond)} + 4H_{(g)} \longrightarrow CH_{4(g)}$
(4) $C_{(graphite)} + 4H_{(g)} \longrightarrow CH_{4(g)}$
CTD099

99. The enthalpy of formation of ammonia is –46.0 kJ mol^{-1}. The enthalpy change for the reaction $2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$ is :
(1) 46.0 kJ mol^{-1} (2) 92.0 kJ mol^{-1}
(3) – 23.0 kJ mol^{-1} (4) – 92.0 kJ mol^{-1}
CTD100

100. Given enthalpy of formation of $CO_2(g)$ and CaO(s) are – 94.0 kJ and – 152 kJ respectively and the enthalpy of the reaction :
$CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$ is 42 kJ. The enthalpy of formation of $CaCO_3(s)$ is
(1) – 42 KJ (2) – 202 KJ
(3) +202 KJ (4) – 288KJ
CTD101

101. Given that standard enthalpy of formation of CH_4, C_2H_4 and C_3H_8 are –17.9, 12.5, –24.8 kCal mol^{-1}. The ΔH for $CH_4 + C_2H_4 \rightarrow C_3H_8$ is :
(1) – 55.2 kCal (2) – 30.2 kCal
(3) 55.2 kCal (4) – 19.4 kCal
CTD102

102. The standard molar heat of formation of ethane, CO_2 and water are respectively –21.1, –94.1 and – 68.3 kCal. The standard molar heat of combustion of ethane will be
(1) –372 kCal (2) –162 kCal
(3) – 240 kCal (4) –183.5 kCal
CTD103

103. The ΔH_f° for $CO_{2(g)}$, $CO_{(g)}$ and $H_2O_{(g)}$ are –393.5, –110.5 and –241.8 kJ mol^{-1} respectively the standard enthalpy change (in kJ) for the reaction
$CO_{2(g)} + H_{2\,(g)} \rightarrow CO_{(g)} + H_2O_{(g)}$ is -
(1) 524.1 (2) 41.2
(3) –262.5 (4) –41.2
CTD104

104. The enthalpies of combustion of carbon and carbon monoxide are –393.5 kJ and –283 kJ, respectively the enthalpy of formation of carbon monoxide is :
(1) –676.5 kJ (2) –110.5 kJ
(3) 110.5 kJ (4) 676.5 kJ
CTD105

105. The standard heat of formation of $CS_2(\ell)$ will be; given that the standard heat of combustion of carbon (s), sulphur(s) and $CS_2(\ell)$ are –393.3, –293.72 and –1108.76 kJ mol^{-1} respectively is
(1) –128.02 kJ $mole^{-1}$ (2) +12.802 kJ mol^{-1}
(3) +128.02 kJ mol^{-1} (4) –12.802 kJ mol^{-1}

CTD106

106. The heat of combustion of $CH_{4(g)}$, $C_{(s)}$ and $H_{2(g)}$ at 25 °C are –212.4 K Cal, –94.0 K Cal and –68.4 K Cal respectively, the heat of formation of CH_4 will be -
(1) +54.4 K Cal (2) –18.4 K Cal
(3) –375.2 K Cal (4) +212.8 K Cal

CTD107

107. Standard enthalpy of formation is zero for.
(1) $C_{diamond}$ (2) $Br_{(g)}$
(3) $C_{graphite}$ (4) $O_{3(g)}$

CTD108

108. The standard heats of formation of $NO_2(g)$ and $N_2O_4(g)$ are 8.0 and 2.0 kCal mol^{-1} respectively the heat of dimerization of NO_2 in kCal is
(1) 10.0 (2) –6.0 (3) –12.0 (4) –14.0

CTD109

109. M is a metal that forms an oxide M_2O

$\frac{1}{2}M_2O \rightarrow M + \frac{1}{4}O_2$ $\Delta H = 120$ kCal

When a sample of metal M reacts with one mole of oxygen what will be the ΔH in that case
(1) 240 kCal (2) – 240 kCal
(3) 480 kCal (4) – 480 kCal

CTD110

HEAT OF COMBUSTION

110. According to equation,
$C_6H_6(\ell) + 15/2\ O_2(g) \rightarrow 6CO_2(g) + 3H_2O(\ell)$; ΔH = – 3264.4 kJ mol^{-1} the energy evolved when 7.8 g benzene is burnt in air will be -
(1) 163.22 kJ (2) 32.64 kJ
(3) 3.264 kJ (4) 326.4 kJ

CTD111

111. Heat of combustion of CH_4,C_2H_6, C_2H_4 and C_2H_2 gases are –212.8, –373.0, –337.0 and –310.5 kCal respectively at the same temperature. The best fuel among these gases is :
(1) CH_4 (2) C_2H_6 (3) C_2H_4 (4) C_2H_2

CTD112

112. Given standard enthalpy of formation of CO (–110 kJ mol^{-1}) and CO_2(–394 kJ mol^{-1}). The heat of combustion when one mole of graphite burns is
(1) – 110 kJ (2) – 284 kJ
(3) – 394 kJ (4) – 504 kJ

CTD113

113. The enthalpy of formation for $C_2H_4(g)$, $CO_2(g)$ and $H_2O(\ell)$ at 25^0C and 1 atm. pressure are 52, – 394 and – 286 kJ $mole^{-1}$ respectively. The enthalpy of combustion of C_2H_4 will be:-
(1) + 1412 kJ $mole^{-1}$ (2) –1412 kJ $mole^{-1}$
(3) + 142.2 kJ $mole^{-1}$ (4) –141.2 kJ $mole^{-1}$

CTD114

114. The combustion of one mole of benzene takes place at 298 K and 1 atm. After combustion, $CO_2(g)$ and $H_2O(l)$ are produced and 3267.0 kJ of heat is liberated. Calculate the standard enthalpy of formation, $\Delta_f H^\circ$ of benzene.
Standard enthalpies of formation of $CO_2(g)$ and $H_2O(l)$ are –393.5 kJ mol^{-1} and –285.83 kJ mol^{-1} respectively.
(1) 48.51 kJ mol^{-1} (2) –48.51 kJ mol^{-1}
(3) –97.02 kJ mol^{-1} (4) 97.02 kJ mol^{-1}

CTD115

115. The heat evolved during the combustion of 112 litre of water gas at STP (mixture of equal volume of H_2 and CO) is : Given
$H_2(g) + ½\ O_2(g) \rightarrow H_2O(g)$; ΔH = –241.8 kJ
$CO(g) + ½\ O_2(g) \rightarrow CO_2(g)$; ΔH = –283 kJ
(1) 241.8 kJ (2) 283 kJ
(3) 1312 kJ (4) 1586 kJ

CTD116

116. A person requires 2870 kCal of energy to lead normal daily life. If heat of combustion of cane sugar is –1349 $kCalmol^{-1}$, then his daily consumption of sugar is :
(1) 728 g (2) 0.728 g
(3) 342 g (4) 0.342 g

CTD117

117. The following are the heats of reactions -
(i) ΔH_f^o of $H_2O_{(\ell)}$ = –68.3 kCal mol^{-1}
(ii) $\Delta H_{comb.}^o$ of C_2H_2 = –337.2 kCal mol^{-1}
(iii) $\Delta H_{comb.}^o$ of C_2H_4 = –363.7 kCal mol^{-1}
Then heat change for the reaction
$C_2H_2 + H_2 \rightarrow C_2H_4$ is -
(1) –716.1 kCal (2) + 337.2 kCal
(3) –41.8 kCal (4) –579.5 kCal

CTD118

118. The heat of combustion of a substance is :-
(1) Always positive
(2) Always negative
(3) Numerically equal to the heat of formation
(4) 1 and 3 both

CTD119

119. The value of ΔH for the combustion of C(s) is –94.4 kCal. The heat of formation of $CO_2(g)$ is :-
(1) –49.5 kCal
(2) –94.4 kCal
(3) –188.0 kCal
(4) More data required

CTD120

120. In the combustion of 0.4 g. of CH_4, 0.25 kCal. of heat is liberated. The heat of combustion of CH_4 is
(1) – 20 kCal (2) – 10 kCal
(3) – 2.5 kCal (4) – 5 kCal

CTD121

121. If $C_6H_{12}O_6(s) + 9O_2(g) \rightarrow 6CO_2(g) + 6H_2O(g)$; ΔH = – 680 kCal The weight of $CO_2(g)$ produced when 170 kCal of heat is evolved in the combustion of glucose is:-
(1) 265 g (2) 66 g (3) 11 g (4) 64 g

CTD122

122. Which of the following equations corresponds to the enthalpy of combustion at 298 K :-
(1) $C_2H_6(g) + 7/2\ O_2(g) \rightarrow 2CO_2(g) + 3H_2O(g)$
(2) $2C_2H_6(g) + 7\ O_2(g) \rightarrow 4CO_2(g) + 6H_2O(g)$
(3) $C_2H_6(g) + 7/2\ O_2(g) \rightarrow 2CO_2(g) + 3H_2O(\ell)$
(4) $2C_2H_6(g) + 7O_2(g) \rightarrow 4CO_2(g) + 6H_2O(\ell)$

CTD123

123. Heat of formation of CO_2 is – 94.0 kCal. What would be the quantity of heat liberated, when 3 g of graphite is burnt in excess of oxygen:-
(1) 23.5 kCal (2) 2.35 kCal
(3) 94.0 kCal (4) 31.3 kCal

CTD124

HEAT OF NEUTRALIZATION

124. The amount of heat liberated when one mole of NH_4OH reacts with one mole of HCl is
(1) 13.7 kCal
(2) More than 13.7 kCal
(3) Less than 13.7 kCal
(4) Cannot be predicted

CTD125

125. If $H^+ + OH^- = H_2O$ + 13.7 kCal, then heat of complete neutralisation of one gram mole of H_2SO_4 with strong base will be :
(1) –13.7 kcal (2) –27.4 kcal
(3) –6.85 kcal (4) –3.425 kCal

CTD126

126. Heat of neutralisation of a strong dibasic acid in dilute solution by NaOH is nearly :
(1) – 27.4 kCal eq^{-1} (2) – 13.7 kCal eq^{-1}
(3) 13.7 kCal eq^{-1} (4) – 13.7 kCal mol^{-1}

CTD127

127. The temperature of a 5 ml of strong acid increases by 5^{0}C when 5 ml of a strong base is added to it. If 10 ml of each are mixed temperature should increase by :
(1) 5^0 C (2) 10^{0}C
(3) 15^{0}C (4) Cannot be known

CTD128

128. The heat of neutralization of HCl by NaOH is –55.9 kJ mol^{-1}. If the heat of neutralization of HCN by NaOH is – 12.1 kJ mol^{-1}. The energy of dissociation of HCN is
(1) –43.8 kJ (2) 43.8 kJ
(3) 68 kJ (4) – 68 kJ

CTD129

129. If water is formed from H^+ ions and OH^- the heat exchange during the reaction :
(1) – 13.7 kCal (2) 13.7 kCal
(3) –63.4 kCal (4) More data required

CTD130

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130. The change in the enthalpy of
$NaOH + HCl \longrightarrow NaCl + H_2O$ is called :
(1) Heat of neutralisation
(2) Heat of reaction
(3) Heat of hydration
(4) Heat of solution

CTD131

HEAT OF HYDROGENATION

131. The heat of combustion of C_2H_4, C_2H_6 and H_2 are –1409.5 kJ, –1558.3 kJ and –285.6 kJ. The heat of hydrogenation of ethene is -
(1) –136.8 kJ (2) –13.68 kJ
(3) 273.6 kJ (4) 1.368 kJ

CTD132

132. The enthalpy of combustion of cyclohexane, cyclohexene and H_2 are respectively –3920, –3800 and –241 kJ mol^{-1}. The heat of hydrogenation of cyclohexene is:-
(1) –121 kJ mol^{-1} (2) 121 kJ mol^{-1}
(3) –242 kJ mol^{-1} (4) 242 kJ mol^{-1}

CTD133

BOND ENERGY/RESONANCE ENERGY

133. Bond energy of a molecule :
(1) Is always negative
(2) Is always positive
(3) Either positive or negative
(4) Depends upon the physical state of the system

CTD134

134. Among the following for which reaction heat of reaction represents bond energy of HCl
(1) $HCl(g) \longrightarrow H^+(g) + Cl^-(g)$
(2) $HCl(g) \longrightarrow \frac{1}{2} H_2(g) + \frac{1}{2} Cl_2(g)$
(3) $2HCl(g) \longrightarrow H_2(g) + Cl_2(g)$
(4) $HCl(g) \longrightarrow H(g) + Cl(g)$

CTD135

135. The bond energies of F_2, Cl_2, Br_2 and I_2 are 155.4, 243.6, 193.2 and 151.2 kJ mol^{-1} respectively. The strongest bond is :
(1) F – F (2) Cl – Cl
(3) Br – Br (4) I – I

CTD136

136. Energy required to dissociate 4g of gaseous hydrogen into free gaseous atoms is 208 kCal at 25^0C. The bond energy of H—H bond will be :
(1) 1.04 kCal (2) 10.4 kCal
(3) 104 kCal (4) 1040 kCal

CTD137

137. Heat evolved in the reaction
$H_2 + Cl_2 \longrightarrow 2HCl$ is 182 kJ. Bond energies of H–H and Cl–Cl are 430 and 242 kJ mol^{-1} respectively. The H–Cl bond energy is :
(1) 245 kJ mol^{-1} (2) 427 kJ mol^{-1}
(3) 336 kJ mol^{-1} (4) 154 kJ mol^{-1}

CTD138

138. The enthalpy change for the reaction
$H_2(g) + C_2H_4(g) \rightarrow C_2H_6(g)$ is ..... The bond energies are in kCal mol^{-1}
H – H = 103, C – H = 99, C – C = 80 & C = C =145
(1) –10 kCal mol^{-1} (2) +10 kCal mol^{-1}
(3) – 30 kCal mol^{-1} (4) +30 kCal mol^{-1}

CTD139

139. Bond dissociation enthalpies of $H_2(g)$ and $N_2(g)$ are 436.0 kJ mol^{-1} and 941.8 kJ mol^{-1} respectively and enthalpy of formation of $NH_3(g)$ is –46 kJ mol^{-1}. What is enthalpy of atomization of $NH_3(g)$?
(1) 390.3 kJ mol^{-1} (2) 1170.9 kJ mol^{-1}
(3) 590 kJ mol^{-1} (4) 720 kJ mol^{-1}

CTD140

140. From the reactions :
$C(s) + 2H_2(g) \rightarrow CH_4(g)$ $\Delta H = -X$ kcal
$C(g) + 4H(g) \rightarrow CH_4(g)$, $\Delta H = -X_1$ kcal
$CH_4(g) \rightarrow CH_3(g) + H(g)$ $\Delta H = +Y$ kcal
Bond energy of C–H bond is –
(1) $\frac{X}{4}$ kCal mol^{-1} (2) Y kCal mol^{-1}
(3) $\frac{X_1}{4}$ kCal mol^{-1} (4) X_1 kCal mol^{-1}

CTD141

141. The enthalpy changes at 298 K in successive breaking of O–H bonds of water are

$H_2O \longrightarrow H(g) + OH(g)$; $\Delta H = 498$ kJ mol^{-1}

$OH(g) \longrightarrow H(g) + O(g)$; $\Delta H = 428$ kJmol^{-1}

the bond enthalpy of O–H bond is

(1) 498 kJ mol^{-1}

(2) 428 kJ mol^{-1}

(3) 70 kJ mol^{-1}

(4) 463 kJ mol^{-1}

CTD142

142. If ΔH_f° of $ICl_{(g)}$, $Cl_{(g)}$, and $I_{(g)}$ is 17.57, 121.34 and 106.96 J mol^{-1} respectively. Then bond dissociation energy of ICl bond is -

(1) 35.15 J mol^{-1} (2) 106.69 J mol^{-1}

(3) 210.73 J mol^{-1} (4) 420.9 J mol^{-1}

CTD143

143. Heat of dissociation of benzene to elements is 5535 kJ mol^{-1}. The bond enthalpies of C–C, C=C and C–H are 347.3, 615.0 and 416.2 kJ respectively. Magnitude of resonance energy of benzene is

(1) 1.51 kJ (2) 15.1 kJ

(3) 151 kJ (4) 1511 kJ

CTD144

SOME OTHER HEAT OF REACTIONS

144. The enthalpy change for the reaction

$2C(\text{graphite}) + 3H_2(g) \longrightarrow C_2H_6(g)$ is called

(1) Enthalpy of formation

(2) Enthalpy of combustion

(3) Enthalpy of hydrogenation

(4) Enthalpy of vaporisation

CTD145

145. $Cl_2(g) \longrightarrow 2Cl(g)$, In this process value of ΔH will be -

(1) Positive

(2) Negative

(3) Zero

(4) Nothing can be predicted

CTD146

146. The magnitude of heat of solution ..... on addition of solvent to solution

(1) Decreases

(2) Increases

(3) Remains constant

(4) Increases or decreases

CTD147

147. If $H_2(g) \longrightarrow 2H(g)$; $\Delta H = 104$ kCal, than heat of atomisation of hydrogen is :

(1) 52 kCal (2) 104 kCal

(3) 208 kCal (4) None of these

CTD148

148. $S_{(\text{rhombic})} + O_{2(g)} \longrightarrow SO_{2(g)}$; $\Delta H = -297.5$ kJ

$S_{(\text{monoclinic})} + O_{2\,(g)} \longrightarrow SO_2$; $\Delta H = -300$ kJ

The data can predict that –

(1) Rhombic sulphur is yellow in colour

(2) Monoclinic sulphur has metallic lusture.

(3) Monoclinic sulphur is more stable

(4) ΔH transition of S_R to S_M is endothermic

CTD149

149. The heat of combustion of yellow phosphorous and red phosphorous are – 9.91 kJ and –8.78 kJ respectively. The heat of transition of yellow phosphorous to red phosphorous is

(1) –18.69 kJ (2) +1.13 kJ

(3) +18.69 kJ (4) –1.13 kJ

CTD150

150. $2CO_{(g)} + O_{2(g)} \longrightarrow 2CO_{2(g)} + X$ kJ

In the above equation X kJ refers to :

(1) Heat of formation of CO_2

(2) Heat of vapourisation

(3) Heat of reaction

(4) Heat of sublimation

CTD151

151. ΔH for the reaction, $I_{(g)} + I_{(g)} \rightarrow I_2(g)$ will be:-

(1) Zero (2) – ve (3) + ve (4) ∞

CTD152

152. Given that :

$A(s) \xrightarrow{T K} A(\ell)$; $\Delta H = x$,

$A(\ell) \xrightarrow{T K} A(g)$; $\Delta H = y$

Calculate enthalpy of sublimation at 'T'K :-

(1) x + y (2) x – y

(3) x or y (4) –(x + y)

CTD153

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HESS LAW

153. The enthalpy change of a reaction does not depend on
(1) State of reactants and products
(2) Nature of reactants and products
(3) Different intermediate reactions
(4) Initial and final enthalpy change of reaction

CTD154

154. From the thermochemical reactions,
$C(graphite) + \frac{1}{2} O_2 \longrightarrow CO$; $\Delta H = -110.5$ kJ
$CO + \frac{1}{2} O_2 \longrightarrow CO_2$; $\Delta H = -283.2$ kJ
the heat of reaction of
$C(graphite) + O_2 \longrightarrow CO_2$ is :
(1) 393.7 kJ (2) – 393.7 kJ
(3) – 172.7 kJ (4) + 172.7 kJ

CTD155

155. If $H_2 + \frac{1}{2} O_2 \longrightarrow H_2O$; $\Delta H = -68.39$ kCal
$K + H_2O + water \longrightarrow KOH(aq) + \frac{1}{2} H_2$; $\Delta H = -48.0$ kCal
$KOH + water \longrightarrow KOH(aq)$ $\Delta H = -14.0$ kCal
the heat of formation of KOH is -
(1) – 68.39 + 48 – 14.0
(2) – 68.39 – 48.0 +14.0
(3) +68.39 – 48.0 + 14.0
(4) + 68.39 + 48.0 – 14.0

CTD156

156. Given $C(s) + O_2(g) \longrightarrow CO_2(g) + 94.2$ kCal
$H_2(g) + \frac{1}{2} O_2(g) \longrightarrow H_2O(\ell) + 68.3$ kCal
$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(\ell) + 210.8$ kCal
The heat of formation of methane in Kcal will be
(1) –45.9 (2) –47.8
(3) –20.0 (4) –47.3

CTD157

157. If, $H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$;
$\Delta H^0 = -44$ kCal
$2Na(s) + 2HCl(g) \longrightarrow 2NaCl(s) + H_2(g)$;
$\Delta H^0 = -152$ kCal
Then, $Na(s) + 0.5\ Cl_2(g) \longrightarrow NaCl(s)$; $\Delta H^0 = ?$
(1) 108 kCal
(2) 196 kCal
(3) – 98 kCal
(4) 54 kCal

CTD158

158. (i) $S(s) + 3/2\ O_2(g) \rightarrow SO_3(g) + 2x$ kCal
(ii) $SO_2(g) + \frac{1}{2} O_2(g) \rightarrow SO_3(g) + y$ kCal
Calculate the heat of formation of SO_2 :
(1) (2x + y) (2) –(2x – y)
(3) x + y (4) 2x / y

CTD159

159. If $S + O_2 \longrightarrow SO_2$; $\Delta H = -298.2$ kJ
$SO_2 + \frac{1}{2} O_2 \longrightarrow SO_3$; $\Delta H = -98.7$ kJ
$SO_3 + H_2O \longrightarrow H_2SO_4$; $\Delta H = -130.2$ kJ
$H_2 + \frac{1}{2} O_2 \longrightarrow H_2O$; $\Delta H = -287.3$ kJ
Then the enthalpy of formation of H_2SO_4 at 298 K is -
(1) –814.4 kJ (2) –650.3 kJ
(3) –320.5 kJ (4) –433.5 kJ

CTD160

160. Given that :
$Zn + \frac{1}{2} O_2 \rightarrow ZnO + 84000$ Cal
$Hg + \frac{1}{2} O_2 \rightarrow HgO + 21700$ Cal
The heat of reaction (ΔH) for,
$Zn + HgO \rightarrow ZnO + Hg$ is :-
(1) 105700 Cal (2) 62300 Cal
(3) –105700 Cal (4) – 62300 Cal

CTD161

161. Given that -
$2C(s) + 2O_2(g) \rightarrow 2CO_2(g)$ $\Delta H = -787$ kJ
$H_2(g) + \frac{1}{2} O_2(g) \rightarrow H_2O(\ell)$ $\Delta H = -286$ kJ
$C_2H_2(g) + \frac{5}{2} O_2(g) \rightarrow 2CO_2(g) + H_2O(\ell)$
$\Delta H = -1310$ kJ Heat of formation of acetylene is :-
(1) + 1802 kJ (2) – 1802 kJ
(3) – 800 kJ (4) + 237 kJ

CTD162

162. The heat of reaction for

$A + \frac{1}{2}O_2 \rightarrow AO$ is –50 kCal and

$AO + \frac{1}{2}O_2 \rightarrow AO_2$ is 100 kCal. The heat of reaction for $A + O_2 \rightarrow AO_2$ is:-

(1) – 50 kCal (2) + 50 kCal

(3) 100 kCal (4) 150 kCal

CTD163

163. $C(s) + O_2(g) \rightarrow CO_2(g) + 94.0$ kCal

$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$, $\Delta H = -67.7$ kCal

from the above reactions find how much heat (kCal mole^{-1}) would be produced in the following reaction : $C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g)$

(1) 20.6 (2) 26.3

(3) 44.2 (4) 161.6

CTD164

164. The enthalpy of vapourisation of liquid water using the data:

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell)$; $\Delta H = -285.77$ kJ mol^{-1}

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$; $\Delta H = -241.84$ kJ mol^{-1}

(1) +43.93 kJ mol^{-1}

(2) –43.93 kJ mol^{-1}

(3) +527.61 kJ mol^{-1}

(4) –527.61 kJ mol^{-1}

CTD165

165. $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell)$;

$\Delta H_{298K} = -68.32$kCal. Heat of vapourisation of water at 1 atm and 25^{0}C is 10.52 kCal. The standard heat of formation (in kCal) of 1 mole of water vapour at 25^{0}C is

(1) 10.52 (2) –78.84 (3) +57.80 (4) –57.80

CTD166

166. The heat of solution of anhydrous $CuSO_4$ and $CuSO_4.5H_2O$ are – 15.89 and 2.80 kCal mol^{-1} respectively. What will be the heat of hydration of anhydrous $CuSO_4$?

(1) –18.69 kCal (2) 18.69 kCal

(3) –28.96 kCal (4) 28.96 kCal

CTD167

167. Which of the following expressions is true:-

(1) $H_f^0(CO,g) = \frac{1}{2}\Delta H_f^0(CO_2,g)$

(2) $\Delta H_f^0(CO,g)$

$= \Delta H_f^0(C,\text{graphite}) + \frac{1}{2}\Delta H_f^0(O_2,g)$

(3) $\Delta H_f^0(CO,g) = \Delta H_f^0(CO_2,g) - \frac{1}{2}\Delta H_f^0(O_2,g)$

(4) $\Delta H_f^0(CO,g)$

$= \Delta H_{comb}^0(C,\text{graphite}) - \Delta H_{comb}^0(CO,g)$

CTD168

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EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	4	4	3	3	1	4	1	1	1	4	2	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	4	1	1	3	1	3	2	2	4	2	2	4	2	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	1	2	3	4	2	2	2	3	1	1	1	2	3	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	4	3	1	1	2	3	3	1	3	2	2	1	4	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	4	3	2	2	1	4	3	2	3	3	4	1	2	2	2
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	4	1	4	1	4	1	1	2	3	1	3	4	1	2	4
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	2	4	2	1	2	4	4	2	2	4	4	1	2	2	3
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	2	3	4	4	4	1	3	2	1	3	1	3	2	2	2
Question	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Answer	2	3	1	3	2	2	1	2	1	1	1	1	2	4	2
Question	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Answer	3	2	3	2	3	4	3	3	1	1	3	2	4	4	3
Question	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Answer	2	1	3	2	2	3	3	2	1	4	4	2	2	1	4
Question	166	167													
Answer	1	4													

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. For the reaction : $X_2O_4(\ell) \longrightarrow 2XO_2(g)$

$\Delta U = 2.1$ kCal, $\Delta S = 20$ cal K^{-1} at 300 K

Hence ΔG is :-

(1) 2.7 kCal (2) – 2.7 kCal

(3) 9.3 kCal (4) – 9.3 kCal

CTD182

AIPMT 2015

2. Which of the following statements is **correct** for a reversible process in a state of equilibrium ?

(1) $\Delta G = 2.30\ RT \log K$

(2) $\Delta G^\circ = -2.30\ RT \log K$

(3) $\Delta G^\circ = 2.30\ RT \log K$

(4) $\Delta G = -2.30\ RT \log K$

CTD183

Re-AIPMT 2015

3. The heat of combustion of carbon to CO_2 is -393.5 kJ mol^{-1}. The heat exchange in the formation of 35.2 g of CO_2 from carbon and oxygen gas is:

(1) –630 kJ (2) –3.15 kJ

(3) –315 kJ (4) +315 kJ

CTD184

NEET-I 2016

4. The correct thermodynamic conditions for the spontaneous reaction at all temperatures is

(1) $\Delta H > 0$ and $\Delta S > 0$

(2) $\Delta H > 0$ and $\Delta S < 0$

(3) $\Delta H < 0$ and $\Delta S > 0$

(4) $\Delta H < 0$ and $\Delta S < 0$

CTD185

NEET-II 2016

5. For a sample of perfect gas when its pressure is changed isothermally from p_i to p_f, the entropy change is given by

(1) $\Delta S = nRT \ln \left(\frac{p_f}{p_i}\right)$ (2) $\Delta S = RT \ln \left(\frac{p_i}{p_f}\right)$

(3) $\Delta S = nR \ln \left(\frac{p_f}{p_i}\right)$ (4) $\Delta S = nR \ln \left(\frac{p_i}{p_f}\right)$

CTD186

NEET(UG) 2017

6. For a given reaction, $\Delta H = 35.5$ kJ mol^{-1} and $\Delta S = 83.6$ $JK^{-1}mol^{-1}$. The reaction is spontaneous at : (Assume that ΔH and ΔS do not vary with temperature)

(1) T > 425 K (2) All temperatures

(3) T > 298 K (4) T < 425 K

CTD187

7. A gas is allowed to expand in a well insulated container against a constant external pressure of 2.5 atm from an initial volume of 2.50 L to a final volume of 4.50 L. The change in internal energy ΔU of the gas in joules will be:-

(1) –500 J (2) –505 J

(3) +505 J (4) 1136.25 J

CTD188

NEET (UG) 2018

8. The bond dissociation energies of X_2, Y_2 and XY are in the ratio of 1 : 0.5 : 1. ΔH for the formation of XY is –200 kJ mol^{-1}. The bond dissociation energy of X_2 will be

(1) 200 kJ mol^{-1}

(2) 100 kJ mol^{-1}

(3) 800 kJ mol^{-1}

(4) 400 kJ mol^{-1}

CTD189

NEET(UG) 2019

9. Under isothermal condition, a gas at 300 K expands from 0.1L to 0.25L against a constant external pressure of 2 bar. The work done by the gas is :- [Given that 1L bar = 100 J]

(1) –30 J (2) 5kJ

(3) 25 J (4) 30 J

CTD190

NEET(UG) 2019 (Odisha)

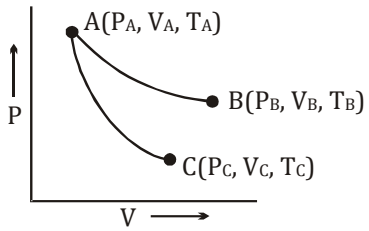
10. An ideal gas expands isothermally from 10^{-3} m^3 to 10^{-2} m^3 at 300 K against a constant pressure of 10^5 Nm^{-2}. The work, in the process is :-

(1) +270 kJ (2) –900 J

(3) +900 kJ (4) –900 kJ

CTD191

11. Reversible expansion of an ideal gas under isothermal and adiabatic conditions are as shown in the figure.



AB → Isothermal expansion

AC → Adiabatic expansion

Which of the following options is **not** correct ?

(1) $\Delta S_{isothermal} > \Delta S_{adiabatic}$

(2) $T_A = T_B$

(3) $W_{isothermal} > W_{adiabatic}$

(4) $T_c > T_A$

CTD192

NEET (UG) 2020

12. The correct option for free expansion of an ideal gas under adiabatic condition is :

(1) $q > 0, \Delta T > 0$ and $w > 0$

(2) $q = 0, \Delta T = 0$ and $w = 0$

(3) $q = 0, \Delta T < 0$ and $w > 0$

(4) $q < 0, \Delta T = 0$ and $w = 0$

CTD193

13. Hydrolysis of sucrose is given by the following reaction.

Sucrose + $H_2O \rightleftharpoons$ Glucose + Fructose

If the equilibrium constant (K_c) is 2×10^{13} at 300K, the value of $\Delta_r G^1$ at the same temperature will be:

(1) $-8.314\ J\ mol^{-1}\ K^{-1} \times 300\ K \times \ln(4 \times 10^{13})$

(2) $-8.314\ J\ mol^{-1}\ K^{-1} \times 300\ K \times \ln(2 \times 10^{13})$

(3) $8.314\ J\ mol^{-1}\ K^{-1} \times 300\ K \times \ln(2 \times 10^{13})$

(4) $8.314\ J\ mol^{-1}\ K^{-1} \times 300\ K \times \ln(3 \times 10^{13})$

CTD194

14. For the reaction $2Cl(g) \rightarrow Cl_2(g)$, the **correct** option is:

(1) $\Delta_r H < 0$ and $\Delta_r S < 0$

(2) $\Delta_r H > 0$ and $\Delta_r S > 0$

(3) $\Delta_r H > 0$ and $\Delta_r S < 0$

(4) $\Delta_r H < 0$ and $\Delta_r S > 0$

CTD195

NEET (UG) 2020 (Covid-19)

15. If for a certain reaction $\Delta_r H$ is 30 kJ mol^{-1} at 450 K, the value of $\Delta_r S$ (in $JK^{-1}\ mol^{-1}$) for which the same reaction will be spontaneous at the same temperature is

(1) 70 (2) –33 (3) 33 (4) –70

CTD196

16. At standard conditions, if the change in the enthalpy for the following reaction is –109 kJ mol^{-1} $H_{2(g)} + Br_{2(g)} \rightarrow 2HBr_{(g)}$

Given that bond energy of H_2 and Br_2 is 435 kJ mol^{-1} and 192 kJ mol^{-1}, respectively, what is the bond energy (in kJ mol^{-1}) of HBr?

(1) 368 (2) 736 (3) 518 (4) 259

CTD197

NEET (UG) 2021

17. Which one among the following is the correct option for right relationship between C_P and C_V for one mole of ideal gas ?

(1) $C_P + C_V = R$ (2) $C_P - C_V = R$

(3) $C_P = RC_V$ (4) $C_V = RC_P$

CTD198

18. For irreversible expansion of an ideal gas under isothermal condition, the correct option is :

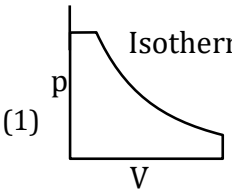
(1) $\Delta U = 0, \Delta S_{total} = 0$ (2) $\Delta U \neq 0, \Delta S_{total} \neq 0$

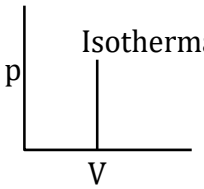
(3) $\Delta U = 0, \Delta S_{total} \neq 0$ (4) $\Delta U \neq 0, \Delta S_{total} = 0$

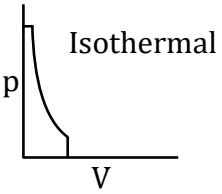
CTD199

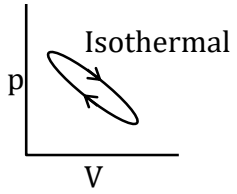
NEET (UG) 2022

19. Which of the following p-V curve represents maximum work done ?

(1) 

(2) 

(3) 

(4) 

CTD200

20. One mole of an ideal gas at 300 K is expanded isothermally from 1 L to 10 L volume. ΔU for this process is (Use R = 8.314 J K^{-1} mol^{-1})

(1) 1260 J (2) 2520 J

(3) 5040 J (4) 0 J

CTD201

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NEET (UG) 2023

21. The equilibrium concentrations of the species in the reaction A + B $\rightleftharpoons$ C + D are 2, 3, 10 and 6 mol L^{-1}, respectively at 300 K. ΔG^0 for the reaction is
(R = 2 cal/mol K)
(1) –137.26 cal (2) –1381.80 cal
(3) –13.73 cal (4) 1372.60 cal

CTD224

22. Which amongst the following options is the **correct** relation between change in enthalpy and change in internal energy?
(1) $\Delta H = \Delta U + \Delta n_g RT$ (2) $\Delta H - \Delta U = -\Delta nRT$
(3) $\Delta H + \Delta U = \Delta nR$ (4) $\Delta H = \Delta U - \Delta n_g RT$

CTD225

NEET (UG) 2024

23. Match List I with List II.

List-I (Process)	List-II (Conditions)
A. Isothermal process	I. No heat exchange
B. Isochoric process	II. Carried out at constant temperature
C. Isobaric process	III. Carried out at constant volume
D. Adiabatic process	IV. Carried out at constant pressure

Choose the correct answer from the options given below :
(1) A-IV, B-III, C-II, D-I (2) A-IV, B-II, C-III, D-I
(3) A-I, B-II, C-III, D-IV (4) A-II, B-III, C-IV, D-I

CTD268

24. In which of the following processes entropy increases ?
A. A liquid evaporates to vapour
B. Temperature of a crystalline solid lowered from 130 K to 0K.
C. $2NaHCO_{3(s)} \rightarrow Na_2CO_{3(s)} + CO_{2(g)} + H_2O_{(g)}$
D. $Cl_{2(g)} \rightarrow 2\,Cl_{(g)}$
Choose the correct answer from the options given below :
(1) A and C (2) A, B and D
(3) A, C and D (4) C and D

CTD269

25. The work done during reversible isothermal expansion of one mole of hydrogen gas at 25°C from pressure of 20 atmosphere to 10 atmosphere is :
(Given R = 2.0 cal K^{-1} mol^{-1})
(1) 0 calorie (2) –413.14 calories
(3) 413.14 calories (4) 100 calories

CTD270

NEET (UG) 2024 (Re-examination)

26. Choose the correct statement for the work done in the expansion and heat absorbed or released when 5 litres of an ideal gas at 10 atmospheric pressure isothermally expands into vacuum until volume is 15 litres :
(1) Both the heat and work done will be greater than zero
(2) Heat absorbed will be less than zero and work done will be positive.
(3) Work done will be zero and heat will also be zero
(4) Work done will be greater than zero and heat will remain zero.

CTD271

27. For an endothermic reaction :
A. q_p is negative. B. $\Delta_r H$ is positive.
C. $\Delta_r H$ is negative. D. q_p is positive.
Choose the **correct** answer from the options given below :
(1) B and D (2) C and D
(3) A and B (4) A and C

CTD272

28. For the following reaction at 300 K
$A_2(g) + 3B_2(g) \rightarrow 2AB_3(g)$
the enthalpy change is + 15 kJ,
then the internal energy change is :
(1) 19988.4 J (2) 200 J
(3) 1999 J (4) 1.9988 kJ

CTD273

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	3	3	4	1	2	3	4	2	4	2	2	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28		
Answer	1	2	3	1	4	2	1	4	3	2	3	1	1		

Exercise – III (Analytical Questions) Master Your Understanding

1. Given below are two statements :
Statement I : A process is said to be reversible, if a change is brought out in such a way that the process could, at any moment, be reversed by an infinitesimal change.
Statement II : A reversible process proceeds by a series of equilibrium states such that system and surroundings are near equilibrium with each other.
In the light of the above statements, choose the most appropriate answer from the options given below :
(1) Both statement I and statement II are incorrect.
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct.
(4) Both statement I and statement II are correct.

CTD202

2. **Statement I :** Combustion of all organic compounds is an exothermic reaction.
Statement II : The enthalpy of formation of all elements in their stable and standard states is zero.
(1) Both statement I and statement II are incorrect.
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct.
(4) Both statement I and statement II are correct.

CTD203

3. Given below are two statements:
Statement I : $C_{p,m} - C_{v,m} = R$ is applicable for all gases.
Statement II : $C_{p,m} - C_{v,m} = R$ is applicable for ideal gas only.
In the light of the above statements, choose the most appropriate answer from the options given below :
(1) Both statement I and statement II are incorrect.
(2) Statement I is correct but statement II is incorrect.
(3) Statement I is incorrect but statement II is correct.
(4) Both statement I and statement II are correct.

CTD204

4. Which of the following is correct -
(1) ΔU is calculated at constant volume and pressure
(2) ΔH is calculated at constant volume only
(3) ΔS is calculated at constant volume only
(4) ΔG calculated at constant temperature and pressure is used to decide spontaneity.

CTD206

5. Which of the following is not correct –
(1) $(\Delta G)_{T,P}$ is zero at equilibrium
(2) $(\Delta G)_{T,P} > 0$ for a spontaneous reaction
(3) $(\Delta G)_{T,P} < 0$ for a spontaneous reaction
(4) $(\Delta G)_{T,P} > 0$ for a nonspontaneous reaction

CTD207

6. Match List – I with List- II.

List – I		List - II	
(a)	Entropy of vapourisation	(i)	ΔS = negative
(b)	ΔU in reversible adiabatic expansion of an ideal gas	(ii)	Decreases
(c)	Crystalline solid state	(iii)	Lowest entropy
(d)	A liquid freezes	(iv)	$\frac{\Delta H_{vap}}{T_b}$

(1) (a) – (iv), (b) – (ii), (c) – (iii), (d) – (i)
(2) (a) – (iii), (b) – (ii), (c) – (iv), (d) – (i)
(3) (a) – (iv), (b) – (i), (c) – (iii), (d) – (ii)
(4) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)

CTD208

7. Predict in which of the following entropy increases.
(a) A liquid crystallizes into a solid
(b) Temperature of a crystalline solid is raised from 0 K to 115 K
(c) $2NaHCO_3(s) \rightarrow Na_2CO_3(s) + CO_2(g) + H_2O(g)$
(d) $H_2(g) \rightarrow 2H(g)$
The correct one is –
(1) a, b (2) b, c, d
(3) a, b, c (4) a, b, c and d

CTD209

8. In a given process for an ideal gas w = 0 and q < 0 for gas.
The correct one is :
(1) Temperature will decrease.
(2) Volume will increase.
(3) Pressure remains constant
(4) Temperature will increase

CTD210

9. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): ΔH_f° for $H_2(g)$, and $Cl_2(g)$ is zero.
Reason (R): ΔH_f° for every element in its stable and standard state is zero.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**
(2) **(A)** is correct but **(R)** is not correct.
(3) **(A)** is not correct but **(R)** is correct.
(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

CTD211

10. Given below are two statements:
Statement I : All natural processes are spontaneous.
Statement II : All natural processes are irreversible.
(1) Both statement I and statement II are incorrect.
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct.
(4) Both statement I and statement II are correct.

CTD212

11. Which of the following pairs for a chemical reaction is certain to result in a spontaneous reaction.
(1) Exothermic and decreasing disorderness.
(2) Exothermic and increasing disorderness.
(3) Endothermic and increasing disorderness.
(4) Endothermic and decreasing disorderness.

CTD214

12. Given below are two statements:
Statement I : In an isolated system there is no exchange of energy but exchange of matter is possible between system and the surroundings.
Statement II : The presence of reactants in a thermous flask is an example of an isolated system.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both statement I and statement II are incorrect.
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct.
(4) Both statement I and statement II are correct.

CTD215

13. Given below are two statements:
Statement I : Natural processes are irreversible and spontaneous in nature.
Statement II : Decrease in enthalpy is a contributory factor for spontaneity.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) Both statement I and statement II are incorrect.
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct.
(4) Both statement I and statement II are correct.

CTD216

14. Given below are two statements:

Statement I : If both ΔH° and ΔS° are positive then reaction will be spontaneous at high temperature.

Statement II : All process with positive entropy change are spontaneous.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both statement I and statement II are incorrect.

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct.

(4) Both statements I and statements II are correct.

CTD218

15. Match List – I with List – II.

List – I		List - II	
(a)	$p_{ext.} = 0$	(i)	Adiabatic change
(b)	$q = p_{ext.}(v_f - v_i)$	(ii)	free expansion of an ideal gas
(c)	$q = 2.303\ nRT \log\left(\frac{v_f}{v_i}\right)$	(iii)	isothermal irreversible change
(d)	$\Delta U = W$	(iv)	isothermal reversible change

Choose the correct answer from the options given below:

(1) (a) – (ii), (b) – (iv), (c) – (iii), (d) – (i)

(2) (a) – (ii), (b) – (iii), (c) – (iv), (d) – (i)

(3) (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii)

(4) (a) – (i), (b) – (ii), (c) – (iii), (d) – (iv)

CTD220

16. Match List – I with List – II.

List – I		List - II	
(a)	$H_2O(s) \xrightarrow[1atm]{0^\circ C} H_2O(l)$	(i)	ΔG = –ve, ΔS = +ve
(b)	$H_2O(s) \xrightarrow[1atm]{10^\circ C} H_2O(l)$	(ii)	ΔG = +ve, ΔS = +ve
(c)	$H_2O(s) \xrightarrow[1atm]{-10^\circ C} H_2O(l)$	(iii)	ΔG = 0, ΔS = +ve
(d)	On addition of HCl in Ag^+ solution	(iv)	ΔG = –ve, ΔS = –ve

Choose the correct answer from the options given below:

(1) (a) – (i), (b) – (ii), (c) – (iii), (d) – (iv)

(2) (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii)

(3) (a) – (iii), (b) – (i), (c) – (ii), (d) – (iv)

(4) (a) – (i), (b) – (iii), (c) – (ii), (d) – (iv)

CTD221

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	3	4	2	1	2	1	4	4	2	3	4	2	2
Question	16														
Answer	3														

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Thermodynamics

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (2)

It is the branch of science which deals with the energy changes taking place in all physical and chemical process.

2. Ans. (3)

Thermo flask is made up of insulating material, so there is no exchange of heat, so it is an example of isolated system.

3. Ans. (4)

Intensive quantities are those which is independent from the mass of the matter.

Extensive are those which depend on the mass of the matter.

Enthalpy, Volume are Extensive quantity

Temperature, refractive index are intensive.

4. Ans. (4)

Mass, Energy, Enthalpy all are extensive property.

5. Ans. (3)

In adiabatic process, q = 0

We know from 1st Law-

$\Delta U = q + W$

$\Delta U = W$

$q = 0$

6. Ans. (3)

In adiabatic process, work is independent of path

$\Delta U = q + W$ (q=0)

$\Delta U = W$

Internal energy is state function, depends only upon initial and final state.

7. Ans. (1)

In adiabatic process, q = 0

From 1st Law

$\Delta U = q + W$

$\Delta U = W$

We know $\rightarrow \Delta U = nC_V\Delta T$

$W = nC_V\Delta T$

When gas is compressed, work is positive, which means, temperature will increase.

So final temperature will be higher than initial temperature.

8. Ans. (4)

Heat supplied at constant pressure.

$q_p = \Delta H$ (Enthalpy change)

Heat supplied at constant volume

$q_v = \Delta E$ (Internal energy change)

ΔH, ΔE, Enthalpy all are state function, depends only upon initial and final state.

9. Ans. (1)

The quantities which doesn't depend on mass of the matter in intensive,

boiling point, EMF of cell, pH doesn't depend upon mass of matter,

10. Ans. (1)

We Know in reversible process, where P_{ext} is variable

$$W = -\int P.dV$$

11. Ans. (1)

In Isothermal expansion, $\Delta U = 0$

From 1st Law-

$\Delta U = q + W$

$q = -W$

$W = -q$

$W = -(-100\ Cal)$

$W = 100\ Cal$ ($\therefore$ 1Cal = 4.184J)

$W = 100 \times 4.184J$

$W = 418.4\ J$

12. Ans. (4)

Temperature is intensive property (doesn't depend on mass of matter)

Volume is extensive property/depends upon mass off matter)

13. Ans. (2)

From 1st Law-

$\Delta U = q + W$

For isothermal process, $\Delta U = 0$, $q = -W$

For adiabatic process, $q = 0$, $\Delta U = W$

14. Ans. (1)

From theory part we have study that in adiabatic expansion, temperature decreases, and in adiabatic compression, temperature increases.

→ In isothermal, temperature is constant throughout the process.

15. Ans. (4)

In reversible process,

→ takes in multisteps.

→ Driving force is infinitesimally greater than apposing force.

→ Work obtained is maximum.

16. Ans. (4)

Both q & w are path function.

Path function → Which depends upon path as well as initial and final state of the system.

17. Ans. (4)

Work done by system (W) = –300 J

Heat is supplied (q) = 100 Cal ($\therefore$ 1 Cal = 4.2J)

= 420 J

From 1st Law-

$\Delta U = q + W$

$\Delta U = 420 - 300$

$\Delta U = 120$ J

18. Ans. (1)

Internal Energy E_1,

q = –450J

W = 600J

From 1st Law

$\Delta E = -450$J

From 1st Law-

$\Delta E = q + W$

$(E_2 - E_1) = -450 + 600$

$E_2 = 150 + E_1$

19. Ans. (1)

Work done by system W = –8J

Heat is supplied q = 40J

From 1st Law-

$\Delta U = q + W$

$\Delta U = 40 - 8$

$\Delta U = 32$J

20. Ans. (3)

Internal Energy change = ?

In reversible isothermal expansion

In isothermal, temperature remains constant so internal energy change is zero

21. Ans. (1)

$\Delta H = \Delta E + P\Delta V$

At constant temperature $\Delta E = 0$

At constant pressure, $W = P\Delta V$

22. Ans. (3)

Heat of reaction at constant pressure $(q_p) = \Delta H$

Heat of reaction at constant volume $(q_v) = \Delta E/\Delta U$

$\Delta H = \Delta U + \Delta ngRT$

$\Delta H - \Delta U = (-3)RT$ $(\Delta ng = 12 - 15 = -3)$

$\Delta H - \Delta U = (-3)(8.314)(298K)$

$= -7.43$ KJ

23. Ans. (2)

$$A_{(g)} + 3B_{(g)} \longrightarrow 3C_{(g)} + 3D_{(g)}$$

$\Delta E = 17$ kCal, T = 300K (273+27=300K)

$R = 2$Cal K^{-1} mol^{-1}

We know-

$\Delta H = \Delta E + \Delta ngRT$

$\Delta H = 17 + ((2) \times 2 \times 300) \times 10^{-3}$

$= 17 + 1.2$

$\Delta H = 18.2$ kCal

24. Ans. (2)

In reaction

$\Delta ng = 2 - (3) = -1$

$\Delta H = \Delta E + \Delta ngRT$

$\Delta H = \Delta E + (-1)RT$

$\Delta H = \Delta E - RT$

$\Delta H < \Delta E$

25. Ans. (4)

$$Ag_2O(s) \longrightarrow 2Ag(s) + \frac{1}{2} O_2(g)$$

$$\Delta ng = \frac{1}{2}$$

$\Delta H = \Delta E + \Delta ngRT$

$\Delta H = \Delta E + (0.5)RT$

$\Delta H > \Delta E$

26. Ans. (2)

$2CO_{(g)} + 1O_{2(g)} \longrightarrow 2CO_2(g)$

$\Delta ng = 2-3 = -1$

$\Delta H = \Delta E + \Delta ngRT$

$\Delta H = \Delta E - RT$

$\Delta H < \Delta E$

27. Ans. (2)

Combustion of isobutane

$$C_4H_{10} + \frac{13}{2}O_2 \longrightarrow 4CO_2 + 5H_2O$$

$$\Delta ng = \frac{3}{2} = 1.5$$

$\Delta H = \Delta E + \Delta ngRT$

$\Delta H = \Delta E + 1.5RT$

$\Delta H > \Delta E$

28. Ans. (4)

$\Delta H = nC_p\Delta T$ Isothermal, T → constant

$\Delta H = (1)C_p(0)$ $\Delta T = 0$

$\Delta H = 0$

29. Ans. (2)

$CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$

T = 977°C

T = 977+273

= 1250K

$\Delta H = 174$ KJ/mol

$\Delta H = \Delta E + \Delta ngRT$

$174 \text{ KJ/mol} = \Delta E + [(1) \times 8.314 \times 1250] \times 10^{-3}$

$\Delta E = 174 - 10.39$

$\Delta E = 163.6$ KJ

30. Ans. (1)

Heat of reaction at constant P(q_p) = ΔH = ?

Heat of reaction at constant

V (q_v) = ΔE = 67.71kCal

We know-

$\Delta H = \Delta E + \Delta ngRT$

$\Delta H = (67.71 \times 10^3) + (1-1.5)RT$

(T = 273 + 17 = 290 K)

(R = 2)

$\Delta H = (-67.7) + ((-0.5) \times 290) \times (2)$

$= -67.7 - 0.29$

$\Delta H = -68$ Kcal

31. Ans. (1)

Enthalpy of vaporisation (ΔH_{vap}) and internal energy ΔE then-

$\Delta H = \Delta E + \Delta ngRT$

For vaporisation of water

$H_2O(l) \longrightarrow H_2O(g)$

$\Delta ng = 1-0$

$= 1$

$\Delta H_{vap} = \Delta E + \Delta ngRT$ T = 100°C

= 273 + 100

T = 373 K

$40.63 = \Delta E + RT$

$\Delta E = 40.63 - (8.314 \times 373) \times 10^{-3}$

$\Rightarrow 40.63 - 3.10$

$\Delta E = 37.53$ KJ mol^{-1}

32. Ans. (1)

$S(s) + O_2(g) \longrightarrow SO_2(g)$

$\Delta H = \Delta E + \Delta ngRT$

$\Delta ng = 1-1 = 0$

$\Delta H = \Delta E$

33. Ans. (2)

$$Co(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$$

$\Delta H = \Delta E + \Delta ngRT$

$$\Delta ng = 1 - \left(1 + \frac{1}{2}\right)$$

$$= \frac{-1}{2}$$

$$\Delta H = \Delta E - \frac{RT}{2}$$

$\Delta H < \Delta E$

34. Ans. (3)

Combustion of sucrose

$C_{12}H_{22}O_{11(s)} + 12O_{2(g)} \longrightarrow 12CO_{2(g)} + 11H_2O_{(l)}$

$\Delta ng = 12-12=0$

$\Delta H = \Delta E$

35. Ans. (4)

$\Delta H \neq \Delta E$

$\Delta ng \neq 0$

(i) $\Delta ng = 0$

(ii) $\Delta ng = 0$

(iii) $\Delta ng = 0$

(iv) $\Delta ng = 2 - 4 = -2$

36. Ans. (2)

Combustion of ethanol

$C_2H_5OH(l) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$

$\Delta ng = -1$

$\Delta H = \Delta E + \Delta ngRT$ $\quad T = 27°C$

$T = 273+27=300K$

$\Delta H = -670.48 + (-1) \times 8.314 \times 300$

$\Delta H = -671.08$ Kcal

37. Ans. (2)

$\Delta H - \Delta E = \Delta ngRT$

$\underset{(g)}{CH_4} + 2\underset{(g)}{O_2} \longrightarrow \underset{(g)}{CO_2} + 2\underset{(l)}{H_2O}$

$\Delta ng = -2$

$\Delta H - \Delta E = (-2)\ (2) \times 298$ Cal

38. Ans. (2)

$\Delta H = \Delta E + \Delta ngRT$

When $\Delta ng < 0$, then $\Delta H < \Delta E$

$\Delta ng = 2-3 = -1$

39. Ans. (3)

$2x_{(s)} + 2y_{(s)} \longrightarrow 2C_{(l)} + D_{(g)}$

$q_p = -28$kCal mol^{-1} $\quad q_v = ?$

$q_p = q_v + \Delta ngRT$

$-28 = q_v + (1) \times 2 \times 300$

$q_v = -28-0.6 = -28.6$ kCal mol^{-1}

40. Ans. (1)

For reversible expansion

n = 1 mol

$V_1 = 10L$, $\quad V_2 = 20L$

For isothermal reversible expansion

1 J = 10^7erg

$$W = -2.303\ nRT \log_{10}\left(\frac{V_2}{V_1}\right)$$

$$W = -2.303 \times 1 \times 8.314 \times 298 \times 10^7 \log \frac{2}{1}$$

41. Ans. (1)

n = 2 mol

Vaccum, $\quad$ (in vaccum work done is zero)

W = 0

42. Ans. (1)

n = 1 mol , V = 3 dm^3 , constant P = 1 atm

1 dm^3 = 1L

$V_1 = 3L$, $\quad V_2 = 13$ L

Work in irreversible expansion-

$W = -P_{ext}(\Delta V)$

$= -1(13-3) = -10$L atm

43. Ans. (2)

Gaseous moles are increasing, so entropy increases.

(ii) gaseous moles are increasing

(i), (iii), (iv) gaseous moles are same.

44. Ans. (3)

In reversible adiabatic process, entropy change of system as well as surrounding is zero.

$\Delta S_T = 0$

45. Ans. (4)

Entropy means-

disorderness, randomness

46. Ans. (3)

Solid is converted into gas, so Δs will be positive distance between molecules is increasing.

47. Ans. (4)

(i) $Br_2(l) \longrightarrow Br_2(g)$

Liquid to gas, entropy increases.

(ii) $C(s) + H_2O(g) \longrightarrow Co(g) + H_2(g)$

Moles of gas molecules increases, so entropy increase

(iii) $N_2(g)$ (10 atm) $\longrightarrow N_2$(g, 1atm)

Pressure is decreasing, so entropy increases.

(iv) Fe(at 400K) $\longrightarrow$ Fe (300 K)

Temperature is decreasing, so entropy will decrease.

48. Ans. (3)

(i) Liquid to solid, ΔS = –ve

(ii) 3 mole gas to 2 mole gas, ΔS = –ve

(iii) Liquid to gas, ΔS = +ve

(iv) 4 mole gas to 2 mol gas, ΔS = –ve

49. Ans. (1)

When egg is boiled, denaturation of protein occurs so entropy increases.

50. Ans. (1)

$H_2(g) + Cl_2(g) \longrightarrow 2HCl$

$$\Delta S^0_{reaction} = \sum S^0_{Product} - \sum S^0_{reactant}$$

$$\Delta S^0 = \left[2 \times S^0_{HCl}\right] - \left[S^0_{H_2} + S^0_{Cl_2}\right]$$

$$\Delta S^0 = [2 \times 0.19] - [0.13 + 0.22]$$

$\Delta S^0_{reaction} = 0.03$ KJ K^{-1}mol^{-1}

$= 30$ J K^{-1} mol^{-1}

51. Ans. (2)

Graphite has more entropy than diamond, because diamond has highly ordered structure, so it has less entropy.

52. Ans. (3)

When two gases are mixed, gaseous molecules increases, so randomness will increase, entropy increases.

53. Ans. (3)

ΔH_{vap} = ?

B.pt. (T) = 79.5°C $\Delta S = 109.8 JK^{-1}mol^{-1}$

= 273 + 79.5

T = 352.5K

We know-

$$\Delta S_{vap} = \frac{\Delta H_{vap}}{T}$$

$109.8 \times 352.5 = \Delta H_{vap}$

$\Delta H_{vap} = 38.70$ KJ mol^{-1}

54. Ans. (1)

$\Delta H = 900 J/g$, T = 100°C

T = 373 K

For 1 g water heat = 900J

For 18 g/1mol water heat = 900×18 J

= 16200 J

We know

$$\Delta S = \frac{\Delta H}{T} = \frac{16200J}{373K} = 43.4 JK^{-1}mol^{-1}$$

55. Ans. (3)

n = 5 mol, $V_1 = 8 dm^3$ T = 27°C

$V_2 = 80\ dm^3$ T = 300 K

ΔS for isothermal-

$$\Delta S = nR \ln\left(\frac{V_2}{V_1}\right)$$

$$\Delta S = 5 \times 8.314 \times \ln\left(\frac{80 dm^3}{8 dm^3}\right)$$

$\Delta S = 2.303 \times 5 \times 8.314 \times 1$

$\Delta S = 95.73\ JK^{-1}$

56. Ans. (2)

For spontaneous irreversible process

$\Delta S_T > 0$, So entropy of system & surrounding increases.

(refer through notes)

57. Ans. (2)

According to 2nd Law of thermodynamics, during an natural or spontaneous thermodynamic process, total entropy or entropy of universe increases.

58. Ans. (1)

$H_2(g) + Br_2(g) \rightarrow 2HBr(g)$ $\Delta S^0 = 20.1 JK^{-1}$

$S^0_{H_2} = 130.6\ J\ mol^{-1}K^{-1}$, $S^0_{HBr} = 198.5\ J\ mol^{-1}K^{-1}$

$$\Delta S^0 = \left[2S^0_{HBr}\right] - \left[S^0_{H_2} + S^0_{Br_2}\right]$$

$20.1 = [2 \times 198.5] - [130.6 + S^0_{Br_2}]$

$S^0_{Br_2} = 246.3\ JK^{-1}\ mol^{-1}$

59. Ans. (4)

(i) Solid → Liquid (entropy increases)
ΔS = +ve

(ii) Expansion of gas (S↑ = ΔS = +ve)

(iii) Crystal dissolve (in dissolution S↑)

(iv) Polymerisation (In polymerisation, mole decreases, S↓ ΔS = –ve)

60. Ans. (2)

$$\Delta H^0_{formation} = 0$$

$\Delta S^0 \neq 0$

Free energy of formation for element in standard state = 0

61. Ans. (4)

Entropy of an adiabatic reversible process is constant,

Whereas $\Delta S = 0$

62. Ans. (3)

In adiabatic reversible process, $\Delta U \neq 0$

$\Delta T \neq 0$

but entropy remain constant, $\Delta S = 0$

63. Ans. (2)

$\Delta H = -11.7 \times 10^3\ J\ mol^{-1}$

$\Delta S = -105\ J\ mol^{-1}\ K^{-1}$ T = 25°C

= 25 + 273

= 298 K

We know-

$\Delta G = \Delta H - T\Delta S$

$\Delta G = -11.7 \times 10^3 - (298)(-105)$

$-11700 + 31290$

$\Delta G = 19590\ J\ mol^{-1}K^{-1}$

$\Delta G > 0$

64. Ans. (2)

$\Delta H > 0, \quad \Delta S > 0$

For spontaneously $\Delta G < 0$

We know-

$\Delta G = \Delta H - T\Delta S$

$\downarrow \quad \downarrow$

+ve +ve

$T\Delta S > \Delta H$

65. Ans. (1)

For equilibrium ($\Delta G = 0$)

$\Delta H = T\Delta S$

$30.5 \text{ KJ mol}^{-1} = T \times 0.066 \text{ KJ mol}^{-1}$

$T = 462.12 \text{ K}$

66. Ans. (4)

$H_2O(l) \rightleftharpoons H_2O(g)$ at T = 100°C P = 1 atm

373K

Liquid to gas, $\Delta S = +ve, \Delta H \neq 0$

$\Delta H = \Delta U + \Delta ngRT$

$\Delta U = \Delta H - \Delta ngRT$

$\Delta U \neq 0$

But we know at equilibrium, $\Delta G = \Delta H - T\Delta S$

$\Delta G = 0$

$\Delta H = T\Delta S$

67. Ans. (3)

$\Delta H = 30.56 \text{ KJ mol}^{-1} \quad \Delta S = 66 \text{ JK}^{-1} \text{ mol}^{-1}$

$\Delta G = 0$

$\Delta H = T\Delta S$

$30.56 \times 10^3 = T \times 66$

$T = \frac{30.56 \times 10^3}{66} = 463K$

68. Ans. (2)

$\frac{1}{2}A_2(g) + \frac{3}{2}B_2(g) \rightarrow AB_3(g) \qquad \Delta H = -20KJ$

$S^0_{A_2} = 60, \quad S^0_{B_2} = 40, \quad S^0_{AB_3} = 50$

For equilibrium –

$\Delta G = 0$

$\Delta H^0 = T\Delta S^0$

$\Delta S^0 = \left[\sum S^0_{product}\right] - \left[\sum S^0_{reactant}\right]$

$\Delta S^0 = \left[S^0_{AB_3}\right] - \left[\frac{1}{2}S^0_{A_2} + \frac{3}{2}S^0_{B_2}\right]$

$\Delta S^0 = [50] - \left[\frac{60}{2} + 40 \times \frac{3}{2}\right]$

$\Delta S^0 = [50] - [90] = -40 \text{ JK}^{-1} \text{ mol}^{-1}$

$\Delta H = T\Delta S$

$-20 \times 10^3 = T \times (-40)$

$T = \frac{20000}{40} = 500 \text{ K}$

69. Ans. (3)

Precipitation of AgCl by Ag^+ ions and HCl

It is a spontaneous process, as soon as the Ag^+ ions added in HCl, precipitate formed.

70. Ans. (3)

Ice melting at 1 atm, 283K

Ice force at 273K or 0°C, at 283 K, it starts melting itself.

This process is spontaneous.

71. Ans. (4)

$A + B \longrightarrow C + D + q$

(Exothermic reaction)

$\Delta S = +ve, \quad \Delta H = -ve$

$\Delta G = \Delta H - T\Delta S$

$\downarrow \quad \downarrow$

$-ve \quad +ve$

Which means at any temperature

$\Delta G < 0$

So reaction is spontaneous.

72. Ans. (1)

We know

$\Delta G = \Delta G^0 + RT\ln Q$

At equilibrium

$\Delta G = 0, \ Q = K_{eq}$

$\Delta G^0 = -RT\ln K_{eq}$

Standard free energy change ΔG^0 is related so K_{eq}.

73. Ans. (2)

Vant Hoff equation

$\Delta G^0 = -RT\log_e K_p$

$-\Delta G^0 = RT\ln K_p$

74. Ans. (2)

$\Delta G^0 > 0$

We know

$\Delta G^0 = -RT\ln K_p$

For $\Delta G^0 > 0$

$\ln K_p < 0$

$K_p < 1$

75. Ans. (2)

$K_p = 10$

$\Delta G^0 = -RT\ln K_p$ $\quad R = 8\ JK^{-1}\ mol^{-1}$

$\Delta G^0 = -8 \times 300 \times \ln(10)$ $\quad T = 300K$

$\Delta G^0 = -5.527\ KJ\ mol^{-1}$

76. Ans. (4)

Evaporation of liquid

$H_2O(l) \longrightarrow H_2O(g)$

(Endothermic reaction)

ΔS = +ve (liquid to gas conversion)

Free energy change (ΔG) decreases

$\Delta G = \Delta H - T\Delta S\uparrow$

ΔG decreases

77. Ans. (1)

$CO_2(s) \longrightarrow CO_2(g)$

Endothermic process (ΔH = +ve)

Entropy will increase due to solid to gas conversion.

78. Ans. (4)

All CO, CO_2, ice are exceptions of 3rd Law of thermodynamics. (As per theory)

79. Ans. (1)

$\Delta G = -26$ kCal, $\quad \Delta S = -60\ Cal\ K^{-1}$

$\Delta H = ?$ $\quad T = 27°C$

$= 300\ K$

We know-

$\Delta G = \Delta H - T\Delta S$

$-26 \times 10^3 = \Delta H - (300)\ (-60)$

$\Delta H = -26000 - 18000$

$\Delta H = -44$ kCal

80. Ans. (4)

For spontaneously

$\Delta G < 0$

(i) $\Delta G = \Delta H - T\Delta S$

$\downarrow \quad \downarrow$

$-ve \quad +ve$

$\Delta G < 0$ (always spontaneous)

(ii) $\Delta G = \Delta H - T\Delta S$

$\downarrow \quad \downarrow$

$-ve \quad -ve$

Can be spontaneous when $\Delta H > T\Delta S$

(iii) $\Delta G = \Delta H - T\Delta S$

$\downarrow \quad \downarrow$

$+ve \quad +ve$

Spontaneous when $T\Delta S > \Delta H$

(iv) $\Delta G = \Delta H - T\Delta S$

$\downarrow \quad \downarrow$

$+ve \quad -ve$

$\Delta G > 0$ (always non spontaneous)

81. Ans. (1)

Formation of water

$H_2(g) + O_2(g) \longrightarrow H_2O(l)$

$\Delta H < 0$

$H_P - H_R < 0$

$H_P < H_R$

Chemical energy of H_2 & O_2 is more than that of water.

82. Ans. (1)

In Exothermic Reaction

$\Delta H < 0$

$H_P < H_R$

83. Ans. (2)

Thermochemical equation tells the physical state doesn't tell about the spontaneously.

84. Ans. (3)

$C(s) + O_2(g) \longrightarrow CO_2(g)$ $\quad \Delta H = -94$ Kcal

exothermic reaction $\quad$ It tells about phase.

85. Ans. (1)

High bond energy of nitrogen molecule.

(As per theory)

86. Ans. (3)

Diamond → Graphite

$\Delta H = -453.5$ Cal

Diamond releases heat to make graphite so energy of graphite is Low. So graphite is more stable than diamond.

87. Ans. (4)

Max. Energy, least stability

64.8 kCal

88. Ans. (1)

Explosive compound NCl_3 so, formation energy is +ve reaction is endothermic.

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89. Ans. (2)

$C(s) + \frac{1}{2}O_2(g) \longrightarrow CO(g)$ $\Delta H = -26.4$ kCal

CO is exothermic compound, ΔH = –ve

90. Ans. (4)

(i) ΔH = +ve (Endo)

(ii) $\Delta H = \Delta E + \Delta ngRT$

↓ ↓

+ve +ve

ΔH = +ve (Endo)

(iii) 180.4 KJ energy is given (Endo)

(iv) ΔE = –ve

$\Delta ng = -1$

$\Delta H = \Delta E + \Delta ngRT$ ΔH =–ve

↓ ↓ (Exo)

–ve –ve

91. Ans. (2)

$3O_2 \rightarrow 2O_3$ ΔH = +ve

Energy of O_3 > Energy of O_2

Stability of O_3 < Stability of O_2

92. Ans. (4)

P(White) → P(Red) ΔH = –ve

Energy of P(Red) < Energy of P(White)

Stability of P(Red) > Stability of P(White)

93. Ans. (2)

Kirchoff's equation tells about temperature. (As per theory)

94. Ans. (1)

$$\Delta C_p = \frac{\Delta H_2 - \Delta H_1}{T_2 - T_1}$$

$$O = \frac{\Delta H_2 - \Delta H_1}{100K}$$

$\Delta H_2 = \Delta H_1 = -3.57$ KJ

95. Ans. (2)

From 2nd reaction

HCl(*l*) → HCl(g) by giving some heat so some heat is used to convert HCl(l) into HCl(g) due to which heat evolved in reaction is less.

X < y

96. Ans. (4)

$(\Delta H_f)_{Compound}$ can be +ve or –ve

ΔH_f (element) = 0

97. Ans. (4)

$H_2(s) + I_2(g) \longrightarrow 2HI$, ...(1) $\Delta H = 12.4$ kCal

$\Delta H_f(HI) \Rightarrow$

$\frac{1}{2}H_2 + \frac{1}{2}I_2 \longrightarrow HI$ ...(2)

So reaction (2) $= \frac{1}{2} \Rightarrow \frac{12.4}{2} = 6.2$kCal

98. Ans. (2)

For heat of formation, 1 mole compound is formed and reactants are taken as reference state.

99. Ans. (2)

$\Delta H_f(NH_3) = -46.0$ KJ/mol

$\frac{1}{2}N_2 + \frac{3}{2}H_2 \longrightarrow NH_3$ ...(1)

Given

$2NH_3 \rightarrow N_2 + 3H_2$ ...(2)

$\Rightarrow$ Eq. (2) = – Eq(1) × 2

$\Delta H = (-46) \times 2$

$= 92$ KJ mol^{-1}

100. Ans. (4)

(ΔH_g) $CO_2 = -94$KJ $\Delta H_f(CaO)) = -152$ KJ

$C + O_2 + CO_2$ ... (1)

$Ca + \frac{1}{2}O_2 \longrightarrow CaO$ ... (2)

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g) \Rightarrow \Delta H = 42$ KJ ...(3)

$\Delta H_f(CaCO_3) \rightarrow Ca + C + \frac{3}{2}O_2 \rightarrow CaCO_3$

(1) + (2) – (3)

$\Delta H = -94 - 152 = 42$

$\Delta H \Rightarrow -288$ KJ

101. Ans. (4)

$\Delta H_f^0 (CH_4) = -17.9$

$\Delta H_g^0 (C_2H_4) = 12.5$

$\Delta H_g^0 (C_3H_8) = -24.8$

ΔH for $CH_4 + C_2H_4 \rightarrow C_3H_8$ is

$$\Delta H = \Sigma(\Delta H_f)_{C_3H_8} - \Sigma(\Delta H_f)_{reactant}$$

$= (-24.8) - (-17.9 + 12.5)$

$\Delta H = -19.4$ kCal

102. Ans. (1)

$\Delta H_f(C_2H_6) = -21.1$

$\Delta H_f(CO_2) = -94.1$

$\Delta H_f(H_2O) = -68.3$ kCal

$\Delta H_C(C_2H_6)$

$C_2H_6 + \frac{7}{2}O_2 \longrightarrow 2CO_2 + 3H_2O$

$\Delta H_c(C_2H_6)$

$= \left[2\Delta H_f(Co_2) + 3 \times \Delta H_f(H_2O)\right] - \left[\Delta H_f(C_2H_6)\right]$

$= [2 \times (-94.1) + 3 \times (-68.3)] - [-21.1]$

$= -372$ K Cal

103. Ans. (2)

$\Delta H_f^0 = (CO_2) = -393.5$

$\Delta H_f^0 (CO) = -110.5$

$\Delta H_f^0 (H_2O) = -241.8$

$C + O_2 \rightarrow CO_2$ -393.5 ... (1)

$C + \frac{1}{2}O_2 \rightarrow CO$ -110.5 ... (2)

$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ -241.8 ... (3)

$CO_2 + H_2 \rightarrow CO + H_2O$ ...(4) $\Delta H = ?$

Eq. (4) = (2) + (3)–(1)

= 41.2 KJ/mol

104. Ans. (2)

$\Delta H_C(C) = -393.5$

$C + O_2 \rightarrow CO_2$

$\Delta H_C = -393.5$ ... (1)

$\Delta H_C(CO) = -283$

$CO + \frac{1}{2}O_2 \rightarrow CO_2$

$\Delta H_C = -283$ ... (2)

$\Delta H_f(CO) \Rightarrow C + \frac{1}{2}O_2 \rightarrow CO$ ... (3)

Eq. (3) = – Eq(2) + Eq(1)

+283 –393.5

= –110.5 KJ

105. Ans. (3)

$\Delta H_C(C) = -393.3$ $\Delta H_C(S) = -293.72$

$\Delta H_C(CS_2) = -1108.76$

$\Delta H_f(CS_2)$

$C + 2S \longrightarrow CS_2$

$\Delta H_f(CS_2) = (\Delta H_C)_{reactant} - (\Delta H_C)_{product}$

$= [(-393.3) + (-293.72) - (-1108.76)]$

$= -980.74 + 1108.76$

$= 128.02$ KJ mol^{-1}

106. Ans. (2)

$C + 2H_2 \rightarrow CH_4$

$\Delta H = (\Delta H_C)_{reactant} - (\Delta H_C)_{product}$

$= [(-94) + (2) \times (-68.4)] - [-212.4]$

$= -230.8 + 212.4$

$\Delta H = -18.4$ kCal.

107. Ans. (3)

For reference state of element, standard heat of formation is zero.

$\Delta H_f(C_{graphite}) = 0$

108. Ans. (4)

$\Delta H_f(NO_2) = 8.0$, $\Delta H_f(N_2O_4) = 2$ kCal mol^{-1}

Dimerization of NO_2

$2NO_2 \longrightarrow N_2O_4$

$\Delta H = \Delta H_f(N_2O_4) - 2 \times \Delta H_f(NO_2)$

$= 2–16$

$= -14.0$ kCal mol^{-1}

109. Ans. (4)

Given-

$\frac{1}{2}M_2O \longrightarrow M + \frac{1}{4}O_2$ ...(1) $\Delta H = 120$ kCal

$4M + O_2 \rightarrow 2M_2O$ $(\Delta H = (1) \times 4)$

$\Delta H = -480$ kCal

110. Ans. (4)

7.8 g

$n = \frac{7.8}{78} = 0.1\,mol$

For 1 mol, $\Delta H = -3264.4$ KJ mol^{-1}

0.1 mol, $\Delta H = -3264.4 \times 0.1$

$= -326.4$ KJ mol^{-1}

111. Ans. (1)

$\text{Calorific value} = \frac{\text{Heat of combustion}}{\text{Molar mass}}$

$CH_4 = \frac{212.8}{16} = 13.3$ ← best fuel

$C_2H_6 = \frac{373}{30} = 12.43$

$C_2H_4 = \frac{337}{28} = 12.03$

$C_2H_2 = \frac{310.5}{26} = 11.94$

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112. Ans. (3)

$\Delta H_f(CO) = -110$ KJ mol^{-1}

$C + \frac{1}{2}O_2 \rightarrow CO$

$\Delta H_f(CO_2) = -394$ KJ mol^{-1}

$C + O_2 \rightarrow CO_2$

$\Delta H_C(C) = \Delta H_f(CO_2) = -394$ KJ mol^{-1}

113. Ans. (2)

$\Delta H_f(C_2H_4) = 52$,

$2C + 2H_2 \rightarrow C_2H_4$ ...(1)

$\Delta H_f(CO_2) = -394$,

$C + O_2 \rightarrow CO_2$ ...(2)

$\Delta H_f(H_2O) = -286$

$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ ...(3)

$\Delta H_C(C_2H_4) = C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$

$\Delta H_C(C_2H_4) = \Delta H_f(\text{product}) - \Delta H_f(\text{reactant})$

$= [2\times(-394)+2(-286)]-[52]$

$= (-788-572)-(52)$

$= -1412$ KJ mol^{-1}

114. Ans. (1)

$C_6H_6 + \frac{1}{2}SO_2 \rightarrow 6CO_2 + 3H_2O$ $\Delta H = -3267$KJ .... (1)

$\Delta H_f(CO_2) = -393.5$

$C + O_2 \rightarrow CO_2$ ... (2)

$\Delta H_f(H_2O) = -285.83$

$H_2 + \frac{1}{2}O_2 + H_2O$ ... (3)

$\Delta H_f(C_6H_6)$

$6C + 3H_2 \rightarrow C_6H_6$ ... (4)

Eq. (4) = ((2)×6−(1)+(3)×3)

$\Delta H = (-2361) + (-857.49) - (-3267)$

$= 48.51$

115. Ans. (3)

112L

$n = \frac{112}{22.4} \Rightarrow 5$

Equal volume of H_2 & CO

56L H_2 & 56L CO

$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$, $\Delta H = -241.8$ KJ

$\Delta H = -241.8 \times 2.5 = -604.5$ KJ

$CO + \frac{1}{2}O_2 \rightarrow CO_2$; ΔH ; -283 KJ

$\left(n = \frac{56}{22.4} = 2.5\text{mol}\right)$

n= 2.5 mol, $\Delta H = -283\times2.5 = -707.5$KJ

Total heat evolved = −707.5−604.5=1312 KJ

116. Ans. (1)

Energy required = 2870 kCal

ΔH_C(cane sugar) = −1349 kCal/mol

Daily consumption of sugar cane = $\frac{2870}{1349}$ = 2.12

So, weight of $C_{12}H_{22}O_{11}$ = 2.12 × 342 = 728 g

117. Ans. (3)

$C_2H_2 + H_2 \rightarrow C_2H_4$

ΔH_f^0 (H_2O) = −68.3

$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ $\Delta H_f = -68.3$

$\Delta H_C[H_2] = -68.3$

$\Delta H_C^0(C_2H_2) = -337.2$

$\Delta H_C^0(C_2H_4) = -363.7$

Heat change of reaction:-

(ΔH_C) reactant − (ΔH_C) product

= [−337.2 − 68.3] − [−363.7]

= [−41.8] kCal

118. Ans. (2)

Heat of combustion of a substance is always :-

Always negative.

119. Ans. (2)

$\Delta H_C(C) = -94.4$ kCal

$C(s) + O_2 \rightarrow CO_2$

$\Delta H_f(CO_2)$

$C + O_2 \rightarrow CO_2$

$\Delta H_f(CO_2) = \Delta H_C(C) = -94.4$ kCal

120. Ans. (2)

$\Delta H_C(CH_4)$ → For 1 mol combustion of CH_4

n of CH_4 = $\frac{0.4\text{ g}}{16} \Rightarrow \frac{1}{40}$mol

for $\frac{1}{40}$ mol, Heat liberated = 2.25 kCal

for 1 mol $\Rightarrow \frac{0.25}{1/40}$ = 0.25 × 40 = −10 kCal

121. Ans. (2)

−680 kCal heat evolved when $C_6H_{12}O_6$ = 1 mol

for 170 kCal heat evolved mole of $C_6H_{12}O_6 = \frac{1}{680}\times 170 = \frac{1}{4}$ mol

$\frac{1}{4}\left(C_6H_{12}O_6 + 9O_2 \rightarrow 6CO_2 + 6H_2O\right)$

CO_2 mol = $\frac{6}{4} = \frac{3}{2}$ = 1.5 mol

Weight of CO_2 = 1.5 × 44 = 66 g

122. Ans. (3)

For combustion, 1 mole reactant taken and $CO_2(g)$ & $H_2O(l)$ will be formed.

123. Ans. (1)

$\Delta H_f(CO_2)$ = −94 kCal

$C + O_2 \rightarrow CO_2$ ΔH = −94 kCal

$C + O_2 \rightarrow CO_2$

mass = 3g

$n = \frac{3}{12} = \frac{1}{4}$ mol

for 1 mol C, ΔH = −94kCal

for $\frac{1}{4}$ mol C, $\Delta H = \frac{-94}{4}$ = −23.5

124. Ans. (3)

1 mol NH_4OH + 1 mol HCl

When acid and base both are strong heat liberate = 13.7 kCal

One is weak then heat is less than 13.7 kCal

125. Ans. (2)

Given $H^+ + OH^- = H_2O$ + 13.7 kCal

1 mol H_2SO_4 with strong base neutralisation

2mol H^+ + 2mol OH^- = −13.7 × 2

$\Rightarrow$ −27.4 Kcal

126. Ans. (2)

Valency factor of NaOH = 1

Valency factor of dibasic acid = 2

Thus, only 1 gm equivalent of dibasic acid is neutralised by 1gm equivalent of NaOH, for reaction

$H^+ + OH^- \rightarrow H_2O$, ΔH = −13.7 KJ/mol

Hence the heat of neutralisation of a strong dibasic acid in dilute solution by NaOH = −13.7 kCal/eq

127. Ans. (1)

When 5 ml strong acid reacting with 5 ml strong base heat evolved = ΔH

$\Delta H = ms\Delta T$

$\Delta T = \frac{\Delta H}{ms}$

When 10 ml of each are mixed

$\Delta T = \frac{2\Delta H}{2ms} = \frac{\Delta H}{ms}$

So temp rise will be same.

128. Ans. (2)

$HCl + NaOH \rightarrow NaCl + H_2O$ ... (1)

ΔH_{neut} = −55.9 KJ/mol

$HCN + NaOH \rightarrow NaCN + H_2O$ ... (2)

ΔH_{neut} = −12.1 KJ/mol

(2) − (1)

$HCN + NaCl + NaOH + H_2O \rightarrow NaCN + H_2O + HCl + NaOH$

ΔH = −12.1 + 55.9

ΔH = 43.8 KJ/mol

Energy of dissociation for 1 mole of HCN

129. Ans. (1)

$H^+ + OH^-$, heat exchange during reaction

= −13.7 kCal

130. Ans. (1)

$NaOH + HCl \rightarrow NaCl + H_2O$

Heat of neutralisation.

131. Ans. (1)

$\Delta H_C(C_2H_4)$, $\Delta H_C(C_2H_6)$, $\Delta H_C(H_2)$

$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$ ... (1)

$C_2H_6 + \frac{7}{2}O_2 \rightarrow 2CO_2 + 3H_2O$ ... (2)

$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ ... (3)

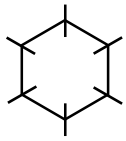
$C_2H_4 + H_2 \rightarrow C_2H_6$

(1) + (3) − (2)

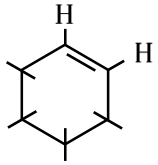
= (−1409.5) + (−285.6) − (−1558.3)

= −136.8 KJ

132. Ans. (1)



$C_6H_{12} + 9O_2 \rightarrow 6CO_2 + 6H_2O$ ... (1)



$C_6H_{10} + \frac{17}{2} O_2 \rightarrow 6CO_2 + 5H_2O$ ... (2)

$H_2 + \frac{1}{2} \rightarrow H_2O$ ... (3)

ΔH_{Hyd} (Cyclohexene)

$C_6 + H_{10} + H_{12} \rightarrow C_6H_{12}$ ... (4)

For eq. (4) $\Rightarrow$ (1) + (3) + (2)

= (+3920) + (–241) – 3800

= –121 KJ/mol

133. Ans. (2)

Bond energy of a molecule is +ve

134. Ans. (4)

Dissociate 1 mol gaseous bond into separate gaseous atoms.

$HCl(g) \rightarrow H(g) + Cl(g)$

135. Ans. (2)

Bond energy for dissociation ↑, Bond strongest.

136. Ans. (3)

For 1 mol bond energy of H_2 = ?

Gaseous $H(H_2)$

For 2 mol/energy = 208 kCal

Mole = $\frac{4}{2}$ = 2 mol

For 1 mol energy = 104 Kcal

137. Ans. (2)

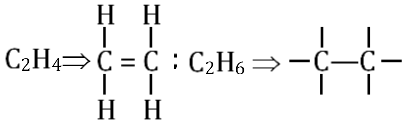
Bond energy of HCl = ?

$\Delta H_{reaction} = (BE)_R - (BE)_P$

182 = [430 + 242] – [2HCl]

HCl = 245 KJ/mol

138. Ans. (3)



$\Delta H_r = (BE)_R - (BE)_P$

$\Delta H_r = [103 + (145+4\times99)] - [80+6\times99]$

$\Delta H_r = -30$ kCal mol^{-1}

139. Ans. (2)

$\Delta H_{BDE}\ H_2 = 436$

$\Delta H_{BDE}\ N_2 = 941.8$

$N_2 + 3H_2 \rightarrow 2NH_3$

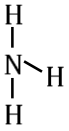
$\Delta H = 2\times\Delta H_f = 2\times(-46 KJ/mol)$

$\Rightarrow$ –92 KJ/mol

$\Delta H = \sum BE_R - \sum BE_P$

$-92 = [941.8 + (3\times436)] - [6x]$

x = 390.3 KJ/mol



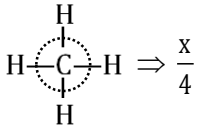
(Let BE of NH = x)

140. Ans. (3)

ΔH = +ve Endo

Average bond energy of C-H bond

$= \frac{\text{dissociation energy of } CH_4}{\text{number of bonds broken}}$



(C + 4H)

141. Ans. (4)

$H-OH \rightarrow H(g) + OH(g)$ $\Delta H = 498$

$OH(g) \rightarrow O(g) + H(g)$ $\Delta H = 428$

Two bond O–H broken so bond enthalpy of O–H bond

$= \frac{498+428}{2} = 463$ KJ mol^{-1}

142. Ans. (3)

ICl(g) + I(g) + Cl(g)

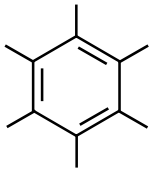
$\Delta H_{BDE}(ICl) = \sum \Delta H_f \text{product} - \sum \Delta H_f \text{ Reactant}$

= (121.34 + 106.96) – (17.57)

= 210.73 J mol^{-1}

143. Ans. (3)

Bond energy of formation of because



3C = C + 3C–C + 6C–H bond

$\Rightarrow$ 6(615) + 3(347.3) + 6(416.2)

$\Rightarrow$ 5384.1 KJ mol^{-1}

Resonance energy = $\Delta H_{dissociation} - \Delta H_f$

= 5535–5384.1

= 151 KJ mol^{-1}

144. Ans. (1)

Enthalpy of formation (Element in standard/reference state)

145. Ans. (1)

$Cl_2(g) \rightarrow 2Cl(g)$

For dissociation energy should be given generally positive.

146. Ans. (3)

Addition of solvent to solution.

Magnitude of react of solution depends upon the reaction (specific heat)

It doesn't depend on solute or solvent. So the specific enthalpy remain constant.

147. Ans. (2)

$H_2(g) \rightarrow 2H(g)$ $\Delta H = 104$ kCal

Atomisation of H_2 is H–H, $\Delta H = 104$ kCal is correct answer.

148. Ans. (4)

From $S_R \rightarrow S_M$

$S_R - S_M$ = +ve (Endothermic)

149. Ans. (4)

$\Delta H_4 - \Delta H_R$

$\Rightarrow$ –9.91 + 8.78

$\Rightarrow$ –1.13 KJ

150. Ans. (3)

Heat of reaction

151. Ans. (2)

For addition of I(g) in $I_2(g)$, Heat will be released so ΔH = –ve

152. Ans. (1)

$A(s) \xrightarrow{T} A(l)$; $\Delta H = x$ ... (1)

$A(l) \xrightarrow{T} A(g)$; $\Delta H = y$ ... (2)

$A(s) \longrightarrow A(g)$ ... (3)

Eq. (3) = Eq.(1) + Eq.(2)

$\Delta H = x + y$

153. Ans. (3)

Different intermediate reactions.

Enthalpy change depends upon nature, states of reactant products.

154. Ans. (2)

$C(g) + \frac{1}{2} O_2 \rightarrow CO$, $\Delta H = -110.5$ ...(1)

$CO + \frac{1}{2} O_2 \rightarrow CO_2$ $\Delta H = -283.2$ ...(2)

$C_{(graphite)} + O_2 \rightarrow CO_2$ ... (3)

Eq. (3) = (1) + (2)

= –110.5 – 283.2

ΔH = –393.7 KJ

155. Ans. (2)

$H_2 + \frac{1}{2} O_2 \rightarrow H_2O$, $\Delta H = -68.39$ ... (1)

$K + H_2O$ + water $\rightarrow$ KOH(aq.) + $\frac{1}{2} H_2$;

$\Delta H = -48$kCal ... (2)

KOH + water $\rightarrow$ KOH(aq)

$\Delta H = -14$ kCal ... (3)

$K + \frac{1}{2} O_2 + \frac{1}{2} H_2 \longrightarrow KOH$ ... (4)

Eq. (4) = (2) + (1) – (3)

= –68.39–48+14

156. Ans. (3)

$C + O_2 \rightarrow CO_2$ $\Delta H = -94.2$ ...(1)

$H_2 + \frac{1}{2} O_2 \rightarrow CO_2$ $\Delta H = -68.3$ ...(2)

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

$\Delta H = -210.8$ ...(3)

$C_{(s)} + 2H_2 \rightarrow CH_4$ ...(4)

Eq. (4) = (1) + (2×(2))–(3)

ΔH = –20 kCal

157. Ans. (3)

$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

$\Delta H^0 = -44kCal$ ...(1)

$2Na + 2HCl \rightarrow NaCl + H_2$

$\Delta H^0 = -152$ kCal ...(2)

$Na(s) + 0.5Cl_2(g) + NaCl(s)$ $\Delta H^0 = ?$ ... (3)

Eq. (3) = $\frac{Eq.(1)}{2} + \frac{Eq.(2)}{2} \Rightarrow \frac{-44}{2} - \frac{-152}{2}$

= –98 kCal

158. Ans. (2)

$S + \frac{3}{2}O_2 \rightarrow SO_3$ $\Delta H = -2x$ ... (1)

$SO_2 + \frac{1}{2}O_2 \rightarrow SO_3$ $\Delta H = -y$ ... (2)

$S + O_2 \rightarrow SO_2$ ... (3)

Eq.(3) = Eq.(1)–(2)

$\Rightarrow -2x + y$

$= -(2x-y)$

159. Ans. (1)

$S + O_2 \rightarrow SO_2$ $\Delta H = -298.2$ ... (1)

$SO_2 + \frac{1}{2}O_2 \rightarrow SO_3$ $\Delta H = -98.7$ ... (2)

$SO_3 + H_2O \rightarrow H_2SO_4$ $\Delta H = -130.2$ ... (3)

$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ $\Delta H = -287.3$ ... (4)

$H_2 + S + 2O_2 \rightarrow H_2SO_4$ ... (5)

Eq.(5) = (1) + (2) + (3) + (4)

$\Delta H = -298.2 - 98.7 - 130.2 - 287.3$

$\Delta H = -814.4$ KJ

160. Ans. (4)

$Zn + \frac{1}{2}O_2 \rightarrow ZnO$ $\Delta H = -84000$ Cal ...(1)

$Hg + \frac{1}{2}O_2 \rightarrow HgO$ $\Delta H = -21700$ Cal ...(2)

$Zn + HgO \rightarrow ZnO + Hg$ ... (3)

Eq(3) = Eq(1)–(2)

= –84000 + 21700 = –62300 Cal

161. Ans. (4)

$2C + 2O_2 \rightarrow 2CO_2$ $\Delta H = -787$ ... (1)

$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ $\Delta H = -286$ ... (2)

$C_2H_5 + \frac{5}{2}O_2 \rightarrow 2CO_2 + H_2O$

$\Delta H = -1310$KJ ... (3)

$2C + H_2 \rightarrow C_2H_2$ ... (4)

Eq.(4) = (1) + (2) – (3)

$\Delta H = +237$ KJ

162. Ans. (2)

$A + \frac{1}{2}O_2 \rightarrow AO$ $\Delta H = -50$ kCal ...(1)

$AO + \frac{1}{2}O_2 \rightarrow AO_2$ $\Delta H = 100$ kCal ...(2)

$A + O_2 \rightarrow AO_2$ ... (3)

Eq. (3) = Eq(1) + (2)

= –50+100

= 50 Kcal

163. Ans. (2)

$C + O_2 \rightarrow CO_2$ $\Delta H = -94$ kCal ... (1)

$CO + \frac{1}{2}CO_2$ $\Delta H = -67.7$ kCal ... (2)

$C + \frac{1}{2}O_2 \rightarrow CO$ ... (3)

Eq.(3) = Eq.(1)–(2)

$\Delta H = -94 + 67.7$

$\Delta H = -26.3$

26.3 kCal heat will be produced.

164. Ans. (1)

$H_2O(l) \rightarrow H_2O(g)$

$\Delta H = (-241.84) - (-285.77)$

$\Delta H = 43.93$ KJ mol^{-1}

165. Ans. (4)

$H_2 + \frac{1}{2}O_2 \rightarrow H_2O(l)$ $\Delta H = -68.32$ kCal ... (1)

$H_2O(l) \rightarrow H_2O(g)$ $\Delta H = 10.52$ kCal ... (2)

$H_2 + \frac{1}{2}O_2 \rightarrow H_2O(g)$ ... (3)

Eq.(3) = Eq.(1) + (2)

= –68.32 + 10.52

= –57.8 Kcal

166. Ans. (1)

$CuSO_{4(s)} \xrightarrow{H_2O} CuSO_{4(aq)}$

$\Delta H = -15.89$ ...(1)

$CuSO_4 \cdot 5H_2O_{(s)} \xrightarrow{H_2O} CuSO_{4(aq)}$

$\Delta H = 2.8$ Kcal ...(2)

$CuSO4_{(s)} \xrightarrow{H_2O} CuSO_4 \cdot 5H_2O_{(s)}$ ...(3)

Eq.(3) = Eq.(1) – (2)

$\Delta H = (-15.89) - (2.8)$

$\Delta H = -18.69$ K cal

167. Ans. (4)

$C + O_2 \rightarrow CO_2$ $\Delta H_{comb}(C)$ ... (1)

$CO + \frac{1}{2}O_2 \rightarrow CO_2$ $\Delta H_{comb}(CO)$ ... (2)

$\Delta H_g(CO)$

$C + \frac{1}{2}O_2 \rightarrow CO$ ... (3)

Eq.(3) = Eq.(1) – (2)

$\Delta H_g^0(CO, g)$

$= \Delta H_{comb}^0(C, \text{graphite}) - \Delta H_{comb}^0(CO, g)$

Exercise-II (Previous Year Questions)

1. Ans. (2)

$X_2O_{4(\ell)} \longrightarrow 2 \times O_{2(g)}$

$\Delta U = 2.1$ K cal, $\Delta S = 20$ cal K^{-1} $T = 300K$

$\Delta H = \Delta U + \Delta_{ng}RT$

$\Delta H = 2.1 + \frac{(2)\times(2)(300)}{1000}$

$\Delta H \Rightarrow 3.3$ K cal mol^{-1}

$\Delta G = \Delta H - T\Delta S$

$\Delta G = 3.3 - \frac{(300)(20)}{1000}$

$\Delta G = 3.3 - 6$

$\Delta G \Rightarrow -2.7$ Kcal mol^{-1}

2. Ans. (2)

$\Delta G = \Delta G^0 + 2.303\ RT\ \log K$

At Equilibrium

$\Delta G = 0$

$\Delta G^0 = -2.303\ RT\ \log K$

3. Ans. (3)

$C + O_2 \rightarrow CO_2$ $\Delta H_{comb.} = -393.5$ KJ mol^{-1}

$C_{(graphite)} + O_2 \rightarrow CO_2$ $\Delta H_f = ?$

$\Delta H_{comb.} = \Delta H_f = -393.5$ KJ mol^{-1}

for 1 mol CO_2 formation = 44 g

for 44g CO_2, $\Delta H = -393.5$

35.2 g CO_2, $\Delta H = \frac{-393.5}{44} \times 35.2$

$\Delta H = -315$ KJ

4. Ans. (3)

For spontaneous reaction $\Delta G < 0$

From $\Delta G = \Delta H - T\Delta S$

For always $\Delta G < 0$

$\Delta H < T\Delta S$

$\Delta S > 0$, $\Delta H < 0$

5. Ans. (4)

$\Delta S = nc_v \ln \frac{T_1}{T_2} + nR \ln \frac{p_1}{p_2}$

For isothermal $T_1 = T_2$

$\Delta S = nR \ln \frac{p_1}{p_2}$

6. Ans. (1)

$\Delta H = 35.5$ KJ mol^{-1}, $\Delta S = 83.6$ J K^{-1} mol^{-1}

For spontaneity $\Delta G < 0$

$\Delta G = \Delta H - T\Delta S$

$\Delta H - T\Delta S < 0$

$\Delta H < T\Delta S$

$T > \frac{\Delta H}{\Delta S} = \frac{35.5\times10^3}{83.6} = 424.64$

$T > 425$ K

7. Ans. (2)

$P_{ext} = 2.5$ atm, $V_i = 2.5$ L, $V_f = 4.5$ L

$\Delta U = ?$

From 1st law –

$\Delta U = q + w$ $(q = 0)$

$\Delta U = -P\Delta V = -2.5$ atm $(4.5 - 2.5)$ L

$\Rightarrow -5$ atm L 1 L. atm = 101.3J

$= -5 \times 101.3$ J

$\Delta U = -506.5 \simeq 505$ J

8. Ans. (3)

$\Delta H_f = -200$ KJ mol–1

ΔH_{BDE} $X_2 : Y_2 : XY$

$1 : 0.5 : 1$

Let $a : a/2 : a$

$$\frac{1}{2}X_2 + \frac{1}{2}Y_2 \longrightarrow XY$$

$\Delta H_{reaction} = \Sigma(\Delta H_{BE})_R - \Sigma(\Delta H_{BE})_P$

-200 KJ mol^{-1}

$$\Rightarrow \left[\frac{1}{2}\Delta H(x-x) + \frac{1}{2}\Delta H(y-y)\right] - \left[\Delta H_{xy}\right]$$

$$-200 = \left[\frac{1}{2}(a) + \frac{1}{2}\frac{(a)}{2}\right] - [a] \Rightarrow a = 800$$

9. Ans. (4)

$T = 300K$, $V_1 = 0.1$ L, $V_2 = 0.25$ L, $P_{ext} = 2$ bar

$w = ?$

$w = -P\Delta V = -2bar\ (0.25 - 0.1) = -30J$

10. Ans. (2)

$V_1 = 10^{-3}m^3$, $V_2 = 10^{-2}m^3$, $T = 300$ K,

$P = 10^{-5}$ N/m^2

$w = -P\Delta V$

$\Rightarrow -10^5\ (10^{-2} - 10^{-3})$

$\Rightarrow -10^5 \times (0.009)$

$\Rightarrow -900$ J

11. Ans. (4)

Constant temp. throughout the process. So, $T_A = T_C$

12. Ans. (2)

For free expansion of ideal gas, adiabatic

$w = 0, \quad \Delta T = 0, \quad q = 0$

13. Ans. (2)

$Kc = 2 \times 10^{13}$, $\Delta G^0 = ?$

$\Delta G^0 = -RT\ln K$

$= -8.314 \times 300 \times \ln(2\times10^{13})$

14. Ans. (1)

$2Cl\ (g) \longrightarrow Cl_2\ (g)$

For association, ΔH exothermic

$\Delta H < 0, \Delta S < 0$

n is decreasing

15. Ans. (1)

$\Delta H = 30$ KJ mol^{-1} $\quad T = 450$ K, $\Delta S = ?$

For spontaneity $\Delta G < 0$

$\Delta H < T\Delta S$

$$T > \frac{\Delta H}{\Delta S}$$

$$\Delta S > \frac{\Delta H}{T} \Rightarrow \frac{36\times10^3}{450}$$

$\Delta S > 66.66$

$\Delta S = 70$

16. Ans. (1)

$\Delta H = -109$ KJ mol–1

$H_2 + Br_2 \rightarrow 2HBr$ $\quad \Delta H_{BE}(H_2) = 435$ KJ mol^{-1}

$\Delta H_{reaction} = \Sigma(\Delta H_{BE})_R - \Sigma(\Delta H_{BE})_P$

$\Delta H_{BE}(Br_2) = 192$ KJ mol^{-1}

$-109 \Rightarrow [435 + 192] - [2 \times \Delta H_{HBr}]$

$\Delta H_{HBr} \Rightarrow 368$ KJmol^{-1}

17. Ans. (2)

For 1 mol of an ideal gas, $C_p - C_v = R$

(As per theory)

18. Ans. (3)

For isothermal, $\Delta U = 0$

$\Delta S_T \neq 0$

19. Ans. (1)

Work done = area under the curve

$\Rightarrow 1$

20. Ans. (4)

$\Delta U = q + w$

For isothermal

$\Delta U = 0$

21. Ans. (2)

$A + B \rightleftharpoons C + D$

$[A] = 2$ mol L^{-1}

$[B] = 3$ mol L^{-1}

$[C] = 10$ mol L^{-1}

$[D] = 6$ mol L^{-1}

$\Delta G^0 = -2.303\ RT \log K_{eq}$

$$= -2.303RT \log \frac{[C][D]}{[A][B]}$$

$$= -2.303 \times 2 \times 300 \times \log \frac{10\times6}{2\times3}$$

$= -2.303 \times 2 \times 300 \times \log 10$

$= -1381.8$ cal

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22. Ans. (1)

$\Delta H = \Delta U + \Delta n_g RT$

23. Ans. (4)

(A) Isothermal process $\rightarrow$ Temperature is constant(II)

(B) Isochoric process $\rightarrow$ Volume is constant(III)

(C) Isobaric process $\rightarrow$ Pressure is constant(IV)

(D) Adiabatic process $\rightarrow$ No exchange of heat(I)

24. Ans. (3)

(A) When liquid evaporates to vapour, randomness increases hence entropy increases.

(B) Temperature decreases $\Rightarrow$ entropy decreases

(C) $\Delta n_g = 2 - 0 = 2$

Δn_g = + ve $\rightarrow \Delta S$ = + ve $\Rightarrow$ entropy increases

(D) $\Delta n_g = 2 - 1 = 1$

Δn_g = +ve $\rightarrow \Delta S$ = + ve $\Rightarrow$ entropy increases

(A), (C), (D) are correct.

25. Ans. (2)

In Reversible Isothermal process,

$$W = -2.303\ nRT\ \log_{10}\left(\frac{P_1}{P_2}\right)$$

$$= -2.303 \times 1 \times 2 \times 298\ \log_{10}\left(\frac{20}{10}\right)$$

$= -413.14$ calories

26. Ans. (3)

Isothermal process $\Delta U = 0$

$\Delta U = q + w$ ___ (1)

$w = 0$ for vaccum

$\Delta U = q + 0 = 0$

27. Ans. (1)

For an endothermic reaction; q_p = (+) Ve

ΔH = (+) Ve

28. Ans. (1)

$A_2(g) + 3B_2(g) \rightarrow 2AB_3(g)\ \Delta H = +15$ kJ

$\Delta H = \Delta U + \Delta ngRT$

$$15 = \Delta U + (-2) \times \frac{25}{3} \times 10^{-3} \times 300$$

$\Delta U = (15 + 5)$ kJ $\simeq 19988.4$ J

Exercise-III (Analytical Questions)

1. Ans. (4)

A reversible process is a process in which all changes occurup at any part of the system are reversed when small always in variable are carried out in opposite direction.

Process takes place in infinite small steps and takes infinite time to complete process.

So option (4) is correct.

2. Ans. (4)

Combustion is an exothermic process.

The enthalpy of formation any standard stable state/reference state is zero.

(4) is correct

3. Ans. (3)

$C_{p_m} - C_{v_m} = R$ is applicable only for ideal gas and doesn't consider for non-ideal gas.

So (3) option correct

4. Ans. (4)

$\Delta G = \Delta H - T\Delta S$

ΔH is calculated at constant pressure.

and constant temp. to decide spontaneity.

5. Ans. (2)

$(\Delta G)_{T,P} = 0$ is not correct for spontaneous reaction for spontaneity $\Delta G < 0$

6. Ans. (1)

(i) $\Delta S_{vap} = \frac{\Delta H_{vap}}{T}$

(ii) $\Delta U = q + w \Rightarrow \Delta U = w = nc_v\Delta T$

For expansion w = -ve

So temp. decreases

(iii) Crystalline solid states entropy is lowest.

(iv) Liquid freezes so intermolecular attraction increases, entropy decreases.

(1) Option correct

7. Ans. (2)

(a) liquid $\rightarrow$ Solid

Intermolecular attraction increases, entropy increases.

(b) T $\uparrow$ entropy $\uparrow$

(c) gaseous mole increases, entropy increases

(d) $H_{2(g)} \rightarrow 2H_{(g)}$

gaseous mole increases, entropy increases

(2) correct

8. Ans. (1)

$w = 0, q < 0$

$\Delta U = q + w$

$\Delta U = q$

$q = nc_v \Delta T$

q decreases, temp. decreases

option (1) correct

9. Ans. (4)

(A) $\Delta H_f^0(H_2) = \Delta H_f^0(Cl_2) = 0$

For reference state formation enthalpy is zero.

(R) Reason is correct

(4) option correct

10. Ans. (4)

As per theory all natural process are spontaneous and irreversible.

11. Ans. (2)

For spontaneity $\Delta G < 0$

$\Delta G = \Delta H - T\Delta S < 0$

For $\Delta G < 0$ ΔH = –ve

ΔS = +ve

Option (2) is correct

12. Ans. (3)

In isolated system, matter and energy transfer is not possible between system and surrounding.

Example $\Rightarrow$ Thermoflask

13. Ans. (4)

$\Delta G = \Delta H - T\Delta S$

ΔH is contributory factor for ΔG, and as per theory.

Statement (1) is correct.

14. Ans. (2)

$\Delta G^0 = \Delta H^0 - T\Delta S^0$

+ve +ve

At high temp. ΔG decreases. So spontaneous.

with +ve entropy if $\Delta H^0 > T\Delta S^0$, $\Delta G^0 > 0$, so non spontaneous.

15. Ans. (2)

(2) option is correct as per theory.

16. Ans. (3)

(a) solid $\longrightarrow$ liquid entropy increase, intermolecular attraction decreases, at equilibrium $\Delta G = 0$

(b) $H_2O_{(s)} \xrightarrow[1atm]{10^\circ C} H_2O_{(l)}$

$T\uparrow, S\uparrow$

$\Delta G = \Delta H - T\Delta S$

$\Delta G < 0$

(c) $H_2O_{(s)} \xrightarrow[1atm]{-10^\circ C} H_2O_{(l)}$

From solid to liquid, $\Delta S\uparrow$, $\Delta G = \Delta H - T\Delta S$

At less than zero degree celsius, can't convert solid into liquid without any external help, so ΔG should be +ve.

(d) $Ag^+ + HCl \longrightarrow AgCl_{(ppt)}$

So entropy decreases, process is spontaneous.

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Exercise - I (Conceptual Questions) Build Up Your Understanding

FACTORS AFFECTING RATE OF REACTION

1. In an elementary reaction $A + 2B \rightarrow 2C + D$. If the concentration of A is increased four times and B is decreased to half of its initial concentration then the rate becomes
(1) Twice
(2) Half
(3) Unchanged
(4) One fourth of the rate

CCE001

2. The role of catalyst in a chemical reaction is :-
(1) To help attain equilibrium in a shorter time.
(2) To lower the activation energy.
(3) To shift the equilibrium in such a way as to increase the concentration of the product
(4) Both 1 & 2

CCE002

EQUILIBRIUM AND CHEMICAL PROCESS

3. $x \rightleftharpoons y$ reaction is said to be in equilibrium, when:-
(1) Only 10% conversion x to y takes place.
(2) Complete conversion of x to y takes place
(3) Conversion of x to y is only 50% complete
(4) The rate of change of x to y is just equal to the rate of change of y to x in the system

CCE003

4. In the chemical reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$ at equilibrium, state whether :-
(1) Equal volumes of N_2 & H_2 are reacting
(2) Equal masses of N_2 & H_2 are reacting
(3) The reaction has stopped
(4) The same amount of ammonia is formed as is decomposed into N_2 and H_2 in the same time

CCE004

5. Active mass of 5 g CaO is :-
(1) 56 (2) 1 (3) 3.5 (4) 2

CCE005

6. Ratio of active masses of 22g CO_2, 3g H_2 and 7g N_2 in a gaseous mixture is :-
(1) 22 : 3 : 7 (2) 0.5: 3 : 7
(3) 1 : 3 : 1 (4) 1 : 3 : 0.5

CCE006

7. Which of the following example shows effect of catalyst on reversible reaction
(1) It gives new reaction path with low activation energy.
(2) It shifts equilibrium right side.
(3) It decreases kinetic energy of activated molecules.
(4) It decreases rate of backward reaction.

CCE007

8. In reversible chemical reaction equilibrium will establish when –
(1) Reactant is completely converted into product
(2) Rate of forward and backward reaction is equal
(3) Minimum yield of product
(4) concentration of reactant and product is equal

CCE008

LAW OF MASS ACTION

9. In a chemical equilibrium, the rate constant for the backward reaction is 7.5×10^{-4} and the equilibrium constant is 1.5. The rate constant for the forward reaction is:-
(1) 2×10^{-3} (2) 5×10^{-4}
(3) 1.12×10^{-3} (4) 9.0×10^{-4}

CCE009

10. The equilibrium concentration of B for the reversible reaction $A \rightleftharpoons B$ can be evaluated by the expression:-
(1) $K_c[A]_e^{-1}$ (2) $\frac{k_f}{k_b}[A]_e^{-1}$
(3) $k_f k_b^{-1}[A]_e$ (4) $k_f k_b [A]_e^{-1}$

CCE010

11. In this reaction $Ag^+ + 2NH_3 \rightleftharpoons Ag(NH_3)_2^+$ at 298K molar concentration of Ag^+, $Ag(NH_3)_2^+$ and NH_3 is 10^{-1}, 10^{-1}, and 10^3. The value of K_C at 298K for this equilibrium :-
(1) 10^{-6} (2) 10^6
(3) 2×10^{-3} (4) 2×10^6

CCE011

12. At 1000 K, the value of K_P for the reaction :
$A(g) + 2B(g) \rightleftharpoons 3C(g) + D(g)$
is 0.05 atm. The value of K_C in terms of R would be :-
(1) 20000 R (2) 0.02 R
(3) 5×10^{-5} R (4) $5 \times 10^{-5} \times R^{-1}$
CCE012

13. For the reaction $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$ the partial pressure of CO and CO_2 are 2.0 and 4.0 atm respectively at equilibrium. The K_P for the reaction is
(1) 0.5 (2) 4.0 (3) 8.0 (4) 1
CCE013

14. For which reaction is $K_p = K_c$:-
(1) $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$
(2) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
(3) $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$
(4) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$
CCE014

15. For the reaction
$CuSO_4.5H_2O_{(s)} \rightleftharpoons CuSO_4.3H_2O_{(s)} + 2H_2O_{(g)}$
Which one is correct representation :–
(1) $K_p = p^2_{H_2O}$ (2) $K_c = [H_2O]^2$
(3) $K_p = K_c (RT)^2$ (4) All of these
CCE015

16. $\log \frac{K_p}{K_c} + \log RT = 0$ is true relationship for the following reaction:-
(1) $PCl_5 \rightleftharpoons PCl_3 + Cl_2$
(2) $2SO_2 + O_2 \rightleftharpoons 2SO_3$
(3) $N_2 + 3H_2 \rightleftharpoons 2NH_3$
(4) (2) and (3) both
CCE016

17. For which reaction at 298 K, the value $\frac{K_p}{K_c}$ of is maximum and minimum respectively:-
(a) $N_2O_4 \rightleftharpoons 2NO_2$
(b) $2SO_2 + O_2 \rightleftharpoons 2SO_3$
(c) $X + Y \rightleftharpoons 4Z$
(d) $A + 3B \rightleftharpoons 7C$
(1) d, c (2) d, b (3) c, b (4) d, a
CCE017

18. Consider the two gaseous equilibrium involving SO_2 and the corresponding equilibrium constants at 299 K
$SO_2(g) + \frac{1}{2} O_2(g) \rightleftharpoons SO_3(g)$; K_1
$4SO_3(g) \rightleftharpoons 4SO_2(g) + 2O_2(g)$; K_2
The value of the equilibrium constants are related by :-
(1) $K_2 = \frac{1}{(K_1)^4}$ (2) $K_2 = K_1^4$
(3) $K_2 = \left(\frac{1}{K_1}\right)^{\frac{1}{4}}$ (4) $K_2 = \frac{1}{K_1}$
CCE018

19. For the reactions :- $A \rightleftharpoons B$; $K_C = 2$,
$B \rightleftharpoons C$; $K_C = 4$, $C \rightleftharpoons D$; $K_C = 6$
K_C for the reaction $A \rightleftharpoons D$:-
(1) 12 (2) 4/3
(3) 24 (4) 48
CCE019

20. Which Oxide of Nitrogen is most stable :-
(1) $2NO_{2(g)} \rightleftharpoons N_{2(g)} + 2O_{2(g)}$
$K = 6.7 \times 10^{16}$ mol L^{-1}
(2) $2NO_{(g)} \rightleftharpoons N_{2(g)} + O_{2(g)}$
$K = 2.2 \times 10^{30}$
(3) $2N_2O_{5(g)} \rightleftharpoons 2N_{2(g)} + 5O_{2(g)}$
$K = 1.2 \times 10^{34}$ mol^5 L^{-5}
(4) $2N_2O_{(g)} \rightleftharpoons 2N_{2(g)} + O_{2(g)}$
$K = 3.5 \times 10^{33}$ mol L^{-1}
CCE020

21. The equilibrium constant in a reversible reaction at a given temperature:-
(1) Depends on initial concentration of the reactants.
(2) Depends on the concentration of the products at equilibrium.
(3) Does not depend on the initial concentrations.
(4) It is not characteristic of the reaction.
CCE021

22. Which one of the following statements is correct about equilibrium constant:-
(1) Equilibrium constant of a reaction changes with temperature.
(2) Equilibrium constant of a reaction depends upon the concentration of reactants with which we start.
(3) Equilibrium constant of a reaction, $3Fe(s) + 4H_2O(g) \rightleftharpoons Fe_3O_4(s) + 4H_2(g)$ is same whether, the reaction is carried out in an open vessel or a closed vessel.
(4) Equilibrium constant of a reaction becomes double if the reaction is multiplied by 2 throughout.

CCE022

23. For a reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$, the value of K_c does not depends upon :-
(a) Initial concentration of the reactants
(b) Pressure
(c) Temperature
(d) Catalyst
(1) Only c (2) a, b, c
(3) a, b, d (4) a, b, c, d

CCE023

24. For any reversible reaction if concentration of reactants increases then value of equilibrium constant :-
(1) Depends on amount of concentration
(2) Unchanged
(3) Decreases
(4) Increases

CCE024

25. Effect of increasing temperature on equilibrium constant is given by

$$\log K_2 - \log K_1 = \frac{-\Delta H^\circ}{2.303R}\left[\frac{1}{T_2} - \frac{1}{T_1}\right]$$

then for an endothermic reaction the false statement is:-
(1) $\left[\frac{1}{T_2} - \frac{1}{T_1}\right]$ = positive (2) $\log K_2 > \log K_1$
(3) ΔH = positive (4) $K_2 > K_1$

CCE025

26. The equilibrium constant for the reaction $Br_2 \rightleftharpoons 2Br$ at 500 K and 700 K are 1×10^{-10} and 1×10^{-5} respectively. The reaction is:-
(1) Endothermic (2) Exothermic
(3) Fast (4) Slow

CCE026

27. In an experiment the equilibrium constant for the reaction $A + B \rightleftharpoons C + D$ is K when the initial concentration of A and B each is 0.1 mol L^{-1} Under the similar conditions in an another experiment if the initial concentration of A and B are taken 2 and 3 mol L^{-1} respectively then the value of equilibrium constant will be:-
(1) $\frac{K}{2}$ (2) K
(3) K^2 (4) $\frac{1}{K}$

CCE027

28. In system $A(s) \rightleftharpoons 2B(g) + 3C(g)$ at equilibrium if concentration of 'C' is doubled then concentration of B at equilibrium will be :-
(1) Double its original concentration
(2) Half its original concentration
(3) $2\sqrt{2}$ its original concentration
(4) $\frac{1}{2\sqrt{2}}$ its original concentration

CCE028

29. For the reaction, $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ equilibrium constant, K_P changes with :-
(1) Temperature
(2) Total pressure
(3) Catalyst
(4) Amount of H_2 and I_2 present

CCE029

30. The equilibrium constant (K_P) for the reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ is 16. If the volume of the container is reduced to one-half its original volume, the value of K_P for the reaction at the same temperature will be :-
(1) 32 (2) 64
(3) 16 (4) 4

CCE030

31. The equilibrium constant of the reaction $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ is 64. If the volume of the container is reduced to one fourth of its original volume, the value of the equilibrium constant will be

(1) 16 (2) 32 (3) 64 (4) 128

CCE031

32. If some He gas is introduced into the equilibrium $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ at constant pressure and temperature then equilibrium constant of reaction :

(1) Increases

(2) Decreases

(3) Unchanged

(4) Nothing can be said

CCE032

33.

	List X		List Y
(A)	Active mass	(I)	$\Delta n = 0$
(B)	Dynamic nature	(II)	Molar concentration
(C)	A + heat $\rightleftharpoons$ B	(III)	Vant hoff's equation
(D)	$\log (K_{p2}/K_{p1})$	(IV)	adaptation if temperature increases $= \frac{\Delta H^\circ}{2.303R}\left[\frac{1}{T_1}-\frac{1}{T_2}\right]$
(E)	$2A(g)+B(g) \rightleftharpoons 3C(g)$	(v)	Chemical equilibrium

Correct match list X and Y

(1) A - (V), B - (II), C - (III), D - (I), E - (IV)

(2) A - (V), B - (IV), C - (III), D - (II), E - (I)

(3) A - (II), B - (V), C - (IV), D - (III), E - (I)

(4) None of these

CCE033

DEGREE OF DISSOCIATION AND APPLICATION OF LAW OF MASS ACTION

34. For the reaction : $P \rightleftharpoons Q + R$. Initially 2 mol of P was taken. Up to equilibrium 0.5 mol of P was dissociated. What would be the degree of dissociation :-

(1) 0.5 (2) 1

(3) 0.25 (4) 4.2

CCE034

35. The dissociation of CO_2 can be expressed as $2CO_2 \rightleftharpoons 2CO + O_2$. If the 2 mol of CO_2 is taken initially and 40% of the CO_2 is dissociated completely. What is the total number of moles at equilibrium:-

(1) 2.4 (2) 2.0 (3) 1.2 (4) 5

CCE035

36. In $A_3(g) \rightleftharpoons 3A\ (g)$ reaction, the initial concentration of A_3 is "a" mol L^{-1} If x is degree of dissociation of A_3. The total number of moles at equilibrium will be:-

(1) $a - \frac{ax}{3}$ (2) $\frac{a}{3} - x$

(3) $\left(\frac{a-ax}{2}\right)$ (4) a + 2 ax

CCE036

37. In the reaction $2P(g) + Q(g) \rightleftharpoons 3R(g) + S(g)$. If 2 mol each of P and Q taken initially in a 1 L flask. At equilibrium which is true:-

(1) [P] < [Q] (2) [P] = [Q]

(3) [Q] = [R] (4) None of these

CCE037

38. The reaction $A + B \rightleftharpoons C + D$ is studied in a one litre Vessel at 250°C. The initial concentration of A was 3n and of B was n. After equilibrium was attained then equilibrium concentration of C was found to be equal to equilibrium concentration of B. What is the concentration of D at equilibrium :-

(1) $\frac{n}{2}$ (2) $\left(3n-\frac{n}{2}\right)$

(3) $\left(n+\frac{n}{2}\right)$ (4) n

CCE038

39. $X_2 + Y_2 \rightleftharpoons 2XY$ reaction was studied at a certain temperature. In the beginning 1 mole of X_2 was taken in a one litre flask and 2 moles of Y_2 was taken in another 2litre flask. What is the equilibrium concentration of X_2 and Y_2 ? (Given equilibrium concentration of [XY] = 0.6 mol L^{-1}).

(1) $\left(\frac{1}{3}-0.3\right),\left(\frac{2}{3}-0.3\right)$

(2) $\left(\frac{1}{3}-0.6\right),\left(\frac{2}{3}-0.6\right)$

(3) (1 – 0.3), (2 – 0.3)

(4) (1 – 0.6), (2 – 0.6)

CCE039

40. In a 20 litre vessel initially 1 - 1 mole CO, H_2O, CO_2 is present, then for the equilibrium of $CO + H_2O \rightleftharpoons CO_2 + H_2$ following is true:-
(1) H_2, more than 1 mole
(2) CO, H_2O, H_2 less than 1 mole
(3) CO_2 & H_2O both more than 1 mole
(4) All of these

CCE040

41. 4 mole of PCl_5 are heated at constant temperature in closed container. If degree of dissociation for PCl_5 is 0.5 then calculate total number of moles at equilibrium
(1) 4.5 (2) 6 (3) 3 (4) 4

CCE041

42. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$
In above reaction, at equilibrium condition mole fraction of PCl_5 is 0.4 and mole fraction of Cl_2 is 0.3. Then find out mole fraction of PCl_3
(1) 0.3 (2) 0.7 (3) 0.4 (4) 0.6

CCE042

43. If 8 mole of PCl_5 is heated in a closed vessel of 10 L capacity and 25% of it dissociates into PCl_3 and Cl_2 at the equilibrium then value of Kp will be [Given : Total equilibrium pressure is P]
(1) P/30 (2) P/15
(3) 2/3P (4) 3/2P

CCE043

44. In the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ the partial pressure of PCl_3, Cl_2 and PCl_5 are 0.3, 0.2 and 0.6 atm respectively at equilibrium. If partial pressure of PCl_3 and Cl_2 was twice at new equilibrium, what will be the partial pressure of PCl_5 is in atm at new equilibrium condition :-
(1) 0.3 (2) 1.2
(3) 2.4 (4) 0.15

CCE044

45. 'a' mol of PCl_5, undergoes, thermal dissociation as: $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, the mole fraction of PCl_3 at equilibrium is 0.25 and the total pressure is 2.0 atm. The partial pressure of Cl_2 at equilibrium is:-
(1) 2.5 (2) 1.0
(3) 0.5 (4) None

CCE045

46. In a 0.25 L tube dissociation of 4 mol of NO takes place. If its degree of dissociation is 10%. The value of K_P for reaction $2\,NO \rightleftharpoons N_2 + O_2$ is :-
(1) $\frac{1}{(18)^2}$ (2) $\frac{1}{(8)^2}$
(3) $\frac{1}{16}$ (4) $\frac{1}{32}$

CCE046

47. In a chemical equilibrium $A + B \rightleftharpoons C + D$ when one mole each of the two reactants are mixed, 0.4 mol each of the products are formed. The equilibrium constant is :-
(1) 1 (2) 0.36 (3) 2.25 (4) $\frac{4}{9}$

CCE047

48. K_C for the esterification reaction :
$CH_3COOH(\ell) + C_2H_5OH(\ell) \rightleftharpoons CH_3COOC_2H_5(\ell) + H_2O(\ell)$
is 4. If 4 mol each of acid and alcohol are taken initially, what is the equilibrium concentration of the acid :-
(1) $\frac{2}{3}$ (2) $\frac{4}{3}$ (3) $\frac{3}{4}$ (4) $\frac{3}{2}$

CCE048

49. Evaluate K_P for the reaction : $H_2 + I_2 \rightleftharpoons 2HI$. If 2 moles each of H_2 and I_2 are taken initially. At equilibrium moles of HI are 2.
(1) 2.5 (2) 4 (3) 0.25 (4) 1.0

CCE049

50. 4 moles of A are mixed with 4 moles of B, when 2 mol of C are formed at equilibrium, according to the reaction, $A + B \rightleftharpoons C + D$. The equilibrium constant is :-
(1) 4 (2) 1 (3) $\sqrt{2}$ (4) $\sqrt{4}$

CCE050

51. Two moles of ammonia is introduced in a evacuated 500 mL vessel at high temperature. The decomposition reaction is :
$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$
At the equilibrium NH_3 becomes 1 mole then the K would be :-
(1) 0.42 (2) 6.75 (3) 1.7 (4) 1.5

CCE051

52. 4.5 mol each of hydrogen and iodine heated in a sealed 10 litre vessel. At equilibrium, 3 mol of HI were found. The equilibrium constant for $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$ is:-

(1) 1 (2) 10 (3) 5 (4) 0.33

CCE052

53. AB dissociates as $2AB\ (g) \rightleftharpoons 2A\ (g) + B_2\ (g)$

When the initial pressure of AB is 500 mm, the total pressure becomes 625 mm when the equilibrium is attained. Calculate K_P for the reaction assuming volume remains constant.

(1) 500 (2) 125 (3) 750 (4) 375

CCE053

LE-CHATLIER'S PRINCIPLE

54. Cis -2- pentene $\rightleftharpoons$ Trans -2- pentene for the above equilibrium the value of standard free energy change at 400 K is –3.67 kJ mol^{-1}. If excess of trans -2- pentene is added to the system then :-

(1) Additional trans -2- pentene will form

(2) Excess of cis -2- pentene will form

(3) Equilibrium will proceed in the forward

(4) Equilibrium will remain unaffected

CCE054

55. When $NaNO_3$ is heated in a closed vessel, O_2 is liberated and $NaNO_2$ is left behind. At equilibrium –

(1) Addition of $NaNO_3$ favours forward reaction

(2) Addition of $NaNO_2$ favours reverse reaction

(3) Increasing pressure favours reverse reaction.

(4) Decreasing temperature favours forward reaction.

CCE055

56. In manufacture of NO, the reaction of N_2 and O_2 to form NO is favourable if :-

(1) Pressure is increased

(2) Pressure is decreased

(3) Temperature is increased

(4) Temperature is decreased

CCE056

57. In which of the following equilibrium reactions, the equilibrium would shift to right side, if total pressure is decreased :-

(1) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (2) $H_2 + I_2 \rightleftharpoons 2HI$

(3) $N_2O_4 \rightleftharpoons 2NO_2$ (4) $H_2 + Cl_2 \rightleftharpoons 2HCl$

CCE057

58. The oxidation of SO_2 by O_2 to SO_3 is exothermic reaction. The concentration of SO_2 will be minimum if :-

(1) Temperature is increased and pressure is kept constant

(2) Temperature is reduced and pressure is increased

(3) Both temperature and pressure are increased

(4) Both temperature and pressure are decreased

CCE058

59. For the manufacture of ammonia by the reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3 + 21.9$ kCal, the favourable conditions are :-

(1) Low temperature, low pressure & catalyst

(2) Low temperature, high pressure & catalyst

(3) High temperature, low pressure & catalyst

(4) High temperature, high pressure & catalyst

CCE059

60. In the reaction $2A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + 362$ kCal. Which combination of pressure and temperature gives the highest yield of C at equilibrium:-

(1) 1000 atm and 500°C

(2) 500 atm and 500°C

(3) 1000 atm and 50°C

(4) 500 atm and 100°C

CCE060

61. Does Le chatelier's principle predict a change of equilibrium concentration for the following reaction if the gas mixture is compressed

$N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$

(1) Yes, backward reaction is favoured

(2) Yes, forward reaction is favoured

(3) No change

(4) No information

CCE061

62. The reaction in which yield of production cannot be increased by the application of high pressure is :-
(1) $PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$
(2) $N_2(g) + O_2(g) \rightleftharpoons 2NO\ (g)$
(3) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
(4) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

CCE062

63. In a vessel containing SO_3, SO_2 and O_2 at equilibrium, some helium gas is introduced so that the total pressure increases while temperature and volume remain constant. According to Le-Chatelier principle, the dissociation of SO_3,
(1) Increases (2) Decreases
(3) Remains unaltered (4) None of these

CCE063

PHYSICAL EQUILIBRIUM

64. For the equilibrium reaction, $H_2O\ (\ell) \rightleftharpoons H_2O_{(g)}$, What happens, if pressure is applied:-
(1) More water evaporates
(2) The boiling point of water is increased
(3) No effect on boiling point
(4) None of the above

CCE064

65. On cooling of following system at equilibrium $CO_{2(s)} \rightleftharpoons CO_{2(g)}$:-
(1) There is no effect on the equilibrium state
(2) More gas is formed
(3) More gas is solidifies
(4) None of above

CCE065

CALCULATION OF DEGREE OF DISSOCIATION BY V.D. METHOD

66. Vapour density of PCl_5 is 104.25 at T°C. Then degree of dissociation of PCl_5 is. (Mw = 208.5)
(1) 20% (2) 0% (3) 30% (4) 15%

CCE066

67. When heating PCl_5 then it decompose PCl_3 and Cl_2 in form of gas, The vapour density of gas mixture is 70.2 and 57.9 at 200° C and 250°C. The degree of dissociation of PCl_5 at 200°C and 250°C is
(1) 48.50% & 80% (2) 60% & 70%
(3) 70% & 80% (4) 80% & 90%

CCE067

68. The equation $\alpha = \frac{D-d}{(n-1)d}$ is correctly matched for Where D = Theoretical vapour density
d = Observed vapour density
(1) $A \rightleftharpoons \frac{nB}{2} + \frac{nC}{3}$
(2) $A \rightleftharpoons \frac{nB}{3} + \left(\frac{2n}{3}\right)C$
(3) $A \rightleftharpoons \left(\frac{n}{2}\right)B + \left(\frac{n}{4}\right)C$
(4) $A \rightleftharpoons \left(\frac{n}{2}\right)B + C$

CCE068

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	4	4	2	4	1	2	3	3	1	4	4	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	1	4	1	3	1	3	2	1	1	2	4	1	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	3	3	3	1	4	1	1	1	2	2	1	2	3	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	4	2	2	2	2	1	2	2	3	3	3	2	2	3
Question	61	62	63	64	65	66	67	68							
Answer	1	2	3	2	3	2	1	2							

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. For the reversible reaction :

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + Heat$

The equilibrium shifts in forward direction :

(1) By increasing the concentration of $NH_3(g)$

(2) By decreasing the pressure

(3) By decreasing the concentrations of $N_2(g)$ and $H_2(g)$

(4) By increasing pressure and decreasing temperature

CCE074

2. For a given exothermic reaction, K_p and K'_p are the equilibrium constants at temperatures T_1 and T_2, respectively where $T_2 > T_1$. Assuming that heat of reaction is constant in temperature range between T_1 and T_2, it is readily observed that:-

(1) $K_p > K'_p$ (2) $K_p < K'_p$

(3) $K_p = K'_p$ (4) $K_p = \frac{1}{K'_p}$

CCE075

AIPMT-2015

3. If the value of an equilibrium constant for a particular reaction is 1.6×10^{12}, then at equilibrium the system will contain :-

(1) Mostly reactants

(2) Mostly products

(3) Similar amounts of reactants and products

(4) All reactants

CCE076

RE-AIPMT-2015

4. If the equilibrium constant for $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is K, the equilibrium constant for $\frac{1}{2} N_2(g) + \frac{1}{2} O_2(g) \rightleftharpoons NO(g)$ will be:-

(1) K (2) K^2

(3) $K^{1/2}$ (4) $\frac{1}{2}K$

CCE077

NEET(UG) 2017

5. The equilibrium constant of the following are:

$N_2 + 3H_2 \rightleftharpoons 2NH_3$; K_1

$N_2 + O_2 \rightleftharpoons 2NO$; K_2

$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$; K_3

The equilibrium constant (K) of the reaction :

$2NH_3 + \frac{5}{2}O_2 \overset{K}{\rightleftharpoons} 2NO + 3H_2O$, will be :

(1) $K_2 K_3^3 / K_1$ (2) K_2K_3/K_1

(3) $K_2^3K_3 / K_1$ (4) $K_1K_3^3 / K_2$

CCE078

6. A 20 litre container at 400 K contains $CO_2(g)$ at pressure 0.4 atm and an excess of SrO (neglect the volume of solid SrO). The volume of the container is now decreased by moving the movable piston fitted in the container. The maximum volume of the container, when pressure of CO_2 attains its maximum value, will be :-

(Given that : $SrCO_3(s) \rightleftharpoons SrO(s) + CO_2(g)$, Kp = 1.6atm)

(1) 10 litre (2) 4 litre

(3) 2 litre (4) 5 litre

CCE079

NEET(UG) 2018

7. Which one of the following conditions will favour maximum formation of the product in the reaction,

$A_2(g) + B_2(g) \rightleftharpoons X_2(g)$ $\Delta_r H = -X$ kJ ?

(1) Low temperature and high pressure

(2) Low temperature and low pressure

(3) High temperature and high pressure

(4) High temperature and low pressure

CCE080

NEET(UG) 2021

8. Equivalent amounts of H_2 and I_2 are heated in a closed container till equilibrium is obtained. If 80% of the hydrogen can be converted to HI, then K_c at this temperature

(1) 64 (2) 16 (3) 0.25 (4) 4

CCE081

9. Choose the correct option for the total pressure (in atm.) in a mixture of 4 g O_2 and 2 g H_2 confined in a total volume of one litre at 0°C is:

[Given R = 0.082 L atm $mol^{-1}K^{-1}$, T=273K]

(1) 2.518 (2) 2.602

(3) 25.18 (4) 26.02

CCE082

NEET(UG) 2022

10. Which one is **not** correct mathematical equation for Dalton's Law of partial pressure ? Here p = total pressure of gaseous mixture

(1) $p = n_1\frac{RT}{V} + n_2\frac{RT}{V} + n_3\frac{RT}{V}$

(2) $p_i = \chi_i p$, where p_i = partial pressure of ith gas χ_i = mole fraction of ith gas in gaseous mixture

(3) $p_i = \chi_i p_i^o$, where χ_i, = mole fraction of ith gas in gaseous mixture p_i^o = pressure of ith gas in pure state

(4) $p = p_1 + p_2 + p_3$

CCE083

11. $3O_2(g) \rightleftharpoons 2O_3(g)$

for the above reaction at 298 K, K_C is found to be 3.0×10^{-59}. If the concentration of O_2 at equilibrium is 0.040 M then concentration of O_3 in M is

(1) 1.9×10^{-63} (2) 2.4×10^{31}

(3) 1.2×10^{21} (4) 4.38×10^{-32}

CCE084

12. K_P for the following reaction is 3.0 at 1000 K.

$CO_2(g) + C(s) \rightleftharpoons 2CO(g)$

What will be the value of K_C for the reaction at the same temperature ?

(Given - R = 0.083 L bar K^{-1} mol^{-1})

(1) 0.36 (2) 3.6×10^{-2}

(3) 3.6×10^{-3} (4) 3.6

CCE085

NEET(UG) 2023

13. For a certain reaction, the rate = k $[A]^2$ [B], when the initial concentration of A is tripled keeping concentration of B constant, the initial rate would

(1) increase by a factor of six

(2) increase by a factor of nine

(3) increase by a factor of three

(4) decrease by a factor of nine

CCE101

NEET(UG) 2024

14. In which of the following equilibria, K_p and K_c are **NOT** equal ?

(1) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

(2) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$

(3) $CO_{(g)} + H_2O_{(g)} \rightleftharpoons CO_{2(g)} + H_{2(g)}$

(4) $2BrCl_{(g)} \rightleftharpoons Br_{2(g)} + Cl_{2(g)}$

CCE137

15. For the reaction $2A \rightleftharpoons B + C$, $K_c = 4 \times 10^{-3}$. At a given time, the composition of reaction mixture is:

$[A] = [B] = [C] = 2 \times 10^{-3}$ M.

Then, which of the following is correct ?

(1) Reaction is at equilibrium.

(2) Reaction has a tendency to go in forward direction.

(3) Reaction has a tendency to go in backward direction

(4) Reaction has gone to completion in forward direction.

CCE138

16. Consider the following reaction in a sealed vessel at equilibrium with concentrations of $N_2 = 3.0 \times 10^{-3}$ M, $O_2 = 4.2 \times 10^{-3}$ M and NO = 2.8×10^{-3} M.

$2NO_{(g)} \rightleftharpoons N_{2(g)} + O_{2(g)}$

If 0.1 mol L^{-1} of $NO_{(g)}$ is taken in a closed vessel, what will be degree of dissociation (α) of $NO_{(g)}$ at equilibrium?

(1) 0.00889 (2) 0.0889

(3) 0.8889 (4) 0.717

CCE139

NEET(UG) 2024 (Re-examination)

17. At a given temperature and pressure, the equilibrium constant values for the equilibria are given below :

$3A_2 + B_2 \rightleftharpoons 2A_3B, \quad K_1$

$A_3B \rightleftharpoons \frac{3}{2}A_2 + \frac{1}{2}B_2, \quad K_2$

The relation between K_1 and K_2 is :

(1) $K_1^2 = 2K_2$ (2) $K_2 = \frac{K_1}{2}$

(3) $K_1 = \frac{1}{\sqrt{K_2}}$ (4) $K_2 = \frac{1}{\sqrt{K_1}}$

CCE140

18. For the reaction in equilibrium

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g), \Delta H = -Q$

Reaction is favoured in forward direction by :

(1) use of catalyst

(2) decreasing concentration of N_2

(3) low pressure, high temperature and high concentration of ammonia

(4) high pressure, low temperature and higher concentration of H_2

CCE141

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	1	2	3	1	4	1	1	3	3	4	2	2	1	3
Question	16	17	18												
Answer	4	4	4												

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Exercise – III (Analytical Questions) Master Your Understanding

1. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion (A) : For any chemical reaction at a particular temperature the equilibrium constant is fixed and is a characteristic property.

Reason (R) : Equilibrium constant is independent of temperature.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is not correct but **(R)** is correct.

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

CCE086

2. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion (A) : In the dissociation of PCl_5 at constant pressure and temperature addition of helium at equilibrium increases the dissociation of PCl_5.

Reason (R) : Equilibrium constant changes only with change in temperature.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is not correct but **(R)** is correct.

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

CCE087

3. Given below are two statements:

Statement I : Solubility of a gas in liquid increases with increase in pressure of the gas in equilibrium with solution

Statement II : The dissolution of a gas in liquid is an exothermic process.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement I and statement II are incorrect.

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct.

(4) Both statements I and statements II are correct.

CCE089

4. Given below are two statements:

Statement I : The effect of temperature on equilibrium constant is given by vant Hoff's equation

Statement II : Vant Hoff's equation is :-

$$\log \frac{K_1}{K_2} = \frac{\Delta H^0}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct and statement II is incorrect.

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct.

(4) Both statements I and statements II are correct.

CCE092

5. Given below are two statements:
Statement I : $K_P = K_C$ (Correct for all reactions)
Statement II : The value of K_{eq} increase when concentration of the reactants are increased.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement I and statement II are incorrect.
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct.
(4) Both statements I and statements II are correct.

CCE093

6. Given below are two statements:
Statement I : There is no effect on equilibrium constant if inert gas is added to the reaction.
$A(g) + B(g) \rightleftharpoons C(g)$
Statement II : Equilibrium constant changes only with temperature.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement I and statement II are incorrect.
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct.
(4) Both statements I and statements II are correct.

CCE094

7. For a reaction at equilibrium.
(A) $\Delta G = 0$ (B) $r_{net} = 0$ (C) $r_f = r_b$ (D) $E_{Cell} = 0$
The correct one is :-
(1) C is correct only
(2) A, B, C and D are correct
(3) A, B and C is correct
(4) B, C and D is correct

CCE095

8. If a chemical reaction is at equilibrium then out of the following :-
(A) $\Delta G = 0$ (B) $K_P = Q_P$
(C) $K_C = 1$ (D) $\Delta G^\circ = 1$
The correct option is :-
(1) A and B (2) B, C and D
(3) A, C and D (4) A, B, C and D

CCE097

9. Identify the set of correct statement-
(a) If $K_C > 10^3$; products predominate over reactant
(b) If $K_C < 10^{-3}$; reactant predominate over product.
(c) If K_C is in the range of 10^{-3} to 10^3 ; appreciable concentration of both reactant and product are present.
(d) If $Q_C > K_C$ net reaction goes from left to right.
Choose the correct answer from the options given below-
(1) a, b only (2) c, d only
(3) a, b, c only (4) a, b, c, d only

CCE098

10. For the reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ the value of K_P is 50 at 400 K and 1700 at 500 K.
(a) The reaction is endothermic
(b) The reaction is exothermic
(c) If $NO_2(g)$ and $N_2O_4(g)$ are mixed at 400 K at partial pressure 20 bar and 2 bar respectively, more $N_2O_4(g)$ will be formed
(d) K is independent of temperature
Choose the correct answer from the options given below-
(1) a, b only (2) a, c only
(3) c, d only (4) b, d only

CCE099

11. If $\Delta G^\circ < 0$ then which is/are true :-
(1) $-\frac{\Delta G^\circ}{RT} > 0$ (2) $e^{-\Delta G^\circ/RT} > 1$
(3) $K > 1$ (4) All of these

CCE100

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	2	1	4	2	1	4	2	1	3	2	4

Chemical Equilibrium

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (3)

For an elementary reaction :

$r = K[A]\,[B]^2$ ...(I)

$r^1 = K[4A]\,[B/2]^2$ ...(II)

Divide (I) by (II)

$$\frac{r}{r^1} = \left(\frac{1}{4}\right) \times (4)$$

$$r = r^1$$

2. Ans. (4)

Catalyst reduces the time taken to attain equilibrium by lowering activation energy.

3. Ans. (4)

At equilibrium :- rate of forward reaction = rate of backward reaction

4. Ans. (4)

5. Ans. (2)

Active mass of pure solid = 1

6. Ans. (4)

$$[CO_2] : [H_2] : [N_2] = \frac{22}{44} : \frac{3}{2} : \frac{7}{28}$$

$$= \frac{1}{2} : \frac{3}{2} : \frac{7}{4}$$

$$= 1 : 3 : 0.5$$

7. Ans. (1)

Catalyst provides an alternate path with lower activation energy.

8. Ans. (2)

Rate of forward and backward reaction is equal

9. Ans. (3)

$$K_{eq.} = \frac{k_f}{k_b}$$

$$K_f = k_{eq.} \times k_b$$

$$= 1.5 \times 7.5 \times 10^{-4}$$

$$= 11.25 \times 10^{-4}$$

$$= 1.12 \times 10^{-3}$$

10. Ans. (3)

$$K_{eq.} = \frac{k_f}{k_b} = \frac{[B]}{[A]}$$

$$[B] = \frac{k_f}{k_b} . [A]$$

$$= k_f . k_b^{-1} [A]$$

11. Ans. (1)

$$K_C = \frac{[Ag(NH_3)_2^+]}{[Ag^+][NH_3]^2}$$

$$= \frac{\left(10^{-1}\right)}{\left(10^{-1}\right)\left(10^3\right)^2} = 10^{-6}$$

12. Ans. (4)

$K_p = K_C\,(RT)^{\Delta ng}$

$\Delta n_g = (3 + 1) - (1 + 2) = 1$

$K_p = K_C\,(RT)^1$

$$K_C = \frac{k_p}{(RT)} = \frac{.05}{(R \times 1000)}$$

$= 5 \times 10^{-5}\ R^{-1}$

13. Ans. (4)

$$K_p = \frac{(P_{CO})^2}{(P_{CO_2})} = \frac{(2)^2}{(4)} = 1$$

14. Ans. (3)

For $\Delta n_g = 0 \Rightarrow K_P = K_C$

15. Ans. (4)

$K_P = P_{H_2O}^2$

$K_C = [H_2O]^2$

$K_P = K_C(RT)^2$

16. Ans. (2)

$$\log \frac{k_p}{k_c} + \log(RT) = 0$$

$$\log \frac{k_p}{k_c} = -\log(RT)$$

$$\frac{k_p}{k_c} = (RT)^{-1}$$

$k_P = k_C(RT)^{-1}$

$\Delta n_g = -1$

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

$\Delta n_g = -1$

17. Ans. (2)

$k_P = k_C(RT)^{\Delta ng}$

$\frac{k_p}{k_c} = (RT)^{\Delta ng}$

for $\frac{k_p}{k_c} = \text{minimum} \Rightarrow \Delta ng \rightarrow \text{minimum}$

for $\frac{k_p}{k_c} = \text{maximum} \Rightarrow \Delta ng \rightarrow \text{maximum}$

18. Ans. (1)

Reverse (1) & multiply by 4.

$K_2 = \frac{1}{(k_1)^4}$

19. Ans. (4)

Add all given reactions to get K of target equation:

Hence K of A $\rightleftharpoons$ D is $2 \times 4 \times 6 = 48$

20. Ans. (1)

Stability $\propto$ Conc.

For reactant to be stable ; conc. of reactant should be more.

Hence K should be less.

So minimum K is of first reaction.

21. Ans. (3)

Equilibrium constant depends only on temperature.

22. Ans. (1)

Equilibrium constant of a reaction changes with temperature.

23. Ans. (3)

Equilibrium constant of a reaction changes with temperature.

24. Ans. (2)

Equilibrium constant of a reaction changes with temperature.

25. Ans. (1)

$\left[\frac{1}{T_2} - \frac{1}{T_1}\right] \Rightarrow \left[\frac{T_1 - T_2}{T_1 T_2}\right] < 0.$

26. Ans. (1)

On $\uparrow$sing Temperature, value of K is increasing. Hence reaction is endothermic.

27. Ans. (2)

Equilibrium constant depends only on temperature.

28. Ans. (4)

$K = [B]^2 [C]^3$

$K = [B^1]^2 [C^1]^3$

Given : $[C^1] = [2C]$

$[B]^2 [\not{C}]^3 = [B^1]^2 [2\not{C}]^3$

$[B^1]^2 = \frac{[B]^2}{8} \Rightarrow [B^1] = \frac{[B]}{2\sqrt{2}}$

29. Ans. (1)

Equilibrium constant depends only on temperature.

30. Ans. (3)

Equilibrium constant depends only on temperature.

31. Ans. (3)

Equilibrium constant depends only on temperature.

32. Ans. (3)

Equilibrium constant depends only on temperature.

33. Ans. (3)

(a) Active mass $\rightarrow$ Concentration.

(b) Dynamic nature $\rightarrow$ Chemical equilibrium.

(c) A + heat $\rightleftharpoons$ B $\rightarrow$ forward reaction increases on $\uparrow$sing temperature.

(d) $\log\left(\frac{k_2}{k_1}\right) = \frac{\Delta H}{2.303}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

$\Rightarrow$ Vant Hoff's equation.

(e) $2A_{(g)} + B_{(g)} \rightleftharpoons 3C_{(g)} \rightarrow \Delta n_g = 0$

34. Ans. (3)

	P	$\rightleftharpoons$	Q	+	R
t = 0	2		0		0
t = eq.	2 – x		x		x
Given			x	=	0.5

$\alpha = \frac{x}{2} = \frac{0.5}{2} = 0.25$

35. Ans. (1)

	$2CO_2$	$\rightleftharpoons$	2CO	+	O_2
t = 0	2mol		0		0
t = eq.	2 – 0.8		0.8		0.4
	= 1.2		0.8		0.4

40% of 2 = 0.8

Moles total at equilibrium

= 1.2 + 0.8 + 0.4 = 2.4 moles.

36. Ans. (4)

$A_3(g) \rightleftharpoons 3A(g)$

t = 0 a 0

t = eq. a – y 3y

$x = \frac{y}{a}$

$y = ax$

total moles at equilibrium

= a – y + 3y

= a + 2y

= a + 2 ax

37. Ans. (1)

$2P_{(g)} + Q_{(g)} \rightleftharpoons 3R_{(g)} + S_{(g)}$

t = 0 2mol 2mol 0 0

t = eq. 2 – 2x 2 – x 3x x

At eq. [Q] > [P]

38. Ans. (1)

$A + B \rightleftharpoons C + D$

t = 0 3n n 0 0

t = eq. 3n–x n – x x x

Given $[C]_{eq.} = [B]_{eq.}$

x = n – x

x = n/2

[D] = x = n/2

39. Ans. (1)

$X_2 + Y_2 \rightleftharpoons 2xy$

t = 0 1mol 2mole 0

t = eq. 1 – x 2 – x 2x

Given [xy] = 0.6

$\frac{2x}{3} = 0.6$

x = 0.9

$[X_2] = \frac{1-x}{3} = \frac{1-0.9}{3} = \frac{0.1}{3}$

$[Y_2] = \frac{2-x}{3} = \frac{2-0.9}{3} = \frac{1.1}{3}$

40. Ans. (2)

$CO + H_2O \rightleftharpoons CO_2 + H_2$

t = 0 1 1 1 0

t = eq. 1 – x 1 – x 1 + x x

x < 1 always

Hence n_{CO} & n_{H_2O} & n_{H_2} always less than 1 mole.

41. Ans. (2)

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

t = 0 4mol 0 0

t = eq. 4 – x x x

Given $\alpha = 0.5 = \frac{x}{4}$

x = 2

total moles = 4 – x + x + x

= 4 + x

= 4 + 2 = 6 moles

42. Ans. (1)

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

At = equilibrium $n_{PCl_3} = n_{Cl_2}$

Hence $X_{PCl_3} = X_{Cl_2}$

So, $X_{PCl_3} = 0.3$

43. Ans. (2)

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

t = 0 8 0 0

t = eq. 8 – 25% of 8 2 2

8 – 2 = 6

$K_P = \frac{P_{PCl_3} \times P_{Cl_2}}{P_{PCl_5}}$

$= \frac{\frac{2}{10} \times P \times \frac{2}{10} \times P}{\frac{6}{10} \times P}$

$= \frac{P}{15}$

44. Ans. (3)

$K_P = \frac{P_{PCl_3} \times P_{Cl_2}}{P_{PCl_5}}$

$K_P = \frac{0.3 \times 0.2}{0.6} = 0.1$

$K_P = \frac{P^1_{PCl_3} \times P^1_{Cl_2}}{P^1_{PCl_5}}$

$0.1 = \frac{0.6 \times 0.4}{P^1_{PCl_5}}$

$P^1_{PCl_5} = 2.4$

45. Ans. (3)

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

At equilibrium $n_{PCl_3} = n_{Cl_2}$

Hence $X_{PCl_3} = X_{Cl_2}$

Also; $P_{PCl_3} = P_{Cl_2}$

$P_{PCl_3} = X_{PCl_3} . P_T$

$= 0.25 \times 2$

Hence $P_{Cl_2} = 0.5$

46. Ans. (1)

	$2NO$	$\rightleftharpoons$	N_2	+	O_2
t = 0	4mole		0		0
t = eq.	4 – 2x		x		x

Gives $\alpha = 0.1 = \frac{2x}{4}$

$\Rightarrow x = 0.2$

$K_P = \frac{P_{N_2} \times P_{O_2}}{(P_{NO})^2} = \frac{0.2 \times 0.2}{(3.6)^2}$

$= \left(\frac{1}{18}\right)^2$

47. Ans. (4)

	A	+	B	$\rightleftharpoons$	C	+	D
t = 0	1mole		1mole		0		0
t = eq.	1 – x		1 – x		x		x

Gives x = 0.4

$K_C = \frac{[C]\,[D]}{[A]\,[B]} = \frac{\frac{1.4}{V} \times \frac{0.4}{V}}{\frac{0.6}{V} \times \frac{0.6}{V}}$

$K_C = \frac{4}{9}$

48. Ans. (2)

$CH_3COOH_{(\ell)} + C_2H_5OH_{(\ell)} \rightleftharpoons CH_3COOC_2H_{5(\ell)} + H_2O_{(\ell)}$

t = 0	4	4	0	0
t = eq.	4 – x	4 – x	x	x

$K_C = \frac{\frac{x}{v} \times \frac{x}{v}}{\left(\frac{4-x}{v}\right)\left(\frac{4-x}{v}\right)}$

$4 = \frac{x^2}{(4-x)^2}$

$2 = \frac{x}{4-x}$

$8 - 2x = x$

$3x = 8$

$X = \frac{8}{3}$

[Acid] $= 4 - x$

$= 4 - \frac{8}{3}$

$= \frac{4}{3}$

49. Ans. (2)

	H_2	+	I_2	$\rightleftharpoons$	$2HI$
t = 0	2		2		0
t = eq.	2 – x		2 – x		2x

Gives $n_{HI} = 2$

$2x = 2$

$x = 1$

$K = \frac{(2)^2}{(1)\,(1)} = 4$

50. Ans. (2)

	A	+	B	$\rightleftharpoons$	C	+	D
t = 0	4		4		0		0
t = eq.	4 – x		4 – x		x		x

Gives x = 2

$K = \frac{\left(\frac{2}{v}\right)\left(\frac{2}{v}\right)}{\left(\frac{2}{v}\right)\left(\frac{2}{v}\right)} = 1$

51. Ans. (2)

	$2NH_3(g)$	$\rightleftharpoons$	$N_2(g)$	+	$3H_2(g)$
t = 0	2mole		0		0
t = eq.	2 – 2x		x		3x

Gives $n_{NH_3} = 1$mole

$2 - 2x = 1$

$x = \frac{1}{2} = 0.5$

$K = \frac{\left(\frac{0.5}{1/2}\right)\left(\frac{3 \times 0.5}{1/2}\right)^3}{\left(\frac{2 - 2 \times 0.5}{1/2}\right)^2} = 6.75$

52. Ans. (1)

	H_2 +	I_2 ⇌	2HI
t = 0	4.5	4.5	0
t = eq.	4.5 – x	4.5 – x	2x

Gives $n_{HI} = 3$mole

$2x = 3$

$x = 1.5$

$$K = \frac{\left(\frac{2\times1.5}{V}\right)^2}{\left(\frac{3}{V}\right)\left(\frac{3}{V}\right)} = 1$$

53. Ans. (2)

	$2AB(g)$ ⇌	$2A(g)$ +	$B_2(g)$
t = 0	500	0	0
t = eq.	500 – 2P	2P	P

Gives $P_T = 500 - 2P + 2P + P = 625$

$P = 125$

$$K_p = \frac{(P_A)^2 . (P_{B_2})}{(P_{AB})^2}$$

$$= \frac{(2\times125)^2 (125)}{(250)^2} = 125$$

54. Ans. (2)

On ↑Sing concentrations of product at equilibrium, reaction shifts backward & more reactant will be formed.

55. Ans. (3)

$$NaNO_3(s) \rightleftharpoons NaNO_2(s) + \frac{1}{2}O_2(g)$$

On ↑Sing pressure; reaction shifts forward less gaseous moles.

i.e. above reaction will shift backward.

56. Ans. (3)

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

Above reaction is endothermic. Increasing temperature will favours forward reaction.

57. Ans. (3)

On ↑Sing pressure; reaction will shift toward less gaseous moles.

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

58. Ans. (2)

$$SO_{2(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons SO_{3(g)} + \text{Heat}$$

On ↓Sing temperature & ↑Sing pressure Reaction shifts forward.

59. Ans. (2)

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} + 21.9 \text{ KCal}$$

On ↓Sing temperature & ↑sing pressure reaction shifts forward.

60. Ans. (3)

$2A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + 326$Kcal

On ↓Sing temperature & ↑sing pressure reaction shifts forward.

61. Ans. (1)

On compressing V↓ ⇒ P↑ Hence on ↑sing pressure equation shifts towards less gas moles.

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$

Hence balkward reaction is favoured.

62. Ans. (2)

For $\Delta ng = 0$ equation is unaffected by pressure.

63. Ans. (3)

On adding inert gas at constant volume. Equation is unaffected.

64. Ans. (2)

On ↑sing pressure ; equilibrium shifts towards less volume.

$V_{H_2O(\ell)} < V_{H_2O(g)}$

Hence B.P. ↑ses.

65. Ans. (3)

Heat + $CO_2(s) \rightleftharpoons CO_2(g)$

On cooling temperature ↓ses. Reaction shifts backward.

66. Ans. (2)

$$\alpha = \frac{D_T - D_0}{(n-1)D_0}$$

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

$$D_T = \frac{208.5}{2} = 104.25$$

$d = 104.25$

$n = 2$

$$\alpha = \frac{104.25 - 104.25}{(2-1)\times104.25} = 0$$

67. Ans. (1)

$$\alpha = \frac{D_T - D_0}{(n-1)D_0}$$

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

$n = 2$

$$\alpha_{200^\circ C} = \frac{104.25-70.2}{(2-1)\times 70.2} = 48.50\%$$

$$\alpha_{250^\circ C} = \frac{104.25-57.9}{(2-1)\times 57.9} = 80\%$$

68. Ans. (2)

Choose that options for which sum of stoichiometric coefficient of gas product = n

$$A \rightleftharpoons \frac{nB}{3} + \left(\frac{2n}{3}\right)C$$

$$\Rightarrow \quad \frac{n}{3} + \frac{2n}{3} = n$$

Exercise-II (Previous Year Questions)

1. Ans. (4)

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + Heat$

By decreasing temperature & increasing pressure forward reaction Proceeds.

2. Ans. (1)

For a exothermic reaction on increasing temperature K_p decreases

3. Ans. (2)

When $K_{eq.} > 10^3$ product dominates over reactant.

4. Ans. (3)

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

Multiply above reaction by $\frac{1}{2}$ to get target reaction hence $K' = (K)^{\frac{1}{2}}$

5. Ans. (1)

Reverse (I)

ADD in (II)

Multiply (III) × 3

$$\frac{1}{K_1} \times K_2 \times (K_3)^3$$

6. Ans. (4)

$P_1 \quad V_1 \quad = \quad P_2 \quad V_2$

$0.4 \times 20 \quad = \quad 1.6 \times V_2$

$V_2 = 5$ litre

7. Ans. (1)

$A_2(g) + B_2(g) \rightleftharpoons X_2(g)$

On decreasing temperature & increasing pressure forward reaction proceeds.

8. Ans. (1)

	H_2	+	I_2	$\rightleftharpoons$	$2HI$
t = 0	a		a		0
t = eq.	a – 0.8a		a – 0.8a		2 × 0.8a
	0.2a		0.2a		1.6a

$$K_c = \frac{[HI]^2}{[H_2]\,[I_2]} = \frac{1.6\cancel{a}\times 1.6\cancel{a}}{0.2\cancel{a}\times 0.2\cancel{a}} = 64$$

9. Ans. (3)

$PV = nRT$

$n_T = n_{O_2} + n_{H_2}$

$$= \frac{4}{32} + \frac{2}{2} = \frac{1}{8} + 1 = \frac{9}{8}$$

$$P = \frac{nRT}{V} = \frac{\frac{9}{8}\times .0821\times 273}{1} = 25.18 \text{ atm}$$

10. Ans. (3)

According to dalton's law :

$P_A = X_A . P_T$

11. Ans. (4)

$$K_c = \frac{[O_3]^2}{[O_2]^3}$$

$$3 \times 10^{-59} = \frac{[O_3]^2}{[0.040]^3}$$

$[O_3] = 4.38 \times 10^{-32}$

12. Ans. (2)

$K_p = K_c (RT)^{\Delta ng}$

$\Delta ng = 2 - 1 = 1$

$$K_c = \frac{Kp}{(RT)} = \frac{3}{(.083)(1000)} = 3.6 \times 10^{-2}$$

13. Ans. (2)

Rate = k $[A]^2$ [B]

If [A] is tripled and [B] is kept constant.

$r^1 = k [3A]^2 [B]$

$r^1 = 9k [A]^2 [B]$

$r^1 = 9r$

Increased by a factor of nine

14. Ans. (1)

$K_p = K_c(RT)^{\Delta n_g}$

when $\Delta n_g = 0$ then $K_p = K_c$

$\Delta n_g \neq 0$ $K_p \neq K_c$

(1) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

$\Delta n_g \neq 0$

$K_p \neq K_c$

15. Ans. (3)

$2A \rightleftharpoons B + C$; $K_c = 4 \times 10^{-3}$

At a given time $\Rightarrow$ [A] = [B] = [C] = 2×10^{-3} M

$$Q_c = \frac{[B][C]}{[A]^2}$$

$Q_c = 1$

as $Q_c > K_c$

Reaction has a tendency to go in backward direction.

16. Ans. (4)

	$2NO(g)$	$\rightleftharpoons$	$N_2(g)$	+	$O_2(g)$
$t=t_{eq}$	$(2.8\times10^{-3})M$		$(3\times10^{-3})M$		$(4.2\times10^{-3})M$

$$K_C = \frac{3\times10^{-3}\times4.2\times10^{-3}}{(2.8\times10^{-3})^2} = \frac{3\times4.2}{2.8\times2.8} = 1.60$$

	$2NO(g)$	$\rightleftharpoons$	$N_2(g)$	+	$O_2(g)$
$t=t_{eq}$	$(C-C\alpha)$		$\frac{C\alpha}{2}$		$\frac{C\alpha}{2}$

$$K_C = \frac{(C\alpha)^2}{4C^2(1-\alpha)^2}$$

$$\left(\frac{\alpha}{1-\alpha}\right)^2 = 4\times1.6$$

$$\frac{\alpha}{1-\alpha} = 2\times1.26$$

$$\frac{\alpha}{1-\alpha} = 2.52$$

$\Rightarrow \alpha = 2.52 - 2.52\,\alpha$

$$\Rightarrow \alpha = \frac{2.52}{3.52} = 0.717$$

17. Ans. (4)

For a reaction

$3A_2 + B_2 \rightleftharpoons 2A_3B$, K_1 ....(I)

$A_3B \rightleftharpoons \frac{3}{2}A_2 + \frac{1}{2}B_2$, K_2 ....(II)

On multiplying by (2) in equation (II)

$2A_3B \rightleftharpoons 3A_2 + B_2$, K_2^2 .....(III)

from (I) & (III)

$$K_2^2 = \frac{1}{K_1}$$

$$K_2 = \frac{1}{\sqrt{K_1}}$$

18. Ans. (4)

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $\Delta H = -Q$

for forward direction high pressure, low temperature, and higher concentration of H_2.

Exercise-III (Analytical Questions)

1. Ans. (2)

Equilibrium constant only on depends on temperature.

2. Ans. (1)

According to Le-chatelier principle on addition of helium gas at constant pressure and temperature dissociation of PCl_5 is increases.

3. Ans. (4)

Dissolution of gas in liquid in exothermic phenomena.

According to Le-chatelier principle

T ↑ gas dissolution ↓

vise-versa

4. Ans. (2)

$$\log\frac{K_2}{K_1} = \frac{\Delta H^0}{2.303R}\left[\frac{T_2 - T_1}{T_1T_2}\right]$$

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5. **Ans. (1)**
Statement-I $\rightarrow$ incorrect
$K_p = K_c$ when $RT = 1$ or $\Delta n_g = 0$
$K_{eq.} \rightarrow$ only depends on temperature

6. **Ans. (4)**
Equilibrium constant only depends on temperature.

7. **Ans. (2)**
At equilibrium
$r_f = r_b$
$\Delta G = 0$
$r_{net} = 0$
$\Delta G = -nFE_{cell}$
$E_{cell} = 0$

8. **Ans. (1)**
At equilibrium
$\Delta G = 0$
$K_p = Q_p$

9. **Ans. (3)**
$K_c > 10^3$ product dominates
$K_c < 10^{-3}$ reactant dominates
$10^{-3} < K_c < 10^3$ both reactant & product are in appreciable concentration.
$Q_c > K_c$ reaction shifts backward.

10. **Ans. (2)**
On increasing temperature if value of K increases reaction is endothermic.

11. **Ans. (4)**
$\Delta G^0 < 0$

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Exercise - I (Conceptual Questions) Build Up Your Understanding

INTRODUCTION

1. The formula weight of H_2SO_4 is 98. The weight of the acid in 400mL of 0.1 M solution is:-
(1) 2.45 g (2) 3.92 g (3) 4.90 g (4) 9.8 g
CIE001

2. Normality of 2M sulphuric acid is:-
(1) 2 N (2) 4N (3) N/2 (4) N/4
CIE002

3. If pH = 3.31, then find out $[H^+]$ (Approx)
(1) 3.39×10^{-4} (2) 5×10^{-4}
(3) 3.0×10^{-3} (4) None
CIE003

4. If $[OH^-] = 5.0 \times 10^{-5}$ M then pH will be :-
(1) 5 – log 5 (2) 9 + log 5
(3) log 5 – 5 (4) log 5 – 9
CIE004

5. Basicity of H_3PO_3 and H_3PO_2 are respectively:-
(1) 1 and 2 (2) 2 and 3
(3) 3 and 2 (4) 2 and 1
CIE005

6. Find out pH of solution having 2×10^{-3} moles of OH^- ions in 2 litre solution :-
(1) pH = 3 (2) pH = 3 + log2
(3) pH = 3 – log2 (4) pH = 11
CIE006

7. pH of tomato juice is 4.4. Then concentration of H_3O^+ will be :-
(1) 39×10^{-4} (2) 3.9×10^{-5}
(3) 3.9×10^{-4} (4) 3.9×10^{5}
CIE007

8. 8 g NaOH is dissolved in one litre of solution, the molarity of the solution is:-
(1) 0.2 M (2) 0.4 M (3) 0.02 M (4) 0.8 M
CIE008

9. The amount of acetic acid present in 100 mL of 0.1M solution is :-
(1) 0.30 g (2) 3.0 g (3) 0.60 g (4) None
CIE009

10. The number of milli equivalents of acid in 100 mL of 0.5N HCl solution is:-
(1) 50 (2) 100 (3) 25 (4) 200
CIE010

11. If the molar concentration of $MgCl_2$ is 1.5×10^{-3} mol L^{-1}, the concentration of chloride ions in g ion L^{-1} is:-
(1) 3.0×10^{-3} (2) 6.0×10^{-3}
(3) 0.3×10^{-3} (4) 0.6×10^{-6}
CIE011

OSTWALD'S DILUTION LAW

12. Degree of dissociation of 0.1 N CH_3COOH is :- (Dissociation constant = 1×10^{-5})
(1) 10^{-5} (2) 10^{-4} (3) 10^{-3} (4) 10^{-2}
CIE012

13. The degree of dissociation of acetic acid is given by the expression $\alpha = 0.1 \times C^{-1}$ (where C = concentration of the acid) What is the pH of the solution :-
(1) 1 (2) 2 (3) 3 (4) 4
CIE013

14. Ostwald's dilution law is not applicable for strong electrolytes because:-
(1) Strong electrolytes are completely ionised
(2) Strong electrolytes are volatile
(3) Strong electrolytes are unstable
(4) Strong electrolytes often contain metal ions
CIE014

15. The degree of ionisation of a compound depends upon :
(1) Size of the solute molecules
(2) Nature of the solute molecules
(3) Nature of the container taken
(4) The amount of current passed
CIE015

16. Find out K_a for 10^{-2} M HCN acid, having pOH = 10 :-
(1) $K_a = 10^{-4}$ (2) $K_a = 10^{-2}$
(3) $K_a = 10^{-5}$ (4) None of these
CIE016

17. Which of the following will occur if a 1.0 M solution of a weak acid is diluted to 0.01 M at constant temperature:-
(1) Percentage ionisation will increase
(2) $[H^+]$ will decrease to 0.01M
(3) K_a will increase
(4) pH will decrease by 2 units
CIE017

18. The extent of ionisation of weak electrolytes increases :-
(1) With the increase in concentration of solute
(2) On decreasing the temp. of solution
(3) On addition of excess of water to the solution
(4) On stirring the solution vigorously

CIE018

19. If K_a of HCN = 4×10^{-10}, then the pH of 2.5×10^{-1} molar HCN (aq) is:-
(1) 4.2 (2) 4.7 (3) 0.47 (4) 5.0

CIE019

20. The molarity of nitrous acid solution at which its pH becomes 2.($K_a = 4.5 \times 10^{-4}$) :-
(1) 0.3333 (2) 0.4444
(3) 0.6666 (4) 0.2222

CIE020

EXPLANATION OF WATER

21. Ionic product of water will increase, if :-
(1) Decrease the pressure
(2) Add H^+
(3) Add OH^-
(4) Increase the temperature

CIE021

22. For water at 25° C, 2×10^{-7} moles per litre is the correct answer for which one of the following
(1) $[H^+] + [OH^-]$ (2) $[H^+]^2$
(3) $[OH^-]^2$ (4) $[H^+] - [OH^-]$

CIE022

23. At 25^0C, the dissociation constant for pure water is given by :-
(1) $(55.4 \times 10^{14})^{-1}$ (2) 1×10^{-14}
(3) $\frac{1\times10^{-14}}{18}$ (4) None of these

CIE023

24. Ionic product of water is equal to :-
(1) Dissociation constant of water × $[H_2O]$
(2) Dissociation constant of water × $[H^+]$
(3) Product of $[H_2O]$ and $[H^+]$
(4) Product of $[OH^-]^2$ and $[H^+]$

CIE024

25. For pure water, addition of H^+ and OH^- ion concentrations at 90^0C is :-
(1) 10^{-14} (2) 10^{-12}
(3) 2×10^{-6} (4) 2×10^{-7}

CIE025

26. At a certain temperature, pure water has $[H_3O^+] = 10^{-6.7}$ mol L^{-1} .What is the value of K_W at this temperature :-
(1) 10^{-6} (2) 10^{-12}
(3) 10^{-67} (4) $10^{-13.4}$

CIE026

27. At 373 K, temp. the pH of pure H_2O can be:-
(1) < 7 (2) > 7 (3) = 7 (4) = 0

CIE027

28. Which of the following is a true statement :
(1) The ionisation constant and ionic product of water are same.
(2) Water is a strong electrolyte.
(3) The value of ionic product of water is less than that of its ionisation constant.
(4) At 298K, the number of H^+ ions in a litre of water is 6.023×10^{16}.

CIE028

HYDROLYSIS OF SALTS

29. At 90^0 C, the pH of 0.1M NaCl aqueous solution is:-
(1) < 7 (2) > 7 (3) 7 (4) 0.1

CIE034

30. What will be the pH of 1.0 M ammonium formate solution, If $K_a = 1 \times 10^{-4}$ acid $K_b = 1 \times 10^{-5}$:-
(1) 6.5 (2) 7.5 (3) 8.0 (4) 9.0

CIE035

31. Which salt will not undergo hydrolysis :-
(1) KCl (2) Na_2SO_4 (3) NaCl (4) All

CIE036

32. $HCOO^- + H_2O \rightleftharpoons HCOOH + OH^-$
Degree of hydrolysis for above reaction is given by:-
(1) $h = \sqrt{K_h}$ (2) $h = \sqrt{\frac{K_h}{C}}$
(3) $h = \sqrt{\frac{K_h}{V}}$ (4) $K_h = \sqrt{hc}$

CIE037

33. The pH of aqueous solution of sodium acetate is
(1) 7 (2) Very low
(3) > 7 (4) < 7
CIE038

34. If pK_b for CN^- at 25°C is 4.7. The pH of 0.5M aqueous NaCN solution is :-
(1) 12 (2) 10 (3) 11.5 (4) 11
CIE039

35. The highest pH value is of :-
(1) 0.1 M NaCl
(2) 0.1 M NH_4Cl
(3) 0.1 M CH_3COONa
(4) 0.1 M CH_3COONH_4
CIE040

36. pH of K_2S solution is:-
(1) 7 (2) Less than 7
(3) More than 7 (4) 0
CIE041

37. For anionic hydrolysis, pH is given by:-
(1) $pH = \frac{1}{2}pK_W - \frac{1}{2}pK_b - \frac{1}{2}\log C$
(2) $pH = \frac{1}{2}pK_W + \frac{1}{2}pK_a - \frac{1}{2}pK_b$
(3) $pH = \frac{1}{2}pK_W + \frac{1}{2}pK_a + \frac{1}{2}\log C$
(4) None of above
CIE042

38. Ionisation constant of a weak acid is 10^{-4}. Find out equilibrium constant for the reaction of this weak acid with strong base :-
(1) 10^{-10} (2) 10^{10} (3) 10^{-9} (4) 10^{9}
CIE043

39. Hydroxyl ion concentration [OH⁻] in the case of sodium acetate can be expressed as (where K_a is dissociation constant of CH_3COOH and C is the concentration of sodium acetate):-
(1) $[OH^-] = (CK_w . K_a)^{1/2}$
(2) $[OH^-] = C.K_w\sqrt{K_a}$
(3) $[OH^-] = \left(\frac{C.K_w}{K_a}\right)^{1/2}$
(4) $[OH^-] = C. K_a. K_w.$
CIE044

40. Consider :-
(a) $FeCl_3$ in water - Basic
(b) NH_4Cl in water - Acidic
(c) Ammonium acetate in water - Acidic
(d) Na_2CO_3 in water - Basic
Which is/are not correctly matched:-
(1) b and d (2) b only
(3) a and c (4) d only
CIE045

41. Which of the following salts undergoes hydrolysis in water:-
(1) Na_3PO_4 (2) CH_3COONa
(3) $NaNO_3$ (4) Both (1) and (2)
CIE046

42. A salt 'X' is dissolved in water of pH = 7. The resulting solution becomes alkaline in nature. The salt is made up of:-
(1) A strong acid and strong base
(2) A strong acid and weak base
(3) A weak acid and weak base
(4) A weak acid and strong base
CIE047

SOLUBILITY & SOLUBILITY PRODUCT (K_{SP})

43. The solubility product of sparingly soluble Uni- univalent salt (AB type) is defined as the product of ionic concentration in a:-
(1) 1 M solution
(2) Concentration solution
(3) Very dilute solution
(4) Saturated solution
CIE048

44. If solubility of salts M_2X, QY_2 and PZ_2 are equal, then the relation between their K_{sp} will be :-
(1) $K_{sp}(M_2X) > K_{sp}(QY_2) > K_{sp}(PZ_2)$
(2) $K_{sp}(M_2X) = K_{sp}(QY_2) < K_{sp}(PZ_2)$
(3) $K_{sp}(M_2X) > K_{sp}(QY_2) = K_{sp}(PZ_2)$
(4) $K_{sp}(M_2X) = K_{sp}(QY_2) = K_{sp}(PZ_2)$
CIE049

45. The expression of solubility product of mercurous iodide is :-
(1) $[2\ Hg^+]^2 \times 2\ [I^-]^2$ (2) $[Hg^{++}]^2 \times [\ 2I^-]^2$
(3) $[\ Hg_2^{2+}\ ] \times [I^-]^2$ (4) $[Hg^{2+}]^2 \times [I^-]^2$
CIE050

46. At 25°C, the K_{sp} value of AgCl is 1.8×10^{-10}. If 10^{-5} moles of Ag^+ are added to solution then K_{sp} will be :-
(1) 1.8×10^{-15} (2) 1.8×10^{-10}
(3) 1.8×10^{-5} (4) $18 \times 10^{+10}$
CIE051

47. At 25°C, the volume of water required to dissolve 1g $BaSO_4$ ($K_{sp} = 1.0 \times 10^{-10}$) will be (Molecular weight of $BaSO_4$ = 233) :-
(1) 820 L. (2) 1 L. (3) 205 L. (4) 430 L.
CIE052

48. Concentration of Ag^+ ions in saturated solution of Ag_2CrO_4 at 20°C is 1.5×10^{-4} mol L^{-1}. At 20°C, the solubility product of Ag_2CrO_4 is :-
(1) 3.3750×10^{-12} (2) 1.6875×10^{-10}
(3) 1.68×10^{-12} (4) 1.6875×10^{-11}
CIE053

49. If the concentration of CrO_4^{2-} ion in a saturated solution of silver chromate will be 2×10^{-4} M, solubility product of silver chromate will be -
(1) 4×10^{-8} (2) 8×10^{-12}
(3) 32×10^{-12} (4) 6×10^{-12}
CIE054

50. If the solubility of AgCl (formula mass=143) in water at 25°C is 1.43×10^{-4} g/100 mL of solution then the value of K_{sp} will be :–
(1) 1×10^{-5} (2) 2×10^{-5}
(3) 1×10^{-10} (4) 2×10^{-10}
CIE055

51. The solubility product of As_2S_3 is given by the expression :-
(1) $K_{sp} = [As^{3+}]\ [S^{-2}]$ (2) $K_{sp} = [As^{3+}]^1\ [S^{-2}]^1$
(3) $K_{sp} = [As^{3+}]^3\ [S^{-2}]^2$ (4) $K_{sp} = [As^{3+}]^2\ [S^{-2}]^3$
CIE056

52. If the solubility of $PbBr_2$ is 'S' g molecules per litre, considering 100% ionisation its solubility product is :-
(1) $2S^3$ (2) $4S^2$ (3) $4S^3$ (4) $2S^4$
CIE057

53. If the solubility of lithium sodium hexafluoro aluminate $Li_3Na_3(AlF_6)_2$ is 'S' mol L^{-1}. Its solubility product is equal to :-
(1) S^8 (2) $12\ S^3$
(3) $18S^3$ (4) $2916\ S^8$
CIE058

54. One litre of saturated solution of $CaCO_3$ is evaporated to dryness, when 7.0 g of residue is left. The solubility product for $CaCO_3$ is:-
(1) 4.9×10^{-3} (2) 4.9×10^{-5}
(3) 4.9×10^{-9} (4) 4.9×10^{-7}
CIE059

APPLICATION OF SOLUBILITY PRODUCT(K_{SP})

55. Solubility of AgBr will be minimum in :-
(1) Pure water (2) 0.1 M $CaBr_2$
(3) 0.1 M NaBr (4) 0.1 M $AgNO_3$
CIE060

56. In which of the following, the solution of AgSCN will be unsaturated :-
(1) $[Ag^+]\ [SCN^-] = K_{sp}$ (2) $[Ag^+]\ [SCN^-] < K_{sp}$
(3) $[Ag^+]\ [SCN^-] > K_{sp}$ (4) $[Ag^+]\ [SCN^-]^2 < K_{sp}$
CIE061

57. If 's' and 'S' are solubility and solubility product of a sparingly soluble binary electrolyte respectively then :–
(1) $s = S$ (2) $s = S^2$
(3) $s = S^{1/2}$ (4) $s = \frac{1S}{2}$
CIE062

58. The solubility product of CuS , Ag_2S and HgS are 10^{-37}, 10^{-44} and 10^{-54} respectively. The solubility of these sulphides will be in the order
(1) $HgS > Ag_2S > CuS$ (2) $Ag_2S > HgS > CuS$
(3) $CuS > Ag_2S > HgS$ (4) $Ag_2S > CuS > HgS$
CIE063

59. If the maximum concentration of $PbCl_2$ in water is 0.01 M at 298 K, its maximum concentration in 0.1 M NaCl will be:-
(1) 4×10^{-3} M (2) 0.4×10^{-4} M
(3) 4×10^{-2} M (4) 4×10^{-4} M
CIE064

60. M_2SO_4 (M^+ is a monovalent metal ion) has a K_{sp} of 1.2×10^{-5} at 298 K. The maximum concentration of M^+ ion that could be attained in a saturated solution of this solid at 298 K is:-
(1) 3.46×10^{-3} M (2) 2.89×10^{-2} M
(3) 2.8×10^{-3} M (4) 7.0×10^{-3} M
CIE065

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61. Which of the following has maximum solubility
(K_{sp} value is given in brackets) :-
(1) HgS (1.6×10^{-54}) (2) $PbSO_4$ (1.3×10^{-8})
(3) ZnS (7.0×10^{-26}) (4) AgCl (1.7×10^{-10})

CIE066

62. Maximum soluble is :- (K_{sp} is given)
(1) CuS (8.5×10^{-36}) (2) CdS (3.6×10^{-28})
(3) ZnS (1.2×10^{-28}) (4) MnS (1.4×10^{-10})

CIE067

63. In which of the following, the solubility of AgCl will be maximum :-
(1) 0.1 M $AgNO_3$ (2) Water
(3) 0.1 M NaCl (4) 0.1 M KCl

CIE068

64. The solubility product of three sparingly soluble salts are given below :

No.	Formula	Solubility product
1	PQ	4.0×10^{-20}
2	PQ_2	3.2×10^{-14}
3	PQ_3	2.7×10^{-35}

The correct order of decreasing molar solubility is:-
(1) 1, 2, 3 (2) 2, 1, 3
(3) 3, 2, 1 (4) 2, 3, 1

CIE069

65. The K_{sp} value for $Gd(OH)_3$ is 2.8×10^{-23}. Find the pH of saturated solution of $Gd(OH)_3$:-
(1) 6.08 (2) 5.08 (3) 8.47 (4) 4.08

CIE070

66. If the solubility product of $AgBrO_3$ and Ag_2SO_4 are 5.5×10^{-5} and 2×10^{-5} respectively, the relationship between their solubilities can be correctly represented as :-
(1) $S_{AgBrO_3} > S_{Ag_2SO_4}$ (2) $S_{AgBrO_3} = S_{Ag_2SO_4}$
(3) $S_{AgBrO_3} < S_{Ag_2SO_4}$ (4) Unpredictable

CIE071

67. Solubility product of $Mg(OH)_2$ is 1×10^{-11}. At what pH, precipitation of $Mg(OH)_2$ will begin from
0.1 M Mg^{2+} solution :-
(1) 9 (2) 5 (3) 3 (4) 7

CIE072

68. A solution, containing 0.01 M Zn^{+2} and 0.01 M Cu^{2+} is saturated by passing H_2S gas. The S^{-2} concentration is 8.1×10^{-21} M, K_{sp} for ZnS and CuS are 3.0×10^{-22} and 8.0×10^{-36} respectively. Which of the following will occur in the solution:-
(1) ZnS will precipitate
(2) CuS will precipitate
(3) Both ZnS and CuS will precipitate
(4) Both Zn^{2+} and Cu^{2+} will remain in the solution

CIE073

69. What will happen if the pH of the solution of 0.001 M $Mg(NO_3)_2$ solution is adjusted to pH = 9 ($K_{sp}Mg(OH)_2 = 8.9 \times 10^{-12}$)
(1) Precipitation will take place
(2) Precipitation will not take place
(3) Solution will be saturated
(4) None of these

CIE074

70. The solubility product constant K_{sp} of $Mg(OH)_2$ is 9.0×10^{-12}. If a solution is 0.010 M with respect to Mg^{2+} ion, what is the maximum hydroxide ion concentration which could be present without causing the precipitation of $Mg(OH)_2$:-
(1) 1.5×10^{-7} M (2) 3.0×10^{-7} M
(3) 1.5×10^{-5} M (4) 3.0×10^{-5} M

CIE075

71. When HCl gas is passed through a saturated solution of common salt, pure NaCl is precipitated because:-
(1) The impurities dissolve in HCl
(2) HCl is slightly soluble in water
(3) The ionic product $[Na^+] \times [Cl^-]$ exceeds the solubility product of NaCl
(4) The solubility product of NaCl is lowered by Cl^- from aq. HCl

CIE076

FEW IMPORTANT POINTS

72. Two monobasic weak acids have the same moles of H^+ ions. What is the relationship between dissociation constant and dilution:-
(1) $Ka_1V_1 = Ka_2V_2$ (2) $Ka_1V_2 = Ka_2V_1$
(3) $[Ka_1V_1]^{\frac{1}{2}} = Ka_2V_2$ (4) $Ka_1V_1 = [Ka_2V_2]^{\frac{1}{2}}$

CIE077

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73. Two solutions having same concentration of H^+ ions are called:-
(1) Isotonic solutions
(2) Isohydric solutions
(3) Hypotonic solutions
(4) Hypertonic solutions
CIE078

74. The pH of a 0.1 M formic acid 0.1% dissociated is equal to 4. What will be the pH of another weak monobasic acid (same concentration) which is 1% dissociated
(1) 2 (2) 3 (3) 1 (4) 4
CIE079

pH

75. pH of water is 7. When any substance Y is dissolved in water then pH becomes 13. Substance Y is a salt of :-
(1) Strong acid and strong base
(2) Weak acid and weak base
(3) Strong acid and weak base
(4) Weak acid and strong base
CIE080

76. Minimum pH is shown by aqueous solution of:-
(1) 0.1 M $BaCl_2$ (2) 0.1 M $Ba(NO_3)_2$
(3) 0.1 M $BeCl_2$ (4) 0.1 M $Ba(OH)_2$
CIE081

77. Given :-
(a) 0.005 M H_2SO_4 (b) 0.1 M Na_2SO_4
(c) 10^{-2} M NaOH (d) 0.01 M HCl
Choose the correct code having same pH :-
(1) a, c, d (2) b, d (3) a, d (4) a, c
CIE082

78. What is H^+ ion concentration of 5×10^{-3} M H_2CO_3 solution having a 10% dissociation:-
(1) 10^{-3} (2) 10^{-2} (3) 10^{-1} (4) 5×10^{-2}
CIE083

79. A metal hydroxide of molecular formula $M(OH)_4$ is 50% ionised. Its 0.0025M solution will have the pH :-
(1) 12 (2) 2 (3) 4 (4) 11.7
CIE084

80. In the following solutions, the conc. of different acids are given, which mixture of the acid has highest pH :-
(1) $\frac{M}{10}H_2SO_4$, $\frac{M}{20}HNO_3$, $\frac{M}{10}HClO_4$
(2) $\frac{M}{20}H_2SO_4$, $\frac{M}{10}HNO_3$, $\frac{M}{20}HClO_4$
(3) $\frac{M}{20}H_2SO_4$, $\frac{M}{10}HNO_3$, $\frac{M}{40}HClO_4$
(4) $\frac{M}{20}H_2SO_4$, $\frac{M}{5}HNO_3$, $\frac{M}{5}HClO_4$
CIE085

81. If 100 mL of pH = 3 and 400 mL of pH = 3 solutions are mixed, what will be the final pH of mixture
(1) 3.2 (2) 3.0 (3) 3.5 (4) 2.8
CIE086

82. 10^{-6} M HCl is diluted 100 times. Its pH is :-
(1) 6.0 (2) 8.0 (3) 6.95 (4) 9.5
CIE087

83. pH of 0.001M acetic acid would be :-
(1) 2 (2) > 3 (3) 7 (4) 14
CIE088

84. At 90°C, the pH of 0.001M KOH solution will be
(1) 3 (2) 11 (3) 5 (4) 9
CIE089

85. The pH of solution is increased from 3 to 6. Its H^+ ion concentration will be :-
(1) Reduced to half
(2) Doubled
(3) Reduced by 1000 times
(4) Increased by 1000 times
CIE090

86. A solution has pOH equal to 13 at 298 K. The solution will be:-
(1) Highly acidic (2) Highly basic
(3) Moderately basic (4) Unpredictable
CIE091

87. The pH of the solution containing 10 mL of a 0.1M NaOH and 10 mL of 0.05M H_2SO_4 would be
(1) Zero (2) 1 (3) >7 (4) 7
CIE092

88. In a solution of pH = 5, more acid is added in order to reduce the pH upto 2. The increase in hydrogen ion concentration is:-
(1) 100 times (2) 1000 times
(3) 3 times (4) 5 times
CIE093

89. The hydrogen ion concentration in a given solution is 6×10^{-4} M. Its pH will be :-
(1) 6 (2) 3.22 (3) 4 (4) 2.
CIE094

90. The pOH of a solution is 10.0. The hydrogen ion concentration will be :–
(a) 10^{-10} (b) $\frac{Kw}{10^{-10}}$ (c) $\frac{Kw}{10^{-8}}$ (d) 10^{-4}
(1) a, d (2) b, c (3) a, b, c (4) None
CIE095

91. An aqueous solution whose pH = 0 is :-
(1) Basic (2) Acidic
(3) Neutral (4) Amphoteric
CIE096

92. Following five solution of KOH were prepare as–
First → 0.1 moles in 1 L
Second → 0.2 moles in 2 L
Third → 0.3 moles in 3 L
Fourth → 0.4 moles in 4 L
Fifth → 0.5 moles in 5 L
The pH of resultant solution is :–
(1) 2 (2) 1 (3) 13 (4) 7
CIE097

93. The pH of a 0.02 M ammonia solution which is 5% ionised will be :-
(1) 2 (2) 11 (3) 5 (4) 7
CIE098

94. For $\frac{N}{10}$ H_2SO_4, pH value is :-
(1) 1 (2) 0.586 (3) 0.856 (4) None
CIE099

95. An aqueous solution of HCl is 10^{-9} M HCl. The pH of the solution should be:-
(1) 9 (2) Between 6 and 7
(3) 7 (4) Unpredictable
CIE100

96. How many moles of HCl must be removed from 1 litre of aqueous HCl solution to change its pH from 2 to 3 :-
(1) 1 (2) 0.02 (3) 0.009 (4) 0.01
CIE101

97. 8 g NaOH and 4.9 g H_2SO_4 are present in one litre of the solution. What is its pH
(1) 1 (2) 13 (3) 12 (4) 2
CIE102

98. Calculate pH of a solution whose 100 mL contains 0.2 g NaOH dissolved in it :-
(1) 10.699 (2) 11.699 (3) 12.699 (4) 13.699
CIE103

99. What is the quantity of NaOH present in 250 cc of the solution, so that it gives a pH = 13 :-
(1) 10^{-13} g (2) 10^{-1} g
(3) 1.0 g (4) 4.0 g
CIE104

100. 0.001 mol of the strong electrolyte $M(OH)_2$ has been dissolved to make a 20 mL of its saturated solution. Its pH will be : - [$K_w = 1 \times 10^{-14}$]
(1) 13 (2) 3.3 (3) 11 (4) 9.8
CIE105

101. Choose the wrong statement :-
(1) For a neutral solution : $[H^+] = [OH^-] = \sqrt{K_w}$
(2) For an acidic solution :
$[H^+] > \sqrt{K_w}$ & $[OH^-] < \sqrt{K_w}$
(3) For a basic solution :
$[H^+] < \sqrt{K_w}$ & $[OH^-] > \sqrt{K_w}$
(4) For a neutral solution :
$[H^+] = [OH^-] = 10^{-7}$ M (at all temperatures)
CIE106

102. The pH of 0.1 M solution of the following salts increases in order :-
(1) NaCl < NH_4Cl < NaCN < HCl
(2) NaCN < NH_4Cl < NaCl < HCl
(3) HCl < NaCl < NaCN < NH_4Cl
(4) HCl < NH_4Cl < NaCl < NaCN
CIE107

BUFFER SOLUTIONS AND INDICATOR

103. In a buffer solution the ratio of concentration of NH_4Cl and NH_4OH is 1 : 1. When it changes to 2 : 1, pH of buffer :-
(1) Increases (2) Decreases
(3) No effect (4) None of these
CIE108

104. To 50 mL of 0.05M formic acid, how much volume of 0.10M sodium formate must be added to get a buffer solution of pH = 4.0 ? (pK_a of the acid is 3.8)
(1) 50 mL (2) 4 mL
(3) 39.6 mL (4) 100 mL
CIE109

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105. In the volumetric estimation of HCl, if we make use of phenolphthalein as an indicator, which base is unsuitable for the titration :-
(1) NaOH (2) RbOH
(3) KOH (4) NH_4OH

CIE110

106. In a mixture of weak acid and its salt, the ratio of concentration of acid to salt is increased ten-fold. The pH of the solution :-
(1) Decreases by one
(2) Increases by one-tenth
(3) Increases by one
(4) Increases ten-fold

CIE111

107. pK_b for NH_4OH at certain temperature is 4.74. The pH of basic buffer containing equimolar concentration of NH_4OH and NH_4Cl will be:-
(1) 7.74 (2) 4.74 (3) 2.37 (4) 9.26

CIE112

108. What is the suitable indicator for titration of NaOH and oxalic acid:-
(1) Methyl orange (2) Methyl red
(3) Phenolphthalein (4) Starch solution

CIE113

109. Phenolphthalein does not act as an indicator for the titration between :-
(1) KOH and H_2SO_4
(2) NaOH and CH_3COOH
(3) Oxalic acid and $KMnO_4$
(4) $Ba(OH)_2$ and HCl

CIE114

110. Which can act as buffer :-
(1) NH_4OH + NaOH
(2) HCOOH + CH_3COONa
(3) 40 mL 0.1 M NaCN + 20 mL of 0.1 M HCl
(4) None of them

CIE115

111. The buffer solution play an important role in:-
(1) Increasing the pH value
(2) Decreasing the pH value
(3) Keeping the pH constant
(4) Solution will be neutral

CIE116

112. K_a for HCN is 5 x 10^{-10} at 25^0C. For maintaining a constant pH of 9, the volume of 5M KCN solution required to be added to 10mL of 2M HCN solution is-
(1) 4 mL (2) 7.95 mL
(3) 2 mL (4) 9.3 mL

CIE117

113. Buffering action of a mixture of CH_3COOH and CH_3COONa is maximum when the ratio of salt to acid is equal to -
(1) 1.0 (2) 100.0 (3) 10.0 (4) 0.1

CIE118

114 The pink colour of phenolphthalein in alkaline medium is due to -
(1) Negative ion (2) Positive ion
(3) OH^- ions (4) Neutral form

CIE119

115. Which indicator works in the pH range 8 – 9.8
(1) Phenolphthalein (2) Methyl orange
(3) Methyl red (4) Litmus

CIE120

116. A basic buffer will obey the equation pOH – pK_b = 1 only under condition:-
(1) [Conjugate acid] : [base] = 1 : 10
(2) [Conjugate acid] = [base]
(3) [Conjugate acid] : [base] = 10 : 1
(4) None of these

CIE121

117. For weak acid-strong base titration, the indicator used is :-
(1) Potassium dichromate
(2) Methyl orange
(3) Litmus
(4) Phenolphthalein

CIE122

118. For which of the following titration, methyl orange is a best indicator :-
(1) CH_3COOH + NaOH
(2) $H_2C_2O_4$ + NaOH
(3) HCl + NaOH
(4) CH_3COOH + NH_4OH

CIE123

119. The total number of different kind of buffers obtained during the titration of H_3PO_4 with NaOH are :-
(1) 3 (2) 1 (3) 2 (4) 0

CIE124

120. A certain acidic buffer solution contains equal concentration of X^- and HX. The K_b for X^- is 10^{-10}. The pH of the buffer is :-
(1) 4 (2) 7 (3) 10 (4) 14
CIE125

121. When 1.0 mL of dil. HCl acid is added to 100 mL of a buffer solution of pH 4.0. The pH of the solution
(1) Becomes 7 (2) Does not change
(3) Becomes 2 (4) Becomes 10
CIE126

122. The pH of blood is maintained by CO_2 and H_2CO_3 in the body and chemical constituents of blood. This phenomenon is called :-
(1) Colloidal (2) Buffer action
(3) Acidity (4) Salt balance
CIE127

123. Phenolphthalein is not a good indicator for titrating
(1) NaOH against oxalic acid
(2) NaOH against HCl
(3) NaOH against H_2SO_4
(4) HCl against NH_4OH
CIE128

124. Which of the following solutions does not act as buffer :–
(1) $H_3PO_4 + NaH_2PO_4$
(2) $NaHCO_3 + H_2CO_3$
(3) $NH_4Cl + HCl$
(4) $CH_3COOH + CH_3COONa$
CIE129

125. On addition of NaOH to CH_3COOH solution, 60% of the acid is neutralised. If pK_a of CH_3COOH is 4.7 then the pH of the resulting solution is :-
(1) More than 4.7 but less than 5.0
(2) Less than 4.7 but more than 4.0
(3) More than 5.0
(4) Remains unchanged
CIE130

126. 500 mL of 0.2 M acetic acid are added to 500 mL of 0.30 M sodium acetate solution. If the dissociation constant of acetic acid is 1.5×10^{-5} then pH of the resulting solution is:-
(1) 5.0 (2) 9.0 (3) 3.0 (4) 4.0
CIE131

127. Half of the formic acid solution is neutralised on addition of a KOH solution to it. If K_a (HCOOH) = 2×10^{-4} then pH of the solution is : (log 2 = 0.3010)
(1) 3.6990 (2) 10.3010
(3) 3.85 (4) 4.3010
CIE132

128. A solution contains 0.2M NH_4OH and 0.2M NH_4Cl. If 1.0 mL of 0.001 M HCl is added to it. What will be the $[OH^-]$ of the resulting solution
$[K_b = 2 \times 10^{-5}]$:-
(1) 2×10^{-5} (2) 5×10^{-10}
(3) 2×10^{-3} (4) None of these
CIE133

129. Henderson equation pH – pK_a = 1 will be applicable to an acidic buffer when :-
(1) [Acid] = [Conjugate base]
(2) [Acid] × 10 = [Conjugate base]
(3) [Acid] = [Conjugate base] × 10
(4) None of these
CIE134

130. 0.05 M ammonium hydroxide solution is dissolved in 0.001 M ammonium chloride solution. What will be the OH^- ion concentration of this solution: $K_b(NH_4OH) = 1.8 \times 10^{-5}$
(1) 3.0×10^{-3} (2) 9.0×10^{-4}
(3) 9.0×10^{-3} (4) 3.0×10^{-4}
CIE135

131. When 0.02 moles of NaOH are added to a litre of buffer solution, its pH changes from 5.75 to 5.80. What is its buffer capacity :-
(1) 0.4 (2) 0.05 (3) –0.05 (4) 2.5
CIE136

132. Calculate the pH of a buffer prepared by mixing 300 cc of 0.3 M NH_3 and 500 cc of 0.5 M NH_4Cl. K_b for $NH_3 = 1.8 \times 10^{-5}$:-
(1) 8.1187 (2) 9.8117
(3) 8.8117 (4) None of these
CIE137

133. Calculate the ratio of pH of a solution containing 1 mole of CH_3COONa + 1 mole of HCl per litre and of other solution containing 1 mole CH_3COONa + 1mole of acetic acid per litre :-
(1) 1 : 1 (2) 2 : 1 (3) 1 : 2 (4) 2 : 3
CIE138

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134. When 20 mL of $\frac{M}{20}$ NaOH are added to 10 mL of $\frac{M}{10}$ HCl, the resulting solution will:-
(1) Turn blue litmus red
(2) Turn phenolphthalein solution
(3) Turn methyl orange red
(4) Will have no effect on either red or blue litmus

CIE139

135. 10 mL of a solution contains 0.1 M NH_4Cl + 0.01 M NH_4OH. Which addition would not change the pH of solution :–
(1) Adding 1 mL water
(2) Adding 5 mL of 0.1 M NH_4Cl
(3) Adding 5 mL of 0.1 M NH_4OH
(4) Adding 10 mL of 0.1 M NH_4Cl

CIE140

136. $\frac{N}{10}$ acetic acid was titrated with $\frac{N}{10}$ NaOH. When 25%, 50% and 75% of titration is over then the pH of solution will be :- [$K_a = 10^{-5}$]
(1) 5 + log 1/3, 5, 5 + log 3
(2) 5 + log 3, 4, 5 + log 1/3
(3) 5 – log 1/3, 5, 5 – log 3
(4) 5 – log 1/3, 4, 5 + log 1/3

CIE141

ACID AND BASE

137. The conjugate acid of O^{-2} is :-
(1) O_2^+ (2) H^+ (3) H_3O^+ (4) OH^-

CIE142

138. Ionization constant of AOH and BOH base are K_{b_1} and K_{b_2}. Their relation is $pK_{b_1} < pK_{b_2}$. Conjugate of following base, does not show maximum pH :
(1) AOH (2) BOH
(3) Both of them (4) None of these

CIE143

139. Select the species which can function as lewis base, bronsted acid and bronsted base:-
(a) H_2O (b) NH_4^+ (c) N^{-3}
Correct code is :-
(1) Only a (2) a, b (3) a, c (4) b, c

CIE144

140. Which ion does not show acid behaviour :-
(1) $\left[Al(H_2O)_6\right]^{+3}$ (2) $\left[Fe(H_2O)_6\right]^{+3}$
(3) HPO_4^{-2} (4) ClO_3^-

CIE145

141. An example of Lewis acid is:-
(1) CaO (2) CH_3NH_2
(3) SO_3 (4) None of these

CIE146

142. In the reaction $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$ water behaves as :-
(1) Acid (2) Base
(3) Neutral (4) Both acid & Base

CIE147

143. Which acts as Lewis base in the reaction
$BCl_3 + :PH_3 \rightarrow Cl_3B \leftarrow PH_3$
(1) PH_3 (2) BCl_3
(3) Both 1 & 2 (4) None

CIE148

144. Which acts as Lewis acid in the reaction
$SnCl_2 + 2Cl^- \longrightarrow [SnCl_4]^{-2}$
(1) Cl^- (2) $SnCl_2$ (3) $SnCl_4$ (4) None

CIE149

145. The conjugate base of $(CH_3)_2\overset{+}{N}H_2$ is :-
(1) CH_3NH_2 (2) $(CH_3)_2N^+$
(3) $(CH_3)_2N$ (4) $(CH_3)_2NH$

CIE150

146. Which equilibrium can be described as Lewis acid base reaction but not Bronsted acid base reaction:-
(1) $H_2O + CH_3COOH \rightleftharpoons H_3O^+ + CH_3COO^-$
(2) $2NH_3 + H_2SO_4 \rightleftharpoons 2NH_4^+ + SO_4^{2-}$
(3) $NH_3 + CH_3COOH \rightleftharpoons NH_4^+ + CH_3COO^-$
(4) $Cu^{+2} + 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{2+}$

CIE151

147. Conjugate base of hydrazoic acid is :-
(1) HN_3^- (2) N_3^- (3) N^{3-} (4) N_2^-

CIE152

148. NH_3 gas dissolves in water to give NH_4OH, in this reaction, water acts as :-
(1) An acid (2) A base
(3) A salt (4) A conjugate base

CIE153

149. When ammonia is added to water, it decreases the concentration of which of the following ion
(1) OH^- (2) H_3O^+ (3) NH^+_4 (4) None
CIE154

150. The strongest acid among the following is -
(1) $ClO_3(OH)$ (2) $ClO_2(OH)$
(3) $SO(OH)_2$ (4) $SO_2(OH)_2$
CIE155

151. Which of the following is not a Bronsted acid:-
(1) $CH_3NH_4^+$ (2) CH_3COO^-
(3) H_2O (4) HSO_4^-
CIE156

152. Which of the following example behave as a Lewis acid BF_3, $SnCl_2$, $SnCl_4$:-
(1) $SnCl_2$, $SnCl_4$ (2) BF_3, $SnCl_2$
(3) Only BF_3 (4) BF_3, $SnCl_2$, $SnCl_4$
CIE157

153. In the reaction
$HNO_3 + H_2O \rightleftharpoons H_3O^+ + NO_3^-$, the conjugate base of HNO_3 is :-
(1) H_2O (2) H_3O^+
(3) NO_3^- (4) H_3O^+ and NO_3^-
CIE158

154. The conjugate base of the weak acid in the reaction $HBr + H_2O \rightleftharpoons H_3O^+ + Br^-$ is
(1) HBr (2) H_2O (3) Br^- (4) H_3O^+
CIE159

155. In the reaction, $AlCl_3 + Cl^- \rightarrow [AlCl_4]^-$, $AlCl_3$ acts as :-
(1) Salt (2) Lewis base
(3) Bronsted acid (4) Lewis acid
CIE160

156. Mg^{2+} is _____ than Al^{3+} :-
(1) Stronger Lewis acid
(2) Stronger Lewis base
(3) Weaker Lewis acid
(4) Weaker Lewis base
CIE161

157. The two Bronsted bases in the reaction
$HC_2O_4^- + PO_4^{3-} \rightleftharpoons HPO_4^{2-} + C_2O_4^{2-}$ are
(1) $HC_2O_4^-$ and PO_4^{2-} (2) HPO_4^{2-} and $C_2O_4^{2-}$
(3) PO_4^{3-} and $C_2O_4^{-2}$ (4) $HC_2O_4^-$ and HPO_4^{2-}
CIE162

158. The compound HCl behaves as ---- in the reaction, $HCl + HF \rightleftharpoons H_2^+Cl + F^-$
(1) Strong acid (2) Strong base
(3) Weak acid (4) Weak base
CIE163

159. Which of the following is not a lewis base:-
(1) NH_3 (2) O^{2-} (3) H_2O (4) I^+
CIE164

160. Which of the following is Bronsted Lowry acid:-
(1) SO_4^{-2} (2) H_3O^+ (3) OH^- (4) Cl^-
CIE165

161. The conjugate base for bicarbonate ion is:-
(1) CO_3^{2-} (2) HCO_3^-
(3) CO_2 (4) H_2CO_3
CIE166

162. HCl does not behave as acid in :-
(1) NH_3 (2) C_2H_5OH
(3) H_2O (4) C_6H_6
CIE167

163. Which of the following is a base according to Bronsted-Lowry concept :-
(1) I^- (2) H_3O^+ (3) HCl (4) NH_4^+
CIE168

164. In which of the following reactions NH_3 acts as acid
(1) $NH_3 + HCl \rightarrow NH_4Cl$
(2) $NH_3 + H^+ \rightarrow NH_4^+$
(3) $NH_3 + Na \rightarrow NaNH_2 + \frac{1}{2} H_2$
(4) NH_3 cannot act as acid
CIE169

165. According to Bronsted concept, the acids in the following reaction $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$ are :-
(1) NH_3 and NH_4^+ (2) H_2O and OH^-
(3) H_2O and NH_4^+ (4) NH_3 and OH^-
CIE170

166. CH_3COO^- ion is a :-
(1) Weak conjugate base
(2) Strong conjugate base
(3) Weak conjugate acid
(4) Strong conjugate acid
CIE171

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167. Which of the following is strongest conjugate base
(1) ClO_4^- (2) HCO_3^- (3) Cl^- (4) HSO_4^-
CIE172

168. Which of the following species can act as Lewis base :-
(1) Cu^{2+} (2) $AlCl_3$ (3) NH_3 (4) BF_3
CIE173

169. A compound having the formula NH_2CH_2COOH may behave :-
(1) Only as an acid
(2) Only as a base
(3) Both as an acid and base
(4) Neither acid nor base
CIE174

170. Which of the following can act both as Bronsted acid and Bronsted base :-
(1) Na_2CO_3 (2) O^{2-}
(3) CO_3^{-2} (4) NH_3
CIE175

171. The strongest conjugate base is :-
(1) NO_3^- (2) Cl^-
(3) SO_4^{2-} (4) CH_3COO^-
CIE176

172. Aluminium chloride is :-
(1) Bronsted Lowry acid
(2) Arrhenius acid
(3) Lewis acid
(4) Lewis base
CIE177

173. Water is a :-
(1) Protogenic solvent
(2) Protophilic solvent
(3) Amphiprotic solvent
(4) Aprotic solvent
CIE178

174. Ammonium ion is :-
(1) A conjugate acid
(2) A conjugate base
(3) Neither an acid nor a base
(4) Both an acid and a base
CIE179

175. Species which do not act both as Bronsted acid and base is :-
(1) $(HSO_4)^{-1}$ (2) Na_2CO_3
(3) NH_3 (4) OH^{-1}
CIE180

176. Which one of the following is strong Lewis base & Bronsted acid & bronsted base:-
(1) NH_3 (2) PH_3
(3) CH_4 (4) BH_3
CIE181

177. Which of the following is not a correct statement
(1) Arrhenius theory of acids-bases is capable of explaining the acidic or basic nature of the substances in the solvents other than water
(2) Arrhenius theory does not explain acidic nature of $AlCl_3$
(3) The aqueous solution of Na_2CO_3 is alkaline although it does not contain OH^- ions
(4) Aqueous solution of CO_2 is acidic although it does not contain H^+ ions
CIE182

178. For the reaction $NH_4^+ + S^{-2} \rightleftharpoons NH_3 + HS^-$,$NH_3$ and S^{-2} are a group of :-
(1) Acids (2) Bases
(3) Acid-base pair (4) None of these
CIE183

179. According to Arrhenius theory, acids are substances that dissociate in water to give ...(X) ... ions and bases are substances that produce ...(Y).... ions. Here, (X) and (Y) refer to
(1) Hydronium, Hydroxyl
(2) Hydroxyl, Hydrogen
(3) Hydrogen, Hydroxyl
(4) Hydroxyl, Hydronium
CIE184

180. 'An acid is substance that is capable of donating a proton (H^+) and base is a substance capable of accepting a proton (H^+). The above statement is justified by
(1) Arrhenius concept
(2) Bronsted-Lowry theory
(3) Lewis concept
(4) All of these
CIE185

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EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	2	2	4	4	2	1	3	1	1	4	1	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	1	3	4	4	4	1	1	1	3	4	1	4	1	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	2	3	3	3	3	3	2	3	3	4	4	4	4	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	4	3	3	3	4	3	4	1	2	2	3	4	4	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	4	2	4	3	3	1	2	2	4	3	2	2	2	4
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	3	3	1	4	3	2	3	2	4	3	1	4	2	2	4
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	2	3	2	1	2	3	2	3	3	1	4	4	2	3	4
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	1	4	3	3	3	3	3	1	1	1	3	4	3	1	1
Question	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Answer	2	2	4	3	1	1	1	1	2	2	1	3	3	4	1
Question	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Answer	1	4	2	1	4	3	1	1	2	4	4	2	1	2	1
Question	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Answer	2	4	3	2	4	3	3	4	4	2	1	4	1	3	3
Question	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Answer	2	2	3	3	4	4	3	3	1	2	1	1	2	3	2

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Which of the following salts will give highest pH in water ?
(1) KCl (2) NaCl (3) Na_2CO_3 (4) $CuSO_4$
CIE196

AIPMT 2015

2. The K_{sp} of Ag_2CrO_4, AgCl, AgBr and AgI are respectively, 1.1×10^{-12}, 1.8×10^{-10}, 5.0×10^{-13}, 8.3×10^{-17}. Which one of the following salts will precipitate last if $AgNO_3$ solution is added to the solution containing equal moles of NaCl, NaBr, NaI and Na_2CrO_4 ?
(1) AgCl (2) AgBr
(3) Ag_2CrO_4 (4) AgI
CIE197

Re-AIPMT 2015

3. Which one of the following pairs of solution is not an acidic buffer ?
(1) H_2CO_3 and Na_2CO_3
(2) H_3PO_4 and Na_3PO_4
(3) $HClO_4$ and $NaClO_4$
(4) CH_3COOH and CH_3COONa
CIE198

4. What is the pH of the resulting solution when equal volumes of 0.1 M NaOH and 0.01 M HCl are mixed?
(1) 7.0 (2) 1.04 (3) 12.65 (4) 2.0
CIE199

NEET-I 2016

5. MY and NY_3, two nearly insoluble salts, have the same K_{sp} values of 6.2×10^{-13} at room temperature. Which statement would be **true** in regard to MY and NY_3 ?
(1) The molar solubilities of MY and NY_3 in water are identical.
(2) The molar solubility of MY in water is less than that of NY_3
(3) The salts MY and NY_3 are more soluble in 0.5 M KY than in pure water.
(4) The addition of the salt of KY to solution of MY and NY_3 will have no effect on their solubilities.
CIE200

NEET-II 2016

6. The percentage of pyridine (C_5H_5N) that forms pyridinium ion ($C_5H_5N^+H$) in a 0.10 M aqueous pyridine solution
(K_b for $C_5H_5N = 1.7 \times 10^{-9}$) is
(1) 0.77% (2) 1.6%
(3) 0.0060% (4) 0.013%
CIE201

7. The solubility of AgCl(s) with solubility product
1.6×10^{-10} in 0.1 M NaCl solution would be
(1) 1.6×10^{-11} M (2) zero
(3) 1.26×10^{-5} M (4) 1.6×10^{-9} M
CIE202

NEET(UG) 2017

8. Concentration of the Ag^+ ions in a saturated solution of $Ag_2C_2O_4$ is 2.2×10^{-4} mol L^{-1} Solubility product of $Ag_2C_2O_4$ is :-
(1) 2.66×10^{-12} (2) 4.5×10^{-11}
(3) 5.3×10^{-12} (4) 2.42×10^{-8}
CIE203

NEET(UG) 2018

9. Following solutions were prepared by mixing different volumes of NaOH and HCl of different concentrations :

a. $60mL\frac{M}{10}HCl + 40mL\frac{M}{10}NaOH$

b. $55mL\frac{M}{10}HCl + 45mL\frac{M}{10}NaOH$

c. $75mL\frac{M}{5}HCl + 25mL\frac{M}{5}NaOH$

d. $100mL\frac{M}{10}HCl + 100mL\frac{M}{10}NaOH$

pH of which one of them will be equal to 1 ?
(1) b (2) a (3) d (4) c
CIE204

10. The solubility of $BaSO_4$ in water 2.42×10^{-3} gL^{-1} at 298 K. The value of solubility product (K_{sp}) will be (Given molar mass of $BaSO_4$ = 233 g mol^{-1})
(1) 1.08×10^{-10} mol^2 L^{-2}
(2) 1.08×10^{-12} mol^2 L^{-2}
(3) 1.08×10^{-14} mol^2 L^{-2}
(4) 1.08×10^{-8} mol^2 L^{-2}
CIE205

NEET(UG) 2019

11. pH of a saturated solution of $Ca(OH)_2$ is 9. The solubility product (K_{sp}) of $Ca(OH)_2$ is :-
(1) 0.5×10^{-15} (2) 0.25×10^{-10}
(3) 0.125×10^{-15} (4) 0.5×10^{-10}
CIE206

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12. Which will make basic buffer ?
(1) 50 mL of 0.1 M NaOH + 25 mL of 0.1 M CH_3COOH
(2) 100 mL of 0.1 M CH_3COOH + 100 mL of 0.1M NaOH
(3) 100 mL of 0.1 M HCl + 200 mL of 0.1 M NH_4OH
(4) 100 mL of 0.1 M HCl + 100 mL of 0.1 M NaOH
CIE207

NEET(UG) (Odisha) 2019

13. The pH of 0.01 M NaOH (aq) solution will be
(1) 7.01 (2) 2 (3) 12 (4) 9
CIE208

14. Which of the following cannot act both as Bronsted acid and as Bronsted base ?
(1) HCO_3^- (2) NH_3 (3) HCl (4) HSO_4^-
CIE209

15. The molar solubility of CaF_2 ($K_{sp} = 5.3 \times 10^{-11}$) in 0.1 M solution of NaF will be
(1) 5.3×10^{-11} mol L^{-1} (2) 5.3×10^{-8} mol L^{-1}
(3) 5.3×10^{-9} mol L^{-1} (4) 5.3×10^{-10} mol L^{-1}
CIE210

NEET (UG) 2020

16. Find out the solubility of $Ni(OH)_2$ in 0.1M NaOH. Given that the ionic product of $Ni(OH)_2$ is 2×10^{-15}.
(1) 1×10^8 M (2) 2×10^{-13} M
(3) 2×10^{-8} M (4) 1×10^{-13} M
CIE211

NEET (UG) 2020 (COVID-19)

17. Which among the following salt solutions is basic in nature ?
(1) Ammonium chloride
(2) Ammonium sulphate
(3) Ammonium nitrate
(4) Sodium acetate
CIE212

18. The solubility product for a salt of the type AB is 4×10^{-8}. What is the molarity of its standard solution?
(1) 2×10^{-4} mol/L (2) 16×10^{-16} mol/L
(3) 2×10^{-16} mol/L (4) 4×10^{-4} mol/L
CIE213

NEET (UG) 2021

19. The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T (K). The correct option for the pH of dimethylammonium acetate solution is:
(1) 8.50 (2) 5.50 (3) 7.75 (4) 6.25
CIE214

NEET (UG) 2022

20. The pH of the solution containing 50 mL each of 0.10 M sodium acetate and 0.01 M acetic acid is [Given pK_a of CH_3COOH = 4.57]
(1) 3.57 (2) 4.57 (3) 2.57 (4) 5.57
CIE215

21. 0.01 M acetic acid solution is 1% ionised, then pH of this acetic acid solution is :
(1) 3 (2) 2 (3) 4 (4)1
CIE216

NEET (UG) 2024 (Re-examination)

22. Which indicator is used in the titration of sodium hydroxide against oxalic acid and what is the colour change at the end point ?
(1) Phenolphthalein, pink to yellow
(2) Alkaline $KMnO_4$, colourless to pink
(3) Phenolphthalein, colourless to pink
(4) Methyl orange, yellow to pinkish red colour
CIE274

23. The ratio of solubility of AgCl in 0.1 M KCl solution to the solubility of AgCl in water is :
(Given : Solubility product of AgCl = 10^{-10})
(1) 10^{-4} (2) 10^{-6} (3) 10^{-9} (4) 10^{-5}
CIE275

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	3	3	2	4	4	3	4	1	1	3	3	3	3
Question	16	17	18	19	20	21	22	23							
Answer	2	4	1	3	4	3	3	1							

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Exercise – III (Analytical Questions) Master Your Understanding

1. Given below are two statements:
Statement-I : During hydrolysis of CH_3COONH_4, the ions formed undergo hydrolysis as follow :

$$CH_3COO^- + NH_4^+ + H_2O \rightleftharpoons CH_3COOH + NH_4OH$$

Statement-II : The degree of hydrolysis of CH_3COONH_4 is independent of concentration of solution.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I & statement-II both are correct
(2) Only statement-I is correct
(3) Only statement-II is correct
(4) Statement-I & statement-II both are incorrect

CIE217

2. Arrange the following solutions in the decreasing order of pH :
(A) 0.01 M HCl
(B) 0.01 M NaOH
(C) 0.01 M CH_3COONa
(D) 0.01 M NaCl
(1) (B) > (D) > (C) > (A)
(2) (B) > (C) > (D) > (A)
(3) (A) > (C) < (D) > (B)
(4) (A) > (D) > (C) > (B)

CIE220

3. Given below are two statements:
Statement-I : Solubility of AgBr decreases in the presence of NaBr.
Statement-II : NaBr undergoes hydrolysis in water.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I and statement-II are incorrect.
(2) Statement-I and statement-II are correct.
(3) Statement-I is incorrect but statement-II is correct.
(4) Statement-I is correct but statement-II is incorrect

CIE222

4. Statement-I : Acids turn red litmus paper blue.
Statement-II : Acids taste bitter and feel soapy.
(1) Statement-I and statement-II are incorrect.
(2) Statement-I and statement-II are correct.
(3) Statement-I is incorrect but statement-II is correct.
(4) Statement-I is correct but statement-II is incorrect

CIE223

5. Given below are two statements:
Statement-I : Conjugate acid and conjugate base of NH_3 respectively are NH_4^+ and NH_2^-.
Statement-II : The acid base pair that differs only by one proton is called a conjugate acid base pair.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I and statement-II are incorrect.
(2) Statement-I and statement-II are correct.
(3) Statement-I is incorrect but statement-II is correct.
(4) Statement-I is correct but statement-II is incorrect

CIE224

6. Given below are two statements:
Statement-I : The cations of strong bases and anions of strong acids simply get hydrolyse but do not get hydrated.
Statement-II : The solutions of salts formed from strong acids & bases are neutral.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I and statement-II are incorrect.
(2) Statement-I and statement-II are correct.
(3) Statement-I is incorrect but statement-II is correct.
(4) Statement-I is correct but statement-II is incorrect

CIE225

7. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion : All Arrhenius acids are Bronsted acids.

Reason : All Bronsted acids are Arrhenius acids.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) (A) is correct but (R) is not correct

(2) Both (A) and (R) are correct and (R) is correct explanation of (A)

(3) Both (A) and (R) are correct but (R) is not correct explanation of (A)

(4) (A) is not correct but (R) is correct

CIE227

8. Match the Column – I and Column - II

	Column-I		**Column-II**
	(Species)		**(Conjugate acid)**
(a)	NH_3	(p)	H_2CO_3
(b)	HCO_3^-	(q)	H_2SO_4
(c)	H_2O	(r)	NH_4^+
(d)	HSO_4^-	(s)	H_3O^+

(1) (a) → (r); (b) → (p); (c) → (s); (d) → (q)

(2) (a) → (p); (b) → (r); (c) → (q); (d) → (s)

(3) (a) → (r); (b) → (s); (c) → (p); (d) → (q)

(4) (a) → (q); (b) → (p); (c) → (r); (d) → (s)

CIE228

9. Match the Column – I and Column - II

	Column-I		**Column-II**
(a)	$Al(OH)_3$	(p)	$S=\left(\frac{K_{sp}}{6912}\right)^{1/7}$
(b)	$BaCrO_4$	(q)	$S=\left(\frac{K_{sp}}{4}\right)^{1/3}$
(c)	$Zr_3(PO_4)_4$	(r)	$S=\left(\frac{K_{sp}}{27}\right)^{1/4}$
(d)	Hg_2I_2	(s)	$S=(K_{sp})^{1/2}$

(1) (a) → (r); (b) → (s); (c) → (p); (d) → (q)

(2) (a) → (s); (b) → (p); (c) → (r); (d) → (q)

(3) (a) → (p); (b) → (q); (c)→ (s); (d) → (r)

(4) (a) → (q); (b) → (r); (c) → (p); (d) → (s)

CIE229

10. If concentrations of two acids are same, their relative strengths can be compared by :

(a) $\frac{\alpha_1}{\alpha_2}$ (b) $\frac{[H^+]_1}{[H^+]_2}$

(c) $\sqrt{\frac{K_1}{K_2}}$ (d) $\frac{K_1}{K_2}$

(1) a, c, d are correct (2) a, b, c are correct

(3) b, c, d are correct (4) All are correct

CIE230

11. Which can act as a buffer ?

(a) $CH_3COOH + NaOH$; if $[CH_3COOH] > [NaOH]$

(b) $HCl + CH_3COONa$; if $[CH_3COONa] > [HCl]$

(c) HCl + NaCl

(d) NH_4CN

(1) a, b, d (2) b, c, d (3) a, c, d (4) c, d

CIE231

12. Given below are two statements:

Statement-I : pH of a 1.0×10^{-8} M solution of HCl is 8 at 25°C.

Statement-II : HCl is a strong acid and negative log of H^+ concentration is called pH.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement-I and statement-II are incorrect.

(2) Statement-I and statement-II are correct.

(3) Statement-I is incorrect but statement-II is correct.

(4) Statement-I is correct but statement-II is incorrect

CIE233

13. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .

Assertion : Water act as base in the reaction

$HCl(aq.) + H_2O(\ell) \rightleftharpoons H_3O^+(aq.) + Cl^-(aq.)$

Reason : According to Bronsted-lowry proton acceptors are known as base.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

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(1) (A) is correct but (R) is not correct
(2) Both (A) and (R) are correct and (R) is correct explanation of (A)
(3) Both (A) and (R) are correct but (R) is not correct explanation of (A)
(4) (A) is not correct but (R) is correct

CIE234

14. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)** .
Assertion : pH of NH_4Cl solution in water is less than 7 at 25°C.
Reason : Ammonium hydroxide is a strong base.
In the light of the above statements, choose the **most appropriate** answer from the options given below :
(1) (A) is correct but (R) is not correct
(2) Both (A) and (R) are correct and (R) is correct explanation of (A)
(3) Both (A) and (R) are correct but (R) is not correct explanation of (A)
(4) (A) is not correct but (R) is correct

CIE235

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Answer	1	2	4	1	2	3	1	1	1	2	1	3	2	1	

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Ionic Equilibrium

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (2)

$$M = \frac{w \times 1000}{M_w \times V_{ml}}$$

$$0.1 = \frac{w \times 1000}{98 \times 400}$$

$w = 3.92g$

2. Ans. (2)

$N = n_f \times M$

$N = 2 \times 2 = 4N$

3. Ans. (2)

$pH = 3.31$

$[H^+] = 10^{-pH} = 10^{-3.31}$

$[H^+] = 5 \times 10^{-4}$

4. Ans. (2)

$[OH^-] = 5 \times 10^{-5}M$

$pOH = -\log(OH^-)$

$= -\log(5 \times 10^{-5})$

$= 5 - \log 5$

$pH = 14 - pOH$

$= 14 - (5 - \log 5)$

$= 9 + \log 5$

5. Ans. (4)

Basicity is equal to number of replaceable H–atom attached to more electronegative atoms.

For H_3PO_3 Basicity is 2

H_3PO_2 Basicity is 1

6. Ans. (4)

$$[OH^-] = \frac{n}{V_{(L)}} = \frac{2 \times 10^{-3}}{2} = 10^{-3}M$$

$pOH = -\log[OH^-] = -\log(10^{-3})$

$= 3$

$pH = 14 - pOH$

$= 14 - 3 = 11$

7. Ans. (2)

$pH = 4.4$

$[H^+] = 10^{-pH} = 10^{-4.4}$

$[H^+] = 3.9 \times 10^{-5}$

8. Ans. (1)

$$M = \frac{n}{V_{(L)}} = \frac{w}{M_w \times V_{(L)}}$$

$$= \frac{8}{40 \times 1} = 0.2M$$

9. Ans. (3)

$$M = \frac{w \times 1000}{M_w \times V_{(ml)}}$$

$$M = \frac{w \times 1000}{60 \times 400} = 0.1$$

$w = 0.60g$

10. Ans. (1)

Mili equivalent $= N \times V_{(ml)}$

$= 0.5 \times 100$

$= 50$

11. Ans. (1)

$MgCl_2 \rightarrow Mg^{+2} + 2Cl^-$

C – –

– C 2C

$[Cl^-] = 2C = 2 \times 1.5 \times 10^{-3}$

$= 3 \times 10^{-3}$

12. Ans. (4)

$$\alpha = \sqrt{\frac{k_a}{c}} = \sqrt{\frac{10^{-5}}{0.1}}$$

$\alpha = 10^{-2}$

13. Ans. (1)

$[H^+] = C\alpha$

$[H^+] = 0.1$

$pH = -\log[H^+]$

$= -\log(0.1)$

$= 1$

14. Ans. (1)

Ostwald's dilution law is not applicable for strong electrolyte because strong electrolytes are completely ionised.

15. Ans. (2)

The degree of ionisation of a compound depends upon nature of solute molecules.

16. Ans. (4)

pOH = 10

pH = 14 – pOH = 4

$[H^+] = 10^{-4}M = \sqrt{K_a \times C}$

$[H^+] = \sqrt{K_a \times C}$

$10^{-4} = \sqrt{K_a \times 10^{-2}}$

$K_a = 10^{-6}$

17. Ans. (1)

On dilution, 1.0M to 0.01M, percentage ionisation will increase.

18. Ans. (3)

On adding water, extent of ionisation of weak electrolytes increases.

19. Ans. (4)

$[H^+] = \sqrt{K_a \times C}$

$= \sqrt{4\times10^{-10}\times2.5\times10^{-1}}$

$= \sqrt{10^{-10}}$

$[H^+] = 10^{-5}$

pH = 5

20. Ans. (4)

pH = 2

$[H^+] = 10^{-2}M$

$[H^+] = \sqrt{K_a \times C}$

$10^{-2} = \sqrt{4.5\times10^{-4}\times C}$

C = 0.2222 M

21. Ans. (4)

$K_w = 10^{-14}$ at (25°C)

On increasing temperature, K_w will increase for example at 90°C, $K_w = 10^{-12}$

22. Ans. (1)

at 25°C

$[H^+] = 10^{-7}$ M

$[OH^+] = 10^{-7}$ M

So, $[H^+] + [OH^-]$

$= 2 \times 10^{-7}$

23. Ans. (1)

$$K_d = \frac{[H^+][OH^-]}{[H_2O]}$$

$$= \frac{10^{-14}}{55.5} = (55.5 \times 10^{14})^{-1}$$

24. Ans. (1)

$H_2O \rightleftharpoons H^+ + OH^-$

$$K_{dissociation} = \frac{[H^+][OH^-]}{[H_2O]}$$

$K_d \times [H_2O] = [H^+] \times [OH^-]$

$= K_w$

So, ionic product of water

= dissociation constant of water × H_2O

25. Ans. (3)

At, 90°C, $[H^+] = 10^{-6}M$

$[OH^-] = 10^{-6}M$

$[H^+] + [OH^-] = 10^{-6} + 10^{-6}$

$= 2 \times 10^{-6}$

26. Ans. (4)

For pure water $[H^+] = [OH^-]$

$= 10^{-6.7}$

So, $K_w = [H^+] \times [OH^-]$

$= 10^{-6.7} \times 10^{-6.7}$

$= 10^{-13.4}$

27. Ans. (1)

At 25°C, $[H^+] = 10^{-7}M$,

= pH = 7

On in Creasing temp, $[H^+]\uparrow$

So pH will $\downarrow$

At 373 K,

pH will be less than 7.

28. Ans. (4)

$$[H^+] = 10^{-7}M = \frac{N}{N_A \times V_{(L)}}$$

$N = 10^{-7} \times 6 \times 10^{23} \times 1$

$= 6 \times 10^{16}$

39. Ans. (1)

pH of 0.1 M NaCl will be less than 7 at 90°C

30. Ans. (1)

$\because K_a = 10^{-4}$

$pK_a = 4$

$K_b = 10^{-5}$

$pK_b = 5$

pH of 1M ammonium formate will be

$$= 7 + \frac{1}{2}pK_a - \frac{1}{2}pK_b$$

$$= 7 + \frac{1}{2}[4 - 5]$$

$= 7 – 0.5 = 6.5$

31. Ans. (1)

KCl is a salt of SA + SB, so it will not undergo hydrolysis.

32. Ans. (2)

$HCOO^- + H_2O \rightleftharpoons HCOOH + OH^-$

C	–	–	–
C – Ch	–	Ch	Ch

$K_h = \frac{[HCOOH][OH^-]}{[HCOO^-]} = \frac{Ch.\,Ch}{C - Ch} = Ch^2$

$h = \sqrt{\frac{K_h}{C}}$

33. Ans. (3)

pH of CH_3COONa will be more than 7.

34. Ans. (3)

pK_b of CN^- is 4.7

pK_a of HCN will be $14 - pK_b$

$= 14 - 4.7 = 9.3$

pH of 0.5 M NaCN is

$= 7 + \frac{1}{2} PK_a + \frac{1}{2} \log C$

$= 7 + \frac{1}{2}(9.3) + \frac{1}{2}\log(0.5)$

$= 7 + \frac{1}{2}(9.3) + (-\frac{1}{2}\log 2)$

$= 7 + \frac{1}{2}[(9.3 - \log 2)]$

$= 7 + \left(\frac{1}{2} \times 9\right) = 11.5$

35. Ans. (3)

Salt	pH
0.1 M NaCl	7
0.1 M NH_4Cl 0.1 M CH_3COONa	< 7 > 7
0.1 M CH_3COONH_4	May be less than or slightly more than 7

So, highest pH will be shown by 0.1 M CH_3COONa

36. Ans. (3)

K_2S is a salt of SB + WA,

So, pH will be more than 7.

37. Ans. (3)

For anionic hydrolysis, (WA + SB) type salt,

$pH = 7 + \frac{1}{2} pK_a + \frac{1}{2} \log C$

or

$\frac{1}{2} pK_w + \frac{1}{2} pK_a + \frac{1}{2} \log C$

38. Ans. (2)

Equilibrium constant for the reaction of WA + SB will be

$\frac{1}{K_H} = \frac{1}{(K_w / K_a)} = \frac{K_a}{K_w} = \frac{10^{-4}}{10^{-14}} = 10^{10}$.

39. Ans. (3)

$CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$

C	–	–	–
C – Ch	–	Ch	Ch

$[OH^-] = Ch$

$K_h = \frac{K_w}{K_a} = Ch^2$

$h = \sqrt{\frac{K_w}{K_a \times C}}$

$[OH^-] = Ch = C\sqrt{\frac{K_w}{K_a \times C}}$

$= \sqrt{\frac{K_w \times C}{K_a}}$

40. Ans. (3)

$FeCl_3$ will be acidic & Ammonium acetate in water may be acidic or basic

41. Ans. (4)

Na_3PO_4 & CH_3COONa salt undergoes hydrolysis in water.

42. Ans. (4)

If resulting solution is Alkaline in nature. The salt is made up of WA + SB.

43. Ans. (4)

The solubility product of sparingly soluble AB type salt is defined as the product of ionic concentration a saturated solution.

44. Ans. (4)

K_{sp}	Salt
$4s^3$	M_2X
$4s^3$	QY_2
$4s^3$	PZ_2

If s is equal for M_2X, QY_2 & PZ_2 then K_{sp} will also be equal.

45. Ans. (3)

$Hg_2I_2 \rightarrow Hg_2^{+2} + 2I^-$

$K_{sp} = \left[Hg_2^{+2}\right]\left[I^-\right]^2$

46. Ans. (2)

K_{sp} is only affected by temperature.

47. Ans. (4)

$K_{sp} = 10^{-10} = s^2$

$s = 10^{-5}$ M

$\because BaSO_4 \rightleftharpoons Ba^{+2} + SO_4^{-2}$

– s s

$M = \frac{n}{V_{(L)}} = 10^{-5}$

$\Rightarrow \frac{w}{M_w \times V_{(L)}} = 10^{-5}$

$\Rightarrow \frac{1}{233 \times V_{(L)}} = 10^{-5}$

$V_{(L)} = 430$ L

48. Ans. (3)

$Ag_2CrO_4 \rightleftharpoons 2Ag^+ + CrO_4^{-2}$

2s s

$K_{sp} = [Ag^+]^2 [CrO_4^{-2}]$

$= (2s)^2 (s)$

$= 4s^3$

$[Ag^+] = 1.5 \times 10^{-4} = 2s$

$s = \frac{1.5 \times 10^{-4}}{2}$

$K_{sp} = 4\left(\frac{1.5}{2} \times 10^{-4}\right)^3$

$= \frac{27}{16} \times 10^{-12}$

$= 1.68 \times 10^{-12}$

49. Ans. (3)

$Ag_2CrO_4 \rightleftharpoons 2Ag^+ + CrO_4^{-2}$

2s s

$[CrO_4^{-2}] = s = 2 \times 10^{-4}$

$K_{sp} = 4s^3 = 4(2 \times 10^{-4})^3$

$= 32 \times 10^{-12}$

50. Ans. (3)

$s = \frac{n}{V_{(ml)}} \times 1000$

$= \frac{1.43 \times 10^{-4} \times 1000}{143 \times 100}$

$= 10^{-5}$M

$AgCl \rightleftharpoons Ag^+ + Cl^-$

– s s

$K_{sp} = [Ag^+] [Cl^-]$

$= s^2$

$K_{sp} = (10^{-5})^2 = 10^{-10}$

51. Ans. (4)

$As_2S_3 \rightleftharpoons 2As^{+3} + 3S^{-2}$

– 2s 3s

$K_{sp} = [As^{+3}]^2 [S^{-2}]^3$

52. Ans. (3)

$PbBr_2 \rightleftharpoons Pb^{+2} + 2Br^-$

s 2s

$K_{sp} = [Pb^{+2}] [Br^-]^2$

$= s(2s)^2$

$= 4s^3$

53. Ans. (4)

$K_{sp} = (3)^3(3)^3(2^2)s^{3+3+2}$

$= 2916s^8$

54. Ans. (1)

$CaCO_3 \rightleftharpoons CaO + CO_2$

– s s

$[CaCO_3] = \frac{7}{100 \times 1} = 7 \times 10^{-2}$ M = s

$K_{sp} = s^2 = (7 \times 10^{-2})^2$

$= 4.9 \times 10^{-3}$

55. Ans. (2)

AgBr in 0.1 M $CaBr_2$

$AgBr \rightleftharpoons Ag^+ + Br^-$

– s' s'

$CaBr_2 \rightleftharpoons Ca^{+2} + 2Br^-$

– 0.1 0.2

$[Br^-] = s' + 0.2 \approx 0.2$

K_{sp} of AgBr $= [Ag^+] [Br^-]$

$K_{sp} = (s') (0.2)$

$s' = \frac{K_{sp}}{0.2}$

56. Ans. (2)

$AgSCN \rightleftharpoons Ag^+ + SCN^-$

If solution is unsaturated

$Q < K_{sp}$

$[Ag^+]\,[SCN^-] < K_{sp}$

57. Ans. (3)

$AB \rightleftharpoons A^+ + B^-$

– s s

$K_{sp} = s = [A^+]\,[B^-]$

$S = s \cdot s$

$S = s^2$

$s = S^{\frac{1}{2}}$

58. Ans. (4)

	K_{sp}	
CuS	$K_{sp} = s_1^2$	$s_1^2 = 10^{-37}$
Ag_2S	$K_{sp} = 4s_2^3$	$4s_2^3 = 10^{-44}$
HgS	$K_{sp} = s_3^2$	$s_3^2 = 10^{-54}$

Order :- $s_2 > s_1 > s_3$

$Ag_2S > CuS > HgS$

59. Ans. (4)

$PbCl_2 \rightleftharpoons Pb^{+2} + 2Cl^-$

– s 2s

(s) = 0.01 M

$K_{sp} = 4s^3 = 4(0.01)^3$

$= 4 \times 10^{-6}$

Solubility of $PbCl_2$ in 0.1 M NaCl will be

$s' = \frac{K_{sp}}{(0.1)^2} = \frac{4\times10^{-6}}{(0.1)^2}$

$= 4 \times 10^{-4}$

60. Ans. (2)

$M_2SO_4 \rightleftharpoons 2M^+ + SO_4^{-2}$

– 2s s

$K_{sp} = [M^+]^2\,[SO_4^{-2}]$

$= (2s)^2\,(s)$

$= 4s^3 = 1.2 \times 10^{-5}$

$= s^3 = 3 \times 10^{-6}$

$s = (3)^{\frac{1}{3}} \times 10^{-2}$

$[M^+] = 2s = 2(3)^{\frac{1}{3}} \times (10^{-2})$

$= 2.89 \times 10^{-2}M$

61. Ans. (2)

	K_{sp}	Solubility
HgS	$s^2 = 1.6 \times 10^{-54}$	$s \approx 10^{-27}$
$PbSO_4$	$s^2 = 1.3 \times 10^{-8}$	$s \approx 10^{-4}$
ZnS	$s^2 = 7 \times 10^{-26}$	$s \approx 10^{-13}$
AgCl	$s^2 = 1.7 \times 10^{-10}$	$s \approx 10^{-5}$

Hence $PbSO_4$ has maximum solubility.

62. Ans. (4)

	$K_{sp} = s^2$	Solubility
CuS	8.5×10^{-36}	$s \approx 10^{-18}$
CdS	3.6×10^{-28}	$s \approx 10^{-14}$
ZnS	1.2×10^{-28}	$s \approx 10^{-14}$
MnS	1.4×10^{-10}	$s \approx 10^{-5}$

Solubility of MnS in maximum.

63. Ans. (2)

Solubility of AgCl will be maximum in water. In $AgNO_3$, NaCl & KCl, due to common ion solubility of AgCl will decrease.

64. Ans. (4)

	K_{sp}	
PQ	$s_1^2 = 4 \times 10^{-20}$	$s_1 = 2 \times 10^{-10}$
PQ_2	$4s_2^3 = 3.2 \times 10^{-14}$	$s_2 = 2 \times 10^{-5}$
PQ_3	$27s_3^4 = 2.7 \times 10^{-35}$	$s_3 = 10^{-9}$

$s_2 > s_3 > s_1$

65. Ans. (3)

$Gd(OH)_3 \rightleftharpoons Gd^{+3} + 3OH^-$

s 3s

$K_{sp} = 27s^4 = 2.8 \times 10^{-23}$

$s \approx 10^{-6}$

$[OH^-] = 3s = 3 \times 10^{-6}$

$pOH = 6 - \log 3$

$pH = 14 - pOH$

$= 14 - (6 - \log 3)$

$= 8 + \log 3$

$= 8.47$

66. Ans. (3)

K_{sp} of $AgBrO_3 = s_1^2 = 5.5 \times 10^{-5}$

$= s_1 = 7.5 \times 10^{-3}$

K_{sp} of $Ag_2SO_4 = 4s_2^3 = 2 \times 10^{-5}$

$= s_2 \simeq 10^{-2}$

$s_2 > s_1$

Hence $s_{Ag_2SO_4} > s_{AgBrO_3}$

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67. Ans. (1)

For precipitation to begin

$Q > K_{sp}$

$Mg(OH)_2 \rightleftharpoons Mg^{+2} + 2OH^-$

0.1 M

$[Mg^{+2}]\,[OH^-]^2 \geq K_{sp}$

$(0.1)\quad (OH^-)^2 \geq 10^{-11}$

$[OH^-]^2 \geq \dfrac{10^{-11}}{0.1}$

$[OH^-]^2 \geq 10^{-10}$

$[OH^-] > 10^{-5}$

$pOH \leq 5$

$pH \geq 9$

68. Ans. (2)

For ZnS

$ZnS \rightleftharpoons Zn^{+2} + S^{-2}$

$Q = [Zn^{+2}]\,[S^{-2}]$

$= [0.01]\,[8.1 \times 10^{-21}]$

K_{sp} of $ZnS = 3 \times 10^{-22}$

$Q < K_{sp}$ No precipitation similarly for CuS

$CuS \rightleftharpoons Cu^{+2} + S^{-2}$

$Q = [Cu^{+2}]\,[S^{-2}]$

$Q = (0.01)\,(8.1 \times 10^{-21})$

$= 8.1 \times 10^{-23}$

K_{sp} of $CuS = 8 \times 10^{-36}$

$Q > K_{sp}$

Hence CuS will precipitate.

69. Ans. (2)

$Mg(OH)_2 \rightleftharpoons Mg^{+2} + 2OH^-$

$pH = 9$

$pOH = 14 - 9 = 5$

$[OH^-] = 10^{-5}$

$[Mg^{+2}] = 0.001$ M (from $Mg(NO_3)_2$)

$Q = [Mg^{+2}]\,[OH^-]^2$

$= [0.001]\,(10^{-5})^2$

$= 10^{-13}$

K_{sp} of $Mg(OH)_2 = 8.9 \times 10^{-12}$

$Q < K_{sp}$

$\therefore$ Precipitation will not take place.

70. Ans. (4)

To prevent precipitation

$Q \leq K_{sp}$

$Mg(OH)_2 \rightleftharpoons Mg^{+2} + 2OH^-$

$Q = [Mg^{+2}]\,[OH^-]^2 \leq 9 \times 10^{-12}$

$(0.01)\,(OH^-)^2 \leq 9 \times 10^{-12}$

$[OH^-]^2 \leq 9 \times 10^{-10}$

$[OH^-] \leq 3 \times 10^{-5}$

So, maximum concentration of OH^- is 3×10^{-5}

71. Ans. (3)

In the presence of HCl, pure NaCl is precipitated because due to common ion $[Cl^-]$, concentration of $[Cl^-]$ increases. & hence ionic product Q exceeds the solubility product of NaCl.

72. Ans. (2)

$[H^+]_1 = [H^+]_2$

$\sqrt{K_{a_1} C_1} = \sqrt{K_{a_2} C_2}$

$K_{a_1} C_1 = K_{a_2} C_2$

$\boxed{C \propto \dfrac{1}{V}}$

So; $K_{a_1} V_2 = K_{a_2} V_1$

73. Ans. (2)

Solutions having same concentration of H^+ ions are called isohydric solutions.

74. Ans. (2)

$[H^+] = C\alpha$

$= (0.1)\left(\dfrac{1}{100}\right)$

$= 10^{-3}$

$pH = 3$

75. Ans. (4)

If pH of Y is more than 7 when it is dissolved in water then Y is a salt of weak acid & strong base.

76. Ans. (3)

$BeCl_2$:- Salt of WB + SA

$pH < 7$

$BaCl_2$:- Salt of SB + SA

$pH = 7$

$Ba(NO_3)_2$:- Salt of SB + SA

$pH = 7$

$Ba(OH)_2$:- strong base

$pH > 7$

So minimum pH will be shown by 0.1 M $BeCl_2$

77. Ans. (3)

0.005 M $H_2SO_4 \Rightarrow [H^+] = 0.005 \times 2$
$= 0.01$ N

0.01 M HCl $\Rightarrow [H^+] = 0.01 \times 1$
$= 0.01$ N

$[H^+]$ for both is same & hence pH will also be same.

78. Ans. (1)

$H_2CO_3 \rightleftharpoons 2H^+ + CO_3^{-2}$ ($\because \alpha = 10\% = 0.1$)

$C - C\alpha \quad 2C\alpha \quad C\alpha$

$[H^+] = 2C\alpha$
$= 2 \times 5 \times 10^{-3} \times (0.1) = 10^{-3}$

79. Ans. (4)

$M(OH)_4 \rightleftharpoons M^{+4} + 4OH^-$ ($\because \alpha = 50\% = 0.5$)

$C - C\alpha \quad C\alpha \quad 4C\alpha$

$[OH^-] = 4C\alpha = 4 \times 0.0025 \times 0.5$
$= 5 \times 10^{-3}$

$pOH = 3 - \log 5$

$pH = 14 - pOH$
$= 14 - (3 - \log 5)$
$= 11 + \log 5 = 11.7$

80. Ans. (3)

$\frac{M}{20}$ H_2SO_4, $\frac{M}{10}$ HNO_3, $\frac{M}{40}$ $HClO_4$ will have highest pH.

$$[H^+] = \frac{\sum NV}{\sum V} = \frac{\left[2 \times \frac{1}{20} \times 1\right] + \left[\frac{1}{10} \times 1\right] + \left[\frac{1}{40} \times 1\right]}{1+1+1}$$

$$= \frac{\frac{1}{10} + \frac{1}{10} + \frac{1}{40}}{3} = \frac{9}{120}$$

$$[H^+] = \frac{3}{40}$$

$$pH = \log[H^+] = -\log\left[\frac{3}{40}\right]$$

$= \log 40 - \log 3$
$= 1.6 - 0.48$
$= 1.12$

81. Ans. (2)

$pH = 3, \Rightarrow [H^+]_1 = N_1 = 10^{-3}$

$pH = 3 \Rightarrow [H^+]_2 = N_2 = 10^{-3}$

$$N_{mixture} = \frac{N_1V_1 + N_2V_2}{V_1 + V_2} = [H^+]_{mixture}$$

$$= \frac{10^{-3} \times 100 + 10^{-3} \times 400}{500}$$

$$\frac{5 \times 10^{-3}}{5} = 10^{-3} = [H^+]$$

$pH = -\log[H^+] = 3$

82. Ans. (3)

10^{-6} M = N,

$N_2 = \frac{N_1}{100}$ [on dilution, concentration decreases]

$N_2 = 10^{-8}$N

pH of 10^{-8} M HCl {since its concentration is less than 10^{-7} M, $[H^+]$ ions from water must be added}

$[H^+] = 10^{-8} + 10^{-7}$
$= 10^{-8}[1 + 10]$
$= 11 \times 10^{-8}$

$pH = -\log [H^+] = 8 - \log 11 = 6.95$

83. Ans. (4)

For 0.001M KOH $\Rightarrow [OH^-] = 10^{-3}$M,

$pOH = -\log[OH^-] = 3$

at 90°C pH + pOH = 12

$pH = 12 - pOH = 12 - 3 = 9$

84. Ans. (2)

pH of 0.001 M acetic acid would be more than 3; since it is a weak acid, 100% dissociation will not take place.

85. Ans. (3)

$pH = 3, [H^+] = 10^{-3}M = N_1$

$pH = 6, [H^+] = 10^{-6}M = N_2 \Rightarrow N_2 = \frac{1}{1000} \times N_1$

if pH is changed from 3 to 6, $[H^+]$ ion concentration reduced by 1000 times.

86. Ans. (1)

$pOH = 13$

$pH = 14 - pOH$ at 298 K

$pH = 14 - 13 = 1$

so, solution will be highly acidic.

87. Ans. (4)

Mixture of strong acid & strong base :-

$\Sigma N_AV_A = 10 \times 0.05 \times 2 = 1$

$\Sigma N_BV_B = 10 \times 0.01 = 1$

If $\Sigma N_AV_A = \Sigma N_BV_B$: solution will be neutral, pH will be 7.

88. Ans. (2)

$pH = 5, [H^+] = 10^{-5}N = N_1$

$pH = 2 \Rightarrow [H^+] = 10^{-2}N = N_2$

$\frac{N_2}{N_1} = \frac{10^{-2}}{10^{-5}} = 10^3$

$N_2 = 1000N_1$

Increase in Hydrogen ion concentration is 1000 times.

89. Ans. (2)

$[H^+] = 6 \times 10^{-4}$

$pH = -\log[H^+] = 4 - \log 6 = 3.22$

90. Ans. (1)

$pOH = 10 \Rightarrow [OH^-] = 10^{-10}$

$pH = 14 - pOH = 4 \Rightarrow [H^+] = 10^{-4}M$

$K_w = 10^{-14} \Rightarrow [H^+][OH^-] = K_w \Rightarrow [H^+] = \frac{K_w}{[OH^-]}$

So, $[H^+] = \frac{K_w}{10^{-10}} = 10^{-4}$

91. Ans. (2)

$pH = 0, [H^+] = 10^{-pH} = 1$ M.

So solution is acidic

92. Ans. (3)

$N_{resultant} = \frac{\text{Total g. eq of KOH}}{\text{Total volume}}$ n_f of KOH = 1

$= \frac{0.1+0.2+0.3+0.4+0.5}{1+2+3+4+5} = 0.1$

$[OH^-] = 0.1$ N

$pOH = -\log(0.1) = 1$

$pH = 14 - pOH = 13$

93. Ans. (2)

pH of 0.02M NH_4OH, $\alpha = 5\%$

$NH_4OH \rightleftharpoons NH_4^+ + OH^-$

C 0 0

$C - C\alpha$ $C\alpha$ $C\alpha$

$[OH^-] = C\alpha$

$= 0.02 \times \frac{5}{100} = 10^{-3}M$

$pOH = -\log[OH^-] = 3$

$pH = 14 - 3 = 11$

94. Ans. (1)

$pH = -\log[H^+]$

$= -\log\left(\frac{1}{10}\right) = 1$

95. Ans. (2)

10^{-9}M HCl is very dilute solution, so $[H^+]$ ion form water must be considered.

$[H^+] = [H^+]_{HCl} + [H^+]_{water}$

$= 10^{-9} + 10^{-7}$

$= 101 \times 10^{-9}$

$pH = 9 - \log(101) = 9 - 2.004 = 6.99$

so pH is between 6 and 7.

96. Ans. (3)

$pH = 2, [H^+] = 10^{-2}M$

$pH = 3; [H^+] = 10^{-3}M$

Number of moles of HCl, which must be removed from 1L solution will be

$= M_1V_1 - M_2V_2$

$= 10^{-2} \times 1 - 10^{-3} \times 1$

$= (10^{-2})\left[\left(1 - \frac{1}{10}\right)\right] = 0.009$

97. Ans. (2)

$N_{NaOH} = \frac{8}{40 \times 1} = 0.2$ N

$N_{H_2SO_4} = \frac{4.9}{49 \times 1} = 0.1$ N

$N_AV_A = 0.1 \times 1 = 0.1$

$N_BV_B = 0.2 \times 1 = 0.2$ $V_A + V_B = 1L = V_T$

$N_BV_B > N_AV_A$

$[OH^-] = \frac{N_BV_B - N_AV_A}{V_A + V_B} = \frac{0.2 - 0.1}{1} = \frac{0.1}{1}$

$[OH^-] = 0.1$ N

$pOH = -\log[OH^-] = 1$

$pH = 14 - pOH = 14 - 1 = 13$

98. Ans. (3)

$[NaOH] = [OH^-] = \frac{w \times 1000}{E_w \times V_{ml}} = \frac{0.2 \times 1000}{40 \times 100}$

$= 0.05M = 5 \times 10^{-2}m$

$pOH = -\log[OH^-] = 2 - \log 5$

$pH = 14 - pOH = 14 - (2 - \log 5)$

$= 12.699$

99. Ans. (3)

$pH = 13$

$pOH = 14 - pH = 14 - 13 = 1$

$[OH^-] = 10^{-1}M$

$M = \frac{w \times 1000}{M_w \times V_{ml}}$

$10^{-1} = \frac{w \times 1000}{40 \times 250}$

$w = 1g$

100. Ans. (1)

$M(OH)_2 \rightleftharpoons M^{+2} + 2OH^-$

0.001 – –

– 0.001 $2 \times 0.001 = 0.002$ mole

$[OH^-] = \frac{n}{V_{ml}} \times 1000$

$= \frac{0.002}{20} \times 1000$

$[OH^-] = 0.1$

$pOH = -\log[OH^-] = 1$

$pH = 14 - pOH$

$= 14 - 1 = 13$

101. Ans. (4)

$[H^+] = [OH^-] = 10^{-7}M$ (at all temperature) is wrong statement

102. Ans. (4)

0.1M HCl; pH = 1

0.1M NaCl pH = 7

0.1M NH_4Cl pH < 7

0.1M NaCN pH > 7

So, pH order

$HCl < NH_4Cl < NaCl < NaCN$

103. Ans. (2)

$pOH = pK_b + \log \frac{[Salt]}{[Base]}$

when $\frac{[Salt]}{[Base]} = 1 : 1$

$pOH = pK_b$ ...(1)

when $\frac{[Salt]}{[Base]} = 2 : 1$

$pOH = pK_b + \log 2$ ...(2)

pOH increases by log2 ; so pH of buffer will decrease.

104. Ans. (3)

$pH = pK_a + \log \left[\frac{Salt}{Acid}\right]$

let V_{ml} of sodium formate is added to 50 ml of 0.05M formic acid.

$[Salt] = \frac{V \times 0.1}{[50+V]}$

$[Acid] = \frac{50 \times 0.05}{(50+V)}$

$pH = pK_a + \log\left[\frac{0.1 \times V}{50 \times 0.05}\right]$

$4 = 3.8 + \log\left[\frac{V}{25}\right]$

$\log\left(\frac{V}{25}\right) = 0.2$

$\frac{V}{25} = 10^{0.2}$

$V_{(ml)} = 39.6ml$

105. Ans. (4)

Phenolphthalein as an indicator is used when base is strong. So it is unsuitable for NH_4OH.

106. Ans. (1)

$pH = pK_a + \log \frac{[Salt]}{[Acid]}$

let initially $\frac{[Salt]}{[Acid]} = 1$

$pH = pK_a$

if $\frac{[Salt]}{[Acid]} = \frac{1}{10}$

$pH = pK_a - \log 10$

$pH = pK_a - 1$

So, pH of the solution decrease by one.

107. Ans. (4)

$pOH = pK_b + \log \frac{[Salt]}{[Base]}$

if [salt] = [base]

then

$pOH = pK_b$

$pOH = 4.74 = pK_b$

$pH = 14 - pOH = 14 - 4.74$

$= 9.26$

108. Ans. (3)

For (SB & WA); Phenolphthalein is used as an indicator.

109. Ans. (3)

Phenolphthalein does not act as an indicator for the titration between oxalic acid & $KMnO_4$.

110. Ans. (3)

$NaCN + HCl \longrightarrow HCN + NaCl$

Mili eq. 40×0.1 20×0.1

$= 4$ $= 2$

$4 - 2 = 2$ – 2 2

(HCN + NaCN) is an Acidic buffer

111. Ans. (3)

Buffer solution resists the change in pH.

112. Ans. (3)

let V_{mL} of KCN is required.

Then [KCN] = $\frac{5\times V}{(10+V)}$

[HCN] = $\frac{10\times 2}{(10+V)}$

$pH = pK_a + \log\left[\frac{[KCN]}{[HCN]}\right]$

$9 = (10 - \log 5) + \log\left(\frac{5V}{20}\right)$

$\log\left(\frac{4}{V}\right) = 0.3 = \log(2)$

$\frac{4}{V} = 2$

V = 2mL

113. Ans. (3)

$pH = pK_a + \log\frac{[Salt]}{[Acid]}$ {∵ max pH = pK_a + 1}

pH of buffer is maximum, when $\frac{[Salt]}{[Acid]} = \frac{10}{1}$

114. Ans. (1)

Pink colour of Phenolphthalein in alkaline medium is due to Negative ion, since in Alkaline medium [H^+] is neutrilised by [OH^-] produced by base.

115. Ans. (1)

Phenolphthalein works in pH range 8 – 9.8.

116. Ans. (3)

$pOH - pK_b = 1$

$pOH = 1 + pK_b$

when $\frac{[Salt]}{[Base]} = 10$

or

$\frac{[Conjugate\ Acid]}{[Base]} = \frac{10}{1}$

117. Ans. (4)

For weak acid – strong base titration, phenolphthalein is used.

118. Ans. (3)

Methyl orange is used for titration of strong acid & weak base, also for strong acid & strong base titration.

119. Ans. (1)

H_3PO_4 + NaOH, total 3 buffer are obtained

(a) $H_3PO_4 + NaH_2PO_4$;

(b) $H_3PO_4 + Na_2HPO_4$;

(c) $H_3PO_4 + Na_3PO_4$;

120. Ans. (1)

If K_b of X^- is 10^{-10}

Then pK_a of HX will be $\frac{10^{-14}}{10^{-10}} = 10^{-4}$

$pH = pK_a + \log\frac{[X^-]}{[HX]}$

if $\frac{[X^-]}{[HX]} = 1$

then $pH = pK_a = 4$

121. Ans. (2)

pH of buffer solution does not change if small amount of acid or base is added.

122. Ans. (2)

Buffer action is to maintain pH if small amount of acid or base is added.

123. Ans. (4)

Phenolphthalein is not a good indicator for titration of SA & WB.

124. Ans. (3)

NH_4Cl + HCl does not Act as buffer because HCl is strong acid.

125. Ans. (1)

$NaOH + CH_3COOH \rightleftharpoons CH_3COONa + H_2O$

initial mole – 100

60%

neutralisation – 100 – 60 = 40 60 60

$[CH_3COONa] = 60$

$[CH_3COOH] = 40$

$pH = pK_a + \log\frac{[Salt]}{[Acid]}$

$pH = 4.7 + \log\left(\frac{60}{40}\right)$

pH = 4.7 + 0.18

= 4.88

126. Ans. (1)

$$[CH_3COOH] = \frac{500\times0.2}{[500+500]} = 0.1M$$

$$[CH_3COONa] = \frac{500\times0.3}{[500+500]} = 0.15M$$

$$pH = pK_a + \log\frac{[CH_3COONa]}{[CH_3COOH]}$$

$$pH = 5 - \log(1.5) + \log\left(\frac{0.15}{0.10}\right)$$

$$pH = 5$$

127. Ans. (1)

At half neutralisation [salt] = [acid]

$$pH = pK_a + \log\frac{[salt]}{[acid]}$$

$$pH = 4 - \log2 + \log(1)$$

$$pH = 4 - \log2 = 3.6990$$

128. Ans. (1)

$$pOH = pK_b + \log\frac{[salt]}{[base]}$$

[salt] = 0.2 M, [base] = 0.2 M

$$pOH = pK_b$$

$$[OH^-] = K_b = 2 \times 10^{-5}$$

129. Ans. (2)

$$pH - pK_a = 1$$

$pH = 1 + pK_a$ is applicable when

[salt] = 10[acid]

Or

[Conjugate base] = 10 × [acid]

130. Ans. (2)

$$[OH^-] = \frac{K_b\times[base]}{[salt]}$$

$$= \frac{1.8\times10^{-5}\times0.05}{0.001} = 9 \times 10^{-4}$$

131. Ans. (1)

Buffer capacity

$$= \frac{\text{number of moles of } OH^- \text{ or } H^+}{\text{Change in pH}}$$

$$= \frac{0.02}{0.05} = 0.4$$

132. Ans. (3)

$$pOH = pK_b + \log\frac{[salt]}{[base]}$$

$$pOH = [5 - \log(1.8)] + \log\frac{[N_SV_S]}{[N_BV_B]}$$

$$= 5 - \log(1.8) + \log\left(\frac{500\times0.5}{300\times0.3}\right)$$

$$= 5 - \log(1.8) + \log\left(\frac{5}{3}\right)^2$$

$$= 5 - \log(1.8) + 2[\log5 - \log3]$$

$$= 5 - \log(1.8) + 2(0.7 - 0.48)$$

$$pH = 14 - pOH$$

$$= 14 - 5 + \log(1.8) - 2\,(0.22)$$

$$pH = 8.81$$

133. Ans. (3)

Case- (I)

$CH_3COONa + HCl \rightleftharpoons CH_3COOH + NaCl$

1	1	–	–
–	–	1	1

pH of 1 mole CH_3COOH in 1 L

$$pH = \frac{1}{2}pK_a - \frac{1}{2}\log C$$

$$= \frac{1}{2}pK_a - \frac{1}{2}\log 1$$

$$(pH)_1 = \frac{1}{2}pK_a \qquad \ldots(1)$$

Case-(II)

1 mole CH_3COONa + 1 mole CH_3COOH is a buffer solution.

$$pH = pK_a + \log\frac{[salt]}{[acid]}$$

$$= pK_a + \log 1$$

$$(pH)_2 = pK_a$$

$$\frac{(pH)_1}{(pH)_2} = \frac{\frac{1}{2}pK_a}{pK_a} = 1:2$$

134. Ans. (4)

20 mL $\frac{M}{20}$ NaOH + 10 mL $\frac{M}{10}$ HCl

will give NaCl solution.

which is neutral (pH = 7)

135. Ans. (1)

On adding 1 mL water, total volume will change but concentration of salt & base will remain same. Hence pH would not change.

136. Ans. (1)

$CH_3COOH + NaOH \rightleftharpoons CH_3COONa + H_2O$

	100	–	–	–
(25% titration)	100–25=75	–	25	25
(50% titration)	100–50=50	–	50	50
(75% titration)	100–75=25	–	75	75

At 25% titration

$[CH_3COOH] = 75$ M $\quad [CH_3COONa] = 25$ M

$$pH = pK_a + \log\frac{[CH_3COONa]}{[CH_3COOH]} = pK_a + \log\left(\frac{25}{75}\right)$$

$$= pK_a + \log\left(\frac{1}{3}\right)$$

Similarly at 50% titration

$pH = pK_a + \log(1) = pk_a$

at 75% titration,

$$pH = pK_a + \log\left(\frac{75}{25}\right) = pK_a + \log 3.$$

137. Ans. (4)

$O^{-2} + H^+ \rightarrow OH^-$

Conjugate base of O^{-2} is OH^-

138. Ans. (2)

If $pK_{b_1} < pK_{b_2}$,

Then BOH is weak base so conjugate acid of BOH will be strong & AOH will be weak, so conjugate of BOH does not show maximum pH

139. Ans. (1)

H_2O can act as Lewis base, Bronsted acid & Bronsted base.

140. Ans. (4)

ClO_3^- does not acidic behaviour.

141. Ans. (3)

SO_3 is Lewis acid.

142. Ans. (1)

H_2O act as Bronsted acid.

143. Ans. (1)

PH_3 will act as Lewis base due to presence of lone pair.

144. Ans. (2)

$SnCl_2$ accepts e^- pair so it act as Lewis acid.

145. Ans. (4)

$$(CH_3)_2\overset{\oplus}{N}H_2 \rightarrow (CH_3)_2NH + H^+$$

Acid (Conjugate base)

146. Ans. (4)

$Cu^{+2} + 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{+2}$

This is Lewis acid base reaction but not Bronsted acid base reaction, because there is no H^+ loss or gain.

147. Ans. (2)

$N_3H \rightleftharpoons N_3^- + H^+$

(Hydrazoic acid) (Conjugate base)

148. Ans. (1)

$NH_3 + H_2O \rightleftharpoons NH_4^+OH^-$

In this reaction H_2O act as an acid.

149. Ans. (2)

$NH_4OH \rightleftharpoons NH_4^+ + OH^-$

$H_2O \rightleftharpoons H^+ + OH^-$

Due to common ion effect, equilibrium shift in backward direction, concentration of H^+ will decrease.

150. Ans. (1)

$ClO_3(OH)$ or $HClO_4$ is strongest acid

151. Ans. (2)

CH_3COO^- is not a Bronsted acid.

152. Ans. (4)

BF_3, $SnCl_2$, $SnCl_4$ All three are Lewis acid due to availability of vacant orbitals.

153. Ans. (3)

NH_3^- is conjugate base of HNO_3.

154. Ans. (2)

$HBr + H_2O \rightleftharpoons H_3O^+ + Br^-$

S. Acid S. Base W. Acid W. Base

So, conjugate of weak acid is H_2O.

155. Ans. (4)

$AlCl_3$ act as Lewis acid.

156. Ans. (3)

Mg^{+2} weaker Lewis acid than Al^{+3} because charge on Mg is less than charge on Al.

157. Ans. (3)

$C_2O_4^{-2}$ & PO_4^{-3} are two Bronsted conjugate base of $HC_2O_4^{-2}$ & HPO_4^{-2} respectively.

158. Ans. (4)

$HCl + HF \rightleftharpoons H_2^+Cl + F^-$

W.B W. A S. A W. B

Here HCl act as weak base.

159. Ans. (4)

I^+ being a cation is not a Lewis base.

160. Ans. (2)

H_3O^+ is a Bronsted Lowry acid.

161. Ans. (1)

Conjugate base for Bicarbonate ion HCO_3^- is CO_3^{-2}.

162. Ans. (4)

HCl does not behave as acid in C_6H_6 (non-polar solvent)

163. Ans. (1)

I^- is a base according to Bronsted – Lowry concept.

$I^- + H^+ \rightarrow HI$

Base C. acid

164. Ans. (3)

$NH_3 + Na \rightarrow NaNH_2 + \frac{1}{2}H_2$

in this reaction NH_3 act as acid because it releases H atom to combine with Na.

165. Ans. (3)

$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$

$\downarrow$ $\downarrow$

Acid Acid

H_2O & NH_4^+ are two acid.

166. Ans. (2)

CH_3COO^- ion is a strong conjugate base of CH_3COOH, which is a weak acid itself.

167. Ans. (2)

HCO_3^- is strongest conjugate base of H_2CO_3 which is a weak acid itself.

168. Ans. (3)

NH_3 is Lewis base because it has electron pair to donate.

169. Ans. (3)

NH_2CH_2COOH may behave both as an acid & base because N atom has e^- pair to donate & –COOH group can release H^+.

170. Ans. (4)

NH_3 can act both as Bronsted acid & Bronsted base.

$NH_3 \longrightarrow NH_2^- + H^+$

Acid

$NH_3 + H^+ \longrightarrow NH_4^+$

Base

171. Ans. (4)

CH_3COOH is a weak acid so CH_3COO^- is strongest conjugate base.

172. Ans. (3)

$AlCl_3$ is Lewis acid because it has vacant p-orbital.

173. Ans. (3)

Water is Amphiprotic solvent because it can accept H^+ or donate H^+.

174. Ans. (1)

NH_4^+ is a conjugate acid of NH_3

175. Ans. (2)

Na_2CO_3 can not act as Bronsted acid & base both.

176. Ans. (1)

NH_3 is strong Lewis base, Bronsted acid & Bronsted base.

177. Ans. (1)

Arrhenius theory explain the acidic or basic nature of the substance in water.

178. Ans. (2)

NH_3 & S^{-2} are conjugated base of NH_4^+ & HS^- respectively.

179. Ans. (3)

As per theory

180. Ans. (2)

As per theory

Exercise-II (Previous Year Questions)

1. Ans. (3)

KCl SASB pH = 7

NaCl SASB pH = 7

Na_2CO_3 WASB pH > 7

$CuSO_4$ SAWB pH < 7

2. Ans. (3)

Higher the solubility, ppt will form last.

3. Ans. (3)

CH_3COOH + CH_3COONa

WA + WASB salt

4. Ans. (3)

NaOH $N_BV_B = 0.1 \times 1 \times 1 = 0.1$

HCl $N_AV_A = 0.01 \times 1 \times 1 = 0.01$

$[OH^-] = \frac{0.1-0.01}{1+1} = \frac{.09}{2} = .045$

pOH = 1.35

pH = 12.65

5. Ans. (2)

MY (AB) $K_{sp} = s^2 = 6.2 \times 10^{-13}$

$s = \sqrt{6.2\times10^{-13}}$

$s \simeq 10^{-7}$

NY_3 (AB_3) $K_{sp} = 27S^4 = 6.2 \times 10^{-13}$

$s \simeq 10^{-3}$

6. Ans. (4)

$K_b = C\alpha^2$

7. Ans. (4)

$AgCl \rightleftharpoons Ag^+ + Cl^-$

s s + 0.1

$K_{sp} = (s)(s + 0.1)$

$s << 0.1$

$K_{sp} = s \times 0.1$

$s = \frac{1.6\times10^{-10}}{10^{-1}} = 1.7 \times 10^{-9}$

8. Ans. (3)

$Ag_2CrO_4 \rightleftharpoons 2Ag^+ + CrO_4^{-2}$

2s s

$[Ag^+] = 2s = 2.2 \times 10^{-4}$

$s = 1.1 \times 10^{-4}$

$K_{sp} = 4s^3 = 4(1.1 \times 10^{-4})^3$

$K_{sp} = 5.3 \times 10^{-12}$

9. Ans. (4)

(c) HCl $N_AV_A = \frac{1}{5}\times1\times75 = 15$

NaOH $N_BV_B = \frac{1}{5}\times1\times25 = 5$

$[H^+] = \frac{15-5}{75+25} = \frac{10}{100} = 10^{-1}$

pH = 1

10. Ans. (1)

$BaSO_4$ (AB)

$K_{sp} = s^2 = \left(\frac{2.42\times10^{-5}}{233}\right)^2 \cong 10^{-10}$

11. Ans. (1)

pH = 9 $Ca(OH)_2 \rightleftharpoons Ca^{+2} + 2OH^-$

s 2s

pOH = 5 $[OH^-] = 2s = 10^{-5}$

$s = 5 \times 10^{-6}$

$[OH^-] = 10^{-5}$ $K_{sp} = 4s^3 = 4(5 \times 10^{-6})^3$

$s = 500 \times 10^{-18}$

12. Ans. (3)

	HCl	+ NH_4OH →	NH_4Cl	+ H_2O
Initial	0.1 × 100	200 × 0.3	–	–
	10	60		
Left.	–	50	10	
	Basic	Buffer		

13. Ans. (3)

$[OH^-] = 10^{-2}$

pOH = 2

pH = 12

14. Ans. (3)

HCl → Can donate H^+ ion but not accept.

15. Ans. (3)

$CaF_2 \rightleftharpoons Ca^{+2} + 2F^-$

s 2s + 0.1

$K_{sp} = (s)(2s + 0.1)^2$

$2s << 0.1$

$5.3 \times 10^{-11} = s \times 10^{-2}$

$s = 5.3 \times 10^{-9}$

16. Ans. (2)

$Ni(OH)_2 \rightleftharpoons Ni^{+2} + 2OH^-$

s 2s + 0.1

$K_{sp} = s(2s + 0.1)^2$

$2s << 0.1$

$K_{sp} = s \times 10^{-2} \Rightarrow s = 2 \times 10^{-13}$

17. Ans. (4)

$NH_4Cl \rightarrow$ SAWB Acidic

$(NH_4)_2SO_4 \rightarrow$ SAWB Acidic

$NH_4NO_3 \rightarrow$ SAWB Acidic

$CH_3COONa \rightarrow$ SBWA Basic

18. Ans. (1)

$K_{sp} = s^2$

$s = \sqrt{K_{sp}} = \sqrt{4\times10^{-8}} = 2 \times 10^{-4}$

19. Ans. (3)

$$pH = 7 + \frac{1}{2} pK_a - \frac{1}{2} pK_b$$

20. Ans. (4)

$$pH = pK_a + \log \frac{(M\times V)_{salt}}{(M\times V)_{acid}}$$

$$= 4.57 + \log \left(\frac{\cancel{50}\times 0.1}{0.01\times \cancel{50}} \right)$$

$pH = 5.57$

21. Ans. (3)

$[H^+] = C\alpha$

$$= 0.01 \times \frac{1}{100} = 10^{-4}$$

$pH = 4$

22. Ans. (3)

Sodium hydroxide against oxalic acid titration; phenolphthalein is used as indicator and it's colour changes from colourless to pink.

23. Ans. (1)

$(k_{sp})AgCl = 10^{-10}$

$AgCl_{(s)} \rightleftharpoons Ag^+_{aq} + Cl^-_{aq}$

$k_{sp} = s^2$

$s = (10^{-10})^{1/2} = 10^{-5}$ mol/L

ln 0.1m KCl solution

$AgCl_{(s)} \rightleftharpoons Ag^+_{aq} + Cl^-_{aq}$, Let new solution is s'

$k_{sp} \rightarrow s'(s' + 0.1) = 10^{-10} = s' (s' + 0.1) = s'^2 + 0.1s'$

$$s' = \frac{10+0}{0.1} = 10^{-9} mol/L$$

$$\text{Ratio } \frac{s'}{s} = \frac{10^{-9}}{10^{-5}} = 10^{-4}$$

Exercise-III (Analytical Questions)

1. Ans. (1)

$K_h = h^2$ for hydrolysis of WA + WB type salt.

$h = \sqrt{K_h}$, degree of hydrolysis is independent of concentration of solution.

2. Ans. (2)

pH = 2 for 0.01 M HCl

pH = 12 for 0.01 M NaOH

pH > 7 for 0.01 M CH_3COONa

pH = 7 for 0.01 M NaCl

order : B > C > D > A

3. Ans. (4)

Due to common ion effect, solubility of AgBr decreases in the presence of NaBr. NaBr does not undergoes hydrolysis in water.

4. Ans. (1)

Acid turns blue litmus paper red & acids are sour in taste.

5. Ans. (2)

$NH_3 + H^+ \rightarrow NH_4^+$

Base Conjugate acid

$NH_3 \rightarrow NH_2^- + H^+$

Acid Conjugate base

6. Ans. (3)

Cations of strong base & Anions of strong acid do not get hydrolysed & salts of strong acids & strong bases are neutral.

7. Ans. (1)

Arrhenius acids releases H^+ in water while Bronsted acid releases H^+ in any solvent

So all Arrhenius acids are Bronsted acids but all Bronsted acids are not Arrhenius acid.

8. Ans. (1)

9. Ans. (3)

$Al(OH)_3$	$K_{sp} = (3^3)s^4$	$s = \left(\frac{K_{sp}}{27}\right)^{\frac{1}{4}}$
$BaCrO_4$	$K_{sp} = s^2$	$s = \sqrt{K_{sp}}$
$Zr_3(PO_4)_4$	$K_{sp} = (3)^3 (4)^2 s^7 = 6912\, s^7$	$s = \left(\frac{K_{sp}}{6912}\right)^{\frac{1}{7}}$
Hg_2I_2	$K_{sp} = 4s^3$	$s = \left(\frac{K_{sp}}{4}\right)^{\frac{1}{3}}$

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10. Ans. (2)

Relative strength of acid can be compared by $\frac{[H^+]_1}{[H^+]_2}$ if concentration are same.

11. Ans. (1)

(a) $CH_3COOH + NaOH \rightleftharpoons CH_3COONa + H_2O$ if $[CH_3COOH] > [NaOH]$, after reaction $[CH_3COOH] + [CH_3COONa]$ will remain in the solution, so it can act as buffer.

(b) Similarly $HCl + CH_3COONa$ can act as buffer if $[CH_3COONa] > [HCl]$

(d) NH_4CN is a salt of WA + WB, so it is also a buffer.

12. Ans. (3)

pH of 10^{-8} M solution of HCl is 6.96.

13. Ans. (2)

In this, H_2O accepts H^+, hence it is a base according to Bronsted.

14. Ans. (1)

Ammonium hydroxide is a weak base & pH of NH_4Cl is less than 7 (since NH_4Cl is a salt of WB + SA).

Exercise - I (Conceptual Questions) Build Up Your Understanding

ELECTROLYTIC CONDUCTANCE

1. Strong electrolytes are those which :
(1) Dissolve readily in water
(2) Conduct electricity
(3) Dissociate into ions even at high concentration
(4) Dissociate into ions at high dilution.

CEL001

2. Molten sodium chloride conducts electricity due to the presence of :
(1) Free electrons
(2) Free ions
(3) Free molecules
(4) free atoms of Na and Cl

CEL002

3. Electrolytic conduction is due to the movement of:
(1) Molecules (2) Atoms
(3) Ions (4) Electrons

CEL003

4. Which of the following solutions of KCl has the lowest value of equivalent conductance ?
(1) 1 M (2) 0.1 M
(3) 0.01 M (4) 0.001 M

CEL004

5. If the specific resistance of a solution of concentration C geq L^{-1} is R, then its equivalent conductance is :
(1) $\frac{100R}{C}$ (2) $\frac{RC}{1000}$
(3) $\frac{1000}{RC}$ (4) $\frac{C}{1000R}$

CEL005

6. The specific conductances in ohm^{-1} cm^{-1} of four electrolytes P, Q, R and S are given in brackets:
P (5.0×10^{-5}) Q (7.0×10^{-8})
R (1.0×10^{-10}) S (9.2×10^{-3})
The one that offers highest resistance to the passage of electric current is
(1) P (2) S
(3) R (4) Q

CEL006

7. The specific conductance of a salt of 0.01M concentration is 1.061×10^{-4} S cm^{-1}. Molar conductance of the same solution will be :
(1) 1.061×10^{-4} S cm^2 mol^{-1}
(2) 1.061 S cm^2 mol^{-1}
(3) 10.61 S cm^2 mol^{-1}
(4) 106.1 S cm^2 mol^{-1}

CEL007

8. Which statement is not correct :–
(1) Conductance of an electrolytic solution increases with dilution
(2) Conductance of an electrolytic solution decreases with dilution
(3) Specific conductance of an electrolytic solution decreases with dilution
(4) Equivalent conductance of an electrolytic solution increases with dilution.

CEL008

9. The resistance of 0.01 N solution of an electrolyte was found to be 210 ohm at 298 K using a conductivity cell of cell constant 0.66 cm^{-1}. The equivalent conductance of solution is :–
(1) 314.28 mho cm^2 eq^{-1}
(2) 3.14 mho cm^2 eq^{-1}
(3) 314.28 mho^{-1} cm^2 eq^{-1}
(4) 3.14 mho^{-1} cm^2 eq^{-1}

CEL009

10. Electrolytic conduction differs from metallic conduction from the fact that in the former
(1) The resistance increases with increasing temperature
(2) The resistance decreases with increasing temperature
(3) The resistance remains constant with increasing temperature
(4) The resistance is independent of the length of the conductor

CEL010

11. The specific conductance of a 0.01 M solution of KCl is 0.0014 ohm^{-1} cm^{-1}at 25° C. Its equivalent conductance (cm^2 ohm^{-1} eq^{-1}) is :–

(1) 140 (2) 14 (3) 1.4 (4) 0.14

CEL011

12. Which one of the following is wrong :–

(1) Specific conductance increases on dilution.

(2) Specific conductance decreases on dilution.

(3) Equivalent conductance increases on dilution.

(4) Molar conductance increases on dilution.

CEL012

KOHLRAUSCH LAW

13. At infinite dilution, the equivalent conductances of CH_3COONa, HCl and CH_3COOH are 91, 426 and 391 mho cm^2 eq^{-1} respectively at 25°C. The eq. conductance of NaCl at infinite dilution will be :

(1) 126 (2) 209 (3) 391 (4) 908

CEL013

14. For HCl solution at 25°C, equivalent conductance at infinite dilution is 425 ohm^{-1} cm^2 eq^{-1}. The specific conductance of a solution of HCl is 3.825 ohm^{-1} cm^{-1}. If the degree of dissociation is 90%, the normality of the solution is :-

(1) 0.90 N (2) 1.0 N (3) 10 N (4) 1.2 N

CEL014

15. The molar conductivities $\wedge^0_{NaOAc}$ and $\wedge^0_{HCl}$ at infinite dilution in water at 25°C are 91.0 and 426.2 S cm^2 mol^{-1} respectively. To calculate $\wedge^0_{HOAc}$ the additional value required is :

(1) $\wedge^0_{NaCl}$ (2) $\wedge^0_{H_2O}$

(3) $\wedge^0_{KCl}$ (4) $\wedge^0_{NaOH}$

CEL015

16. The molar conductance of $AgNO_3$, AgCl and NaCl at infinite dilution are 116.5, 121.6 and 110.3 S cm^2 mol^{-1} respectively. The molar conductance of $NaNO_3$ is :

(1) 111.4 S cm^2 mol^{-1} (2) 105.2 S cm^2 mol^{-1}

(3) 130.6 S cm^2 mol^{-1} (4) 150.2 S cm^2 mol^{-1}

CEL016

17. The conductivity of a saturated solution of $BaSO_4$ is 3.06×10^{-6} ohm^{-1} $cm^{-1}$and its molar conductance is 1.53 ohm^{-1} cm^2 mol^{-1}. The K_{sp} of $BaSO_4$ will be

(1) 4×10^{-12} (2) 2.5×10^{-9}

(3) 2.5×10^{-13} (4) 4×10^{-6}

CEL017

18. Kohlrausch's law states that at :-

(1) Infinite dilution, each ion makes definite contribution to conductance of an electrolyte whatever be the nature of the other ion of the electrolyte.

(2) Infinite dilution, each ion makes definite contribution to equivalent conductance of an electrolyte whatever be the nature of the other ion of the electrolyte.

(3) Finite dilution, each ion makes definite contribution to equivalent conductance of an electrolyte whatever be the nature of the other ion of the electrolyte.

(4) Infinite dilution, each ion makes definite contribution to equivalent conductance of an electrolyte depending on the nature of the other ion of electrolyte.

CEL018

GALVANIC CELL

19. In the galvanic cell

$Cu(s)\,|Cu^{2+}\,(1\,M)\,||\,Ag^{+}\,(1\,M)\,|\,Ag(s)$; the electrons will travel in the external circuit :

(1) From Ag to Cu

(2) From Cu to Ag

(3) Electrons do not travel in the external circuit

(4) In any direction

CEL019

20. The direction of current in the Daniell cell when Zn and Cu electrodes are connected is :

(1) From Cu to Zn in the cell

(2) From Cu to Zn outside the cell

(3) From Zn to Cu outside the cell

(4) In any direction in the cell

CEL020

21. The equation representing the process by which standard reduction potential of zinc can be defined is
(1) $Zn^{2+}(s) + 2e^- \longrightarrow Zn(s)$
(2) $Zn(g) \longrightarrow Zn^{2+}(g) + 2e^-$
(3) $Zn^{2+}(g) + 2e^- \longrightarrow Zn(s)$
(4) $Zn^{2+}(aq.) + 2e^- \longrightarrow Zn(s)$
CEL021

22. A standard hydrogen electrode has zero electrode potential because :
(1) Hydrogen is easiest to oxidize.
(2) This electrode potential is assumed to be zero.
(3) Hydrogen atom has only one electron.
(4) Hydrogen is the lightest element.
CEL022

23. Which is not true for a standard hydrogen electrode ?
(1) The hydrogen ion concentration is 1 M.
(2) Temperature is 25°C.
(3) Pressure of hydrogen is 1 bar.
(4) It contains a metallic conductor which does not adsorb hydrogen.
CEL023

24. E° for the half cell
Zn^{2+} | Zn is –0.76 V. E.M.F. of the cell
Zn | Zn^{2+} (1M) || H^+ (1M) | H_2 (1 atm) is :
(1) –0.76 V (2) +0.76 V
(3) –0.38 V (4) + 0.38 V
CEL024

25. $Cu(s)|Cu^{+2}(1\ M)|Zn^{+2}(1\ M)|Zn(s)$
A cell represented above should have emf.
(1) Positive
(2) Negative
(3) Zero
(4) Cannot be predicted
CEL025

26. Given electrode potentials :
$Fe^{3+} + e \longrightarrow Fe^{2+}$; E° = 0.771 V
$I_2 + 2e \longrightarrow 2I^-$; E° = 0.536 V
$E°_{cell}$ for the cell reaction
$2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$ is -
(1) (2 × 0.771 – 0.536) = 1.006 V
(2) (0.771 – 0.5 × 0.536) = 0.503 V
(3) 0.771 – 0.536 = 0.235 V
(4) 0.536 – 0.771 = –0.235 V
CEL026

27. Which of the following is not an anodic half reaction:-
(1) $Ag^+ \rightarrow Ag - e^-$
(2) $Cu \rightarrow Cu^{2+} + 2e^-$
(3) $Fe^{2+} \rightarrow Fe^{3+} + e^-$
(4) $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$
CEL027

28. Which of the following statements is correct :–
(1) Oxidation occurs at anode in both galvanic and electrolytic cell.
(2) Reduction occurs at anode in both galvanic and electrolytic cell
(3) Reduction occurs at anode in electrolytic cell whereas oxidation occurs at cathode in galvanic cell
(4) Oxidation occurs at anode in electrolytic cell whereas reduction occurs at anode in a galvanic cell,
CEL028

29. Other things being equal, the life of a Daniell cell may be increased by :–
(1) Keeping low temperature
(2) Using large copper electrode
(3) Decreasing concentration of copper ions
(4) Using large zinc electrodes
CEL029

ELECTROCHEMICAL SERIES

30. Zn can not displace following ions from their aqueous solution :-
(1) Ag^+ (2) Cu^{2+} (3) Fe^{2+} (4) Na^+
CEL030

31. The standard reduction potentials at 25 °C for the following half reactions are given against each :-
$Zn^{2+}(aq) + 2e^- \rightarrow Zn(s)$, –0.762 V
$Cr^{3+}(aq) + 3e^- \rightarrow Cr(s)$, –0.740 V
$2H^+ + 2e^- \rightarrow H_2(g)$, 0.00 V
$Fe^{3+} + 2e^- \rightarrow Fe^{2+}$, 0.77 V
Which is the strongest reducing agent ?
(1) Zn (2) Cr
(3) $H_2(g)$ (4) $Fe^{2+}(aq)$
CEL031

32. Using the standard electrode potential values given below, decide which of the statements, I, II, III and IV are correct. Choose the right answer from (1), (2), (3) and (4).

$Fe^{2+} + 2e^- \rightarrow Fe$; $E° = -0.44$ V

$Cu^{2+} + 2e^- \rightarrow Cu$; $E° = +0.34$ V

$Ag^+ + e^- \rightarrow Ag$; $E° = +0.80$ V

I. Copper can displace iron from $FeSO_4$ solution.

II. Iron can displace copper from $CuSO_4$ solution.

III. Silver can displace copper from $CuSO_4$ solution.

IV. Iron can displace silver from $AgNO_3$ solution.

(1) I and II (2) II and III

(3) II and IV (4) I and IV

CEL032

33. The standard electrode potentials for the elements A, B and C are 0.68, –2.50 and 0.50 V respectively. The order of their reducing power is :

(1) A > B > C (2) A > C > B

(3) C > B > A (4) B > C > A

CEL033

34. The oxidation potential of Zn, Cu, Ag, H_2 and Ni electrodes are 0.76 V, –0.34 V, –0.80 V, 0 V, 0.55 V respectively. Which of the following reaction will provide maximum voltage ?

(1) $Zn + Cu^{2+} \longrightarrow Cu + Zn^{2+}$

(2) $Zn + 2Ag^+ \longrightarrow 2Ag + Zn^{2+}$

(3) $H_2 + Cu^{2+} \longrightarrow 2H^+ + Cu$

(4) $H_2 + Ni^{2+} \longrightarrow 2H^+ + Ni$

CEL034

35. The standard reduction potential at 25°C of Li^+/Li, Ba^{2+}/Ba, Na^+/Na and Mg^{2+}/Mg are –3.05V, –2.73V, –2.71V and –2.37V respectively. Which one of the following is the strongest oxidising agent ?

(1) Na^+ (2) Li^+

(3) Ba^{2+} (4) Mg^{2+}

CEL035

36. A gas X at 1 atm is bubbled through a solution containing a mixture of 1 M Y^- and 1 M Z^- at 25°C. If the reduction potential of $(Z/Z^-) > (Y/Y^-) > (X/X^-)$ then :

(1) Y will oxidise X^- and not Z^-

(2) Y will oxidise Z^- and not X^-

(3) Y will oxidise both X^- and Z^-

(4) Y will reduce both X^- and Z^-

CEL036

37. The standard electrode potential of Zn, Ag and Cu electrodes are –0.76 V, 0.80 V and 0.34 V respectively, then :

(1) Ag can oxidise Zn and Cu

(2) Ag can reduce Zn^{2+} and Cu^{2+}

(3) Zn can reduce Ag^+ and Cu^{2+}

(4) Cu can oxidise Zn and Ag

CEL037

38. Standard reduction potentials of four metal electrodes are

A = – 0.250 V, B = – 0.140 V

C = – 0.126 V, D = – 0.402 V

The metal that displaces A from aqueous solution of its compounds is :-

(1) B (2) C

(3) D (4) None of the above

CEL038

39. The following four colourless salt solutions are placed in separate test tubes and a strip of Cu is placed in each solution. Which solution finally turns blue :–

(1) $Zn(NO_3)_2$ (2) $Mg(NO_3)_2$

(3) KNO_3 (4) $AgNO_3$

CEL039

40. Which of the following displacement does not occur

(1) $Zn + 2H^+ \rightarrow Zn^{2+} + H_2 \uparrow$

(2) $Fe + 2Ag^+ \rightarrow Fe^{2+} + Ag \downarrow$

(3) $Cu + Fe^{2+} \rightarrow Cu^{2+} + Fe \downarrow$

(4) $Zn + Pb^{2+} \rightarrow Zn^{2+} + Pb \downarrow$

CEL040

41. On the basis of the following E° values, the strongest oxidizing agent is :-
$[Fe(CN)_6]^{4-} \rightarrow [Fe(CN)_6]^{3-} + e^-$, E° = –0.35 V
$Fe^{2+} \rightarrow Fe^{3+} + e^-$; E° = –0.77 V
(1) Fe^{3+} (2) $[Fe(CN)_6]^{3-}$
(3) $[Fe(CN)_6]^{4-}$ (4) Fe^{2+}

CEL041

NERNST EQUATION

42. $E°(Ni^{2+}/Ni)$ = –0.25 V
$E°(Au^{3+} / Au)$ = 1.50 V
The emf of the voltaic cell.
$Ni \mid Ni^{2+}$ (1.0 M) || Au^{3+} (1.0 M) | Au is :
(1) 1.25 V (2) –1.75 V
(3) 1.75 V (4) 4.0 V

CEL042

43. The emf of the cell Tl(s) | Tl^+ (0.0001 M) || Cu^{2+} (0.01M) | Cu(s) is 0.83 V. The emf of this cell will be increased by :–
(1) Increasing the concentration of Cu^{+2} ions
(2) Decreasing the concentration of Tl^+
(3) Increasing the concentration of both
(4) (1) & (2) both

CEL043

44. Which of the following represents the electrode potential of silver electrode dipped into 0.1 M $AgNO_3$ solution at 25° C ?
(1) $E°_{red}$ (2) $(E°_{red} + 0.059)$
(3) $(E°_{ox} - 0.059)$ (4) $(E°_{red} - 0.059)$

CEL044

45. The electrode potential of a hydrogen electrode dipped in solution of pH = 1 is
(1) 0.059 V (2) 0.00 V
(3) –0.059 V (4) 0.59 V

CEL045

46. Consider the reaction
$Cl_2(g) + 2Br^-(aq) \longrightarrow 2Cl^-(aq) + Br_2(\ell)$
The emf of the cell when
$[Cl^-] = [Br^-]$ = 0.01 M and Cl_2 gas at 0.25 atm pressure will be (E° for the above reaction is = 0.29 V)
(1) 0.54 V (2) 0.272 V
(3) 0.29 V (4) –0.29 V

CEL046

47. The standard emf for the cell reaction
$Zn + Cu^{2+} \longrightarrow Zn^{2+} + Cu$ is 1.10 V at 25°C. The emf for the cell reaction when 0.1 M Cu^{2+} and 0.1 M Zn^{2+} solution are used at 25°C is :
(1) 1.10 V (2) 0.110 V
(3) –1.10 V (4) –0.110 V

CEL047

48. E° for $F_2 + 2e^- \rightarrow 2F^-$ is 2.8 V,
E° for $\frac{1}{2}F_2 + e^- \rightarrow F^-$ is ?
(1) 2.8 V (2) 1.4 V
(3) –2.8 V (4) –1.4 V

CEL048

49. How much will the potential of Zn | Zn^{2+} change if the solution of Zn^{2+} is diluted 10 times
(1) Increases by 0.03 V
(2) Decreases by 0.03 V
(3) Increases by 0.059 V
(4) Decreases by 0.059 V

CEL049

50. How much will the potential of a hydrogen electrode change when its solution initially at pH = 0 is neutralised to pH = 7 ?
(1) Increases by 0.059 V
(2) Decreases by 0.059 V
(3) Increases by 0.41 V
(4) Decreases by 0.41 V

CEL050

51. Which of the following will increase the voltage of the cell with following cell reaction
$Sn(s) + 2Ag^+(aq) \rightarrow Sn^{+2}(aq) + 2Ag(s)$
(1) Increase in the size of silver rod
(2) Increase in the concentration of Sn^{+2} ions
(3) Increase in the concentration of Ag^+ ions
(4) Decrease in the concentration of Ag^+ ions

CEL051

52. E° for the reaction $Fe + Zn^{2+} \rightarrow Zn + Fe^{2+}$ is –0.35 V. The given cell reaction is :
(1) Spontaneous
(2) Non-spontaneous
(3) In equilibrium
(4) Can't say anything

CEL052

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53. For a reaction – $A(s) + 2B^+ \rightleftharpoons A^{2+} + 2B(s)$; K_C has been found to be 10^{12}. The $E°_{cell}$ is:
(1) 0.354 V (2) 0.708 V
(3) 0.0098 V (4) 1.36 V
CEL053

54. The standard electrode potential (E°) for OCl^-/Cl^- and $Cl^- / ½\ Cl_2$ respectively are 0.94 V and –1.36 V. The E° value of $OCl^- / ½\ Cl_2$ will be :
(1) –2.20 V (2) –0.42 V
(3) 0.52 V (4) 1.04 V
CEL054

55. The hydrogen electrode is dipped in a solution of pH = 3 at 25°C. The electrode potential of the half cell would be :
(1) 0.177 V (2) – 0.177 V
(3) 0.087 V (4) 0.059 V
CEL055

56. What is the potential of the cell containing two hydrogen electrodes as represented below
$Pt, H_2(g) \mid H^+(10^{-8}\ M) \parallel H^+(0.001\ M) \mid H_2(g), Pt;$
(1) – 0.295 V (2) – 0.0591 V
(3) 0.295 V (4) 0.0591 V
CEL056

57. Consider the cell $Cu|Cu^{+2} \parallel Ag^+|Ag$. If the concentration of Cu^{+2} and Ag^+ ions becomes ten times, then the emf of the cell will :–
(1) Becomes 10 times
(2) Remains same
(3) Increases by 0.0295 V
(4) Decreases by 0.0295 V
CEL057

58. The emf of the cell
$Ni|Ni^{+2}\ (1.0\ M) \mid\mid Au^{+3}\ (0.1M) |Au$
[$E°$ for $Ni^{+2}|Ni = –0.25$ V,
$E°$ for $Au^{+3}|Au = 1.50$ V] is given as:-
(1) 1.25 V (2) – 1.75 V
(3) 1.78 V (4) 1.73 V
CEL058

59. The equilibrium constant (in approx) of the cell reaction :
$Cu(s) + 2Ag^+(aq.) \rightleftharpoons Cu^{+2}(aq.) + 2Ag(s)$
if $E°_{cell} = 0.465$ V at 298 K is :-
(1) 2.0×10^{10} (2) 3.16×10^{12}
(3) 3.16×10^{15} (4) 4×10^{10}
CEL059

60. The emf of the cell in which the following reaction
$Zn(s) + Ni^{2+}\ (a = 0.1) \rightleftharpoons Zn^{2+}(a = 1.0) + Ni(s)$
occurs, is found to be 0.5105 V at 298 K. The standard e.m.f. of the cell is :-
(1) –0.5105 V (2) 0.5400 V
(3) 0.4810 V (4) 0.5696 V
CEL060

61. If $E^o_{Fe^{+2}/Fe}$ = – 0.441V and $E^o_{Fe^{+3}/Fe^{+2}}$ = 0.771V the standard EMF of the reaction $Fe + 2Fe^{+3} \rightarrow 3Fe^{+2}$ will be :
(1) 0.330 V (2) 1.653 V
(3) 1.212 V (4) 0.111 V
CEL061

ELECTROLYSIS

62. When an electric current is passed through acidified water, 112 mL of hydrogen gas at STP collects at the cathode in 965 s. The current passed in ampere is :
(1) 1.0 (2) 0.5 (3) 0.1 (4) 2.0
CEL062

63. Two electrolytic cells one containing acidified ferrous chloride and another acidified ferric chloride are connected in series. The ratio of iron deposited at cathodes in the two cells when electricity is passed through the cells will be :
(1) 3 :1 (2) 2 : 1 (3) 1 : 1 (4) 3 : 2
CEL063

64. A current of 9.65 A flowing for 10 minute deposits 3.0 g of a metal. The equivalent weight of the metal is :
(1) 10 (2) 30 (3) 50 (4) 96.5
CEL064

65. How many coulombs of electric charge are required for the oxidation of 1 mol of H_2O to O_2 ?
(1) 9.65×10^4 C (2) 4.825×10^5 C
(3) 1.93×10^5 C (4) 1.93×10^4 C
CEL065

66. On passing 10800 C through electrolytic solution, 2.977 g of metal (atomic mass 106.4 g mol^{-1}) was deposited, the charge on the metal cation is -
(1) +4 (2) +3 (3) +2 (4) +1
CEL066

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67. On passing electricity through dilute H_2SO_4 solution the amount of substance liberated at the cathode and anode are in the ratio :
(1) 1 : 8 (2) 8 : 1
(3) 16 : 1 (4) 1 : 16
CEL067

68. During electrolysis of fused calcium hydride, the hydrogen is produced at :
(1) Cathode
(2) Anode
(3) Hydrogen is not liberated at all
(4) H_2 produced reacts with oxygen to form water
CEL068

69. A silver cup is plated with silver by passing 965 A current for one second, the mass of Ag deposited is :– (At. wt. of Ag = 107.87)
(1) 9.89 g. (2) 107.87 g.
(3) 1.0787 g. (4) 100.2 g.
CEL069

70. When electricity is passed through molten $AlCl_3$, 13.5 g Al is deposited. The number of Faradays must be :–
(1) 5.0 (2) 1.0 (3) 1.5 (4) 3.0
CEL070

71. A solution of sodium sulphate in water is electrolysed using inert electrodes. The product at the cathode and anode are respectively :–
(1) H_2, SO_2 (2) O_2, H_2
(3) O_2, Na (4) H_2, O_2
CEL071

72. One Faraday of electricity will liberate one mole of the metal from the solution of
(1) Auric chloride ($AuCl_3$)
(2) Silver nitrate
(3) Calcium chloride
(4) Copper sulphate
CEL072

73. When 96500 C of electricity are passed through molten barium chloride solution, the amount of barium deposited will be :-
(1) 0.5 mol (2) 1.0 mol
(3) 1.5 mol (4) 2.0 mol
CEL073

74. A factory produces 40 kg of calcium in two hours by electrolysis. How much aluminium can be produced by the same current in two hours :–
(1) 22 kg (2) 18 kg
(3) 9 kg (4) 27 kg
CEL074

75. What would be the ratio of moles of Ag, Cu, Fe that would be deposited by passage of same quantity of electricity through solutions of salts containing Ag^+, Cu^{+2}, Fe^{+3} :-
(1) 1 : 1 : 1 (2) $1 : \frac{1}{2} : \frac{1}{3}$
(3) $\frac{1}{3} : \frac{1}{2} : 1$ (4) 1 : 2 : 3
CEL075

76. Electrolysis of aq. $CuSO_4$ causes :–
(1) An increase in pH
(2) A decrease in pH
(3) Either decrease or increase in pH
(4) None
CEL076

77. The passage of current liberates H_2 at cathode and Cl_2 at anode. The solution is :–
(1) $CuSO_4$ (aq) (2) $CuCl_2$ (aq.)
(3) NaCl (aq.) (4) Water
CEL077

COMMERCIAL CELLS

78. When lead accumulator is charged, it acts as :
(1) an electrolytic cell
(2) a galvanic cell
(3) a Daniel cell
(4) none of the above
CEL078

79. The thermodynamic efficiency of cell is given by -
(1) $\frac{\Delta H}{\Delta G} \times 100$ (2) $\frac{nFE}{\Delta G} \times 100$
(3) $-\frac{nFE}{\Delta H} \times 100$ (4) Zero
CEL079

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EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	3	1	3	3	3	2	1	2	1	1	1	3	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	4	2	2	2	4	2	4	2	2	3	1	1	4	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	3	4	2	4	1	3	3	4	3	1	3	4	4	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	1	1	1	4	3	2	1	3	2	3	3	4	3	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	3	1	4	3	3	1	1	2	3	3	4	2	1	2	2
Question	76	77	78	79											
Answer	2	3	1	3											

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. When 0.1 mol MnO_4^{2-} is oxidised the quantity of electricity required to completely oxidise MnO_4^{2-} to MnO_4^- is :-
(1) 96500 C (2) 2 × 96500 C
(3) 9650 C (4) 96.50 C
CEL097

2. The weight of silver (at wt. = 108) displaced by a quantity of electricity which displaces 5600 mL of O_2 at STP will be :-
(1) 5.4 g (2) 10.8 g
(3) 54.0 g (4) 108.0 g
CEL098

AIPMT 2015

3. A device that converts energy of combustion of fuels like hydrogen and methane, directly into electrical energy is known as :-
(1) Electrolytic cell (2) Dynamo
(3) Ni-Cd cell (4) Fuel Cell
CEL099

NEET-I 2016

4. The pressure of H_2 required to make the potential of H_2-electrode zero in pure water at 298 K is :-
(1) 10^{-14} atm (2) 10^{-12} atm
(3) 10^{-10} atm (4) 10^{-4} atm
CEL100

NEET-II 2016

5. The molar conductivity of a 0.5 mol/dm^3 solution of $AgNO_3$ with electrolytic conductivity of 5.76×10^{-3} S cm^{-1} at 298 K is
(1) 0.086 S cm^2/mol (2) 28.8 S cm^2/mol
(3) 2.88 S cm^2/mol (4) 11.52 S cm^2/mol
CEL101

6. During the electrolysis of molten sodium chloride, the time required to produce 0.10 mol of chlorine gas using a current of 3 A is
(1) 220 minutes (2) 330 minutes
(3) 55 minutes (4) 110 minutes
CEL102

7. If the $E°_{cell}$ for a given reaction has a negative value, which of the following gives the **correct** relationships for the values of $\Delta G°$ and K_{eq} ?
(1) $\Delta G° < 0$; $K_{eq} > 1$ (2) $\Delta G° < 0$; $K_{eq} < 1$
(3) $\Delta G° > 0$; $K_{eq} < 1$ (4) $\Delta G° > 0$; $K_{eq} > 1$
CEL103

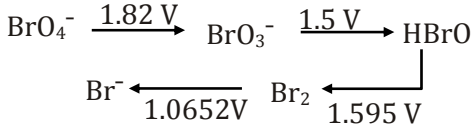
8. The number of electrons delivered at the cathode during electrolysis by a current of 1 A in 60 s is (charge on electron = 1.60×10^{-19} C)
(1) 3.75×10^{20} (2) 7.48×10^{23}
(3) 6×10^{23} (4) 6×10^{20}
CEL104

NEET(UG) 2017

9. In the electrochemical cell :-
Zn | $ZnSO_4$(0.01M) || $CuSO_4$(1.0 M) | Cu, the emf of this Daniel cell is E_1. When the concentration of $ZnSO_4$ is changed to 1.0M and that of $CuSO_4$ changed to 0.01M, the emf changes to E_2.
Which one of the relationship is correct between E_1 and E_2? (Given, $\frac{RT}{F} = 0.059$)
(1) $E_1 < E_2$ (2) $E_1 > E_2$
(3) $E_2 = 0 \neq E_1$ (4) $E_1 = E_2$ **CEL105**

NEET(UG) 2018

10. Consider the change in oxidation state of Bromine corresponding to different EMF values as shown in the diagram below:



Then the species undergoing disproportionation is:-
(1) BrO_3^- (2) BrO_4^- (3) Br_2 (4) HBrO
CEL106

NEET(UG) 2019

11. For a cell involving one electron $E°_{cell} = 0.59$V at 298 K, the equilibrium constant for the cell reaction is :-
$\left[\text{Given that } \frac{2.303RT}{F} = 0.059\text{V at T} = 298\text{K}\right]$
(1) 1.0×10^2 (2) 1.0×10^5
(3) 1.0×10^{10} (4) 1.0×10^{30}
CEL107

12. For the cell reaction
$2Fe^{3+}(aq) + 2I^-(aq) \rightarrow 2Fe^{2+}(aq) + I_2(aq)$
$E°_{cell} = 0.24$V at 298 K. The standard Gibbs energy $(\Delta_r G°)$ of the cell reaction is :
[Given that Faraday constant F = 96500 C mol^{-1}]
(1) – 46.32 kJ mol^{-1} (2) – 23.16 kJ mol^{-1}
(3) 46.32 kJ mol^{-1} (4) 23.16 kJ mol^{-1}
CEL108

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NEET(UG) (Odisha) 2019

13. Following limiting molar conductivities are given as $\lambda^0_{m(H_2SO_4)} = x\ S cm^2\ mol^{-1}$

$\lambda^0_{m(K_2SO_4)} = y\ S cm^2\ mol^{-1}$

$\lambda^0_{m(CH_3COOK)} = z\ S cm^2\ mol^{-1}$

λ^0_m (in S cm^2 mol^{-1}) for CH_3COOH will be-

(1) x – y + 2 z (2) x + y – z

(3) x – y + z (4) $\frac{(x-y)}{2}+z$

CEL109

14. The standard electrode potential (E°) values of $Al^{3+}|Al$, $Ag^+|Ag$, $K^+|K$ and $Cr^{3+}|Cr$ are –1.66 V, 0.80 V, –2.93 V and –0.74 V respectively. The correct decreasing order of reducing power of the metal is :

(1) Ag > Cr > Al > K (2) K > Al > Cr > Ag

(3) K > Al > Ag > Cr (4) Al > K > Ag > Cr

CEL110

NEET (UG) 2020

15. The number of Faradays(F) required to produce 20 g of calcium from molten $CaCl_2$ (Atomic mass of Ca = 40 g mol^{-1}) is :

(1) 4 (2) 1 (3) 2 (4) 3

CEL111

16. On electrolysis of dil. sulphuric acid using Platinum (Pt) electrode, the product obtained at anode will be:

(1) SO_2 gas (2) Hydrogen gas

(3) Oxygen gas (4) H_2S gas

CEL112

NEET (UG) 2020 (COVID-19)

17. Identify the reaction from following having top position in EMF series (Std. red. potential) according to their electrode potential at 298 K.

(1) $Mg^{2+} + 2e^- \rightarrow Mg(s)$ (2) $Fe^{2+} + 2e^- \rightarrow Fe(s)$

(3) $Au^{3+} + 3e^- \rightarrow Au(s)$ (4) $K^+ + 1e^- \rightarrow K(s)$

CEL113

18. In a typical fuel cell, the reactants (R) and product (P) are :-

(1) R = $H_{2(g)}$, $O_{2(g)}$; P = $H_2O_{2(\ell)}$

(2) R = $H_{2(g)}$, $O_{2(g)}$; P = $H_2O_{(\ell)}$

(3) R = $H_{2(g)}$, $O_{2(g)}$, $Cl_{2(g)}$; P = $HClO_{4(aq)}$

(4) R = $H_{2(g)}$, $N_{2(g)}$; P = $NH_{3(aq)}$

CEL114

NEET (UG) 2021

19. The molar conductance of NaCl, HCl and CH_3COONa at infinite dilution are 126.45, 426.16 and 91.0 S cm^2 mol^{-1} respectively. The molar conductance of CH_3COOH at infinite dilution is. Choose the right option for your answer.

(1) 201.28 S cm^2 mol^{-1} (2) 390.71 S cm^2 mol^{-1}

(3) 698.28 S cm^2 mol^{-1} (4) 540.48 S cm^2 mol^{-1}

CEL115

20. The molar conductivity of 0.007 M acetic acid is 20 S cm^2 mol^{-1}. What is the dissociation constant of acetic acid ? Choose the correct option.

$$\left[\begin{array}{l}\Lambda^o_{H^+} = 350\ S cm^2 mol^{-1} \\ \Lambda^o_{CH_3COO^-} = 50\ S cm^2 mol^{-1}\end{array}\right]$$

(1) 1.75×10^{-4} mol L^{-1} (2) 2.50×10^{-4} mol L^{-1}

(3) 1.75×10^{-5} mol L^{-1} (4) 2.50×10^{-5} mol L^{-1}

CEL116

NEET (UG) 2022

21. At 298 K, the standard electrode potentials of Cu^{2+}/Cu, Zn^{2+}/Zn, Fe^{2+}/Fe and Ag^+/Ag are 0.34V, –0.76V, –0.44V and 0.80V, respectively. On the basis of standard electrode potential, predict which of the following reaction can not occur ?

(1) $CuSO_4(aq) + Fe(s) \rightarrow FeSO_4(aq) + Cu(s)$

(2) $FeSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Fe(s)$

(3) $2CuSO_4(aq) + 2Ag(s) \rightarrow 2Cu(s) + Ag_2SO_4(aq)$

(4) $CuSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Cu(s)$

CEL117

22. Given below are half cell reactions :

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$,

$E^\circ_{Mn^{2+}/MnO_4^-} = -1.510V$

$\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$, $E^\circ_{O_2/H_2O} = +1.223V$

Will the permanganate ion, MnO_4^- liberate O_2 from water in the presence of an acid ?

(1) No, because E°_{cell} = –0.287 V

(2) Yes, because E°_{cell} = +2.733 V

(3) No, because E°_{cell} = –2.733 V

(4) Yes, because E°_{cell} = +0.287 V

CEL118

23. Two half cell reactions are given below :

$Co^{3+} + e^- \rightarrow Co^{2+}$, $E^o_{Co^{2+}/Co^{3+}} = -1.81\ V$

$2Al^{3+} + 6e^- \rightarrow 2\ Al(s)$, $E^o_{Al/Al^{3+}} = +1.66\ V$

The standard EMF of a cell with feasible redox reaction will be :

(1) +7.09 V (2) +0.15 V

(3) +3.47 V (4) –3.47 V

CEL119

24. Standard electrode potential for the cell with cell reaction
$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$ is 1.1 V. Calculate the standard gibbs energy change for the cell reaction. (Given F = 96487 C mol^{-1})
(1) –200.27 kJ mol^{-1} (2) –212.27 kJ mol^{-1}
(3) –212.27 J mol^{-1} (4) –200.27 J mol^{-1}

CEL120

NEET (UG) 2023

25. The conductivity of centimolar solution of KCl at 25°C is 0.0210 ohm^{-1} cm^{-1} and the resistance of the cell containing the solution at 25°C is 60 ohm. The value of cell constant is-
(1) 3.28 cm^{-1}
(2) 1.26 cm^{-1}
(3) 3.34 cm^{-1}
(4) 1.34 cm^{-1}

CEL141

26. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R :**
Assertion A : In equation $\Delta_r G = -nFE_{cell}$, value of $\Delta_r G$ depends on n.
Reasons R : E_{cell} is an intensive property and $\Delta_r G$ is an extensive property.
In the light of the above statements, choose the **correct** answer from the options given below :
(1) Both **A** and **R** are true and **R** is **NOT** the correct explanation of **A**.
(2) **A** is true but **R** is false
(3) **A** is false but R is true
(4) Both **A** and **R** are true and **R** is the correct explanation of **A.**

CEL142

NEET (UG) 2024

27. Match List-I with List-II.

List-I (Conversion)	List-II (Number of Faraday required)
(A) 1 mol of H_2O to O_2	(I) 3F
(B) 1 mol of MnO_4^- to Mn^{2+}	(II) 2F
(C) 1.5 mole of Ca from molten $CaCl_2$	(III) 1F
(D) 1 mol of FeO to Fe_2O_3	(IV) 5F

(1) A-II, B-IV, C-I, D-III (2) A-III, B-IV, C-I, D-II
(3) A-II, B-III, C-I, D-IV (4) A-III, B-IV, C-II, D-I

CEL159

28. Mass in grams of copper deposited by passing 9.6487 A current through a voltmeter containing copper sulphate solution for 100 seconds is:
(Given : Molar mass of Cu : 63 g mol^{-1}, 1F = 96487 C)
(1) 3.15 g (2) 0.315 g
(3) 31.5 g (4) 0.0315 g

CEL160

NEET (UG) 2024 (Re-examination)

29. From the following select the one which is **not** an example of corrosion.
(1) Rusting of iron object
(2) Production of hydrogen by electrolysis of water
(3) Tarnishing of silver
(4) Development of green coating on copper and bronze ornaments

CEL161

30. The standard cell potential of the following cell
$Zn|Zn^{2+}(aq)||Fe^{2+}(aq)|Fe$
is 0.32 V. Calculate the standard Gibbs energy change for the reaction :
$Zn(s) + Fe^{2+}(aq) \rightarrow Zn^{2+}(aq) + Fe(s)$
(Given : 1 F = 96487C)
(1) –61.75 kJ mol^{-1} (2) +5.006 kJ mol^{-1}
(3) –5.006 kJ mol^{-1} (4) +61.75 kJ mol^{-1}

CEL162

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	4	1	4	4	3	1	2	4	3	1	4	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	2	2	3	3	4	3	2	2	4	1	2	2	1

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Exercise - III (Analytical Questions) Master Your Understanding

1. Given below are two statements:
Statement-I : In electrochemical cells, Gibbs energy of spontaneous redox reaction is converted into electrical work.
Statement-II : Gibb's energy can be calculated by using $\Delta G = -nFE$ & it is an intensive property.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement-I and II are incorrect.
(2) Both statement-I and II are correct.
(3) Statement-I is incorrect but statement-II is correct.
(4) Statement-I is correct but statement-II is incorrect

CEL121

2. Given below are two statements:
Statement-I : Sodium can be obtained by the electrolysis of aqueous solution NaCl using platinum electrodes.
Statement-II : Copper metal can not reduce iron(II) ions in acidic solutions.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement-I and II are incorrect.
(2) Both statement-I and II are correct.
(3) Statement-I is incorrect but statement-II is correct.
(4) Statement-I is correct but statement-II is incorrect

CEL124

3. Given below are two statements:
Statement-I : The cell potential becomes half if the cell reaction is divided by 2 throughout.
Statement-II : Cell voltage is an extensive quantity.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement-I and II are incorrect.
(2) Both statement-I and II are correct.
(3) Statement-I is incorrect but statement-II is correct.
(4) Statement-I is correct but statement-II is incorrect

CEL128

4. Given below are two statements:
Statement-I : Cathode is negative terminal in an electrolytic cell where as anode is negative terminal in a galvanic cell.
Statement-II : In a Daniel cell, electrons flow from zinc electrode to copper electrode in outer circuit.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement-I and II are incorrect.
(2) Both statement-I and II are correct.
(3) Statement-I is incorrect but statement-II is correct.
(4) Statement-I is correct but statement-II is incorrect

CEL129

5. Given below are two statements:
Statement-I : The electrolysis of molten sodium hydride liberates H_2 gas at the anode.
Statement-II : In electrolytic refining of an impure metal, the pure metal is obtained at anode.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement-I and II are incorrect.
(2) Statement-I is incorrect but statement-II is correct.
(3) Statement-I is correct but statement-II is incorrect
(4) Both statement-I and II are correct.

CEL130

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6. Given below are two statements. Identify the correct choice from the following.

Statement-I : Molar conductivity increases with decreases in concentration for weak electrolytes.

Statement-II : Number of ions increases and number of ions per unit volume decreases due to dilution.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I is correct and Statement-II is correct and is the correct explanation of Statement-I.

(2) Statement-I is correct and Statement-II is correct but not the correct explanation of Statement-I.

(3) Statement-I is correct and Statement-II is incorrect.

(4) Statement-I is incorrect and Statement-II is correct.

CEL132

7. Given below are two statements. Identify the correct choice from the following.

Statement-I : Increasing the concentration of Cu^{+2} ions increases the emf of Daniel cell.

Statement-II : According to the Nernst equation

$$\text{EMF of cell} = E^0_{Cell} + \frac{0.059}{2}\log\frac{[Cu^{2+}]}{[Zn^{2+}]}$$

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I is correct and Statement-II is correct and is the correct explanation of Statement-I.

(2) Statement-I is correct and Statement-II is correct but not the correct explanation of Statement-I.

(3) Statement-I is correct and Statement-II is incorrect.

(4) Statement-I is incorrect and Statement-II is correct.

CEL133

8. Given below are two statements. Identify the correct choice from the following.

Statement-I : The cell constant of a cell depends upon the nature of material of the electrodes.

Statement-II : The observed conductance depends upon the nature of the electrolyte and the concentration of the solution.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Statement-I is correct and Statement-II is correct and is the correct explanation of Statement-I.

(2) Statement-I is correct and Statement-II is correct but not the correct explanation of Statement-I.

(3) Statement-I is correct and Statement-II is incorrect.

(4) Statement-I is incorrect and Statement-II is correct.

CEL134

9. Given :

$E^0_{Ag^+|Ag} = 0.80V$, $E^0_{Mg^{2+}|Mg} = -2.73V$,

$E^0_{Cu^{2+}|Cu} = 0.34V$, $E^0_{Hg^{2+}|Hg} = 0.79V$

Which of the following statements are correct.

(a) $AgNO_3$ can be stored in copper vessel.

(b) $Mg(NO_3)_2$ can be stored in copper vessel.

(c) $CuCl_2$ can be stored in silver vessel.

(d) $HgCl_2$ can be stored in copper vessel.

(1) a, b are correct

(2) b, c are correct

(3) a, b, d are correct

(4) a, d are correct

CEL138

10. Any redox reaction would occur spontaneously, if

(a) the free energy change (ΔG) is negative.

(b) the ΔG° is positive.

(c) the cell EMF is negative.

(d) the cell EMF is positive.

(1) a, b are correct

(2) b, c, d are correct

(3) a, d are correct

(4) a, b & c are correct

CEL139

11. Mark out the correct statement(s) regarding electrolytic molar conductivity.

(a) It increases as temperature increases.

(b) It experiences resistance due to vibration of ion at the mean position.

(c) It decreases as concentration increases.

(d) It increases with increase in polarity of solvent.

(1) a, b, c, d are correct

(2) a, b, c are correct

(3) a, c, d are correct

(4) a, b, d are correct

CEL140

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	4	3	1	2	3	2	1	4	2	3	3

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Electrochemistry

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (3)

Strong electrolytes are those for which degree of dissociation (α) is 100% at any concentration.

2. Ans. (2)

When a salt like NaCl is molten, in molten state its ions becomes mobile and those mobile ions are charge carrier which are responsible for conduction of electricity.

3. Ans. (3)

In electrolytic conduction charge carrier are mobile ions due to which conduction occurs.

4. Ans. (1)

Equivalent conductance $\left(\lambda_{eq.}\right)=\frac{\kappa\times1000}{N}$

For KCl v.f. = 1, So N = M

$$\lambda_{eq}\propto\frac{1}{N}$$

$N\uparrow\Rightarrow\lambda_{eq}\downarrow$ or $M\uparrow\Rightarrow\lambda_{eq}\downarrow$

5. Ans. (3)

Given : Specific resistance (ρ) = R

Concentration (N) = C g-eq L^{-1}

Now, Specific conductance (κ) = $\frac{1}{\rho}=\frac{1}{R}$

Equivalent conductance $(\lambda_{eq})=\frac{\kappa\times1000}{N}$

$$=\frac{\frac{1}{R}\times1000}{C}$$

$$=\frac{1000}{R.C}$$

6. Ans. (3)

Specific conductance $(\kappa)=\frac{1}{\rho}$

$\kappa\downarrow\Rightarrow\ \rho\uparrow$

$R=\rho.\frac{\ell}{A}$

$\kappa\downarrow\Rightarrow\rho\uparrow\Rightarrow R\uparrow$

7. Ans. (3)

Given :

Specific conductance (κ) = $1.061\times10^{-4}5cm^{-1}$

Now Molar conductance $(\lambda_m)=\frac{\kappa\times1000}{M}$

$$=\frac{1.061\times10^{-4}\times10^{3}}{0.01}$$

= 10.61 S cm^2mol^{-1}

8. Ans. (2)

With dilution, inter ionic interaction decreases so conductance increase.

9. Ans. (1)

Given : Concentration (N) = 0.01 N

Resistance (R) = 210 Ω

Cell constant (G*) = $\frac{\ell}{A}=0.66cm^{-1}$

Now $G=\frac{1}{R}=\frac{1}{120}$

$$\kappa=G*G^{*}=\frac{1}{210}\times0.66$$

$$\lambda_{eq}=\frac{\kappa\times10^{3}}{N}=\frac{\left(\frac{1}{210}\times0.66\right)\times10^{3}}{0.01}$$

= 314.28 mho cm^2eq^{-1}

10. Ans. (2)

In case of electrolytic conduction charge carrier are mobile ions while in case of metallic conduction change carrier ore Kernels, so increasing temp. mobility of ion increases hence resistance decreases.

11. Ans. (1)

Given : specific conductance

$(\kappa)=0.0014\Omega^{-1}cm^{-1}$

Conc. (M) = 0.01M

Now

$$\lambda_{eq}=\frac{\kappa\times1000}{N}$$

n-factor for KCl = 1

So N = M = 0.01

$$=\frac{0.0014\times1000}{0.01}$$

$\lambda_{eq}=140$

12. Ans. (1)

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Specific conductance is conductance due to 1 ml electrolytic solution on dilution no of ions per ml decreases hence specific conductance decreases.

13. Ans. (1)

Given: $\lambda^{\infty}_{e\ CH_3COONa} = 91$ mho cm^2eq^{-1}

$\lambda^{\infty}_{e\ HCl} = 426$ mho cm^2eq^{-1}

$\lambda^{\infty}_{e\ CH_3COOH} = 391$ mho cm^2eq^{-1}

$\lambda^{\infty}_{e\ NaCl} = \lambda^{\infty}_{e\ Na^+} + \lambda^{\infty}_{e\ Cl^-}$

$= \lambda^{\infty}_{e\ CH_3COONa} + \lambda^{\infty}_{e\ HCl} - \lambda^{\infty}_{e\ CH_3COOH}$

= 91 + 426 – 391

= 126

14. Ans. (3)

Given : $\alpha = 90\% = 0.9$

$\lambda^{\infty}_e = 425\Omega^{-1}cm^2eq^{-1}$

$\kappa = 3.825\Omega^{-1}cm^{-1}$

Now, $\alpha = \dfrac{\lambda^c_e}{\lambda^{\infty}_e}$

$\lambda^c_e = \lambda^{\infty}_e \times \alpha$

$\lambda^c_e = 425 \times \dfrac{90}{100}$ ... (1)

Also, $\lambda^c_e = \dfrac{\kappa \times 1000}{N}$ ... (2)

Equation (1) & (2)

$425 \times \dfrac{90}{100} = \dfrac{\kappa \times 1000}{N}$

$425 \times \dfrac{90}{100} = \dfrac{3.825 \times 1000}{N}$

N = 10N

15. Ans. (1)

$\wedge^{o}_{HOAc} = \wedge^{o}_{H^+} + \wedge^{o}_{OAc^-}$

$= \wedge^{o}_{NaOAc} + \wedge^{o}_{HCl} - \wedge^{o}_{NaCl}$

To substract molar conductance of Na^+ and Cl^- We required $\wedge^{0}_{NaCl}$.

16. Ans. (2)

$\wedge^{o}_{M\ NaNO_3} = \wedge^{o}_{M\ Na^+} + \wedge^{o}_{M\ NO_3^-}$

$= \wedge^{o}_{M\ NaCl} + \wedge^{o}_{M\ AgNO_3} - \wedge^{o}_{M\ AgCl}$

= 110.3 + 116.5 – 121.6

= 105.2 $Scm^2\ mol^{-1}$

17. Ans. (4)

Given : $\kappa = 3.06 \times 10^{-6}\Omega^{-1}cm^{-1}$

$\lambda_M = 1.53\ \Omega^{-1}cm^{-1}mol^{-1}$

At saturation,

$\lambda_M = \dfrac{\kappa \times 1000}{S}$

$S = \dfrac{3.06 \times 10^{-6} \times 1000}{1.53} = 2 \times 10^{-3}$ mol / L

$BaSO_4$ is AB type of salt

$K_{sp} = S^2$

$= (2 \times 10^{-3})^2 = 4 \times 10^{-6}$

18. Ans. (2)

Acc. To KOHLRAUSCH'S LAW,

At infinite dilution, each ion make definite contribution to equivalent conductance of an electrolyte whatever be the nature of other ion of the electrolyte.

19. Ans. (2)

In galvanic cell, electron move from anode to cathode in external circuit so current flows from cathode to anode. From representation Cu to Ag.

20. Ans. (2)

In Daniel cell current flows from cathode to anode. So copper to zinc in external circuit.

21. Ans. (4)

Reaction representing standard reduction potential of zinc.

$Zn^{+2}_{(aq)} + 2e^- \rightarrow Zn_{(s)}$

22. Ans. (2)

Zero electrode potential of standard hydrogen electrode is assumed to be zero.

23. Ans. (4)

In construction of S.H.E. a platinum block is used with conducting wire at 1M[H^+], 1 bar pressure and 25°C temp. The use of platinum block is adsorb H_2 gas at surface.

24. Ans. (2)

$\underbrace{Zn|Zn^{+2}(1M)}_{Anode}||\underbrace{H^+(1M)|H_2(1atm)}_{Cathode}$

$E^o_{cell} = \left(E^o_{RP}\right)_{Cathode} - \left(E^o_{RP}\right)_{Anode}$

$\downarrow$

$= E^o_{H^+/H_2} - E^o_{Zn^{+2}/Zn}$

= 0–(–0.76)

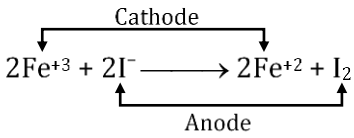
= +0.76 V

25. Ans. (2)

Acc. to representation, it is reverse of Daniel cell so non spontaneous

So E_{cell} = –ive

26. Ans. (3)



$E^{\circ}_{Cell} = E^{\circ}_{Fe^{+3}/Fe^{+2}} - E^{\circ}_{I_2/I^-}$

$= 0.771 - 0.536$

$= 0.235$ V

27. Ans. (1)

$Ag^+ \longrightarrow Ag - e^-$

On rearranging

$Ag^+ + e^- \longrightarrow Ag$

That is reduction reaction.

28. Ans. (1)

At anode always oxidation occurs At cathode always reduction occurs.

29. Ans. (4)

In Daniel cell, Anode is zinc and cathode is copper.

Zinc is consumed and copper is deposited

30. Ans. (4)

Acc. to E.C.S.; $SRP_{Na^+} < SRP_{Zn^{+2}}$

So Na being more reactive, Zn cannot displace Na^+ ion.

31. Ans. (1)

Strong reducing agent will have more standard oxidation potential.

$E^{\circ}_{Zn/Zn^{+2}} > E^{\circ}_{Cr/Cr^{+3}} > E^{\circ}_{H_2/H^+} > E^{\circ}_{Fe^{+2}/Fe^{+3}}$

32. Ans. (3)

More SOP $\Rightarrow$ more reactive metal displaces less reactive.

$E^{\circ}_{Fe/Fe^{+2}} > E^{\circ}_{Cu/Cu^{+2}} > E^{\circ}_{Ag/Ag^+}$

33. Ans. (4)

More SOP $\Rightarrow$ more Reducing power

SOP A < C < B

34. Ans. (2)

Check by option $E_{cell} = (E_{RP})_{Cathode} - (E_{RP})_{Anode}$

1. $\Rightarrow E_{Cu^{+2}/Cu} - E_{Zn^{+2}/Zn}$

   $= +0.34 - (-0.76) = 1.1$ V

2. $\Rightarrow E_{Ag^+/Ag} - E_{Zn^{+2}/Zn}$

   $= +0.80 - (-0.76) = +1.56$ V

3. $\Rightarrow E_{Cu^{+2}/Cu} - E_{H^+/H_2}$

   $= 0.34 - 0 = +0.34$ V

4. $\Rightarrow E_{Ni^{+2}/Ni} - E_{H^+/H_2}$

   $= -0.55 - 0 = -0.55$ V

35. Ans. (2)

SRP $Mg^{+2} > Na^+ > Ba^{+2} > Li^+$

More SRP $\Rightarrow$ strong oxidising Agent

36. Ans. (1)

For non-metal, substance having more SRP have greater tendency to reduce itself and oxidise other, while less SRP substance have greater tendency to oxidise itself and reduce other.

37. Ans. (3)

Lesser SRP substance will oxidise itself and reduce other.

Like Zn will oxidise itself but reduce Ag^+ and Cu^{+2} ions.

38. Ans. (3)

SRP C > B > A > D

Lesser SRP $\Rightarrow$ more reactive metal more reactive metal displace less reactive from its aqueous solution.

39. Ans. (4)

For solution to turn blue copper must get oxidise and convert into Cu^{+2}.

Acc. to E.C.S.

SRP of Cu < SRP of Ag

So, copper will displace Ag^+ ions from its solutions.

40. Ans. (3)

Acc. to E.C.S., metals having lesser SRP are more reactive and displacer less reactive metal ion in their aqueous solutions.

41. Ans. (1)

Strong oxidizing agent will have lesser SOP. Oxidising agent ion will reduce itself and oxidise other.

42. Ans. (3)

Cell is at standard condition.

So $E^{o}_{cell} = \left(E^{o}_{RP}\right)_{cathode} - \left(E^{o}_{RP}\right)_{Anode}$

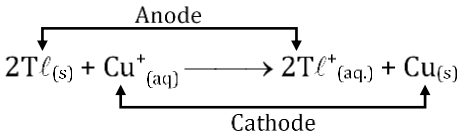
$= E^{o}_{Au^{+3}/Au} - E^{o}_{Ni^{+2}/Ni}$

$= 1.5 - (-0.25)$

$= 1.75\ V$

43. Ans. (4)

Cell reaction :



$$\text{Ecell} = E_{cell} = E^{0}_{cell} - \left\{\frac{0.0591}{n}\log\frac{\left[T\ell^{+}\right]^{2}}{\left[Cu^{+2}\right]}\right\}$$

From relation $[Cu^{+2}]\uparrow$ or $[T\ell^{+}]\downarrow \Rightarrow E_{cell}\uparrow$

44. Ans. (4)

$Ag^{+}_{(aq.)} + e^{-} \rightarrow Ag_{(s)}$

$$E_{cell} = E^{o}_{red} - \frac{0.0591}{n}\log\frac{1}{[Ag^{+}]}$$

$$= E^{o}_{red} - \frac{0.0591}{1}\log\frac{1}{10^{-1}} \quad \because |Ag^{+}| = 10^{-1}$$

$= E^{o}_{red} - 0.0591$

45. Ans. (3)

pH = 1; $[H^{+}] = 10^{-1}M$

$2H^{+}_{(aq.)} + 2e^{-} \longrightarrow H_{2(g)}$

Nernst equation

$$E_{cell} = E^{o}_{red} - \frac{0.0591}{2}\log\frac{p_{H_2}}{[H^{+}]^{2}}$$

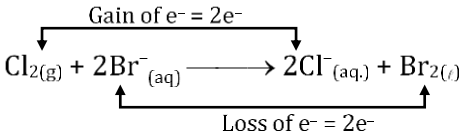
$$\because p_{H_2} = 1; [H^{+}] = 1M; E^{0}_{red} = 0$$

$$E_{cell} = 0 - \frac{0.0591}{2}\log\frac{1}{(10^{-1})^{2}}$$

$E_{cell} = -0.0591\ V$

46. Ans. (2)

Reaction :



Nernst equation

$$E_{cell} = E^{o}_{cell} - \frac{0.0591}{n}\log\frac{\left[Cl^{-}\right]^{2}}{P_{Cl_2} \times \left[Br^{-}\right]^{2}}$$

$E^{o}_{cell} = 0.29V; n = 2; P_{Cl_2} = 0.25;$

$[Cl^{-}] = [Br^{-}] = 0.01M$

$$E_{cell} = 0.29 - \frac{0.0591}{2}\log\frac{(0.01)^{2}}{(0.25)(0.01)^{2}}$$

$= 0.272\ V$

47. Ans. (1)

Reaction $Zn + Cu^{+2} \longrightarrow Cu + Zn^{+2}$

no. of e^{-} transferred = $2e^{-}$

Nernst equation

$$E_{cell} = E^{o}_{cell} - \frac{0.0591}{n}\log\frac{[Zn^{+2}]}{[Cu^{+2}]}$$

$$= 1.1 - \frac{0.0591}{2}\log\frac{(0.1)}{(0.1)}$$

$= 1.1\ V$

48. Ans. (1)

EMF of cell is intensive property does not depend on stochiometric coefficient.

49. Ans. (1)

$Zn \longrightarrow Zn^{+2} + 2e^{-}$

Let $[Zn^{+2}]_i = 1M$

After 10 times dilution $[Zn^{+2}]_f = \frac{1}{10}M$

Nernst equation

$$E_i = E^{o} - \frac{0.0591}{2}\log\left[Zn^{+2}\right]^{1}_{i} = E^{0}$$

$$E_f = E^{o} - \frac{0.0591}{2}\log\left[Zn^{+2}\right]^{1}_{f}$$

$$= E^{0} - \frac{0.0591}{2}\log\left(10^{-1}\right)^{1}$$

$= E^{o} + 0.0295$

$= \approx E^{0} + 0.03$

50. Ans. (4)

$2H^+ + 2e^- \longrightarrow H_2$

$pH_i = 0$; $[H^+]_i = 10^0 = 1$ M

$pH_f = 7$; $[H^+]_f = 10^{-7}$ M

Nernst equation

$$E_{cell} = E^o_{cell} - \frac{0.0591}{2}\log\frac{P_{H_2}}{[H^+]^2}$$

$$\left(\because P_{H_2} = 1atm\right);\ E^0_{cell} = 0$$

$$E_i = 0 - \frac{0.0591}{2}\log\frac{1}{1} = 0$$

$$E_f = 0 - \frac{0.591}{2}\log\frac{1}{(10^{-7})^2} = \frac{-0.0591}{2}\times\log 10^{14}$$

$$= -\frac{0.0591}{2}\times 14$$

$= -0.0591 \times 7$

$= -0.41$ V

51. Ans. (3)

$$Sn_{(s)} + 2Ag^+_{(aq.)} \rightarrow Sn^{+2}_{(aq.)} + 2Ag_{(s)}$$

no. of e⁻ transferred = 2e⁻

$$E_{cell} = E^o_{cell} - \frac{0.0591}{2}\log\frac{\left[Sn^{+2}\right]}{\left[Ag^+\right]^2}$$

$[Ag^+] \uparrow \Rightarrow E_{cell} \uparrow$

52. Ans. (2)

$E^o_{cell} = -0.35V$

$\Delta G^o = -nFE^0_{cell}$

= +ve

non-spontaneous

53. Ans. (1)

$$E^o_{cell} = +\frac{0.0591}{n}\log(K_c)$$

$$= \frac{0.0591}{2}\log 10^{12}$$

= 0.354 V

54. Ans. (3)

$\Delta G^o = -nFE^o_{cell}$

(i) $OCl^- + 2e^- \rightarrow Cl^-$; $\Delta G^o_1 = -2\times F\times 0.94$

(ii) $Cl^- \rightarrow \frac{1}{2}Cl_2 + 1e^-$; $\Delta G^o_2 = -1\times F\times(-1.36)$

(iii) $OCl^- + 1e^- \rightarrow \frac{1}{2}Cl_2$; $\Delta G^o_3 = -1\times F\times E^0_3$

(iii) = (i) + (ii)

$\Delta G^o_3 = \Delta G^o_1 + \Delta G^o_2$

$-1\times F\times E^o_3 = \left[-2\times F\times 0.94\right] + \left[-1\times F\times(-1.36)\right]$

$E^o_3 = 0.52V$

55. Ans. (2)

$2H^+_{(aq.)} + 2e^- \rightarrow H_{2(g)}$

pH = 3; $[H^+] = 10^{-3}$M

Nernst equation

$$E_{cell} = E^o_{cell} - \frac{0.0591}{n}\log\frac{P_{H_2}}{[H^+]^2}$$

n = 2; $pH_2 = 1atm$; $[H^+] = 10^{-3}$M ; $E^o_{cell} = 0$

$$E_{cell} = 0 - \frac{0.0591}{2}\log\frac{1}{(10^{-3})^2}$$

= –0.177V

56. Ans. (3)

Reaction : $H_{2(g)} + \underset{(0.001M)}{2H^+} \longrightarrow \underset{(10^{-8}M)}{2H^+} + H_{2(g)}$

no. of e⁻ transferred = 2e⁻

$$E_{cell} = E^o_{cell} - \frac{0.0591}{n}\log\frac{(10^{-8})^2}{(10^{-3})^2}$$

$E^o_{cell} = 0$; n = 2

$$E_{cell} = 0 - \frac{0.0591}{2}\log\frac{10^{-16}}{10^{-6}}$$

= + 0.295 V

57. Ans. (3)

Cell reaction : $Cu + 2Ag^+ \longrightarrow Cu^{+2} + Ag$

No. of e^- transferred = 2

Initial : $[Cu^{+2}]_i = 1M = [Ag^+]_i$

Final : $[Cu^{+2}]_f = 10M = [Ag^+]_f$

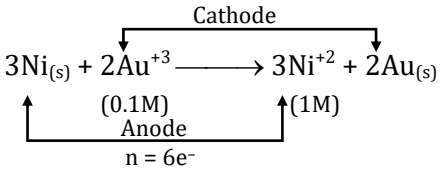
Initial : $E_{cell} = E^o_{cell} - \frac{0.0591}{2}\log\frac{[Cu^{+2}]^1_i}{[Ag^+]^2_i} = E^o_{cell}$

Final :

$$E_{cell} = E^o_{cell} - \frac{0.0591}{2}\log\frac{(10)^1}{(10)^2} = E^o + 0.0295V$$

58. Ans. (4)

Cell Reaction :



$$E_{cell} = E^{o}_{cell} - \frac{0.0591}{6}\log\frac{\left[Ni^{+2}\right]^3}{\left[Au^{+3}\right]^2}$$

$$= \left[E^{o}_{Au^{+3}/Au} - E^{o}_{Ni^{+2}/Ni}\right] - \frac{0.0591}{6}\log\frac{1^3}{(10^{-1})^2}$$

$$= \left[1.5-(-0.25)\right] - \frac{0.0591}{6}\log 10^2$$

= 1.73 V

59. Ans. (3)

At 298K for reaction n = 2

So, $E^{o}_{cell} = \frac{0.0591}{n}\log K_C$

$$\log K_C = \frac{E^{o}_{cell}\times n}{0.0591}$$

$$= \frac{0.465\times 2}{0.0591}$$

$\log K_C = 15.73$

Taking antilog

$K_C = 3.16\times 10^{15}$

60. Ans. (2)

For reaction n = 2

E_{cell} = 0.5105 V

Nernst equation

$$E_{cell} = E^{o}_{cell} - \frac{0.0591}{2}\log\frac{\left[Zn^{+2}\right]^1}{\left[Ni^{+2}\right]^1}$$

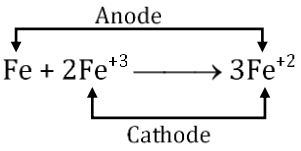
$$E^{o}_{cell} = E_{cell} + \frac{0.0591}{2}\log\frac{[Zn^{+2}]}{[Ni^{+2}]}$$

$$= 0.5105 + \frac{0.0591}{2}\log\frac{(1)}{(0.1)}$$

= 0.54 V

61. Ans. (3)

Cell reaction :



$$E^{o}_{cell} = \left(E^{o}_{RP}\right)_{cathode} - \left(E^{o}_{RP}\right)_{Anode}$$

$$= E^{o}_{Fe^{+3}/Fe^{+2}} - E^{o}_{Fe^{+2}/Fe}$$

= 0.771–(–0.441)

= 1.212V

62. Ans. (1)

$$n_{H_2} = \frac{V_{ml}\text{ at STP}}{22400} = \frac{112}{22400} = \frac{1}{200}$$

$$w_{H_2} = n_{H_2}\times(M_w)_{H_2} = \frac{1}{200}\times 2 = \frac{1}{100}\text{g}$$

Faraday Ist law

$$w_{H_2} = \frac{E_{H_2}}{96500}\times i\times t \qquad \because E_{H_2} = 1$$

$$\frac{1}{100} = \frac{1}{96500}\times i\times 965$$

i = 1 A

63. Ans. (4)

Acc to Faraday IInd law :-

For $FeCl_2$ and $FeCl_3$

$$\frac{w_{Fe^{+2}}}{w_{Fe^{+3}}} = \frac{E_{Fe^{+2}}}{E_{Fe^{+3}}}$$

$$\because E_W = \frac{M_w}{\text{v.f.}}$$

$$= \frac{\frac{56}{2}}{\frac{56}{3}}$$

$$= \frac{3}{2}$$

64. Ans. (3)

Farady Ist law :-

$$w = \frac{E_W}{96500}\times i\times t$$

$$3 = \frac{E_W}{96500}\times 9.65\times(10\times 60)$$

E_W = 50

65. Ans. (3)

$$1\text{mol} \rightarrow \overset{-2}{H_2O} \longrightarrow \overset{0}{O_2} + 2e^-$$

Loss of e^- = 2mol

Charge transfer = 2F ($\because$ 1 mol e^- = 1F)

= 2×96500C

= 1.93×10⁵C

66. Ans. (1)

Faraday Ist law

$$w = \frac{\frac{\text{At wt}}{\text{v.f.}}}{96500} \times Q \qquad (\because Q = i \times t)$$

$$2.977\,g = \frac{\frac{106.4}{\text{v.f.}}}{96500} \times 10800$$

v.f. = +4

67. Ans. (1)

For dil. H_2SO_4

At Cathode : H_2 gas is liberated

At Anode : O_2 gas is liberated

Faraday IInd law :

$$\frac{w_{\text{Cathode}}}{w_{\text{Anode}}} = \frac{E_{H_2}}{E_{O_2}} = \frac{1}{8}$$

68. Ans. (2)

Fused $CaH_2 \longrightarrow Ca^{+2} + 2H^-$

At Cathode : Ca is deposited

At Anode : H_2 is liberated

69. Ans. (3)

Faraday Ist law :

$$w = \frac{E_{Ag}}{96500} \times i \times t$$

$$= \frac{\frac{107.87}{1}}{96500} \times 965 \times 1$$

= 1.0787 g

70. Ans. (3)

Acc. to reaction

1 mol Al deposite an transfer of 3 mol e^- (3 F charge)

1 mol Al ——— 3F charge

$13.5\,g \Rightarrow \frac{1}{2}$ mol Al ——— $\frac{3}{2}$ = 1.5 F charge

71. Ans. (4)

For aq. Na_2SO_4

$SOP_{SO_4^{2}} < SOP_{H_2O}$

So at anode : O_2 will liberate

$SRP_{Na^+} < SRP_{H_2O}$

Cathode : H_2 will liberate

72. Ans. (2)

1 F $\Rightarrow$ 1 mol e^-

So, $AgNO_3$

$$\downarrow$$

$$Ag^+ + \underset{1\text{mol } e^-}{1e^-} \longrightarrow \underset{1\text{mol Ag}}{Ag}$$

73. Ans. (1)

$BaCl_2 \rightarrow Ba^{+2} + 2Cl^-$

96500C $\Rightarrow$ 1 Faraday $\Rightarrow$ 1 mol e^-

$Ba^{+2} + 2e^- \longrightarrow Ba$

$\downarrow \qquad \downarrow$

(2F) 2 mol $e^- \rightarrow$ 1 mole barium

$1F \longrightarrow \frac{1}{2}$ mole barium

74. Ans. (2)

Faraday IInd law

$$\frac{w_{Ca}}{w_{Ac}} = \frac{(E_w)_{Ca}}{(E_w)_{Al}}$$

$$\frac{40 \times 10^3}{w_{Al(g)}} = \frac{\frac{40}{2}}{\frac{27}{3}}$$

w_{Al} = 18000 g

= 18 Kg

75. Ans. (2)

Acc. to Faraday IInd law :

$$w_{Ag} : w_{Cu} : w_{Fe} = E_{w_{Ag}} : E_{w_{Cu}} : E_{w_{Fe}}$$

$$= \frac{108}{1} : \frac{63.5}{2} : \frac{56}{3}$$

Now convert mass in mole = $\frac{w}{\text{At.wt}}$

$$n_{Ag} : n_{Cu} : n_{Fe} = \frac{w_{Ag}}{(\text{At. wt})_{Ag}} : \frac{w_{Cu}}{(\text{At. wt})_{Cu}} : \frac{w_{Fe}}{(\text{At. wt})_{Fe}}$$

$$= \frac{\frac{108}{1}}{108} : \frac{\frac{63.5}{2}}{63.5} : \frac{\frac{56}{3}}{56}$$

$$= \frac{1}{1} : \frac{1}{2} : \frac{1}{3}$$

76. Ans. (2)

For aq. $CuSO_4$

At Cathode : $Cu^{+2} + 2e^- \longrightarrow Cu_{(s)}$

At Anode : $2H_2O_{(\ell)} \longrightarrow O_{2(g)} + 4H^+ + 4e^-$

$[H^+]\uparrow \Rightarrow pH\downarrow$

77. Ans. (3)

For aq. NaCl

At cathode : H_2 gas is liberated

At anode : Cl_2 gas is liberated

78. Ans. (1)

On charging lead accumulator, reverse of galvanic cell happen i.e. becomes electrolytic cell.

79. Ans. (3)

Thermodynamic efficiency of cell :

$= \frac{\Delta G}{\Delta H} \times 100$

$= \frac{-nFE}{\Delta H} \times 100$

Exercise-II (Previous Year Questions)

1. Ans. (3)

Balancing Charge

$MnO_4^{-2} \longrightarrow MnO_4^- + 1e^-$

$\downarrow$

1mol $\longrightarrow$ 1mol e^-

0.1mol $MnO_4^{-2} \longrightarrow$ 0.1 mol e^-

$\Downarrow$

0.1 F

$\Downarrow$

0.1×96500

$= 9650$ C

2. Ans. (4)

$n_{O_2} = \frac{V_{ml}}{22400} = \frac{5600}{22400} = \frac{1}{4}$

$w_{O_2} = n_{O_2} \times (M_w)_{O_2}$

$= \frac{1}{4} \times 32 = 8g$

Faraday IInd law

$\frac{w_{Ag}}{w_{O_2}} = \frac{E_{Ag}}{E_{O_2}} = \frac{\frac{108}{1}}{8}$

$\frac{w_{Ag}}{8} = \frac{108}{8}$

$w_{Ag} = 108$

3. Ans. (4)

Fuel cell converts combustion energy of fuels like H_2 & CH_4 directly into electrical energy.

4. Ans. (1)

In pure water pH = 7 ; $[H^+] = 10^{-7}$

$2H^+ + 2e^- \longrightarrow H_{2(g)}$

$E_{Cell} = E^o_{Cell} - \frac{0.0591}{2} \log \frac{P_{H_2}}{[H^+]^2}$

$0 = 0 - \frac{0.0591}{2} \log \frac{P_{H_2}}{(10^{-7})^2}$

$0 = \log P_{H_2} - \log 10^{-14}$

$\text{Log } P_{H_2} = \log 10^{-14}$

$P_{H_2} = 10^{-14}$ atm

5. Ans. (4)

$\lambda_M = \frac{\kappa \times 1000}{M}$

$= \frac{5.76 \times 1000}{0.5}$

$= 11.52$ S cm²/mol

6. Ans. (4)

Faraday Ist law :

$w_{Cl_2} = \frac{E_{Cl}}{96500} \times i \times t \quad \because \quad w_{Cl_2} = n_{Cl_2} \times (M_w)_{Cl_2}$

$7.1 = \frac{\frac{35.5}{1}}{96500} \times 3 \times t \qquad = 0.1 \times 71$

$t = 6600$ Sec $\qquad = 7.1g$

$t = 110$min.

7. Ans. (3)

Given :- $E^o_{Cell} = -ve$

$\Delta G° = -nF E^o_{Cell}$: $E^o_{Cell} = \frac{0.0591}{n} \log K$

$\Rightarrow$ +ve or > 1 : $\log K < 0$

$K < 1$

8. Ans. (1)

Charge (Q) = i × t

= 1 × 60

= 60 C

no. of e^- = $\frac{Q}{\text{Charge on } 1e^-} = \frac{60}{1.6\times10^{-19}}$

= 3.75×10^{20}

9. Ans. (2)

$Zn + Cu^{+2} \longrightarrow Zn^{+2} + Cu$

(1M) (10^{-2}M)

$E_1 = E° - \frac{0.00591}{2}\log\frac{|Zn^{+2}|^1}{|Cu^{+2}|^1}$

$= E° - \frac{0.0591}{2}\log\frac{10^{-2}}{1} = E° + 0.0591$

$E_2 = E° - \frac{0.0591}{2}\log\frac{1}{10^{-2}} = E° - 0.0591$

$E_1 > E_2$

10. Ans. (4)

$HBrO \xrightarrow{1.5V} Bro_3^-$ ...(i)

$HBrO \xrightarrow{1.595V} Br_2$ ...(ii)

$2HBrO \longrightarrow BrO_3^- + Br_2$

$E^o_{Cell} = \left(E^o_{RP}\right)_C - \left(E^o_{RP}\right)_A$

= −1.5 − (−1.595)

= +0.095V ⇒ + ve Value

Spontaneous reaction

So HBrO disproportionates to BrO_3^- & Br_2

11. Ans. (3)

$E^o_{Cell} = \frac{0.0591}{n}\log K_C$

$\log K_C = \frac{E^o_{Cell}\times n}{0.0591} = \frac{0.59\times1}{0.0591}$

$\log K_C = 10$

$K_C = 10^{10}$

12. Ans. (1)

$\Delta G° = -nF E^o_{Cell}$

$= -2 \times 96500 \times 0.24$

$= -46320$ J/mol

$= -46.32$ KJ/mol

13. Ans. (4)

$\lambda^o_{M_{CH_3COOH}} = \lambda^o_{M_{CH_3COO^-}} + \lambda^o_{M_{H^+}}$

$= \lambda^o_{M_{CH_3COO^-K^+}} + \frac{1}{2}\lambda^o_{M_{H_2SO_4}} - \frac{1}{2}\lambda^o_{M_{K_2SO_4}}$

$= z + \frac{x}{2} - \frac{y}{2}$

14. Ans. (2)

More Reducing Power ⇒ SRP min

K > Al > Cr > Ag

15. Ans. (2)

$CaCl_2 \longrightarrow Ca^{+2} + 2Cl^-$

$Ca^{+2} + 2e^- \longrightarrow Ca_{(s)}$

(2F) 2 mol $e^- \longrightarrow$ 1mol Calcium (40g)

40g Ca ⟶ 2 F Charge

20g Ca ⟶ 1 F charge

16. Ans. (3)

For dil. H_2SO_4

at Anode H_2O get discharge

So product at anode will be O_2 (gas).

17. Ans. (3)

According to ECS ; max SRP will be of

$Au^{+3} + 3e^- \longrightarrow Au(s)$

18. Ans. (2)

Fuel cell :-

Reactant ⇒ $H_{2(g)}$; $O_{2(g)}$

↓

Product ⇒ $H_2O_{(g)}$

19. Ans. (2)

$\lambda^\infty_{M_{CH_3COOH}} = \lambda^\infty_{M_{CH_3COO^-}} + \lambda^\infty_{M_{H^+}}$

$= \lambda^\infty_{M_{CH_3COONa^+}} + \lambda^\infty_{M_{HCl^-}} - \lambda^\infty_{M_{NaCl^-}}$

= 91 + 426.16 −126.45

= 390.71

20. Ans. (3)

$\lambda^o_{M_{CH_3COOH}} = \lambda^o_{M_{CH_3COO^-}} + \lambda^o_{M_{H^+}}$

= 50 + 350 = 400

$\alpha = \frac{\lambda^C_M}{\lambda^o_M} = \frac{20}{400} = \frac{1}{20}$

$\alpha = 5\%$

$K_c = C\alpha^2$

$= 0.007 \times \left(\frac{1}{20}\right)^2$

$= 1.75\times10^{-5}$ mol/L

21. Ans. (3)

For non spontaneous $\Rightarrow E^{o}_{Cell}$ = –ve

$E^{o}_{Cell} = \left(E^{o}_{RP}\right)_C - \left(E^{o}_{RP}\right)_A$

$= E^{o}_{Cu^{+2}/Cu} - E^{o}_{Ag^{+}/Ag}$

= 0.34 – (0.8)

= –0.46 V

22. Ans. (4)

Acc. to question for O_2 to liberate possible reaction at Cathode & Anode will be :

Anode : $H_2O \longrightarrow \frac{1}{2}O_2 + 2H^+ + 2e^-$

Cathode : $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{+2} + 4H_2O$

So $E^{o}_{Cell} = \left(E^{o}_{RP}\right)_C - \left(E^{o}_{RP}\right)_A$

= 1.51 – (1.223)

= +0.287V

= the sport.

So O_2 gas will liberate

23. Ans. (3)

$E^{o}_{Co^{+3}/Co^{+2}} > E^{o}_{Al^{+3}/Al}$

So reaction at cathode & Anode will be:-

Cathode : $Co^{+3} + e \longrightarrow Co^{+2}$

Anode : $Al \longrightarrow Al^{+3} + 3e^-$

$E^{o}_{Cell} = \left(E^{o}_{RP}\right)_C - \left(E^{o}_{RP}\right)_A = 1.81 - (-1.66)$

$= + 3.47V$

24. Ans. (2)

$\Delta G° = -nF E^{o}_{Cell}$

$= -2 \times 96500 \times 1.1$

$= -21227$ J/mol

$= -212.27$ kJ/mol

25. Ans. (2)

Centimolar solution = $\frac{1}{100}M = 0.01M$

Conductivity (k) = 0.0210 ohm^{-1} cm^{-1}

Resistance (R) = 60 ohm

$k = \frac{1}{R}\left(\frac{\ell}{A}\right)$

$\Rightarrow 0.0210 = \frac{1}{60}\left(\frac{\ell}{A}\right) \Rightarrow \frac{\ell}{A} = 1.26 cm^{-1}$

26. Ans. (4)

$\Delta_r G = -nFE_{cell}$

E_{cell} is an intensive property and $\Delta_r G$ is an extensive property as it depends on number of $e^{\ominus}$ transferred in cell reaction.

27. Ans. (1)

(A) $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$

1 mol H_2O will req. 2F charge

(B) $8H^+ + MnO_4^- + 5e^- \rightarrow Mn^{+2} + 4H_2O$

1 mole MnO_4^- will req. 5F charge.

(C) $CaCl_2 + 2e^- \rightarrow Ca + 2Cl^{-1}$

1.5 mole Ca will req. 3F charge.

(D) $2FeO + H_2O \rightarrow Fe_2O_3 + 2H^+ + 2e^-$

1 mole FeO will req. 1 F charge.

(A)-(II), (B)-(IV), (C)-(I), (D)-(III)

28. Ans. (2)

$W = \frac{Eit}{96487}$

$W = \frac{\frac{63}{2} \times 9.6487 \times 100}{96487}$

W = 0.315 g of Cu

29. Ans. (2)

Production of hydrogen by electrolysis of water is not an example of corrosion.

30. Ans. (1)

$Zn|Zn^{2+}(aq)||Fe^{2+}(aq)|Fe \quad E^{0}_{cell} = 0.32$ V

$\Delta G° = -nF E^{0}_{cell}$

$\Delta G° = -2 \times 96487 \times 0.32$ J

$\Delta G° = -61.75$ kJ/mol

Exercise-III (Analytical Questions)

1. Ans. (4)

Statement-I : $W_{NonPV} = -\Delta G = W_{electrical}$

Statement-II : $\Delta G = -nFE_{Cell}$

n $\Rightarrow$ no. of e^- transfer depends on & stoichiometry

E_{Cell} $\Rightarrow$ EMF of Cell $\rightarrow$ intensive

F $\Rightarrow$ 96500 C

G $\Rightarrow$ extensive property

2. Ans. (3)

Statement-I : For electrolysis of aq. NaCl Product obtained are

Anode :- $2Cl^- \rightarrow Cl_{2(g)} + 2e^-$

Cathode :- $2H_2O_{(\ell)} + 2e^- \rightarrow H_{2(g)} + 2OH^-_{(aq.)}$

Statement-II : $E^o_{Cu^{+2}/Cu} > E^o_{Fe^{+2}/Fe}$

Copper is more reactive than iron

3. Ans. (1)

Statement-I : EMF of cell is intensive property

Statement-II : EMF of cell = cell voltage which is intensive

4. Ans. (2)

Statement-I : In electrolytic cell

Cathode is negative

Anode is positive

In galvanic cell

Cathode is positive

Anode is negative

Statement-II : In Daniel cell

Oxidation of Zinc occurs $\Rightarrow$ loss of e^-

Reduction of Copper occurs $\Rightarrow$ gain of e^-

5. Ans. (3)

Statement-I : molten NaH $\Rightarrow Na^+ ; H^-$

Cathode :- $Na^+ + e^- \rightarrow Na_{(s)}$

Anode :- $2H^+ \rightarrow H_{2(g)} + 2e^-$

Statement-II : In electrolytic refining

Impure metal $\Rightarrow$ Anode

Pure metal $\Rightarrow$ Cathode

6. Ans. (2)

On dilution of weak electrolyte

1.dod $\alpha\% \uparrow \Rightarrow$ no. of ions $\uparrow$

2.Inter ionic attraction forces $\downarrow$

Hence molar conductance increases rapidly.

7. Ans. (1)

Daniel Cell

$$Zn_{(s)} + Cu^{+2}_{(aq)} \rightarrow Zn^{+2}_{(aq)} + Cu_{(s)}$$

Nernst equation:

$$E_{cell} \quad E^o_{cell} - \frac{0.0591}{2}\log\frac{|Zn^{+2}|^1}{|Cu^{+2}|^1}$$

$$E_{cell} = E^o_{cell} + \frac{0.0591}{2}\log\frac{|Cu^{+2}|^1}{|Zn^{+2}|^1}$$

$[Cu^{+2}]\uparrow \Rightarrow E_{cell} \uparrow$

8. Ans. (4)

Statement-I :

Cell const. $G^* = \frac{\ell}{A}$

depends on Construction of cell.

Statement-II : Conductance is due to charge corries (ions):-

(i) mobility

(ii) no. of charge carrie (ions)

9. Ans. (2)

Min Mg^{+2}/Mg

Cu^{+2}/Cu

Hg^{+2}/Hg

Ag^+/Ag max

SRP

SRP min $\Rightarrow$ metal reactivity $\uparrow$

* Mg^{+2} can not be displaced by Cu.

* Ag can not displace Cu^{+2}

10. Ans. (3)

Spontaneous Condition :-

$\rightarrow$ ΔG = negative

also $\Delta G = -nFE_{cell}$

$\rightarrow$ So E_{cell} = positive

11. Ans. (3)

(a) λ_m - with temp. - because in increase in K.E. of Ions.

(c) increase in concentration, increases inter ionic attraction forces hence λ_M decreases.

(d) Increases in polarity will increase d.o.d (X)

Which will increases ions no. so increase in molar conductivity.

AIM: FREE EDUCATION TO ALL APUL

Exercise - I (Conceptual Questions) Build Up Your Understanding

CONCENTRATION TERMS

1. 8 g NaOH is dissolved in one litre of solution. Its molarity is :
(1) 0.8 M (2) 0.4 M (3) 0.2 M (4) 0.1 M
CSO001

2. If 18 g of glucose is present in 1000 g of solvent, the solution is said to be :
(1) 1 molar (2) 0.1 molar
(3) 0.5 molar (4) 0.1 molal
CSO002

3. The molarity of a solution of sodium chloride (mol wt. = 58.5) in water containing 5.85 g of sodium chloride in 500 mL of solution is :-
(1) 0.25 (2) 2.0 (3) 1.0 (4) 0.2
CSO003

4. For preparing 0.1 M solution of H_2SO_4 in one litre, we need H_2SO_4 :
(1) 0.98 g (2) 4.9 g (3) 49.0 g (4) 9.8 g
CSO004

5. Mole fraction of glycerine($C_3H_5(OH)_3$) in a solution of 36 g of water and 46 g of glycerine is :
(1) 0.46 (2) 0.36 (3) 0.20 (4) 0.40
CSO005

6. 1000 g aqueous solution of $CaCO_3$ contains 10 g of calcium carbonate, concentration of the solution is :
(1) 10 ppm (2) 100 ppm
(3) 1000 ppm (4) 10,000 ppm
CSO006

7. What is the normality of 1M H_3PO_4 solution ?
(1) 0.5 N (2) 1.0 N (3) 2.0 N (4) 3.0 N
CSO007

8. The molarity of 0.2 N Na_2CO_3 solution will be:
(1) 0.05 M (2) 0.2 M (3) 0.1 M (4) 0.4 M
CSO008

9. Normality of 0.3 M phosphorous acid is:-
(1) 0.15 (2) 0.6 (3) 0.9 (4) 0.1
CSO009

10. The molarity of pure water is :
(1) 100 M (2) 55.5 M (3) 50 M (4) 18M
CSO010

11. Molarity of 720 g of pure water -
(1) 40M
(2) 4M
(3) 55.5M
(4) Can't be determined
CSO011

12. Equal weight of NaCl and KCl are dissolved separately in equal volumes of solutions then molarity of the two solutions will be –
(1) Equal
(2) That of NaCl will be less than that of KCl
(3) That of NaCl will be more than that of KCl Solution
(4) That of NaCl will be half of that of KCl solution
CSO012

13. The mole fraction of oxygen in a mixture of 7g of nitrogen and 8g of oxygen is :
(1) $\frac{8}{15}$ (2) 0.5 (3) 0.25 (4) 1.0
CSO013

14. In a solution of 7.8 g benzene (C_6H_6) and 46.0g toluene ($C_6H_5CH_3$), the mole fraction of benzene is:-
(1) $\frac{1}{6}$ (2) $\frac{1}{5}$ (3) $\frac{1}{2}$ (4) $\frac{1}{3}$
CSO014

15. An X molal solution of a compound in benzene has mole fraction of solute equal to 0.2. The value of X is:-
(1) 14 (2) 3.2 (3) 1.4 (4) 2
CSO015

16. A 500 g tooth paste sample has 0.02 g fluoride concentration. What is the concentration of fluorine in terms of ppm level:-
(1) 250 (2) 40 (3) 400 (4) 1000
CSO016

17. The normality of 10 mL of a '20 V' H_2O_2 solution is
(1) 1.79 (2) 3.58
(3) 60.86 (4) 6.086
CSO017

18. H_2O_2 solution used for hair bleaching is sold as a solution of approximately 5.0 g H_2O_2 per 100 mL of the solution. The molecular mass of H_2O_2 is 34. The molarity of this solution is approximately:-
(1) 0.15 M (2) 1.5 M (3) 3.0 M (4) 3.4 M
CSO018

19. Normality of 10% (w\V) H_2SO_4 solution is nearly
(1) 0.1 (2) 0.2 (3) 0.5 (4) 2
CSO019

20. What volume of 0.1 N HNO_3 solution can be prepared from 6.3 g of HNO_3 ?
(1) 1 litre (2) 2 litre
(3) 0.5 litre (4) 5 litre
CSO020

21. A molal solution is one that contains one mole of a solute in
(1) 1000 g of the solvent
(2) one litre of the solution
(3) one litre of the solvent
(4) 22.4 litres of the solution
CSO021

22. Which of the following statement is true :-
(a) Molarity is the no. of moles of solute dissolved per litre of solvent.
(b) The molarity and normality of a solution of sodium carbonate are same.
(c) Molality (m) of a solution is defined as the number of moles of solute dissolved is 1000 gm of solution
(d) The ratio of mole fraction of solute and solvent is in the ratio of their respective moles.
(1) a & c (2) a & d
(3) b & c (4) Only d
CSO022

23. Two bottles of A and B contain 1M and 1m aqueous solution of sulphuric acid respectively-
(1) A is more concentrated than B
(2) B is more concentrated than A
(3) Concentration of A = concentration of B
(4) It is not possible to compare the concentration
CSO023

24. Molar concentration of a solution in water is :
(1) Always equal to normality
(2) More than molality of the solution
(3) Equal to molality of the solution
(4) Less than the molality of the solution
CSO024

25. The molarity of 98% (w/W) H_2SO_4 (d = 1.8 g mL^{-1}) is :-
(1) 6 M (2) 18 M
(3) 10 M (4) 4 M
CSO025

SOLUBILITY (HENRY'S LAW)

26. Henry's law constant for dissolution of CH_4 in benzene at 298 K is 2×10^5 mm of Hg. Then solubility of CH_4 in benzene at 298 K (in terms of mole fraction) under 760 mm of Hg is :
(1) 1.2×10^{-5} (2) 3.8×10^{-3}
(3) 4×10^{-7} (4) 1×10^{-2}
CSO026

27. Which of the following gas does not obey Henry's law ?
(1) NH_3 (2) H_2
(3) O_2 (4) He
CSO027

VAPOUR PRESSURE (LIQUID-LIQUID MIXTURE)

28. 1 mol of heptane (V. P. = 92 mm of Hg) was mixed with 4 mol of octane (V. P. = 31mm of Hg). The vapour pressure of resulting ideal solution is :
(1) 46.2 mm of Hg (2) 40.0 mm of Hg
(3) 43.2 mm of Hg (4) 38.4 mm of Hg
CSO028

29. At 88°C benzene has a vapour pressure of 900 torr and toluene has a vapour pressure of 360 torr. What is the mole fraction of benzene in the mixture with toluene that will boil at 88°C at 1 atm. pressure, (benzene - toluene form an ideal solution) :
(1) 0.416 (2) 0.588
(3) 0.688 (4) 0.740
CSO029

30. If P_A^0 and P_B^0 are 108 and 36 torr respectively. What will be the mole fraction of A in vapour phase if B has mole fraction of 0.5 in solution :-

(1) 0.25 (2) 0.75

(3) 0.60 (4) 0.35

CSO030

31. What is correct relation between mole fraction in vapour phase (Y_A) of A in terms of X_A. If mole fraction in solution of A is (X_A) (If P_A^0 is vapour pressure of A in pure state)

(1) $(1-X_A)P_A^0$ (2) $\frac{X_A}{1-X_A}P_A^0$

(3) $\frac{1-X_A}{X_A}P_A^0$ (4) $\frac{P_A^0 X_A}{P_S}$

CSO031

IDEAL AND NON-IDEAL SOLUTIONS

32. Which condition is not satisfied by an ideal solution ?

(1) $\Delta H_{mixing} = 0$

(2) $\Delta V_{mixing} = 0$

(3) $\Delta S_{mixing} = 0$

(4) Obeyance of Raoult's law

CSO032

33. Among the following, that does not form an ideal solution is ?

(1) C_6H_6 and $C_6H_5CH_3$

(2) C_2H_5Cl and C_6H_5OH

(3) C_6H_5Cl and C_6H_5Br

(4) C_2H_5Br and C_2H_5I

CSO033

34. An azeotropic mixture of two liquids has boiling point lower than either of them when it :-

(1) Shows a (+ve) deviation from Raoult's law

(2) Shows no deviation from Raoult's law

(3) Shows (+ve) deviation from Henry's law

(4) Shows (–ve) deviation from Henry's law

CSO034

COLLIGATIVE PROPERTIES

35. Which is not a colligative property ?

(1) Osmotic pressure

(2) Relative lowering in vapour pressure

(3) Depression in freezing point

(4) Refractive index

CSO035

36. The lowering of vapour pressure of a solvent by addition of a non-volatile solute to it, is directly proportional to :

(1) Mole fraction of solute

(2) The nature of the solute in the solution

(3) The atmospheric pressure

(4) All

CSO036

37. The relative lowering of vapour pressure is equal to the mole fraction of the nonvolatile solute. This statement was given by :

(1) Raoult (2) Henry

(3) Joule (4) Dalton

CSO037

38. The vapour pressure of a solution having solid as solute and liquid as solvent is :

(1) Directly proportional to mole fraction of the solvent

(2) Inversely proportional to mole fraction of the solvent

(3) Directly proportional to mole fraction of the solute

(4) Inversely proportional to mole fraction of the solute

CSO038

39. If P_0 and P_S are the vapour pressure of solvent and its solution respectively. N_1 and N_2 are the mole fraction of solvent and solute respectively then :

(1) $P_S = \frac{P_0}{N_2}$

(2) $P_0 - P_S = P_0N_2$

(3) $P_S = P_0N_2$

(4) $\frac{(P_0 - P_S)}{P_S} = \frac{N_1}{(N_1 + N_2)}$

CSO039

40. One mole of non volatile solute is dissolved in two mole of water. The vapour pressure of the solution relative to that of water is

(1) $\frac{2}{3}$ (2) $\frac{1}{3}$ (3) $\frac{1}{2}$ (4) $\frac{3}{2}$

CSO040

41. The vapour pressure of a dilute aqueous solution of Glucose is 750 mm Hg at 373 K. The mole fraction of solute is :

(1) $\frac{1}{10}$ (2) $\frac{1}{7.6}$ (3) $\frac{1}{35}$ (4) $\frac{1}{76}$

CSO041

42. The vapour pressure of water at room temperature is 23.8 mm of Hg. The vapour pressure of an aqueous solution of sucrose with mole fraction 0.1 is equal to :

(1) 23.9 mm Hg (2) 24.2 mm Hg
(3) 21.42 mm Hg (4) 31.44 mm Hg

CSO042

43. The vapour pressure of pure A is 10 torr. At the same temperature, when 1 g of B is dissolved in 20 gm of A, its vapour pressure is reduced to 9.0 torr. If the molecular mass of A is 200 amu, then the molecular mass of B is :

(1) 100 amu (2) 90 amu
(3) 75 amu (4) 120 amu

CSO043

44. The boiling point of C_6H_6, CH_3OH, $C_6H_5NH_2$ and $C_6H_5NO_2$ are 80°C, 65°C, 184°C and 212°C respectively. Which will show highest vapour pressure at room temperature :

(1) C_6H_6 (2) CH_3OH
(3) $C_6H_5NH_2$ (4) $C_6H_5NO_2$

CSO044

45. If Raoult's law is obeyed, the vapour pressure of the solvent in a solution is directly proportional to :

(1) Mole fraction of the solvent
(2) Mole fraction of the solute
(3) Mole fraction of the solvent and solute
(4) The volume of the solution

CSO045

46. The vapour pressure of a pure liquid 'A' is 70 torr at 27°C. It forms an ideal solution with another liquid B. The mole fraction of B is 0.2 and total vapour pressure of the solution is 84 torr at 27°C. The vapour pressure of pure liquid B at 27°C is -

(1) 14 (2) 56 (3) 140 (4) 70

CSO046

47. The boiling point of an aqueous solution of a non volatile solute is 100.15 °C. What is the freezing point of an aqueous solution obtained by diluting the above solution with an equal volume of water? The values of K_b and K_f for water are 0.512 and 1.86 K molality^{-1} :

(1) –0.544 °C (2) –0.512 °C
(3) –0.272 °C (4) –1.86 °C

CSO047

48. The molal elevation constant is the ratio of the elevation in B.P. to :

(1) Molarity
(2) Molality
(3) Mole fraction of solute
(4) Mole fraction of solvent

CSO048

49 Elevation in boiling point was 0.52°C when 6 g of a compound X was dissolved in 100 g of water. Molecular weight of X is :

(K_b for water = 0.52 K mol^{-1})

(1) 120 (2) 60 (3) 100 (4) 342

CSO049

50. An aqueous solution containing 1g of urea boils at 100.25°C. The aqueous solution containing 3g of glucose in the same volume will boil at -

(1) 100.75°C (2) 100.5°C
(3) 100°C (4) 100.25°C

CSO050

51. Pure benzene freezes at 5.45°C at a certain place but a 0.374 m solution of tetrachloroethane in benzene freezes at 3.55 °C. The K_f for benzene is-

(1) 5.08 K kg mol^{-1} (2) 508 K kg mol^{-1}
(3) 0.508 K kg mol^{-1} (4) 50.8°C kg mol^{-1}

CSO051

52. An aqueous solution freezes at – 0.186°C (K_f = 1.86 K kg mol^{-1} ; K_b = 0.512 K kg mol^{-1}). What is the elevation in boiling point ?
(1) 0.186 (2) 0.512
(3) $\frac{0.512}{1.86}$ (4) 0.0512

CSO052

53. In the depression of freezing point experiment, it is found that :-
(1) The vapour pressure of solution is more than of pure H_2O.
(2) The vapour pressure of solution is less than that of pure solute
(3) Only solute molecules solidify at freezing point
(4) Only solvent molecules solidify at freezing point

CSO053

54. Molal depression constant of water is 1.86 K kg mol^{-1}. 0.02 mole of urea dissolved in 100g of water will produce a depression in freezing point of :
(1) 0.186°C (2) 0.372°C
(3) 1.86°C (4) 3.72°C

CSO054

55. What would be the freezing point of aqueous solution containing 18 g of $C_6H_{12}O_6$ in 1000g of water ? (K_f = 1.86 K molality^{-1})
(1) –0.186°C (2) –0.372°C
(3) –0.54°C (4) –0.72°C

CSO055

56. A solution of 1.25 g of a non-electrolyte in 20g of water freezes at 271.94 K. If K_f = 1.86K molality^{-1} and F.pt. of water is 273.15 K then the molar mass of solute is :
(1) 207.8 g mol^{-1} (2) 179.79 g mol^{-1}
(3) 209.6 g mol^{-1} (4) 96.01 g mol^{-1}

CSO056

57. If a thin slice of sugar beet is placed in concentrated solution of NaCl then
(1) Sugar beet will lose water from its cells
(2) Sugar beet will absorb water from solution
(3) Sugar beet will neither absorb nor lose water
(4) Sugar beet will dissolve in solution

CSO057

58. In osmosis phenomenon net flow of :
(1) Solvent molecules occurs from higher concentration to lower concentration
(2) Solvent molecules occurs from lower concentrated solution to higher concentrated solution
(3) Solute molecules occurs from higher concentration to lower concentration
(4) Solute molecules occurs from lower concentration to higher concentration

CSO058

59. At constant temperature, the osmotic pressure of a solution is :
(1) Directly proportional to the concentration
(2) Inversely proportional to the concentration
(3) Directly proportional to the square of concentration
(4) Directly proportional to the square root of concentration

CSO059

60. Which inorganic compounds can act as semipermeable membrane ?
(1) Calcium sulphate
(2) Barium oxalate
(3) Nickel phosphate
(4) Copper ferrocyanide

CSO060

61. If 0.1 M solution of glucose and 0.1 M urea solution are placed on two sides of a semipermeable membrane to equal heights, then it will be correct to say that :
(1) There will be no net movement across the membrane
(2) Glucose will flow towards urea solution
(3) Urea will flow towards glucose solution
(4) Water will flow from urea solution towards glucose solution.

CSO061

62. If mole fraction of the solvent in solution decreases then :
(1) Vapour pressure of solution increases
(2) B. P. decreases
(3) Osmotic pressure increases
(4) All are correct

CSO062

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63. The osmotic pressure of a solution increases if :
(1) Temperature is lowered
(2) Volume is increased
(3) Number of solute molecules is increased
(4) None

CSO063

64. Which of the following solutions at the same temperature will be isotonic :
(1) 3.42 g of cane sugar in one litre water and 0.18 g of glucose in one litre water.
(2) 3.42 g of cane sugar in one litre water and 0.18 g of glucose in 0.1 litre water.
(3) 3.42 g of cane sugar in one litre water and 0.585g of NaCl in one litre water.
(4) 3.42 g of cane sugar in one litre water and 1.17 g of NaCl in one litre water.

CSO064

65. If density of 2 molal sucrose solution is 1.4 g mL^{-1} at 25°C, find osmotic pressure.
(1) 4.06 atm (2) 2 atm
(3) 40.7 atm (4) 3.4 atm

CSO065

66. If total concentration of dissolved particles inside red blood cells is 0.3 M (approximate) and the membrane surrounding the cell is semipermeable. Find osmotic pressure inside the cell
(1) 7.34 atm (2) 1.78 atm
(3) 2.34 atm (4) 0.74 atm

CSO066

67. Equal volume of 0.1 M urea and 0.1 M glucose are mixed. The mixture will have :-
(1) Lower osmotic pressure
(2) Same osmotic pressure
(3) Higher osmotic pressure
(4) None of these

CSO067

68. Osmotic pressure of a solution (density is 1g mL^{-1}) containing 3 g of glucose (molecular weight = 180) in 60 g of water at 15°C is :
(1) 0.34 atm (2) 0.65 atm
(3) 6.25 atm (4) 5.57 atm

CSO068

69. Osmotic pressure of a sugar solution at 24°C is 2.5 atm. The concentration of the solution in mole per litre is :
(1) 10.25 (2) 1.025
(3) 1025 (4) 0.1025

CSO069

70. A solution containing 8.6 g urea in one litre was found to be isotonic with 0.5% (wt./vol) solution of an organic non volatile solute. The molecular weight of organic solute is :
(1) 348.9 (2) 34.89 (3) 3489 (4) 861.2

CSO070

71. If 6.84% (w/V) solution of cane-sugar (mol. wt. 342) is isotonic with 1.52% (w/V) solution of thiocarbamide, then the molecular weight of thiocarbamide is :
(1) 152 (2) 76 (3) 60 (4) 180

CSO071

72. A solution containing 500 g of a protein per litre is isotonic with a solution containing 3.42 g of sucrose per litre. The molecular mass of protein is
(1) 5 (2) 146 (3) 34200 (4) 50000

CSO072

73. ___(A)___ injection are dissolved in water containing salts at particular ___(B)___ concentrations that matches ___(C)___ concentration. Here A, B and C refer to :
(1) Intravenous, ionic, blood plasma
(2) Intravenous, blood plasma, ionic
(3) Blood plasma, intravenous, ionic
(4) Blood plasma, ionic, intravenous

CSO073

ABNORMAL COLLIGATIVE PROPERTIES

74. Equimolal solutions of A and B show depression in freezing point in the ratio of 2 : 1. If A remains in normal state in solution, B will be in ..... state in solution :
(1) Normal (2) Associated
(3) Hydrolysed (4) Dissociated

CSO074

75. Vant Hoff factor is :
(1) Less than one in case of dissociation
(2) More than one in case of association
(3) Always less than one
(4) Less than one in case of association
CSO075

76. The vant Hoff factor (i) for a dilute solution of $K_3[Fe(CN)_6]$ is :
(1) 10 (2) 4 (3) 5 (4) 0.25
CSO076

77. The experimental molecular weight of an electrolyte will always be less than its calculated value because the value of vant Hoff factor, 'i' is :
(1) Less than 1 (2) Greater than 1
(3) One (4) Zero
CSO077

78. The vant Hoff factor (i) for a dilute aqueous solution of Glucose is :
(1) Zero (2) 1.0 (3) 1.5 (4) 2.0
CSO078

79. Osmotic pressure of 0.585% w/v NaCl solution at 27°C is.
(1) 2.49 atm (2) 4.92 atm
(3) 1.2 atm (4) 3.8 atm
CSO079

80 The substance A when dissolved in solvent B shows the molecular mass corresponding to A_3. The vant Hoff's factor will be -
(1) 1 (2) 2 (3) 3 (4) $\frac{1}{3}$
CSO080

81. The ratio of the value of any colligative property for KCl solution to that for sugar solution is nearly ..... time :
(1) 1 (2) 0.5 (3) 2 (4) 2.5
CSO081

82. The lowering of vapour pressure of 0.1M aqueous solutions of NaCl, $CuSO_4$ and K_2SO_4 are:
(1) All equal
(2) In the ratio of 1 : 1 : 1.5
(3) In the ratio of 3 : 2 : 1
(4) In the ratio of 1.5 : 1 : 2.5
CSO082

83. The freezing point of 1 molal NaCl solution assuming NaCl to be 100% dissociated in water is: (K_f = 1.86 K Molality^{-1})
(1) –1.86°C (2) –3.72°C
(3) +1.86°C (4) + 3.72°C
CSO083

84. The molal elevation constant of water is 0.51. The boiling point of 0.1 molal aqueous NaCl solution is nearly :
(1) 100.05°C (2) 100.1°C
(3) 100.2°C (4) 101.0°C
CSO084

85. Phenol associates in benzene as

$$C_6H_5OH \rightleftharpoons \frac{1}{2}(C_6H_5OH)_2$$

If degree of association of phenol is 40%, Van't Hoff factor i is :-
(1) 1 (2) 0.8 (3) 1.4 (4) 0.6
CSO085

86. The values of observed and calculated molecular weight of calcium nitrate are respectively 65.6 and 164. The degree of dissociation of calcium nitrate will be :
(1) 25% (2) 50% (3) 75% (4) 60%
CSO086

87. A 0.004M solution of Na_2SO_4 is isotonic with a 0.010M solution of glucose at the 25°C temperature. The degree of dissociation of Na_2SO_4 is
(1) 25% (2) 50% (3) 75% (4) 85%
CSO087

88. A 5.8% (wt./vol.) NaCl solution will exert an osmotic pressure closest to which one of the following :
(1) 5.8% (wt./vol) sucrose solution
(2) 5.8% (wt./vol) glucose solution
(3) 2 M sucrose solution
(4) 1 M glucose solution
CSO088

89. Which salt shows maximum osmotic pressure in its 1M solution :
(1) $AgNO_3$ (2) Na_2SO_4
(3) $(NH_4)_3PO_4$ (4) $MgCl_2$
CSO089

90. Which solution will exert highest osmotic pressure?
(1) 1M glucose solution
(2) 1M urea solution
(3) 1M Alum solution
(4) 1M NaCl solution

CSO090

91. Which is the correct relation between osmotic pressure of 0.1M NaCl solution and 0.1M Na_2SO_4 solution ?
(1) The osmotic pressure of Na_2SO_4 is less than NaCl solution
(2) The osmotic pressure Na_2SO_4 is more than NaCl solution
(3) Both have same osmotic pressure
(4) None of the above

CSO091

92. Which one of the following solutions will have highest osmotic pressure ? (Assume that all the salts are equally dissociated)
(1) 0.1M $Al_2(SO_4)_3$
(2) 0.1M $BaCl_2$
(3) 0.1 M Na_2SO_4
(4) The solution obtained by mixing equal volumes of (2) and (3)

CSO092

93. The following solutions have equal concentrations. Which one will show minimum osmotic pressure?
(1) $BaCl_2$ (2) $AgNO_3$
(3) Na_2SO_4 (4) $(NH_4)_3PO_4$

CSO093

94. The osmotic pressure of equimolar solutions of $BaCl_2$, NaCl, and glucose will be in the order:
(1) Glucose > NaCl > $BaCl_2$
(2) $BaCl_2$ > NaCl > Glucose
(3) NaCl > $BaCl_2$ > Glucose
(4) NaCl > Glucose > $BaCl_2$

CSO094

95. Which one of the following pairs of solutions will be expected to be isotonic under the same temperature ?
(1) 0.1M urea and 0.1 M NaCl
(2) 0.1M urea and 0.2 M $MgCl_2$
(3) 0.1M NaCl and 0.1M Na_2SO_4
(4) 0.1M $Ca(NO_3)_2$ and 0.1M Na_2SO_4

CSO095

96. Two solutions of KNO_3 and CH_3COOH are prepared separately. Molarity of both is 0.1M and osmotic pressures are P_1 and P_2 respectively. The correct relationship between the osmotic pressures is
(1) $P_2 > P_1$ (2) $P_1 = P_2$
(3) $P_1 > P_2$ (4) $\frac{P_1}{P_1+P_2} = \frac{P_2}{P_1+P_2}$

CSO096

97. The correct relationship between the boiling points of dilute solutions of $AlCl_3(T_1)$ and $CaCl_2(T_2)$, having the same molar concentration is :
(1) $T_1 = T_2$ (2) $T_1 > T_2$
(3) $T_2 = T_1$ (4) $T_2 \geq T_1$

CSO097

98. Which of the following 0.1 M aqueous solutions will have the lowest freezing point :
(1) Potassium Sulphate
(2) Sodium Chloride
(3) Urea
(4) Glucose

CSO098

99. Which aqueous solution has minimum freezing point?
(1) 0.01 M NaCl (2) 0.005 M C_2H_5OH
(3) 0.005 M MgI_2 (4) 0.005 M $MgSO_4$

CSO099

100. If α is the degree of dissociation of $K_4[Fe(CN)_6]$, then abnormal mass of complex in the solution will be :-
(1) $M_{normal}\,(1+2\alpha)^{-1}$ (2) $M_{normal}\,(1+3\alpha)^{-1}$
(3) $M_{normal}\,(1+\alpha)^{-1}$ (4) $M_{normal}\,(1+4\alpha)^{-1}$

CSO100

101. Which solution will have the least vapour pressure :
(1) 0.1 M $BaCl_2$ (2) 0.1 M urea
(3) 0.1 M Na_2SO_4 (4) 0.1 M Na_3PO_4

CSO101

102. Which has maximum freezing point :
(1) 1M of NaCl solution (2) 1M of KCl solution
(3) 1M of $CaCl_2$ solution (4) 1M of urea solution

CSO102

103. The freezing point of 1% w/w aqueous solution of calcium nitrate will be :
(1) 0^0C (2) Above 0^0C
(3) 1^0C (4) Below 0^0C
CSO103

104. The following aqueous solution in the correct order of decreasing freezing point is -
(1) 0.2M $BaCl_2$, 0.2M KCl, 0.1M Na_2SO_4
(2) 0.2M KCl, 0.1M Na_2SO_4, 0.2M $BaCl_2$
(3) 0.1M Na_2SO_4, 0.2M KCl, 0.2M $BaCl_2$
(4) 0.1M Na_2SO_4, 0.2M $BaCl_2$, 0.2M KCl
CSO104

105. Which of the following solutions will have the highest boiling point ?
(1) 1% w/w Glucose in water
(2) 1% w/w NaCl in water
(3) 1% w/w $Ca_3(PO_4)_2$ in water
(4) 1% w/w Urea in water
CSO105

106. The freezing point of equimolal aqueous solution will be highest for :
(1) Urea (2) $Ca(NO_3)_2$
(3) $C_6H_{12}O_6$(Glucose) (4) (1) and (3) both
CSO106

107. When mercuric Iodide is added to the aqueous solution of potassium iodide ?
(1) The boiling point does not change
(2) Freezing point is raised
(3) The freezing point is lowered
(4) Freezing point does not change
CSO107

108. The molecular weight of benzoic acid in benzene as determined by depression in freezing point method corresponds to :
(1) Ionisation of benzoic acid
(2) Dimerization of benzoic acid
(3) Trimerization of benzoic acid
(4) Solvation of benzoic acid
CSO108

109. What would be the osmotic pressure of 0.1 M K_2SO_4 solution (90% dissociated) at 27°C :-
(1) 6.89 atm (2) 0.689 atm
(3) 0.344 atm (4) 3.4 atm
CSO109

110. Which one has same Van't Hoff factor i as that of Hg_2Cl_2 :-
(1) NaCl (2) Na_2SO_4
(3) $Al(NO_3)_3$ (4) $Al_2(SO_4)_3$
CSO110

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	4	4	3	4	4	3	2	2	3	3	2	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	2	4	1	1	4	1	4	2	2	1	3	4	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	3	2	1	4	1	1	1	2	1	4	3	2	2	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	3	2	2	4	1	4	4	2	1	4	1	2	1	4
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	3	3	2	3	1	2	3	4	2	2	4	1	2	4
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	2	2	2	2	4	3	2	2	2	2	3	3	3	3	3
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	2	1	2	2	4	3	2	1	1	4	4	4	4	3	2
Question	106	107	108	109	110										
Answer	4	2	2	1	2										

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Of the following 0.10m aqueous solutions, which one will exhibit the largest freezing point depression?
(1) KCl (2) $C_6H_{12}O_6$
(3) $Al_2(SO_4)_3$ (4) K_2SO_4

CSO122

AIPMT 2015

2. The boiling point of 0.2 mol kg^{-1} solution of X in water is greater than equimolal solution of Y in water. Which one of the following statements is true in this case?
(1) Molecular mass of X is greater than the molecular mass of Y.
(2) Molecular mass of X is less than the molecular mass of Y.
(3) Y is undergoing dissociation in water while X undergoes no change.
(4) X is undergoing dissociation in water while Y undergoes no change.

CSO123

3. Which one is not equal to zero for an ideal solution ?
(1) ΔS_{mix}
(2) ΔV_{mix}
(3) $\Delta P = P_{observed} - P_{Raoult}$
(4) ΔH_{mix}

CSO124

4. Which one of the following electrolytes has the same value of van't Hoff's factor (i) as that of the $Al_2(SO_4)_3$ (if all are 100% ionised) ?
(1) $K_3[Fe(CN)_6]$ (2) $Al(NO_3)_3$
(3) $K_4[Fe(CN)_6]$ (4) K_2SO_4

CSO125

Re-AIPMT 2015

5. What is the mole fraction of the solute in a 1.00 m aqueous solution ?
(1) 0.0354 (2) 0.0177
(3) 0.177 (4) 1.770

CSO126

NEET-I 2016

6. Which of the following statement about the composition of the vapour over an ideal a 1 : 1 molar mixture of benzene and toluene is correct? Assume that the temperature is constant at (25°C).
(Given : Vapour Pressure Data at 25°C, benzene = 12.8 kPa, Toluene = 3.85 kPa)
(1) The vapour will contain a higher percentage of benzene
(2) The vapour will contain a higher percentage of toluene
(3) The vapour will contain equal amounts of benezene and toluene
(4) Not enough information is given to make a predication

CSO127

7. At 100°C the vapour pressure of a solution of 6.5g of a solute in 100 g water is 732 mm. If K_b = 0.52°C m^{-1}, the boiling point of this solution will be :-
(1) 101°C (2) 100°C (3) 102°C (4) 103°C

CSO128

8. Consider the following liquid - vapour equilibrium.
Liquid $\rightleftharpoons$ Vapour
Which of the following relations is **correct** ?

(1) $\frac{d\ell nG}{dT^2} = \frac{\Delta H_v}{RT^2}$ (2) $\frac{d\ell nP}{dT} = \frac{-\Delta H_v}{RT}$

(3) $\frac{d\ell nP}{dT^2} = \frac{-\Delta H_v}{T^2}$ (4) $\frac{d\ell nP}{dT} = \frac{\Delta H_v}{RT^2}$

CSO129

NEET-II 2016

9. The van't Hoff factor (i) for a dilute aqueous solution of the strong electrolyte barium hydroxide is
(1) 2 (2) 3 (3) 0 (4) 1

CSO130

10. Which one of the following is **incorrect** for ideal solution ?
(1) $\Delta P = P_{obs} - P_{calculated\ by\ Raoult's\ law} = 0$
(2) $\Delta G_{mix} = 0$
(3) $\Delta H_{mix} = 0$
(4) $\Delta U_{mix} = 0$

CSO131

NEET(UG) 2017

11. If molality of a dilute solution is doubled, the value of molal depression constant (K_f) will be:-
(1) Halved (2) Tripled
(3) Unchanged (4) Doubled
CSO132

12. Which of the following is dependent on temperature?
(1) Molarity (2) Mole fraction
(3) Weight percentage (4) Molality
CSO133

NEET(UG) 2019

13. For an ideal solution, the **correct** option is :-
(1) Δ_{mix} S = 0 at constant T and P
(2) Δ_{mix} V ≠ 0 at constant T and P
(3) Δ_{mix} H = 0 at constant T and P
(4) Δ_{mix} G = 0 at constant T and P
CSO134

NEET(UG)(Odisha) 2019

14. Which of the following statements is correct regarding a solution of two compounds A and B exhibiting positive deviation from ideal behaviour?
(1) Intermolecular attractive forces between A-A and B-B are stronger than those between A-B.
(2) Δ_{mix} H = 0 at constant T and P
(3) Δ_{mix} V = 0 at constant T and P
(4) Intermolecular attractive forces between A-A and B-B are equal to those between A-B.
CSO135

15. The density of 2 M aqueous solution of NaOH is 1.28 g/cm^3. The molality of the solution is [Given that molecular mass of NaOH = 40 g mol^{-1}]
(1) 1.20 m (2) 1.56 m
(3) 1.67 m (4) 1.32 m
CSO136

NEET (UG) 2020

16. The freezing point depression constant (K_f) of benzene is 5.12 K kg mol^{-1}. The freezing point depression for the solution of molality 0.078 m containing a non-electrolyte solute in benzene is (rounded off upto two decimal places) :
(1) 0.60 K (2) 0.20 K
(3) 0.80 K (4) 0.40 K
CSO137

17. The mixture which shows positive deviation from Raoult's law is :-
(1) Chloroethane + Bromoethane
(2) Ethanol + Acetone
(3) Benzene + Toluene
(4) Acetone + Chloroform
CSO138

NEET (UG) 2020 (COVID-19)

18. If 8g of a non-electrolyte solute is dissolved in 114 g of n-octane to reduce its vapour pressure to 80%, the molar mass (in g mol^{-1}) of the solute is
[Given that molar mass of n-octane is 114g mol^{-1}]
(1) 40 (2) 60 (3) 80 (4) 20
CSO139

19. Isotonic solutions have same
(1) Vapour pressure
(2) Freezing temperature
(3) Osmotic pressure
(4) Boiling temperature
CSO140

NEET (UG) 2021

20. The following solutions were prepared by dissolving 10 g of glucose ($C_6H_{12}O_6$) in 250 ml of water (P_1), 10 g of urea (CH_4N_2O) in 250 ml of water (P_2) and 10 g of sucrose ($C_{12}H_{22}O_{11}$) in 250 ml of water (P_3). The right option for the decreasing order of osmotic pressure of these solutions is :
(1) $P_2 > P_1 > P_3$ (2) $P_1 > P_2 > P_3$
(3) $P_2 > P_3 > P_1$ (4) $P_3 > P_1 > P_2$
CSO141

21. The correct option for the value of vapour pressure of a solution at 45°C with benzene to octane in molar ratio 3 : 2 is :
[At 45°C vapour pressure of benzene is 280 mm Hg and that of octane is 420 mm Hg. Assume Ideal solution]
(1) 160 mm of Hg (2) 168 mm of Hg
(3) 336 mm of Hg (4) 350 mm of Hg
CSO142

NEET (UG) 2022

22. In one molal solution that contains 0.5 mole of a solute, there is
(1) 500 g of solvent (2) 100 mL of solvent
(3) 1000 g of solvent (4) 500 mL of solvent
CSO143

23. K_H value for some gases at the same temperature 'T' are given :

gas	K_H/k bar
Ar	40.3
CO_2	1.67
HCHO	1.83×10^{-5}
CH_4	0.413

where K_H is Henry's Law constant in water. The order of their solubility in water is :

(1) $Ar < CO_2 < CH_4 < HCHO$

(2) $Ar < CH_4 < CO_2 < HCHO$

(3) $HCHO < CO_2 < CH_4 < Ar$

(4) $HCHO < CH_4 < CO_2 < Ar$

CSO144

NEET (UG) 2023

24. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Helium is used to dilute oxygen in diving apparatus.

Reasons R : Helium has high solubility in O_2.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**.

(2) **A** is true but **R** is false

(3) **A** is false but **R** is true

(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

CSO165

NEET (UG) 2024

25. The Henry's law constant (K_H) values of three gases (A, B, C) in water are 145, 2×10^{-5} and 35 kbar, respectively. The solubility of these gases in water follow the order :

(1) B > A > C (2) B > C > A

(3) A > C > B (4) A > B > C

CSO190

26. The plot of osmotic pressure (Π) vs concentration (mol L^{-1}) for a solution gives a straight line with slope 25.73 L bar mol^{-1}. The temperature at which the osmotic pressure measurement is done is:

(Use R = 0.083 L bar mol^{-1} K^{-1})

(1) 37°C (2) 310°C

(3) 25.73°C (4) 12.05°C

CSO191

NEET (UG) 2024 (Re-examination)

27. The amount of glucose required to prepare 250 mL of $\frac{M}{20}$ aqueous solution is :

(Molar mass of glucose : 180 g mol^{-1})

(1) 2.25 g (2) 4.5 g (3) 0.44 g (4) 1.125g

CSO192

28. Mass of glucose ($C_6H_{12}O_6$) required to be dissolved to prepare one litre of its solution which is isotonic with 15g L^{-1} solution of urea (NH_2CONH_2) is :

(Given : Molar mass in g mol^{-1}C : 12, H : 1, O : 16, N : 14)

(1) 55 g (2) 15 g (3) 30 g (4) 45 g

CSO193

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	4	1	3	2	1	1	4	2	2	3	1	3	1	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28		
Answer	4	2	1	3	1	3	1	1	2	2	1	1	4		

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Exercise - III (Analytical Questions) Build Up Your Understanding

1. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.

Assertion (A) : Raoult's law is a special case of Henry's law

Reason (R) : According to Raoult's law $P_i = x_i p_i^\circ$ According to Henry's law $P = K_H X$

Here K_H differ from p_i°. In special case K_H becomes equal to p_i°.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**.

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is not correct but **(R)** is correct.

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.

CSO145

2. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.

Assertion (A) : Van't Hoff factor (i) depend on concentration in case of incomplete dissociation.

Reason (R) : For incomplete dissociation of electrolyte, value of Van't Hoff factor(i) increase with decrease in concentration.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is not correct but **(R)** is correct.

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

CSO146

3. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.

Assertion (A) : Cooking time is reduced in pressure cooker.

Reason (R) : Boiling point of liquid inside the pressure cooker is raised.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**.

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is not correct but **(R)** is correct.

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.

CSO147

4. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.

Assertion (A) : Isotonic solutions does not show net phenomenon of osmosis.

Reason (R) : Isotonic solutions have equal osmotic pressure.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**

(2) **(A)** is correct but **(R)** is not correct.

(3) **(A)** is not correct but **(R)** is correct.

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

CSO148

5. Consider following statements :-

(a) Isotonic solution does not show net phenomenon of osmosis

(b) Isotonic solution does not show flow of solvent

(c) Isotonic solution have equal osmotic pressure

(d) Osmotic pressure depend on temperature.

Which is/are correct statements

(1) a, b (2) a, c, d

(3) b, d (4) b, c

CSO151

6. Match the column

	Column-I		Column-II
(I)	Van't Hoff factor for dissociation	(a)	More than 1
(II)	Van't Hoff factor for association	(b)	Less than 1
(III)	Van't Hoff factor for non electrolyte	(c)	May be greater than 1 or less than 1.
(IV)	Van't Hoff factor for electrolyte	(d)	Equal to 1

Which is correct :-

(1) (I) → (a), (II) → (d), (III) → (c), (IV) → (b)

(2) (I) → (d), (II) → (b), (III) → (a), (IV) → (c)

(3) (I) → (b), (II) → (c), (III) → (a), (IV) → (d)

(4) (I) → (a), (II) → (b), (III) → (d), (IV) → (c)

CSO152

7. Consider following statements.

(a) Ideal solutions are one which obeys Raoult's law at all temperature & concentration

(b) Very dilute solution can be treated as ideal solution.

(c) Ideal solution form azeotropic mixture

(d) Azeotropic mixture are not constant boiling mixture.

Which are correct :-

(1) a, b (2) b, c (3) a, c (4) c, d

CSO154

8. Which is correct statement :-

(1) Colligative properties depend on chemical nature of solute.

(2) Colligative properties depend on relative number of solute particles.

(3) Colligative properties are same for electrolyte and nonelectrolyte solute solution.

(4) Osmosis is a colligative property.

CSO157

9. Which is correct statement :-

(1) Solute and solvent molecules pass through SPM.

(2) Only solute molecules pass through SPM.

(3) Only solvent molecules pass through SPM

(4) Neither solute nor solvent pass through SPM

CSO158

10. Formula of Van't Hoff factor will be :-

(1) $i = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$

(2) $i = \frac{\text{Theoretical colligative property}}{\text{Observed colligative property}}$

(3) $i = \frac{\text{Abnormal molar mass}}{\text{Normal molar mass}}$

(4) $i = \frac{\text{No. of mole of particles before association or dissociation}}{\text{No. of mole of particle after association or dissociation}}$

CSO159

11. Which are colligative properties

(1) Boiling point, osmotic pressure

(2) Freezing point and osmosis

(3) Elevation in boiling point and osmotic pressure

(4) Vapour pressure and depression in freezing point

CSO160

12. Consider the following statements.

(a) Molal depression constant depend on the nature of solvent.

(b) Elevation in boiling point for dilute solution is directly proportional to molality.

(c) The unit of molal depression constant is K kg eq^{-1}.

(d) Molality is the function of temperature.

Which are correct statement.

(1) a, b (2) b, c

(3) c, d (4) a, d

CSO161

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12
Answer	4	4	4	4	2	4	1	2	3	1	3	1

Solutions

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (3)

$$M = \frac{n}{V_{(L)}} = \frac{w}{M_w \times V_{(L)}} = \frac{8}{40 \times 1} = 0.2\ M$$

2. Ans. (4)

Molality is defined as no. of moles of solute dissolved in 1000 g of solvent

$$M = \frac{n_B}{w_A(g)} \times 1000$$

$$\frac{18 \times 1000}{180 \times 1000} = 0.1M$$

3. Ans. (4)

$$M = \frac{n}{V_{(ml)}} \times 1000 = \frac{5.85 \times 1000}{58.5 \times 500} = 0.2\ M$$

4. Ans. (4)

$$M = \frac{n}{V_{(ml)}} \times 1000$$

$$0.1 = \frac{w \times 1000}{M_w \times 1000}$$

$$0.1 = \frac{w \times 1000}{98 \times 1000}$$

w = 9.8 g

5. Ans. (3)

$$X_A = \frac{n_A}{n_A + n_B}$$

$$n_{glycerine} = \frac{w}{M_w} = \frac{46}{92} = 0.5 = n_A$$

$$n_{H_2O} = \frac{w}{M_w} = \frac{36}{18} = 2 = n_B$$

$$X_A = \frac{n_A}{n_A + n_B} = \frac{0.5}{2.5} = 0.2$$

6. Ans. (4)

$$ppm = \frac{\text{wt of solute(g)}}{\text{wt of solution(g)}} \times 10^6$$

$$= \frac{10}{1000} \times 10^6 = 10{,}000\ ppm$$

7. Ans. (4)

$$N = \frac{w}{E_w \times V_{(L)}} \text{ or } N = n_f \times M$$

n_f of H_3PO_4 = 3

$N = n_f \times M$

Normality of 1M H_3PO_4 is = 3×1 = 3N

8. Ans. (3)

Normality = $n_f \times M$

$N = 2 \times M$

$0.2 = 2 \times M$

$M = 0.1\ M$

n_f of Na_2CO_3 = 2

$Na_2CO_3 = 2Na^+ + CO_3^{2-}$

n_f = total +ive or total -ive charge = 2

9. Ans. (2)

Phosphorous acid (H_3PO_3) ; $n_f = 2$

$N = n_f \times M$

$N = 2 \times 0.3 = 0.6N$

10. Ans. (2)

Molarity of pure water is 55.5 M

$$M = \frac{n}{V_{(L)}} = \frac{w}{M_w \times V_{(ml)}} \times 1000$$

$$\left(\text{density} = \frac{w}{V_{(ml)}} = 1\right)$$

Density of water is 1 g 1 ml

$$M = \frac{1000}{M_W} = \frac{1000}{18} = 55.5\ M$$

11. Ans. (3)

Molarity of pure water is always 55.5 M

12. Ans. (3)

$w_{NaCl} = w_{KCl}$, but M_w of NaCl < M_w of KCl

$$M = \frac{n}{V_{(L)}} = \frac{w}{M_w \times V_{(L)}}$$

$$M \propto \frac{1}{\text{Mol. wt of solute}}$$

So, $M_{NaCl} > M_{KCl}$

13. Ans. (2)

$$X_A = \frac{n_A}{n_A + n_B}$$

$$n_{O_2} = \frac{w}{M_w} = \frac{8}{32} = \frac{1}{4};\ n_{N_2} = \frac{w}{M_w} = \frac{7}{28} = \frac{1}{4}$$

$$X_{O_2} = \frac{n_{O_2}}{n_{O_2} + n_{N_2}} = \frac{\frac{1}{4}}{\frac{1}{4} + \frac{1}{4}} = 0.5$$

14. Ans. (1)

$$X_A = \frac{n_A}{n_A + n_B}$$

$$n_A = \frac{w}{M_w} = \frac{7.8}{78} = 0.1$$

$$n_B = \frac{w}{M_w} = \frac{46}{92} = 0.5$$

$$X_{Benzene} = \frac{n_{Benzene}}{n_{Benzene} + n_{toluene}} = \frac{0.1}{0.1 + 0.5} = \frac{1}{6}$$

15. Ans. (2)

$$\text{Molality} = \frac{n_B}{w_A} \times 1000$$

Mole traction of solute in benzene = 0.2

$$X_B = \frac{n_B = 0.2}{n_A + n_B}$$

$$X_A = 1 - X_B = 1 - 0.2 = 0.8 = = \frac{n_A}{n_A + n_B}$$

$$\frac{X_B}{X_A} = \frac{n_B}{n_A} = \frac{0.2}{0.8} \qquad n_A = \frac{w_A}{M_A}$$

$$M = \frac{n_B}{w_A} \times 1000 \qquad w_A = n_A \times M_A$$

$$M = \frac{n_B}{78 \times n_A} \times 1000 \qquad = n_A \times 78$$

$$= \frac{0.2}{0.8 \times 78} \times 1000 \qquad M_A = M_w \text{ of benzene}$$

$$= \frac{1000}{78 \times 4} = 3.2 \qquad = 78$$

$$\boxed{X = 3.2}$$

16. Ans. (2)

$$\text{ppm} = \frac{\text{wt of solute(g)}}{\text{wt of sol}^n\text{(g)}} \times 10^6$$

$$= \frac{0.02}{500} \times 10^6$$

= 40 ppm

17. Ans. (2)

$$N = \frac{X}{5.6} = \frac{20}{5.6} = 3.58$$

18. Ans. (2)

$$M = \frac{n}{V_{(ml)}} \times 1000 = \frac{5 \times 1000}{34 \times 100} = 1.5M$$

19. Ans. (4)

$10\%\left(\frac{w}{V}\right)$ H_2SO_4 solution means 10g H_2SO_4 in 100 ml solution

$$N = \frac{w \times 1000}{E_w \times V_{ml}} \qquad \because E_w \text{ of } H_2SO_4 = 49g$$

$$= \frac{10 \times 1000}{49 \times 100} = 2$$

20. Ans. (1)

$$N = \frac{w \times 1000}{E_w \times V_{(ml)}}$$

$$0.1 = \frac{6.3 \times 1000}{63 \times V_{(ml)}}$$

$V_{(ml)}$ = 1000 ml = 1 L

21. Ans. (1)

As per definition, molality is defined as no. of moles of a solute in 1kg solvent.

22. Ans. (4)

As per theory

23. Ans. (1)

$$M = \frac{n_1}{V_{solution}} \qquad m = \frac{n_2}{w_{solution}}$$

$$1 = \frac{n_2}{V} \qquad 1 = \frac{n_2}{w}$$

$w_{solvent} < V_{solution}$ $\quad (d_{water} = 1g/ml)$

$n_1 > n_2$

A is more concentrated than B.

24. Ans. (4)

$M = \frac{n}{V_{solution}}$ $\quad m = \frac{n}{w_{solution}}$

$w_{solvent} < V_{solution}$ $\quad (d_{water} = 1g/ml)$

Hence m > M

25. Ans. (2)

98% (w/W) H_2SO_4 means 98g H_2SO_4 in 100 g solution.

$d_{solution} = 1.8$ g/ml

$\text{density} = \frac{\text{wt. of solution}}{\text{Vol. of solution}}$

$1.8 = \frac{100}{\text{Vol. of Solution}}$

$\text{Vol. of solution} = \frac{100}{1.8}$

$M = \frac{n\times1000}{V_{(ml)}} = \frac{98\times1.8\times1000}{98\times100} = 18\ M$

26. Ans. (2)

$P = K_H X$

$760 = 2\times10^5 X_g$

$X_g = \frac{760}{2\times10^5} = 3.8\times10^{-3}$

27. Ans. (1)

NH_3 reacts with water

28. Ans. (3)

$P_S = P_A^0X_A + P_B^0X_B$

$X_A = \frac{1}{1+4} = \frac{1}{5}$

$X_B = \frac{4}{1+4} = \frac{4}{5}$

$P_S = 92\times\frac{1}{5} + 31\times\frac{4}{5}$

$= \frac{92+124}{5} = \frac{216}{5} = 43.2\ \text{mm Hg}$

29. Ans. (4)

$P_S = P_A^0X_A + P_B^0X_B$ $\quad \because P_S$ = 1atm or 760 torr

$760 = 900\ X_A + 360\ X_B$

$760 = 900\ X_A + 360(1-X_A)$

$760 = 360 + 540\ X_A$

$400 = 540\ X_A$

$X_A = 0.740$

30. Ans. (2)

$Y_A = \frac{P_A^0X_A}{P_A^0X_A + P_B^0X_B}$ $\quad X_A = 0.5 = X_B$

$Y_A = \frac{108\times0.5}{108\times0.5+36\times0.5} = \frac{108}{144} = 0.75$

31. Ans. (4)

$Y_A = \frac{P_A^0X_A}{P_S}$ As per theory

32. Ans. (3)

$\Delta S_{mixing} \neq 0$ (for ideal solution)

Entropy always increases on mixing

33. Ans. (2)

As per theory

34. Ans. (1)

As per definition

35. Ans. (4)

Refractive index is not a colligative property

36. Ans. (1)

$P_S = P_A^0X_A + P_B^0X_B$ (P_B^0 =0 for non-volatile solute)

$P_S = P_A^0 \times X_A$

$P_S = P_A^0(1-X_B)$ $\quad \because X_A = 1-X_B$

$P_S = P_A^0 - P_A^0X_B$

$P_A^0X_B = P_A^0 - P_S = \Delta P$

$\Delta P = P_A^0X_B$

$\boxed{\Delta P \propto X_B}$

37. Ans. (1)

As per theory

38. Ans. (1)

$P_S = P_A^0X_A + P_B^0X_B$

$P_B^0 = 0$; Since solute is solid (non-volatile)

$P_S = P_A^0X_A$

$\boxed{P_S \propto P_A^0}$

39. Ans. (2)

$P_s = P_0\ N_1 + O$

$N_1 + N_2 = 1$

$N_1 = 1 - N_2$

$P_s = P_0\ (1-N2)$

$P_s = P_0 - P_0\ N_2$

$P_0 - P_s = P_0\ N_2$

40. Ans. (1)

$P_S = P_A^0 X_A + P_B^0 X_B$ $\quad X_A = \frac{n_A}{n_A + n_B} = \frac{2}{1+2} = \frac{2}{3}$

$P_S = P_A^0 X_A$ $\quad X_B = \frac{n_B}{n_A + n_B} = \frac{1}{3}$

$P_S = P_A^0 \times \frac{2}{3}$ $\quad P_B^0 = 0$

$\frac{P_S}{P_A^0} = \frac{2}{3}$

41. Ans. (4)

$P_S = P_A^0 X_A + P_B^0 X_B$

At 373K ; P_A^0 = 760 mm Hg for aqueous sol.

$750 = 760\, X_A + 0$

$750 = 760(1 - X_B)$

$750 = 760 - 760 X_B$

$X_B = \frac{10}{760} = \frac{1}{76}$

42. Ans. (3)

$P_S = P_A^0 X_A + P_B^0 X_B$ $\quad X_B = 0.1$

$P_S = 23.8\,(0.9) + 0$ $\quad X_A = 1 - X_B = 0.9$

$P_S = 21.42$ mm Hg

43. Ans. (2)

$\frac{P_A^0 - P_S}{P_S} = \frac{n_B}{n_A} = \frac{w_B \times M_A}{M_B \times w_A}$

$\frac{10-9}{9} = \frac{1 \times 200}{M_B \times 20}$

$\frac{1}{9} = \frac{10}{M_B}$

M_B = 90 amu

44. Ans. (2)

CH_3OH will have highest vapour pressure since its Bpt is lowest.

45. Ans. (1)

As per Raoult's law.

Vapour pressure of each component is directly proportional to its mole fraction.

46. Ans. (3)

$P_S = P_A^0 X_A + P_B^0 X_B$

$84 = 70(0.8) + P_B^0(0.2)$

$84 = 56 + 0.2\, P_B^0$

$28 = 0.2\, P_B^0$

P_B^0 = 140

47. Ans. (3)

$\Delta T_b = K_b.m \,\&\, \Delta T_f = K_f m$

$(\Delta T_b)_1 = K_b \frac{n_B}{w_A}$

$\Delta T_f = K_f \frac{n_B}{(2w_A)}$

$\frac{\Delta T_b}{\Delta T_f} = \frac{K_b(2)}{K_f}$

$\frac{0.15}{\Delta T_f} = \frac{2(0.512)}{1.86}$

$\Delta T_f = 0.272$

$T_f' = -0.272°C$

48. Ans. (2)

$\Delta T_b = K_b m$

$K_b = \frac{\Delta T_b}{m}$

49. Ans. (2)

$\Delta T_b = K_b m$

$0.52 = 0.52 \times \left[\frac{6 \times 1000}{M_X.100}\right]$

$M_X = 60$

50. Ans. (4)

$\Delta T_b = K_b m$

For 1 g area

$0.25 = K_b \times \left[\frac{1}{60.w_A}\right]$ ... (1)

For 3g glucose

$\Delta T_b = K_b \times \frac{3}{180 \times w_A}$ ... (2)

$\frac{eq.(1)}{eq.(2)} \Rightarrow \frac{0.25}{\Delta T_b} = 1$ $\quad \Rightarrow \Delta T_b = 0.25$

$T'_b = 100.25°$ C

51. Ans. (1)

$\Delta T_f = K_f m$

$(5.45 - 3.55) = K_f \times 0.374$

$1.90 = K_f \times 0.374$

$K_f = 5.08\ K\ kg\ mol^{-1}$

52. Ans. (4)

$\Delta T_f = 0.186$

$$\frac{\Delta T_f}{\Delta T_b} = \frac{K_f m}{K_b m}$$

$$\frac{0.186}{\Delta T_b} = \frac{1.86}{0.512}$$

$\Delta T_b = 0.0512$

53. Ans. (4)

As per theory

Only solvent molecules

Solidify at freezing point

54. Ans. (2)

$\Delta T_f = K_f M$

$$= 1.86 \times \left(\frac{0.02}{100} \times 1000 \right)$$

$= 0.372°C$

55. Ans. (1)

$\Delta T_f = K_f m$

$$= 1.86 \times \left(\frac{18}{180} \times \frac{1000}{1000} \right) = 0.186$$

$T_f' = -0.186°C$

56. Ans. (4)

$\Delta T_f = 273.15 - 271.94$

$= 1.21$

$\Delta T_f = K_f m$

$$1.21 = 1.86 \times \left[\frac{1.25}{M_w} \times \frac{1000}{20} \right]$$

$M_W = 96.01 g/mol$

57. Ans. (1)

As per definition of osmosis

58. Ans. (2)

As per definition of osmosis

59. Ans. (1)

$\pi \propto C$

$\pi = CRT$

60. Ans. (4)

Silicates of cobalt & nickel & copper ferrocyanide are inorganic compounds which act as semipermeable membrane.

61. Ans. (1)

If $C_1 = C_2$

No movement across membrane.

62. Ans. (3)

Mole Fraction of solute will increase & hence osmotic pressure will increase.

63. Ans. (3)

$$\pi = \frac{n}{v} RT$$

Osmotic pressure of solution will increase if no. of solute molecule is increases.

64. Ans. (2)

$\pi_1 = C_1 RT$

$$= \frac{3.42}{342 \times 1} RT$$

$= 0.01\ RT$

$\pi_2 = C_2 RT$

$$= \frac{0.18}{180 \times 0.1} \times RT$$

$= 0.01\ RT$

65. Ans. (3)

2 molal sucrose solution means 2 mole sucrose in 1kg solvent

wt of sol. = wt of solute + wt of solvent

$= (2 \times 342) + 1000$

$= 1684\ g$

$$d_{sol.} = \frac{W}{V_{sol}}$$

$$1.4 = \frac{1684}{V_{(ml)}}$$

$$V_{(ml)} = \frac{1684}{1.4}$$

$\pi = CRT$

$$= \left(\frac{n}{V_{(ml)}} \times 1000 \right) \times 0.0821 \times 298$$

$$= \frac{2}{1684} \times 1.4 \times 1000 \times 0.082 \times 298$$

$= 40.7\ atm$

66. Ans. (1)

$\pi = CRT$

$= 0.3 \times (0.0821 \times 298)$

= 7.34 atm

67. Ans. (2)

$n_1 = M_1V_1$ $(V_1=V_2=V)$

$n_2 = M_2V_2$ $M_1 = M_2 = 0.1M$

$n_1 = 0.1\ V$

$n_2 = 0.1\ V$

when equal volume of both are mixed, then

$$\pi = \left(\frac{n_1+n_2}{2V}\right)RT$$

$$= \left(\frac{0.1V+0.1V}{2V}\right)RT$$

$= (0.1)\ RT$

68. Ans. (3)

wt of solution = 3+60

= 63 g

$$d = \frac{w}{V_{(ml)}} = 1g/ml$$

$$\frac{63}{V_{(ml)}} = 1$$

$V_{ml} = 63$ ml

$\because \pi = CRT$

$$\pi = \left(\frac{n}{V_{ml}}\times 1000\right)\times 288\times 0.0821$$

$$= \left(\frac{3\times 1000}{180\times 63}\right)\times 288\times 0.0821 = 6.25 \text{ atm.}$$

69. Ans. (4)

$\pi = CRT$

$2.5 = C(0.0821 \times 297)$

$C = 0.1025$

70. Ans. (2)

$\pi_1 = \pi_2$

$C_1RT = C_2RT$

$C_1 = C_2$

$$\frac{8.6}{60\times 1} = \frac{0.5\times 1000}{M_W\times 100}$$

$M_W = 34.89$ g

71. Ans. (2)

$\pi_1 = \pi_2$

$C_1RT = C_2RT$

$C_1 = C_2$

$$\frac{n_1}{V_1} = \frac{n_2}{V_2}$$

$$\frac{6.84\times 1000}{342\times 100} = \frac{1.52\times 1000}{M_2\times 100}$$

$M_2 = 76$

72. Ans. (4)

$\pi_1 = \pi_2$

$C_1RT = C_2RT$

$C_1 = C_2$

$$\frac{n_1}{V_1} = \frac{n_2}{V_2}$$

$$\frac{500}{M_W\times 1} = \frac{3.42}{342\times 1}$$

$M_2 = 50000$

73. Ans. (1)

Intravenous injection are dissolved in water containing salts at particular ionic concentration that matches plasma concentration.

74. Ans. (2)

$$\frac{(\Delta T_f)_A}{(\Delta T_f)_B} = \frac{(iK_fm)_A}{(iK_fm)_B} = \frac{2}{1}$$

$i_A = 1$ (given)

$$\frac{i_A}{i_B} = \frac{2}{1}$$

$$\frac{1}{i_B} = \frac{2}{1}$$

$$i_B = \frac{1}{2}$$

$i_B < 1$ (Case of Association)

75. Ans. (4)

$i < 1$ (case of Association)

76. Ans. (2)

$$K_3[Fe(CN)_6] \rightarrow 3K^+ + [Fe(CN)_6]^{-3}$$

Total no. of ions = 4

(After dissociation)

77. Ans. (2)

If $i > 1$, experimental molecular weight of an electrolyte will always be less than its calculated value.

78. Ans. (2)

For non-electrolyte Vant Hoff factor is 1

79. Ans. (2)

$\pi = iCRT$

($i = 2$ for NaCl)

$\pi = \left(\frac{2\times0.585\times1000}{58.5\times100}\right)RT = 4.92$ atm

80. Ans. (4)

$i = 1 - \alpha + \frac{\alpha}{n}$

(n = 3) (α = 100% = 1)

$3A \rightarrow A_3$

$i = 1 - 1 + \frac{1}{3} = \frac{1}{3}$

81. Ans. (3)

i = 2 (for KCl)

i = 1 (for sugar)

So, ratio of value of any colligative property will be 2

82. Ans. (2)

$i_{NaCl} : i_{CuSO4} : i_{K2SO4}$

= 2 : 2 : 3

= 1 : 1 : 1.5

83. Ans. (2)

$\Delta T_f = iK_fm$

$= 2\times1.86\times1$

= 3.72

$T_f^1 = -3.72°C$

84. Ans. (2)

$\Delta T_b = iK_bm$

(i = 2 for NaCl)

$\Delta T_b = 2\times0.51\times0.1$

= 0.102

$T_b^1 = 100 + 0.1$

= 100.1°C

85. Ans. (2)

$i = 1 - \alpha + \frac{\alpha}{n}$

α = 0.4, n = 2

i = 0.8

86. Ans. (3)

$i = \frac{\text{calculated } M_w}{\text{observed } M_w}$

$i = \frac{164}{65.6} = 2.5$

$i = 1-\alpha + n\alpha$

for $Ca(NO_3)_2$; n is 3

$i = 1-\alpha + 3\alpha = 1+2\alpha = 2.5$

α = 0.75

degree of dissociation is 75%

87. Ans. (3)

$\pi_{Na_2SO_4} = \pi_{glucose}$

(iCRT) = (iCRT)

$i \times 0.004 = 1\times0.010$

i = 2.5

for Na_2SO_4

$i = 1+2\alpha = 2.5$

α = 0.75

Degree of dissociation is 75%

88. Ans. (3)

$\pi = i$CRT (i = 2 for NaCl)

$= \left(\frac{2\times5.85\times1000\times RT}{58.5\times100}\right)$

= 2RT

Which is isotonic with 2M sucrose solution.

89. Ans. (3)

i = 4 for $(NH_3)_4PO_4$ & $\pi = i$CRT,

hence, $(NH_3)_4PO_4$ will exert maximum osmotic pressure.

90. Ans. (3)

1M Alum will exert highest osmotic pressure because Vant Hoff factor is maximum.

91. Ans. (2)

$\pi_{NaCl} = i$CRT (i = 2)

$\pi_{Na_2SO_4} = i$CRT (i = 3)

$i_{Na_2SO_4} > i_{NaCl}$

Hence osmotic pressure of Na_2SO_4 is more than NaCl solution.

92. Ans. (1)

i = 5 for $Al_2(SO_4)_3$ will have highest osmotic pressure.

93. Ans. (2)

i = 2 for $AgNO_3$

i = 3 for $BaCl_2$

i = 3 for Na_2SO_4

i = 4 for $(NH_4)_3PO_4$

Vant Hoff factor for $AgNO_3$ is minimum.

So it will exert minimum osmotic pressure.

94. Ans. (2)

i = 1 [Glucose]

i = 2 [NaCl]

i = 3 [$BaCl_2$]

Glucose < NaCl < $BaCl_2$

95. Ans. (4)

$\pi_{Ca(NO_3)_2} = iCRT$

= 3 × (0.1)RT

= 0.3RT

$\pi_{Na_2SO_4} = iCRT$

= 3×0.1(RT)

= (0.3RT)

96. Ans. (3)

$i_{KNO_3} = 2$

$i_{CH_3COOH} = \frac{1}{2}$

$\pi_{KNO_3} > \pi_{CH_3COOH}$

$P_1 > P_2$

97. Ans. (2)

$\Delta T_{b(AlCl_3)} = iK_bm$

$\Delta T_{b(CaCl_2)} = iK_bm$

$i_{AlCl_3} > i_{CaCl_2}$

98. Ans. (1)

$\Delta T_f = iK_fm$

$i_{K_2SO_4} = 3$ (maximum)

Hence K_2SO_4 will have lowest Freezing point.

99. Ans. (1)

$\Delta T_f = iK_fm$

$\Delta T_f \propto im$

For 0.01 M NaCl, ΔT_f will be maximum & hence minimum freezing point.

100. Ans. (4)

$i = \frac{M_{normal}}{M_{abnormal}}$

For $K_4[Fe(CN)_6]$

i = 1 +4α

$M_{Abnormal} = \frac{M_{Normal}}{i}$

= $M_{Normal}(1+4\alpha)^{-1}$

101. Ans. (4)

0.1 M Na_3PO_4 will have maximum solute particle & hence will have least vapour pressure.

102. Ans. (4)

$\Delta T_f = iK_fm$

$\Delta T_f = im$

For 1 M area sol. depression in fpt will be least & hence maximum fpt.

103. Ans. (4)

On Adding non volatile solute, fpt of solution is less than pure solvent.

104. Ans. (3)

$fpt \propto \frac{1}{\Delta T_f} \propto \frac{1}{im}$

105. Ans. (2)

$\Delta T_b = iK_bm \Rightarrow \Delta T_b \propto im$

for glucose : (i=1)

$\Delta T_b = 1 \times K_b \times \left(\frac{1\times1000}{180\times100}\right)$

For 1% $\left(\frac{w}{w}\right)$ NaCl (i=2)

$\Delta T_b = 2 \times K_b \times \left(\frac{1\times1000}{55.5\times100}\right)$

For 1% $\left(\frac{w}{w}\right)$ $Ca_3(PO_4)_2$

(i=5)

$\Delta T_b = 5 \times K_b \times \left(\frac{1}{310}\right) \times 1000 \times 100$

For 1% $\left(\frac{w}{w}\right)$ urea (i=1)

$\Delta T_b = \left(\frac{1\times1\times1000}{60\times100}\right)K_b$

For (NaCl), ΔT_b is maximum & hence maximum boiling point.

106. Ans. (4)

$\Delta T_f = iK_fM$

$\Delta T_f \propto i$

For urea (i=1) which is minimum & hence maximum freezing point.

107. Ans. (2)

$HgI_2 + 2KI \rightarrow [K_2HgI_4]$

On adding mercuric iodide to potassium iodide, No of solute particles decreases as it forms complex.
So freezing point is raised.

108. Ans. (2)

Benzoic acid dimerises in benzene.

109. Ans. (1)

$\pi = iCRT$

$= 2.8 \times 0.1 \times 24.6$

$= 6.89$ atm

For K_2SO_4

$i = 1 + 2\alpha = 1+2(0.9)$

$= 2.8$

110. Ans. (2)

For Hg_2Cl_2 $i = 3$

For Na_2SO_4 $i = 3$

$Hg_2Cl_2 \rightarrow Hg_2^{2+} + 2Cl^-$

Exercise-II (Previous Year Questions)

1. Ans. (3)

$\Delta T_f = iK_f m$

$\Delta T_f \propto i$

For $Al_2(SO_4)_3$ i = 5 (maximum)

Hence $Al_2(SO_4)_3$ will exhibit

Largest freezing point depression

2. Ans. (4)

$(\Delta T_b)_x > (\Delta T_b)_y$

$(iK_b m)_x > (iK_b m)_y$

$(i_x > i_y)$

if $i_y = 1$ then $i_x > 1$

So, X is undergoing dissociation in water while y undergoes no change

3. Ans. (1)

$\Delta S_{mix} > 0$ (always)

4. Ans. (3)

$i_{Al_2(SO_4)_3} = 5$

$i_{K_4[Fe(CN)_6]} = 5$

5. Ans. (2)

1 molal aqueous solution means 1 mole solute in 1000 g water

So

$n_B = 1$

$n_A = \frac{1000}{18} = 55.5$

$X_B = \frac{n_B}{n_A + n_B} = \frac{1}{1+55.5} = 0.0177$

6. Ans. (1)

$Y_A = \frac{P_A^0 X_A}{P_A^0 X_A + P_B^0 X_B}; Y_B = \frac{P_B^0 X_B}{P_A^0 X_A + P_B^0 X_B}$

$X_A = \frac{1}{2}$

$X_B = \frac{1}{2}$

$P_A^0 = 12.8$ KPa (Benzene)

$P_B^0 = 3.85$ KPa (toluene)

$Y_A > Y_B$

7. Ans. (1)

$\frac{P_A^0 - P_S}{P_S} = \frac{n_B}{n_A}$

$\frac{760-732}{73.2} = \frac{n_B \times m_A}{w_A} \times \frac{1000}{1000}$

$\frac{28}{732} = m \times \frac{m_A}{1000}$

$\frac{28}{732} = \frac{m \times 18}{1000}$

$m = 2.12$

$\Delta T_b = K_b m = 2.12 \times 0.52$

$= 1.1$

$T_b' = 101°C$

8. Ans. (4)

For liq $\rightleftharpoons$ Vapour

$\frac{d(\ln p)}{dT} = \frac{\Delta Hv}{RT^2}$ is correct.

9. Ans. (2)

$Ba(OH)_2 \rightarrow Ba^{+2} + 2OH^-$

$i = 1+(n-1)\alpha$

$n = 3, \alpha = 1$

$i = 1+2\alpha = 3$

10. Ans. (2)

For spontaneous process

ΔG_{mix} is Always less than zero

11. Ans. (3)

K_f does not depend on molality.

12. Ans. (1)

Molarity is dependent on temperature.

13. Ans. (3)

$\Delta H_{mix} = 0$

14. Ans. (1)

Interaction attractive forces between A-A & B-B are stronger than those between A-B.

15. Ans. (3)

2M aq. Solution of NaOH means 2 mole NaOH in 1L.

$$d = \frac{w}{V_{(ml)}}$$

$$1.28 = \frac{w_{Sol}}{1000}$$

$w_{sol.} = 1280$ g

$w_{solvent} = w_{sol} - w_{solute}$

$= 1280 - 80 = 1200$

$$\text{Molality} = \frac{n_B \times 100}{w_A}$$

$$= \frac{2 \times 1000}{1200} = 1.67\,M$$

16. Ans. (4)

$$\Delta T_f = K_f m$$

$= 5.12 \times 0.078$

$= 0.40$ K

17. Ans. (2)

"Ethanol + Acetone" shows positive deviation from Raoult's law.

18. Ans. (1)

$$\frac{P_A^0 - P_S}{P_A^0} = \frac{n_B}{n_A + n_B}$$

For dil soln.

$n_B < n_A$

$$\frac{P_A^0 - P_S}{P_A^0} = \frac{n_B}{n_A}$$

$$\frac{100-80}{100} = \frac{8 \times 114}{M_W \times 114}$$

$M_W = 40$

19. Ans. (3)

Isotonic solutions have same osmotic pressure.

20. Ans. (1)

$$P_1 = \left(\frac{10 \times 1000}{180 \times 250}\right) RT$$

$$P_2 = \left(\frac{10 \times 1000}{60 \times 250}\right) RT$$

$$P_3 = \left(\frac{10 \times 1000}{342 \times 250}\right) RT$$

$P_2 > P_1 > P_3$

21. Ans. (3)

$$\frac{n_{Benzene}}{n_{octane}} = \frac{3}{2}$$

$$X_{Benzene} = \frac{3}{5}$$

$$X_{octane} = \frac{2}{5}$$

$$P_S = P_A^0 X_A + P_B^0 X_B$$

$$= 280 \times \frac{3}{5} + 420 \times \frac{2}{5}$$

$= 336$ mmHg

22. Ans. (1)

$$M = \frac{n_B}{w_A} \times 1000$$

$$1 = \frac{0.5}{w_A} \times 1000$$

$w_A = 500$ g solvent

23. Ans. (1)

$P = K_H X$

$$X \propto \frac{1}{K_H}$$

24. Ans. (2)

Assertion is true because He has low solubility in blood. (NCERT)

25. Ans. (2)

$$\text{Solubility} \propto \frac{1}{K_H}$$

$KH \downarrow \rightarrow$ Solubility $\uparrow$

B > C > A

26. Ans. (1)

$$\underset{\downarrow \text{bar}}{\pi} = \underset{\downarrow \text{mol L}^{-1}}{C} \quad RT$$

On comparing with **y = mx**

Slope (m) = RT

25.73 = 0.083 × T

T = 310 K = 37°C

27. Ans. (1)

Amount of glucose required = ?

$$n_{glucose} = M \times V_{(\ell)} = \frac{1}{20} \times \frac{250}{1000} = \frac{1}{80} \text{ mol}$$

$$= \frac{1}{80} \times 180 \text{ g}$$

= 2.25 g

28. Ans. (4)

for an isotonic solution

$\pi_1 = \pi_2$

$C_1RT = C_2RT$

$C_1 = C_2$

$$\frac{w}{180 \times 1} = \frac{15}{60}$$

w = 45 g

Exercise-III (Analytical Questions)

1. Ans. (4)

$P = K_H x$ {x is note Fraction & P is partial if $K_H = P_1°$ pressure)

$P = P_1° X_1$

Hence Raoult's law is special case of Henry's law

2. Ans. (4)

$i = 1 + (n - 1)\alpha$

on decreasing concentration $\alpha \uparrow$ & Hence i increases.

3. Ans. (4)

In pressure cooker, at high pressure cooker, Boiling point of liquid inside the pressure cooker is raised ($\because P \propto T$) which helps in fast cooking

4. Ans. (4)

$\pi_1 = \pi_2$ (for isotonic solution)

Hence there is no net movement of solvent particles.

5. Ans. (2)

For isotonic solution there is no net movement. also osmotic pressure depend on temperature.

6. Ans. (4)

i = 1 (for non electrolyte) for electrolyte

i > 1 (for dissociation) for electrolyte

i < 1 (for association) for electrolyte

7. Ans. (1)

Non ideal solution form azeotropic mixture which are constant boiling azeotropes.

8. Ans. (2)

Colligative properties depend on relative number of solute particles not on the nature of solute particle.

9. Ans. (3)

SPM allows only solvent molecules to pass through it.

10. Ans. (1)

$$i = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$$

11. Ans. (3)

Total 4 colligative properties

(1) Relative lowering in V.P

(2) Elevation in Bpt

(3) Depression in fpt

(4) Osmotic pressure

12. Ans. (1)

k_f depend on nature of solvent & elevation in boiling point is directly proportional to molality.

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Exercise - I (Conceptual Questions) Build Up Your Understanding

RATE OF REACTION

1. Consider the chemical reaction :

$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

The rate of this reaction can be expressed in terms of concentration of $N_2(g)$, $H_2(g)$ or $NH_3(g)$. Identify the correct relationship amongst the rate expressions.

(1) Rate $= -\frac{d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$

(2) Rate $= -\frac{d[N_2]}{dt} = -\frac{3d[H_2]}{dt} = \frac{2d[NH_3]}{dt}$

(3) Rate $= \frac{d[N_2]}{dt} = \frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$

(4) Rate $= -\frac{d[N_2]}{dt} = -\frac{d[H_2]}{dt} = \frac{d[NH_3]}{dt}$

CCK001

2. In the formation of sulphur trioxide by the contact process $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$; the rate of reaction is expressed as $-\frac{d[O_2]}{dt} = 2.5 \times 10^{-4}$ mol L^{-1} s^{-1}

The rate of disappearance of (SO_2) will be

(1) 5×10^{-4} mol L^{-1} s^{-1}

(2) -2.25×10^{-4} mol L^{-1} s^{-1}

(3) 3.75×10^{-4} mol L^{-1} s^{-1}

(4) 50.0×10^{-4} mol L^{-1} s^{-1}

CCK002

3. In a catalytic reaction involving the formation of ammonia by Haber's process $N_2 + 3H_2 \rightarrow 2NH_3$ the rate of appearance of NH_3 was measured as 2.5×10^{-4} mole L^{-1} s^{-1}

The rate of disappearance of H_2 will be –

(1) 2.5×10^{-4} mol L^{-1} s^{-1}

(2) 1.25×10^{-4} mol L^{-1} s^{-1}

(3) 3.75×10^{-4} mol L^{-1} s^{-1}

(4) 5×10^{-4} mol L^{-1} s^{-1}

CCK003

4. Which of the following statement is correct for a reaction X + 2Y → Product (p)

(1) The rate of disappearance of X = twice the rate of disappearance of Y.

(2) The rate of disappearance of X = $\frac{1}{2}$ rate of appearance of product (p)

(3) The rate of appearance of product (p) = $\frac{1}{2}$ the rate of disappearance of Y

(4) The rate of appearance of product (p) = $\frac{1}{2}$ the rate of disappearance of X

CCK004

5. For the reaction, $N_2O_5 \longrightarrow 2NO_2 + ½\ O_2$

Given $-\frac{d[N_2O_5]}{dt} = K_1[N_2O_5]$,

$\frac{d[NO_2]}{dt} = K_2[N_2O_5]$, $\frac{d[O_2]}{dt} = K_3[N_2O_5]$

The relation between K_1, K_2 and K_3 is –

(1) $2K_1 = K_2 = 4K_3$

(2) $K_1 = K_2 = K_3$

(3) $2K_1 = 4K_2 = K_3$

(4) None

CCK005

6. Rate of formation of SO_3 according to the reaction $2SO_2+O_2 \rightarrow 2SO_3$ is 1.6×10^{-3} kg L^{-1} min^{-1} Hence rate of decomposition of SO_2 is :-

(1) 1.6×10^{-3} kg L^{-1} min^{-1}

(2) 8.0×10^{-4} kg L^{-1} min^{-1}

(3) 3.2×10^{-3} kg L^{-1} min^{-1}

(4) 1.28×10^{-3} kg L^{-1} min^{-1}

CCK006

7. For a general chemical change 2A + 3B → products, the rate of disappearance of A is r_1 and of B is r_2. The rates r_1 and r_2 are related as :–

(1) $3r_1 = 2r_2$ (2) $r_1 = r_2$

(3) $2r_1 = 3r_2$ (4) $r_1^2 = 2r_2^2$

CCK007

8. For the reaction $2A + B \rightarrow 3C + D$, Which of the following does not express the reaction rate ?

(1) $-\frac{1}{3}\frac{d[C]}{dt}$ (2) $-\frac{d[B]}{dt}$

(3) $\frac{d[D]}{dt}$ (4) $-\frac{1}{2}\frac{d[A]}{dt}$

CCK008

9. Consider the reaction

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

The correct equality relationship between $\frac{d[NH_3]}{dt}$ and $-\frac{d[H_2]}{dt}$ is :

(1) $\frac{d[NH_3]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt}$

(2) $+\frac{d[NH_3]}{dt} = -\frac{2}{3}\frac{d[H_2]}{dt}$

(3) $+\frac{d[NH_3]}{dt} = -\frac{3}{2}\frac{d[H_2]}{dt}$

(4) $\frac{d[NH_3]}{dt} = -\frac{d[H_2]}{dt}$

CCK009

RATE LAW / ORDER / MOLECULARITY

10. For gaseous reaction, rate = k [A] [B]. If volume of container is reduced to $\frac{1}{4}$ of initial, then the rate of the reaction will be..... times of initial:-

(1) $\frac{1}{8}$ (2) 8 (3) $\frac{1}{16}$ (4) 16

CCK010

11. $2A(g) + B(g) \longrightarrow$ Product is an elementary reaction, If pressure of both A and B is increased three times of the initial pressure, the rate of forward reaction will be ------ of the previous rate :-

(1) 9 times (2) 27 times

(3) $\frac{1}{9}$ times (4) $\frac{1}{27}$ times

CCK011

12. The rate of certain hypothetical reaction $A + B + C \rightarrow$ products is given by

$r = \frac{-d[A]}{dt} = K\,[A]^{1/2}\,[B]^{1/3}\,[C]^{1/4}$ The order of the reaction –

(1) 1 (2) $\frac{1}{2}$ (3) 2 (4) $\frac{13}{12}$

CCK012

13. Which of the following rate law has an overall order of 0.5 for reaction involving substances x,y and z ?

(1) Rate = K (C_x) (C_y) (C_z)

(2) Rate = K $(C_x)^{0.5}(C_y)^{0.5}(C_z)^{0.5}$

(3) Rate = K $(C_x)^{1.5}$ $(C_y)^{-1}(C_z)^0$

(4) Rate = K$(C_x)(C_z)^0$ / $(C_y)^2$

CCK013

14. A chemical reaction involves two reacting species. The rate of reaction is directly proportional to square of the concentration of one of them and inversely proportional to the concentration of the other. The order of reaction is –

(1) 1 (2) 2

(3) Zero (4) Unpredictable

CCK014

15. For the reaction $H_2(g) + Br_2(g) \rightarrow 2HBr(g)$, the experimental data suggests, Rate = K $[H_2]$ $[Br_2]^{1/2}$. The order for this reaction is –

(1) 2 (2) $1\frac{1}{2}$ (3) 1 (4) $2\frac{1}{2}$

CCK015

16. Select the rate law that corresponds to the data shown for the following reaction $A + B \rightarrow C$

Exp.	[A]	[B]	Initial rate
1.	0.012	0.035	0.10
2.	0.024	0.070	1.6
3.	0.024	0.035	0.20
4.	0.012	0.070	0.80

(1) Rate = K $[B]^3$ (2) Rate = K$[B]^4$

(3) Rate = K$[A][B]^3$ (4) Rate = K$[A]^2[B]^2$

CCK016

17. Select the rate law that corresponds to data shown for the following reaction 2A+B→ C+D :-

Exp.	[A]	[B]	Initial rate (Mol L^{-1} min^{-1})
1.	0.1	0.1	6×10^{-3}
2.	0.3	0.2	9×10^{-2}
3.	0.3	0.4	3.6×10^{-1}
4.	0.5	0.1	3×10^{-2}

(1) Rate = K $[A]^2[B]$ (2) Rate = K$[A][B]^2$
(3) Rate = K$[A][B]^3$ (4) Rate = K[A][B]

CCK017

18. For a hypothetical reaction ; A + B → C the following data were obtained in three different experiments :–

[A] (mol L^{-1})	[B] (mol L^{-1})	Rate of reaction (Mol L^{-1} min^{-1})
0.01	0.01	1.0 10^{-4}
0.01	0.03	9.0 10^{-4}
0.03	0.03	2.70 10^{-3}

The rate law will be :–
(1) r = K$[A]^2[B]$ (2) r = K$[A][B]^2$
(3) r = K[A][B] (4) None of these

CCK018

19. Calculate the order of the reaction w.r.t. A and B:

[A] (mol L^{-1})	[B] (mol L^{-1})	Rate (mol L^{-1} s^{-1})
0.05	0.05	1.2×10^{-3}
0.10	0.05	2.4×10^{-3}
0.05	0.10	1.2×10^{-3}

(1) 1 and 0 (2) 1 and 1
(3) 0 and 1 (4) None

CCK019

20. For a chemical reaction A + B → product, the order is one with respect to each A and B. Value of x and y from the given data is :–

Rate (mol L^{-1} s^{-1})	[A]	[B]
0.10	0.20 M	0.05 M
0.40	x	0.05 M
0.80	0.40 M	y

(1) 0.20, 0.80 (2) 0.80, 0.40
(3) 0.80, 0.20 (4) 0.40, 0.20

CCK020

21. The rate law for the single step reaction 2A + B → 2C, is given by –
(1) Rate = K[A][B] (2) Rate = K$[A]^2[B]$
(3) Rate = K[2A] [B] (4) Rate = K$[A]^2[B]^0$

CCK021

22. For a reaction of the type A + B → products, it is observed that doubling the concentration of A causes the reaction rate to increase four times but doubling the amount of B does not effect the rate. The rate equation is –
(1) Rate = K [A][B] (2) Rate = K $[A]^2$
(3) Rate = K$[A]^2[B]$ (4) Rate = K$[A]^2[B]^2$

CCK022

23. Point out incorrect statement.
(1) Rate law is an experimental value
(2) Law of mass action is a theoretical proposal
(3) Rate law is more informative than law of mass action for developing mechanism
(4) Rate law is always different from the expression of law of mass action.

CCK023

24. For an elementary process 2X + Y → Z + W, the molecularity is –
(1) 2 (2) 1
(3) 3 (4) Unpredictable

CCK024

25. For a reaction A + B → products, the rate of the reaction was doubled when the concentration of A was doubled, the rate was again doubled when the concentration of both A & B were doubled the order of the reaction with respect to A & B are:–
(1) 1, 1 (2) 2, 0 (3) 1, 0 (4) 0, 1

CCK025

26. For the reaction A + B → products, it is found that the order w.r.t. A is 1 and the order w.r.t. B is $\frac{1}{2}$. When the concentration of both A and B are increased four times, the rate will increase by a factor of :–
(1) 16 (2) 8 (3) 6 (4) 4

CCK026

AIM: FREE EDUCATION TO ALL APUL

27. The rate law for a reaction A + B → product is rate = $K[A]^1[B]^2$. Then which one of the following statement is false :–

(1) If [B] is kept constant while [A] is doubled, the reaction will proceed twice as fast

(2) If [A] is kept constant while [B] is reduced to one quarter, the rate will be halved

(3) If [A] and [B] are both doubled, the reaction will proceed 8 times as fast

(4) This is a third order reaction

CCK027

28. For a chemical reaction A → B, the rate of reaction doubles when the concentration of A is increased 8 times. The order of reaction w.r.t. A is :–

(1) 3 (2) $\frac{1}{2}$

(3) $\frac{1}{3}$ (4) Zero

CCK028

29. The rate constant of a first order reaction depends on the :–

(1) Concentration of the reactant

(2) Concentration of the product

(3) Time

(4) Temperature

CCK029

30. Consider the following reaction,

$KClO_3 + 6FeSO_4 + 3H_2SO_4 \longrightarrow KCl + 3Fe_2(SO_4)_3 + 3H_2O$

Which of the statements regarding this reaction is incorrect?

(1) It is a complex reaction

(2) It is a tenth order reaction

(3) It is completed in several steps

(4) Order of this reaction can be determined experimentally

CCK030

PSEUDO FIRST ORDER REACTION / MECHANISM OF REACTION

31. Following mechanism has been proposed for a reaction $2A + B \rightarrow D + E$

$A + B \rightarrow C + D$ .....(slow)

$A + C \rightarrow E$ .....(fast)

The rate law expression for the reaction is –

(1) $r = K[A]^2[B]$ (2) $r = K[A][B]$

(3) $r = K[A]^2$ (4) $r = K[A][C]$

CCK031

32. The chemical reaction $2O_3 \rightarrow 3O_2$ proceeds as follows

$O_3 \rightleftharpoons O_2 + O$ ..... (fast)

$O + O_3 \rightarrow 2O_2$ ..... (slow)

The rate law expression should be –

(1) $r = K[O_3]^2$ (2) $r = K[O_3]^2[O_2]^{-1}$

(3) $r = K[O_3][O_2]$ (4) Unpredictable

CCK032

33. The hypothetical reaction $A_2 + B_2 \rightarrow 2AB$ follows the mechanism as given below –

$A_2 \rightleftharpoons A + A$ ..... (fast)

$A + B_2 \rightarrow AB + B$ ..... (slow)

$A + B \rightarrow AB$ ..... (fast)

The order of the over all reaction is –

(1) 2 (2) 1 (3) 1½ (4) Zero

CCK033

34. The rate for the reaction

RCl + NaOH (aq) → ROH + NaCl is given by rate = $k_1[RCl]$. The rate of the reaction is –

(1) Doubled on doubling the concentration of NaOH

(2) Halved on reducing the concentration of RCl to half

(3) Decreased on increasing the temperature of reaction

(4) Unaffected by increasing the temperature of the reaction

CCK034

35. For reaction $NO_2 + CO \rightarrow CO_2 + NO$, the rate expression is, Rate = $k[NO_2]^2$

The number of molecules of CO involved in the slowest step will be –

(1) 0 (2) 1 (3) 2 (4) 3

CCK035

36. The rate law of the reaction A + 2B → product is given by $\frac{d[P]}{dt} = K[A]^2[B]$. If A is taken in large excess, the order of the reaction will be –
(1) Zero (2) 1 (3) 2 (4) 3
CCK036

37. The acidic hydrolysis of ester is –
(1) Psuedo Ist order reaction
(2) IInd order reaction
(3) Trimolecular reaction
(4) All of the above
CCK037

38. The reaction 2A + B → P, follows the mechanism

$2A \rightleftharpoons A_2$ (fast)

$A_2 + B \rightarrow P$ (slow)

The order of the reaction is :–
(1) 1.5 (2) 3 (3) 1 (4) 2
CCK038

39. For the reaction $2NO + Cl_2 \rightarrow 2NOCl$ the following mechanism has been proposed

$NO + Cl_2 \rightleftharpoons NOCl_2$ (fast)

$NOCl_2 + NO \rightarrow 2NOCl$ (slow) the rate law for the reaction is :-
(1) Rate = $K[NO]^2[Cl_2]$ (2) Rate = $K[NO][Cl_2]^2$
(3) Rate = $K[NOCl_2]$ (4) Rate = $K[NOCl]^2$
CCK039

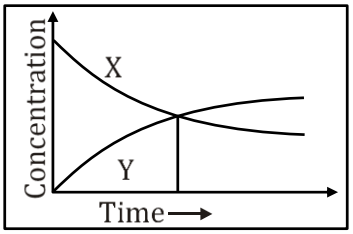
ZERO / FIRST / NTH ORDER REACTIONS

40. The rate constant is numerically the same for three reactions of first, second and third order respectively. Which one is true at a moment for rate of all three reactions if concentration of reactants is same and greater than 1 M.
(1) $r_1 = r_2 = r_3$ (2) $r_1 > r_2 > r_3$
(3) $r_1 < r_2 < r_3$ (4) All
CCK040

41. K for a zero order reaction is 2×10^{-2} mol L^{-1} s^{-1}. If the concentration of the reactant after 25s is 0.5M, the initial concentration must have been.
(1) 0.5 M (2) 1.25 M
(3) 12.5 M (4) 1.0 M
CCK041

42. The decomposition of N_2O_5 occurs as, $2N_2O_5 \rightarrow 4NO_2 + O_2$, and follows first order kinetics; hence
(1) The reaction is bimolecular
(2) The reaction is unimolecular
(3) $t_{1/2} \propto a^0$
(4) $t_{1/2} \propto a^2$
CCK042

43. The accompanying figure depicts the change in concentration of species X and Y for the reaction X → Y as a function of time. The point of intersection of the two curves represents.



(1) $t_{1/2}$
(2) $t_{3/4}$
(3) $t_{2/3}$
(4) Data is insufficient to predict
CCK043

44. If the first order reaction involves gaseous reactants and gaseous products the unit of its rate is -
(1) atm (2) atm s (3) atm s^{-1} (4) atm^2 s^2
CCK044

45. Plot of log (a – x) vs time t is straight line. This indicates that the reaction is of –
(1) Second order (2) First order
(3) Zero order (4) Third order
CCK045

46. The rate constant of a first order reaction is 4×10^{-3} s^{-1}. At a reactant concentration of 0.02 M, the rate of reaction would be–
(1) 8×10^{-5} M s^{-1} (2) 4×10^{-3} M s^{-1}
(3) 2×10^{-1} M s^{-1} (4) 4×10^{-1} M s^{-1}
CCK046

47. In a first order reaction, the concentration of the reactant is decreased from 1.0 M to 0.25 M in 20 min. The rate constant of the reaction would be–
(1) 10 min^{-1} (2) 6.931 min^{-1}
(3) 0.6931 min^{-1} (4) 0.06931 min^{-1}
CCK047

48. In a first order reaction, the value of a/(a – x) was found to be 8 after 10 minute. The rate constant is

(1) $\frac{(2.303\times 3\log 2)}{10}$ (2) $\frac{(2.303\times 2\log 3)}{10}$

(3) 10 × 2.303 × 2log3 (4) 10 × 2.303 × 3log2

CCK048

49. 75 % of a first order reaction was found to complete in 32 min. When will 50 % of the same reaction complete –

(1) 24 min (2) 16 min

(3) 8 min (4) 4 min

CCK049

50. A first order reaction has a half life period of 69.3 s. At 0.10 mol L^{-1} reactant concentration, the rate will be –

(1) 10^{-4} M s^{-1} (2) 10^{-3} M s^{-1}

(3) 10^{-1} M s^{-1} (4) 6.93×10^{-1} M s^{-1}

CCK050

51. For a first order reaction A → products, the rate of reaction at [A] = 0.2 M is 1×10^{-2} mol $L^{-1}min^{-1}$. The half life period for the reaction is -

(1) 832 min. (2) 440 s

(3) 416 min. (4) 14 min.

CCK051

52. 99 % of a first order reaction was completed in 32 min. when will 99.9 % of the reaction complete ?

(1) 50 min. (2) 46 min.

(3) 49 min. (4) 48 min.

CCK052

53. What is the half life of a radioactive substance if 87.5% of any given amount of the substance disintegrate in 40 minutes ?

(1) 160 min (2) 10 min

(3) 20 min (4) 13 min 20 sec.

CCK053

54. For a given reaction of first order, it takes 20 min for the concentration to drop from 1 M to 0.6 M. The time required for the concentration to drop from 0.6 M to 0.36 M will be :

(1) More than 20 min

(2) Less than 20 min

(3) Equal to 20 min

(4) Infinity

CCK054

55. A first order reaction is carried out with an initial concentration of 10 mol per litre and 80% of the reactant changes into the product. Now if the same reaction is carried out with an initial concentration of 5 mol per litre for the same period, the percentage of the reactant changing to the product is.

(1) 40

(2) 80

(3) 160

(4) Cannot be calculated

CCK055

56. The reaction L → M is started with 10 g/L. After 30 minute and 90minute, 5 g/L and 1.25 g/L are left respectively. The order of reaction is

(1) 0 (2) 2 (3) 1 (4) 3

CCK056

57. Doubling the initial concentration of a reactant doubles $t_{½}$ of the reaction, then order of the reaction is–

(1) 3 (2) 2 (3) 1 (4) 0

CCK057

58. The half life period for catalytic decomposition of AB_3 at 50 mm is found to be 4 hrs and at 100 mm it is 2 hrs. The order of reaction is –

(1) 3 (2) 1 (3) 2 (4) 0

CCK058

59. The rate constant for a reaction is 10.8×10^{-5} mol L^{-1} s^{-1} The reaction obeys –

(1) First order (2) Zero order

(3) Second order (4) All are wrong

CCK059

60. A substance 'A' decomposes in solution following the first order kinetics flask I contains 1L of 1M. solution of A and flask II contains. 100 ml of 0.6 M solution. After 8 hr. the concentration of A in flask I become 0.25 M, what will be the time for concentration of A in flask II to become 0.3 M.

(1) 0.4 hr.

(2) 2.4 hr.

(3) 4.0 hr.

(4) Unpredictable as rate constant is not given

CCK060

61. The rate constant (K) for the reaction $2A + B \rightarrow$ product, was found to be 2.57×10^{-5} L mol^{-1} s^{-1} after 15 s, 2.60×10^{-5} L mol^{-1} s^{-1} after 30 s and 2.55×10^{-5} L mol^{-1} s^{-1} after 50 s. The order of reaction is

(1) 2 (2) 3

(3) Zero (4) 1

CCK061

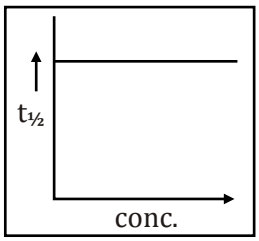
62. The rate constant for a second order reaction is 8×10^{-5} M^{-1} min^{-1} : How long will it take a 1M solution to be reduced to 0.5 M.

(1) 8.665×10^{3} min (2) 8×10^{-3} min

(3) 1.25×10^{4} min (4) 4×10^{-5} min

CCK062

63. A graph between $t_{1/2}$ and concentration for n^{th} order reaction is a straight line. Reaction of this nature is completed 50% in 10 minutes when concentration is 2 mol L^{-1}. This is decomposed 50% in t minutes at 4 mol L^{-1}, n and t are respectively



(1) 0, 20 min. (2) 1, 10 min.

(3) 1, 20 min. (4) 0, 5 min.

CCK063

64. In the case of first order reaction, the ratio of time required for 99.9 % completion to 50 % completion is :–

(1) 2 (2) 5

(3) 10 (4) None

CCK064

65. From different sets of data of $t_{1/2}$ at different initial concentrations say 'a' for a given reaction, the $[t_{1/2} \times a]$ is found to be constant. The order of reaction is :–

(1) 0 (2) 1

(3) 2 (4) 3

CCK065

66. Which of the following statement is not correct for the reaction whose rate is r = k (rate constant)

(1) Rate of reaction is independent of concentration of reactant

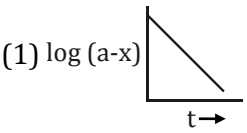
(2) $t_{1/2}$ of reaction does not depend upon concentration of reactant

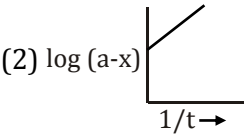
(3) Rate constant is independent of concentration of reactant

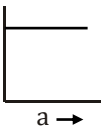
(4) This is zero order reaction

CCK066

67. Which of the following curves represents a I^{st} order reaction :–

(1) 

(2) 

(3) 

(4) 1 & 3 both

CCK067

68. The following data were obtained at a certain temperature for the decomposition of ammonia

p (mm)	50	100	200
$t_{1/2}$	3.64	1.82	0.91

The order of the reaction is :–

(1) 0 (2) 1 (3) 2 (4) 3

CCK068

69. A reaction is found to have the rate constant x s^{-1} , by what factor the rate is increased if initial concentration of reactant is tripled

(1) 3 (2) 9

(3) x (4) Remains same

CCK069

70. Which is incorrect :–

(1) Half life of a first order reaction is independent of initial concentration

(2) Rate of reaction is constant for first order reaction

(3) Unit of K for second order reaction is mol^{-1} L s^{-1}

(4) Half life of zero order is proportional to initial concentration

CCK070

71. Hydrolysis of ester in alkaline medium is :–
(1) First order reaction
(2) Second order reaction
(3) Zero order reaction
(4) Third order reaction

CCK071

72. The expression which gives $\frac{1}{4}$th life of Ist order reaction is :–

(1) $\frac{k}{2.303}\log\frac{4}{3}$ (2) $\frac{2.303}{k}\log 3$

(3) $\frac{2.303}{k}\log\frac{3}{4}$ (4) $\frac{2.303}{k}\log\frac{4}{3}$

CCK072

73. The rate constant of a zero order reaction is 0.2 mol $dm^{-3}h^{-1}$. If the concentration of the reactant after 30 minutes is 0.05 mol dm^{-3}. Then its initial concentration would be :-
(1) 6.05 mol dm^{-3} (2) 0.15 mol dm^{-3}
(3) 0.25 mol dm^{-3} (4) 4.00 mol dm^{-3}

CCK073

74. A reaction is of first order. After 100 minutes, 75 gm of the reactant A are decomposed when 100 gm are taken initially, calculate the time required when 150 gm of the reactant A are decomposed, the initial weight taken is 200 gm :-
(1) 100 minutes (2) 200 minutes
(3) 150 minutes (4) 175 minutes

CCK074

75. For which of the following, the unit of rate and rate constant of the reaction are identical :-
(1) First order reaction
(2) Zero order reaction
(3) Second order reaction
(4) Fractional order of reaction

CCK075

76. A $\rightarrow$ Product, follows the first order kinetics. If the half life period of the reaction at $[A]_{Initial}$ = 0.2 mol L^{-1} is 20 min. then the value of rate constant would be:-
(1) 4 s^{-1} (2) 20 s^{-1}
(3) 4 L mol^{-1} $min.^{-1}$ (4) 0.034 min^{-1}

CCK076

77. Correct statement about first order reaction is:-
(1) $t_{completion}$ = finite
(2) $t_{1/2} \propto \frac{1}{a}$
(3) Unit of k is mole L^{-1} s^{-1}
(4) $t_{1/2} \times k$ = constant (at given temperature)

CCK077

COLLISION THEORY AND FACTORS AFFECTING RATE OF REACTION

78. Activation energy of a reaction is –
(1) The energy released during the reaction
(2) The energy evolved when activated complex is formed
(3) Minimum extra amount of energy needed to overcome the potential barrier of reaction
(4) The energy needed to form one mole of the product

CCK078

79. The minimum energy for molecules to enter into chemical reaction is called.
(1) Kinetic energy (2) Potential energy
(3) Threshold energy (4) Activation energy

CCK079

80. The rate constant k_1 of a reaction is found to be double that of rate constant k_2 of another reaction. The relationship between corresponding activation energies of the two reactions at same temperature (E_1 & E_2) can be represented as
(1) $E_1 > E_2$ (2) $E_1 < E_2$
(3) $E_1 = E_2$ (4) $E_1 = 4E_2$

CCK080

81. At room temperature, the reaction between NO and O_2 to give NO_2 is fast while that between CO and O_2 is slow it is due to –
(1) CO is smaller in size than that of NO
(2) CO is poisonous
(3) The activation energy for the reaction $2NO + O_2 \rightarrow 2NO_2$ is less than $2CO + O_2 \rightarrow 2CO_2$
(4) NO_2 is poisonous

CCK081

82. Chemical reaction occurs as a result of collisions between reacting molecules. Therefore the reaction rate is given by
(1) Total number of collisions occurring in a unit volume per second
(2) Fraction of molecules which possess energy less than the threshold energy.
(3) Total number of effective collisions
(4) Temperature

CCK082

83. An endothermic reaction A → B has an activation energy 15 kcal/mol and the heat of the reaction is 5 kcal/mol. The activation energy of the reaction B → A is –
(1) 20 kcal/mol (2) 15 kcal/mol
(3) 10 kcal/mol (4) Zero

CCK083

84. A large increase in the rate of a reaction for a rise in temperature is due to –
(1) Increase in the number of collisions
(2) Increase in the number of activated molecules
(3) Lowering of activation energy
(4) Shortening of the mean free path

CCK084

85. Rate of which reactions increases with temperature
(1) Of all reactions
(2) Of exothermic reactions
(3) Of endothermic reactions
(4) Can't be predicted

CCK085

86. The rate of a chemical reaction doubles for every 10^0C rise in temperature. If the temperature is increased by 60^0C the rate of reaction increases by :
(1) 20 times (2) 32 times
(3) 64 times (4) 128 times

CCK086

87. According to the Arrhenius equation, a straight line is to be obtained by plotting the logarithm of the rate constant of chemical reaction (log k) against.
(1) T (2) log T
(3) $\frac{1}{T}$ (4) log $\frac{1}{T}$

CCK087

88. Which plot will give the value of activation energy.
(1) k v/s T (2) $\frac{1}{k}$ v/s T
(3) lnk v/s T (4) lnk v/s $\frac{1}{T}$

CCK088

89. Given that k is the rate constant of any reaction at temperature T then the value of $\lim_{T\to\infty} \log k$ is :
(1) A ≅ 2.303 (2) A
(3) 2.303 A (4) log A

CCK089

90. The rate constant, the activation energy and the frequency factor of a chemical reaction at 25^0C are 3×10^{-4} s^{-1}; 104.4 kJ mol^{-1} and 6.0×10^{14} s^{-1} respectively, the value of the rate constant as T→ ∞ is.
(1) 2×10^8 s^{-1} (2) 6×10^{14} s^{-1}
(3) Infinity (4) 3.6×10^{30} s^{-1}

CCK090

91. For an endothermic reaction where ΔH represents the enthalpy of the reaction in kJ/mol ; the minimum value for the energy of activation will be
(1) Less than ΔH (2) Zero
(3) More than ΔH (4) Equal to ΔH

CCK091

92. The rate of reaction increases 2.3 times when the temperature is raised from 300 K to 310 K. If k is the rate constant at 300 K then the rate constant at 310 K will be equal to –
(1) 2 k (2) k (3) 2.3 k (4) $3\ k^2$

CCK092

93. If the concentration units are reduced by n times then the value of rate constant of first order will
(1) Increases by n times
(2) Decreases by factor of n
(3) Remain constant
(4) Decrease 1/n times

CCK093

94. Which is used in the determination of reaction rates.
(1) Reaction Temperature
(2) Reaction Concentration
(3) Specific rate constant
(4) All of these

CCK094

95. The rate constant of a first order reaction depends on the :–
(1) Concentration of the reactant
(2) Concentration of the product
(3) Volume of the reaction container
(4) Temperature

CCK095

96. For the decomposition of $N_2O_5(g)$ it is given that $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$ activation energy = E_a, $N_2O_5(g) \rightarrow 2NO_2(g) + \frac{1}{2}O_2(g)$ activation energy = E_a' then
(1) $E_a = 2E'_a$ (2) $E_a > E'_a$
(3) $E_a < E'_a$ (4) $E_a = E'_a$

CCK096

97. For a reaction in which case the activation energies of forward and reverse reactions are equal :–
(1) $\Delta H = 0$
(2) $\Delta S = 0$
(3) The order is zero
(4) There is no catalyst

CCK097

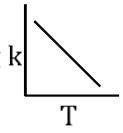
98. The energy of activation of a forward reaction is 50 kcal. The energy of activation of its backward reaction is:–
(1) Equal to 50 kcal.
(2) Greater than 50 kcal.
(3) Less than 50 kcal.
(4) Either greater or less than 50 kcal.

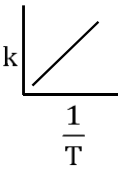
CCK098

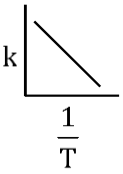
99. An exothermic reaction $X \rightarrow Y$ has an activation energy 30 kJ mol^{-1}. If energy change (ΔH) during the reaction is – 20 kJ, then the activation energy for the reverse reaction is :–
(1) 10 kJ (2) 20 kJ
(3) 50 kJ (4) – 30 kJ

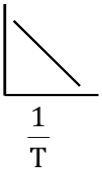
CCK099

100. Which of the following plot is in accordance with the Arrhenius equation :–

(1) 

(2) 

(3) 

(4) 

CCK100

101. The rate of reaction increases with increase in temperature because :–
(1) Collision is increased
(2) Energy of products decreases
(3) Fraction of molecules possessing energy $\geq E_T$ (Threshold energy) increases
(4) Mechanism of a reaction is changed

CCK101

102. For a certain gaseous reaction, rise of temperature from 25° C to 35° C doubles the rate of reaction. What is the value of activation energy :–

(1) $\frac{10}{2.303R \times 298 \times 308}$

(2) $\frac{2.303 \times 10}{298 \times 308R}$

(3) $\frac{0.693R \times 10}{290 \times 308}$

(4) $\frac{0.693R \times 298 \times 308}{10}$

CCK102

103. For producing the effective collisions, the colliding molecules must possess :-
(1) A certain minimum amount of energy
(2) Energy equal to or greater than threshold energy
(3) Proper orientation
(4) Threshold energy as well as proper orientation of collision

CCK103

104. The half life for a reaction ..... of temperature :-
(1) Independent
(2) Increases with increase
(3) Decreases with increase
(4) Increases or decreases with increase

CCK104

105. The activation energy for a chemical reaction depends upon :-
(1) Temperature
(2) Nature of reacting species
(3) Concentration of the reacting species
(4) Collision frequency

CCK105

106. During decomposition of an activated complex
(1) Some energy is absorbed
(2) Some energy is released
(3) No energy change
(4) Both (1) & (2)

CCK106

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	3	3	1	4	1	1	2	4	2	4	3	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	2	2	1	3	2	2	4	3	3	2	2	3	4	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	2	3	2	1	2	1	2	1	3	4	3	1	3	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	4	1	2	2	4	4	4	3	2	3	4	3	2	3
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	3	2	3	3	2	4	3	1	2	2	4	2	1	2
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	4	4	3	3	2	3	3	3	2	1	3	3	4	4	2
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	3	3	3	4	4	4	1	4	3	3	3	4	4	3	2
Question	106														
Answer	2														

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2015

1. The activation energy of a reaction can be determined from the slope of which of the following graphs ?

(1) $\frac{\ln k}{T}$ v / s.T (2) ln k v/s. $\frac{1}{T}$

(3) $\frac{T}{\ln k}$ v / s. $\frac{1}{T}$ (4) ln k v/s. T

CCK124

2. When initial concentration of a reactant is doubled in a reaction, its half-life period is not affected. The order of the reaction is :-

(1) First

(2) Second

(3) More than zero but less than first

(4) Zero

CCK125

Re-AIPMT 2015

3. The rate constant of the reaction A → B is 0.6×10^{-3} M sec^{-1}. If the concentration of A is 5 M, then concentration of B after 20 min is :-

(1) 0.36 M (2) 0.72 M

(3) 1.08 M (4) 3.60 M

CCK126

NEET-I 2016

4. The rate of a first-order reaction is 0.04 mol $L^{-1}s^{-1}$ at 10 s and 0.03 mol L^{-1} s^{-1} at 20 s after initiation of the reaction. The half-life period of the reaction is :

(1) 24.1 s (2) 34.1 s

(3) 44.1 s (4) 54.1 s

CCK127

5. The addition of a catalyst during a chemical reaction alters which of the following quantities ?

(1) Entropy (2) Internal energy

(3) Enthalpy (4) Activation energy

CCK128

NEET-II 2016

6. The decomposition of phosphine (PH_3) on tungsten at low pressure is a first-order reaction. It is because the

(1) Rate is independent of the surface coverage

(2) Rate of decomposition is very slow

(3) Rate is proportional to the surface coverage

(4) Rate is inversely proportional to the surface coverage

CCK129

NEET(UG) 2017

7. Mechanism of a hypothetical reaction $X_2 + Y_2 \rightarrow 2XY$ is given below :

(i) $X_2 \rightleftharpoons X + X$(fast)

(ii) $X + Y_2 \rightarrow XY + Y$ (slow)

(iii) $X + Y \rightarrow XY$ (fast)

The overall order of the reaction will be :

(1) 2 (2) 0 (3) 1.5 (4) 1

CCK130

8. A first order reaction has a specific reaction rate of 10^{-2} s^{-1}. How much time will it take for 20 g of the reactant to reduce to 5 g ?

(1) 138.6 s (2) 346.5 s

(3) 693.0 s (4) 238.6 s

CCK131

NEET(UG) 2018

9. The correct difference between first- and second-order reaction is that

(1) The rate of a first-order reaction does not depend on reactant concentration; the rate of a second-order reaction does depend on reactant concentrations.

(2) The half-life of a first-order reaction does not depend on $[A]_0$; the half-life of a second-order reaction does depend on $[A]_0$

(3) A first-order reaction can be catalyzed; a second-order reaction cannot be catalyzed.

(4) The rate of a first-order reaction does depend on reactant concentrations; the rate of a second-order reaction does not depend on reactant concentrations

CCK132

10. When initial concentration of the reactant is doubled, the half-life period of a zero order reaction
(1) Is halved
(2) Is doubled
(3) Is tripled
(4) Remains unchanged

CCK133

NEET(UG) 2019

11. If the rate constant for a first order reaction is k, then the time (t) required for the completion of 99% of the reaction is given by:-
(1) $t = 0.693/k$ (2) $t = 6.909/k$
(3) $t = 4.606/k$ (4) $t = 2.303/k$

CCK134

12. For the chemical reaction
$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
the **correct** option is :
(1) $-\frac{1}{3}\frac{d[H_2]}{dt} = -\frac{1}{2}\frac{d[NH_3]}{dt}$
(2) $-\frac{d[N_2]}{dt} = 2\frac{d[NH_3]}{dt}$
(3) $-\frac{d[N_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$
(4) $3\frac{d[H_2]}{dt} = 2\frac{d[NH_3]}{dt}$

CCK135

NEET(UG) 2019 (Odisha)

13. A first order reaction has a rate constant of 2.303×10^{-3} s^{-1}. The time required for 40g of this reactant to reduce to 10 g will be-
[Given that $\log_{10} 2 = 0.3010$]
(1) 230.3 s (2) 301 s
(3) 2000 s (4) 602 s

CCK136

14. For a reaction, activation energy $E_a = 0$ and the rate constant at 200K is $1.6\times10^{6} s^{-1}$. The rate constant at 400K will be-
[Given : gas constant R= 8.314 J K^{-1} mol^{-1}]
(1) 3.2×10^{4} s^{-1} (2) 1.6×10^{6} s^{-1}
(3) 1.6×10^{3} s^{-1} (4) 3.2×10^{6} s^{-1}

CCK137

NEET (UG) 2020

15. The rate constant for a first order reaction is 4.606×10^{-3} s^{-1}. The time required to reduce 2.0 g of the reactant to 0.2 g is :
(1) 1000 s (2) 100 s
(3) 200 s (4) 500 s

CCK138

16. An increase in the concentration of the reactants of a reaction leads to change in :
(1) Collision frequency (2) Activation energy
(3) Heat of reaction (4) Threshold energy

CCK139

NEET (UG) 2020 (COVID-19)

17. The half-life for a zero order reaction having 0.02 M initial concentration of reactant is 100 s. The rate constant (in mol L^{-1} s^{-1}) for the reaction is
(1) 1.0×10^{-4} (2) 2.0×10^{-4}
(3) 2.0×10^{-3} (4) 1.0×10^{-2}

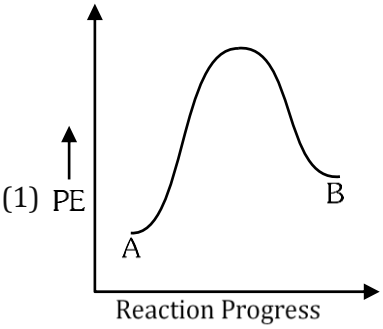
CCK140

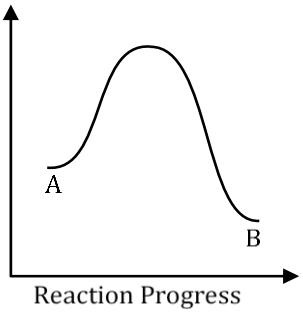
18. In collision theory of chemical reaction, Z_{AB} represents
(1) The fraction of molecules with energies greater than E_a
(2) The collision frequency of reactants, A and B
(3) Steric factor
(4) The fraction of molecules with energies equal to E_a

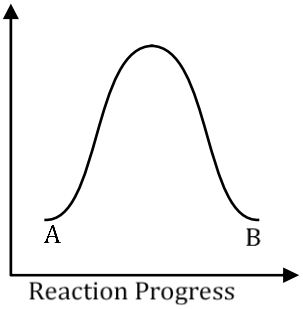
CCK141

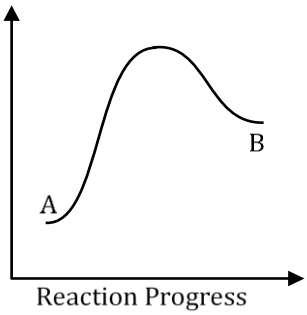
NEET (UG) 2021

19. For a reaction A → B, enthalpy of reaction is –4.2 kJ mol^{-1} and enthalpy of activation is 9.6 kJ mol^{-1}. The correct potential energy profile for the reaction is shown in option.

(1)


(2)


(3)


(4)


CCK142

20. The slope of Arrhenius Plot $\left(\ln k \text{ v/s } \frac{1}{T}\right)$ of first order reaction is -5×10^3 K. The value of E_a of the reaction is. Choose the correct option for your answer.

[Given R=8.314 JK^{-1} mol^{-1}]

(1) 41.5 kJ mol^{-1}

(2) 83.0 kJ mol^{-1}

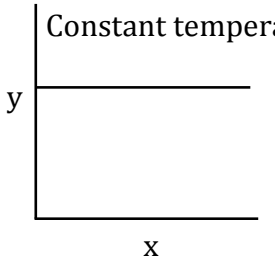
(3) 166 kJ mol^{-1}

(4) –83 kJ mol^{-1}

CCK143

NEET (UG) 2022

21. The given graph is a representation of kinetics of a reaction.



The y and x axes for zero and first order reactions, respectively are

(1) Zero order (y = concentration and x = time), first order (y = rate constant and x = concentration)

(2) Zero order (y = rate and x = concentration), first order ($y = t_{1/2}$ and x = concentration)

(3) Zero order (y = rate and x = concentration), first order (y = rate and $x = t_{1/2}$)

(4) Zero order (y = concentration and x = time), first order ($y = t_{1/2}$ and x = concentration)

CCK144

22. For a first order reaction A → Products, initial concentration of A is 0.1 M, which becomes 0.001 M after 5 minutes. Rate constant for the reaction in min^{-1} is

(1) 0.9212 (2) 0.4606

(3) 0.2303 (4) 1.3818

CCK145

23. The half life of a first order reaction is 2000 years. If the concentration after 8000 years is 0.02 M, then the initial concentration was :

(1) 0.16 M (2) 0.32 M

(3) 0.08 M (4) 0.04 M

CCK146

AIM: FREE EDUCATION TO ALL APUL

24. For a chemical reaction $4A + 3B \rightarrow 6C + 9D$ rate of formation of C is 6×10^{-2} mol L^{-1} s^{-1} and rate of disappearance of A is 4×10^{-2} mol L^{-1} s^{-1}. The rate of reaction and amount of B consumed in interval of 10 seconds, respectively will be :

(1) 1×10^{-2} mol L^{-1} s^{-1} and 30×10^{-2} mol L^{-1}

(2) 10×10^{-2} mol L^{-1} s^{-1} and 10×10^{-2} mol L^{-1}

(3) 1×10^{-2} mol L^{-1} s^{-1} and 10×10^{-2} mol L^{-1}

(4) 10×10^{-2} mol L^{-1} s^{-1} and 30×10^{-2} mol L^{-1}

CCK147

NEET (UG) 2023

25. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : A reaction can have zero activation energy.

Reasons R : The minimum extra amount of energy absorbed by reactant molecules so that their energy becomes equal to threshold value, is called activation energy.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**.

(2) **A** is true but **R** is false

(3) **A** is false but **R** is true

(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

CCK163

NEET (UG) 2024

26. Activation energy of any chemical reaction can be calculated if one knows the value of

(1) rate constant at standard temperature.

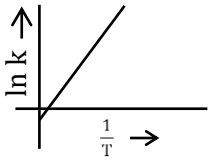
(2) probability of collision.

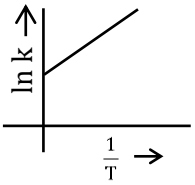
(3) orientation of reactant molecules during collision.

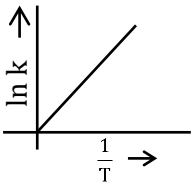
(4) rate constant at two different temperatures.

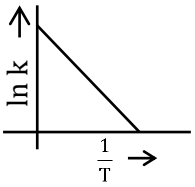
CCK179

27. Which plot of ln k vs $\frac{1}{T}$ is consistent with Arrhenius equation?

(1) 

(2) 

(3) 

(4) 

CCK180

28. The rate of a reaction quadruples when temperature changes from 27°C to 57°C. Calculate the energy of activation.

Given R = 8.314 J K^{-1} mol^{-1}, log 4 = 0.6021

(1) 38.04 kJ/mol (2) 380.4 kJ/mol

(3) 3.80 kJ/mol (4) 3804 kJ/mol

CCK181

NEET (UG) 2024 (Re-examination)

29. Following data is for a reaction between reactants A and B :

Rate mol L^{-1} s^{-1}	[A]	[B]
2×10^{-3}	0.1 M	0.1 M
4×10^{-3}	0.2 M	0.1 M
1.6×10^{-2}	0.2 M	0.2 M

The order of the reaction with respect to A and B, respectively are :

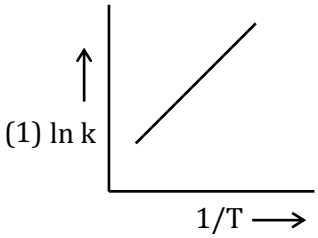
(1) 1, 0 (2) 0, 1

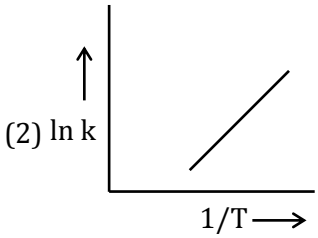
(3) 1, 2 (4) 2, 1

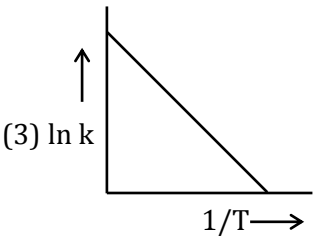
CCK182

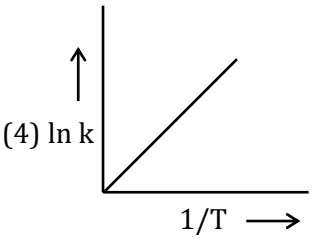
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30. Which of the following plot represents the variation of ln k versus $\frac{1}{T}$ in accordance with Arrhenius equation ?

(1) 

(2) 

(3) 

(4) 

CCK183

31. Rate constants of a reaction at 500 K and 700 K are $0.04s^{-1}$ and $0.14s^{-1}$, respectively, then, activation energy of the reaction is :
(Given : log 3.5 = 0.5441, R = 8.31 J $K^{-1}mol^{-1}$)

(1) 182310 J (2) 18500 J
(3) 18219 J (4) 18030 J

CCK184

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	2	1	4	3	3	1	2	2	3	3	4	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	1	2	2	1	2	1	2	1	3	4	4	1	3	3
Question	31														
Answer	3														

Exercise – III (Analytical Questions) Master Your Understanding

1. Given below are two statements:

Statement-I : Rate of zero order reactions is independent of temperature.

Statement-II : Half-life of zero order reactions is independent of concentration.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I & Statement-II are correct and the Statement-II is a correct explanation of Statement-I.

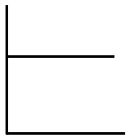
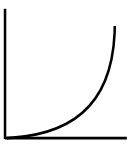
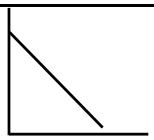
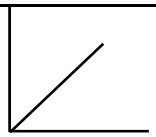
(2) Both Statement-I & Statement-II are correct but Statement-II is not a correct explanation of Statement-I.

(3) Statement-I is correct and Statement-II is incorrect.

(4) Both Statement-I and Statement-II are incorrect.

CCK148

2. Match the following

(a)	$[A]_t$ Vs time for zero order	(p)	
(b)	Half -life Vs Initial concentration for zero order	(q)	
(c)	Rate Vs time for zero order	(r)	
(d)	Rate constant Vs Temperature	(s)	

(1) (a) → (r), (b) → (s), (c) → (p), (d) → (q)

(2) (a) → (s), (b) → (r), (c) → (p), (d) → (q)

(3) (a) → (r), (b) → (s), (c) → (q), (d) → (p)

(4) (a) → (s), (b) → (r), (c) → (q), (d) → (p)

CCK149

3. Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.

Assertion (A) : For a reaction A → P, Rate and Rate constant are numerically equal when [A] = 1M.

Reason (R) : Rate and rate constant are equal for zero order reaction involving single reactant.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both Assertion & Reason are correct and the Reason is a correct explanation of Assertion.

(2) Both Assertion & Reason are correct but Reason is not a correct explanation of Assertion.

(3) Assertion is correct and Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

CCK151

4. Given below are two statements:

Statement-I : Catalyst does not change equilibrium constant of a reaction.

Statement-II : Catalyst decreases the time required to attain equilibrium.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I & Statement-II are correct and the Statement-II is a correct explanation of Statement-I.

(2) Both Statement-I & Statement-II are correct but Statement-II is not a correct explanation of Statement-I.

(3) Statement-I is correct and Statement-II is incorrect.

(4) Both Statement-I and Statement-II is incorrect.

CCK152

5. Consider the following statements :

(a) Order of a reaction is an experimental quantity.

(b) Order is applicable only for elementary reactions.

(c) Rate constant for a first order gaseous reaction will have the unit of atm. s^{-1}.

(d) Half-life period of a reaction decreases with increases in temperature.

Which statements are correct ?

(1) b, c

(2) a, b, d

(3) a, d

(4) c, d

CCK154

6. Which of the following statements are correct according to Arrhenius equation ?

(a) Rate of a reaction increases with increase in temperature.

(b) Rate of a reaction increases with decreases in activation energy.

(c) Rate constant decreases exponentially with increase in temperature.

(d) Rate of reaction decreases with decrease in activation energy.

(1) a, b

(2) b, d

(3) a, b and c

(4) a, d

CCK157

7. According to Maxwell Boltzmann distribution of energy.

(a) The fraction of molecules with most probable kinetic energy decreases at higher temperature.

(b) The fraction of molecules with most probable kinetic energy increases at higher temperature

(c) Most probable kinetic energy increases at higher temperature.

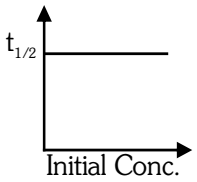
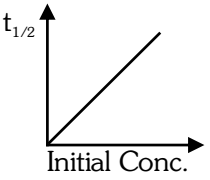
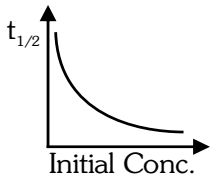
(d) Most probable kinetic energy decreases at higher temperature.

(1) a, b are correct (2) a, c are correct

(3) b, c are correct (4) b, d are correct

CCK161

8. Match the group given in column I with the order of reaction given in column II.

Column-I		Column-II	
(i)		(a)	First order
(ii)		(b)	Zero order
(iii)		(c)	Second order

	(i)	(ii)	(iii)
(1)	a	b	c
(2)	b	c	a
(3)	b	a	c
(4)	c	a	b

CCK162

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	4	1	2	2	3	1	2	1

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Chemical Kinetics

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Ans. (1)

$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

$$\text{rate} = \frac{\text{rod}}{\text{S.C. of reactent}} = \frac{\text{roa}}{\text{S.C. of product}}$$

$$\text{rate} = \frac{\frac{-d(N_2)}{dt}}{1} = \frac{\frac{-d(H_2)}{dt}}{3} = \frac{+\frac{d(NH_3)}{dt}}{2}$$

$$\text{rate} = \frac{-d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2}{dt} = +\frac{1}{2}\frac{d[NH_3]}{dt}$$

2. Ans. (1)

$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$

$$\text{ror} = \frac{\text{rod of } SO_2}{2} = \frac{\text{rod of } O_2}{1} = \frac{\text{roa of } SO_3}{2}$$

Given ror =

$$\text{rod of } O_2 = \frac{-d[O_2]}{dt} = 2.5\times10^{-4}\ \frac{\text{mol}}{\text{L sec}}$$

$$2.5\times10^{-4} = \frac{\text{rod of } SO_2}{2}$$

$$\text{rod of } SO_2 = 5\times10^{-4}\ \frac{\text{mol}}{\text{L sec}}$$

3. Ans. (3)

$N_2 + 3H_2 \longrightarrow 2NH_3$

$$\text{ror} = \frac{\text{rod of } N_2}{1} = \frac{\text{rod of } H_2}{3} = \frac{\text{roa of } NH_3}{2}$$

$$\text{Given roa of } NH_3 = 2.5\times10^{-4}\ \frac{\text{mol}}{\text{L sec}}$$

$$\frac{\text{rod of } H_2}{3} = \frac{2.5\times10^{-4}}{2}$$

$$\text{rod of } H_2 = 3.75\times10^{-4}\ \frac{\text{mol}}{\text{L sec}}$$

4. Ans. (3)

$X + 2y \longrightarrow P$

$$\frac{\text{rod of x}}{1} = \frac{\text{rod of y}}{2} = \frac{\text{rod of P}}{1}$$

5. Ans. (1)

$N_2O_5 \longrightarrow 2NO_2 + \frac{1}{2}O_2$

$$\frac{\frac{-d(N_2O_5)}{dt}}{1} = \frac{\frac{-d(NO_2)}{dt}}{2} = \frac{\frac{-d(O_2)}{dt}}{1/2}$$

$$K_1[N_2O_5] = \frac{1}{2}\times K_2[N_2O_5] = 2K_3[N_2O_5]$$

$$K_1 = \frac{K_2}{2} = 2K_3$$

$2K_1 = K_2 = 4K_3$

6. Ans. (4)

$2SO_2 + O_2 \longrightarrow 2SO_3$

$$\text{rof of } SO_3 = 1.6\times10^{-3}\ \frac{\text{kg}}{\text{L min}}$$

M_w → SO_2=64, SO_3=80

$$\frac{\text{rod of } SO_2}{M_w(SO_2)\times2} = \frac{\text{roa of } SO_3}{M_w/(SO_3)\times2}$$

$$\text{rod of } SO_2 = \text{roa of } SO_3 \times \frac{M_w(SO_2)}{M_w(SO_3)}$$

$$= 1.6\times10^{-3}\times\frac{64}{80} = 1.28\times10^{-3}\ \frac{\text{Kg}}{\text{L min}}$$

7. Ans. (1)

$2A + 3B \longrightarrow$ product

$$\frac{\text{rod of A}}{2} = \frac{\text{rod of B}}{3}$$

Given rod of A = r_1 ; rod of B = r_2

$$\frac{r_1}{2} = \frac{r_2}{3}$$

$3r_1 = 2r_2$

8. Ans. (1)

$2A + B \longrightarrow 3C + D$

ror =

$$\frac{\text{rod of A}}{2} = -\frac{\text{rod of B}}{1} = \frac{\text{roa of C}}{3} = \frac{\text{roa of D}}{1}$$

$$\text{ror} = \frac{1}{2}\left[\frac{-d(A)}{dt}\right] = \frac{-d[B]}{dt} = \frac{1}{3}\left[+\frac{d(C)}{dt}\right] = +\frac{d[D]}{dt}$$

9. Ans. (2)

$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

$$\frac{\text{roa of } NH_3}{2} = \frac{\text{rod of } H_2}{3}$$

$$\text{roa of } NH_3 = \frac{2}{3} \text{ rod of } H_2$$

$$+\frac{d[NH_3]}{dt} = \frac{2}{3}\left\{-\frac{d[H_2]}{dt}\right\}$$

10. Ans. (4)

$r = K(A)\ (B)$

$$C \propto \frac{1}{V}$$

If volume decreases $\frac{1}{4}$ then concentration increases 4 times

Old rate = K(A)[(B] ...(1)

New rate = K{4(A)} {4[B]} ...(2)

$$\frac{\text{New rate}}{\text{Old rate}} = 16$$

New rate = 16 × old rate

11. Ans. (2)

2(A)g + B(g) ⟶ product (Elementary reaction)

Old $r_f = KP_A^2P_B$ ...(1)

New $r_f = K(3P_A)^2(3P_B) = 27KP_A^2P_B$ ...(2)

$$\frac{\text{New } r_f}{\text{Old } r_f} = 27$$

New r_f = 27 × (old r_f) ...(2)

12. Ans. (4)

A + B + C ⟶ product

$$r = \frac{-d[A]}{dt} = K[A]^{\frac{1}{2}}[B]^{\frac{1}{3}}[C]^{\frac{1}{4}}$$

order = SUM of S.C. in rate law

$$\text{Order} = \frac{1}{2}+\frac{1}{3}+\frac{1}{4} = \frac{13}{12}$$

13. Ans. (3)

Overall order

Option 1 = 1 + 1 + 1 = 3

Option 2 = 0.5 + 0.5 + 0.5 = 1.5

Option 3 = 1.5 – 1 + 0 = 0.5

Option 4 = 1 + 0 – 2 = –1

14. Ans. (1)

Let reaction has two reactant A & B.

$$r \propto \frac{[A]^2}{[B]}$$

$$r \propto [A]^2[B]^{-1}$$

Order = 2–1 = 1

15. Ans. (2)

$H_2(g) + Br_2(g) \longrightarrow 2HBr(g)$

Rate = $K[H_2][Br_2]^{1/2}$

$$\text{Order } 1 + \frac{1}{2} = 1\frac{1}{2}$$

16. Ans. (3)

A + B ⟶ C

Let Intial rate (r) = $K[A]^x\ [B]^y$

Exp.1 $0.1 = K[0.012]^x\ [0.035]^y$ ...(1)

Exp.2 $1.6 = K[0.024]^x\ [0.07]^y$ ...(2)

$0.2 = K[0.024]^x\ [0.035]^y$ ...(3)

$0.8 = K[0.012]^x\ [0.07]^y$ ...(4)

$$\frac{(3)}{(1)}\ \frac{0.2}{0.1} = 2 = \left(\frac{0.024}{0.012}\right)^x = 2^x \Rightarrow x = 1$$

$$\frac{(4)}{(1)}\ \frac{0.8}{0.1} = 8 = \left(\frac{0.07}{0.035}\right)^y = 2^y$$

$2^3 = 2^y \Rightarrow y = 3$

$R = K[A]^1\ [B]^3$

17. Ans. (2)

2A + B ⟶ C + D

Let Initial rate (r) = $K[A]^x\ [B]^y$

Exp(1) $7.5\times10^{-3} = K(0.1)^x(0.1)^y$ ...(1)

$9\times10^{-2} = K(0.3)^x(0.2)^y$ ...(2)

$3.6\times10^{-1} = K(0.3)^x(0.4)^y$ ...(3)

$3.6\times10^{-2} = K(0.4)^x(0.1)^y$ ...(4)

$$\frac{(4)}{(1)}\quad \frac{3.0\times10^{-2}}{7.5\times10^{-3}} = 4 = 4^x \Rightarrow \boxed{x = 1}$$

$$\frac{(3)}{(2)}\quad \frac{3.6\times10^{-1}}{9\times10^{-2}} = 4 = 2^y \Rightarrow \boxed{y = 2}$$

$r = K(A)^1\ (B)^2$

18. Ans. (2)

$A + B \longrightarrow C$

Let rate of reaction (r) = $K[A]^x[B]^y$

$1\times10^{-4} = K(0.01)^x (0.01)^y$ ...(1)

$9\times10^{-4} = K(0.01)^x (0.03)^y$ ...(2)

$2.7\times10^{-3} = K(0.03)^x (0.03)^y$ ... (3)

$\frac{(2)}{(1)}$ $9 = 3^y \Rightarrow \boxed{y = 2}$

$\frac{(3)}{(2)}$ $\frac{2.7\times10^{-3}}{9\times10^{-4}} = 3 = 3^x \Rightarrow \boxed{x = 1}$

$r = K[A]^1 [B]^2$

19. Ans. (1)

Let rate (r) = $K[A]^x [B]^y$

$1.2 \times 10^{-3} = K(0.05)^x (0.05)^y$ ...(1)

$2.4 \times 10^{-3} = K(0.1)^x (0.05)^y$ ...(2)

$1.2 \times 10^{-3} = K(0.05)^x (0.1)^y$ ...(3)

$\frac{(2)}{(1)}$ $2 = 2^x \Rightarrow x = 1$

$\frac{(3)}{(1)}$ $1 = 2^y \Rightarrow y = 0$

Order of reaction wrt A and B are 1 and 0

20. Ans. (3)

Order wrt A & B each is 1

Rate = $K[A]^1[B]^1$

$0.1 = K(0.2)\ (0.05)$ ...(1)

$0.4 = K(x)\ (0.05)$ ...(2)

$0.8 = K(0.4)\ (y)$ ...(3)

$\frac{(2)}{(1)}$ $\frac{0.4}{0.1} = \frac{x}{0.2}$

$\boxed{x = 0.8}$

$\frac{(3)}{(1)}$ $\frac{0.8}{0.1} = \frac{(0.4)(y)}{(0.2)(0.05)}$

$\boxed{0.2 = y}$

21. Ans. (2)

$2A + B \longrightarrow 2C$

Single step reaction is also called elementary

Order of reaction wrt A = 2

Order of reaction wrt B = 1

Rate = $K[A]^2 [B]^1$

22. Ans. (2)

Reaction rate depends on [A] and [B]

Rate (r) = $K[A]^x [B]^y$ ...(1)

$4r = K\{2[A]\}^x [B]^y$ ...(2)

$r = K [A]^x \{2[B]\}^y$ ...(3)

$\frac{(2)}{(1)} = 2^2 = 4 = 2^x$

$\boxed{x = 2}$

$\frac{(3)}{(1)} = 2^\circ = 1 = 2^y$

Rate (r) = $K[A]^2 [B]^0$

Rate = $K[A]^2$

23. Ans. (4)

Rate law is experimental concept whereas Law of mass action is theoretical concept.

Experimental concept is always more realistic and more informative.

In Elementary reaction, rate law and law of Mass action give same rate expression.

24. Ans. (3)

For an elementary reaction molecularity is sum of stoichiometry coefficient of reactant.

So molecularity of given reaction is 3.

25. Ans. (3)

Reaction rate depends on concentration of A & B.

Let rate of reaction,(r) = $K[A]^x[B]^y$ ...(1)

$2r = K\{2[A]\}^x[B]^y$ ...(2)

$2r = K\{2(A)^x\}\{2(B)^y\}$ ...(3)

$\frac{(2)}{(1)}$ $2 = 2^x$

$\boxed{x = 1}$

$\frac{(3)}{(2)}$ $2^\circ = 1 = 2^y$

$\boxed{y = 0}$

Order of reaction wrt A = x = 1

Order of reaction wrt B_0 = y = 0

26. Ans. (2)

Old rate $r = K[A]^1[B]^{1/2}$ ...(1)

New rate $r^1 = K\{4[A]\}^1 \{4(B)\}^{1/2}$ ...(2)

$$\frac{r^1}{r} = 4\times(4)^{\frac{1}{2}} = 8$$

$r^1 = 8r$

New rate = 8 × old rate

27. Ans. (2)

Given old rate $(r) = K[A]^1[B]^2$ ...(1)

Option (1)

New rate $(r^1) = K\{2(A)\}(B)^2$ ...(2)

$$\frac{(2)}{(1)} \quad \frac{r^1}{r} = 2 \Rightarrow \boxed{r^1 = 2r}$$

Option (2)

$$\text{New rate } (r^1) = K[A]\left\{\frac{(B)}{4}\right\}^2 = \frac{K(A)(B)^2}{16}$$

...(3)

$$\frac{(3)}{(1)} \quad \frac{r^1}{r} = \frac{1}{16} \Rightarrow \boxed{r^1 = \frac{1}{16}r}$$

Option (3) New rate (r^1)

$= \{2(A)\}\{2(B)\}^2 = 8K(A)(B)^2$ ...(4)

$$\frac{(4)}{(1)} \quad \frac{r^1}{r} = 8 \Rightarrow \boxed{r^1 = 8r}$$

Option (4) order of reaction = 1+ 2 = 3

28. Ans. (3)

Let rate of reaction $(r) = K(A)^x$ ...(1)

$= 2r = K\{8(A)\}^x$ ...(2)

$$\frac{(2)}{(1)} \quad 2 = 8^x = (2^3)^x = 2^{3x}$$

$3x = 1$

$$x = \frac{1}{3}$$

Order of reaction wrt A = $x = \frac{1}{3}$

29. Ans. (4)

$K = A\, e^{-Ea/RT}$

rate constant depends on activation energy and temperature.

30. Ans. (2)

If the given reaction is elementary then its order and molecularity will be 10 but usually maximum molecularity of elementary reaction is 3. So given reaction cant be elementary reaction it is a complex reaction.

31. Ans. (2)

We write the rate law from slow step by considering it elementary reaction.

So rate $(r) = K[A]^1[B]^1$

32. Ans. (2)

$$O_3 \underset{K_{Eq}}{\rightleftharpoons} O_2 + O \text{ (Fast)} \quad ...(1)$$

$$O + O_3 \xrightarrow{K^1} 2O_2 \text{ (slow)} \quad ...(2)$$

* We write rate from slow step by considering it elementary reaction.

$r = K^1[O]\ [O_3]$ ...(3)

Intermediate

* We write equilibrium expression for reversible reaction and eliminate intermediat with the help of this expression.

$$K_C = \frac{[O_2][O]}{[O_3]}$$

$[O] = K_C[O_3][O_2]^{-1}$ ...(4)

Place [O] in equation (3)

$r = K^1\{K_C[O_3][O_2]^{-1}\}\ [O_3]$

$r = K[O_3]^2[O_2]^{-1}$

$\boxed{K^1K_C = K}$

33. Ans. (3)

$$A_2 \underset{K_C}{\rightleftharpoons} A + A \text{----(Fast)} \quad ...(1)$$

$$A + B_2 \xrightarrow{K'} AB + B \text{----(Slow)} \quad ...(2)$$

$$A + B \underset{K''}{\rightleftharpoons} AB \text{----(Fast)} \quad ...(3)$$

$r = K'\ [A]\ [B_2]$ ...(4)

↑

Intermediate

$$K_C = \frac{[A][A]}{[A_2]} = \frac{[A]^2}{[A_2]}$$

$[A]^2 = K_C[A_2]$

$[A] = K_C^{\frac{1}{2}}[A_2]^{\frac{1}{2}}$ ...(5)

Place [A] in equation (4)

$$r = K'\left\{K_C^{\frac{1}{2}}[A_2]^{\frac{1}{2}}\right\}[B_2]$$

$$r = K[A_2]^{\frac{1}{2}}[B_2] \qquad K = K'K_C^{\frac{1}{2}}$$

$$\text{order} = \frac{1}{2} + 1 = 1\frac{1}{2}$$

34. Ans. (2)

$r = K_1[RCl]$

On increasing temperature, rate of reaction increases because effective collision between molecules increases.

* In rate law expression, NaOH is not present so no effect on rate by doubling the concentration of NaOH.

* rate of reaction is directly proportional to [RCl].

As concentration of RCl become half, rate of reaction become half.

35. Ans. (1)

$NO_2 + CO \longrightarrow CO_2 + NO$, Rate $= K(NO_2)^2$

It is not elementary reaction. It is complex reaction

Rate expression Rate $= K(NO_2)^2$ is derived from slowest step.

As there is no CO term in rate expression so CO molecules would not be involved in slowest step.

36. Ans. (2)

$$\text{rate} = \frac{dp}{dt} = K[A]^2[B]$$

If 'A' is taken in large excess then during the reaction its concentration almost remains constant and rate become independent of it.

rate $= \{K[A]^2\}[B]$ $K' = K(A)^2$

(Effective) rate $= K'[B]\uparrow$

Order = 1 New constant

37. Ans. (1)

$CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$

Rate $= K[CH_3COO_2H_5]$

and hydrolysis of ester is 1st order reaction

38. Ans. (2)

$2A \underset{}{\overset{K_C}{\rightleftharpoons}} A_2$ (Fast) ...(1)

$A_2 + B \xrightarrow{K'} P$ (slow) ...(2)

Rate $= K'[A_2][B]$

$\uparrow$

Intermediate

$$K_C = \frac{[A_2]}{[A]^2} \Rightarrow [A_2] = K_C[A]^2 \quad \text{... (4)}$$

Place $[A_2]$ in equation (3)

rate $= K'\{K_C[A]^2\}[B] = K'K_C[A]^2[B]$

order = 2 + 1 = 3

39. Ans. (1)

$NO + Cl_2 \underset{}{\overset{K_C}{\rightleftharpoons}} NOCl_2$ (Fast) ...(1)

$NOCl_2 + NO \xrightarrow{K'} 2NOCl$ (Slow) ...(2)

Rate $= K'[NOCl_2][NO]$

$\uparrow$

Intermediate

$$K_C = \frac{[NOCl_2]}{[NO][Cl_2]}$$

$[NOCl_2] = K_C[NO][Cl_2]$ ...(4)

Place $(NOCl_2)$ in equation (3)

rate $= K'K_C[NO][Cl_2][NO]$

rate $= K[NO]^2[Cl_2]$

$K = K'K_C$

40. Ans. (3)

reaction 1, 1st order

$r_1 = K[A]$

reaction 2, 2nd order

$r_2 = K[A]^2$

reaction 3, 3rd order

$r_3 = K[A]^3$

when [A] > 1M

$r_1 < r_2 < r_3$

41. Ans. (4)

Given 0 order, $K = 2\times10^{-2}\ \frac{\text{mol}}{\text{L sec}}$

t = 25 sec, $(A)_t = 0.5M$, $[A]_0 = ?$

$[A]_t = [A]_0 - Kt$

$0.5 = [A]_0 - 2 \times 10^{-2} \times 25$

$[A]_0 = 1M$

42. Ans. (3)

$2N_2O_5 \longrightarrow 4NO_2 + O_2$, 1st order

This reaction is complex reaction and molecularity is not define for complex reaction. In 1st order reaction half lite $\propto$ (Initial concentration)^0

43. Ans. (1)

	x	⟶	y
Initially	aM		0
At time 't'	a–x		x

Let at time 't' both carve intersect so there concentration will be same.

$a - x = x$

$$x = \frac{a}{2}$$

At point of intersection, half of the reactant converted into product so it is half life so point of intersection of two curve represent $t_{\frac{1}{2}}$.

44. Ans. (3)

A(g) $\longrightarrow$ Product

Rate = $-\frac{dP_A}{dt}$

Unit = $\frac{atm}{sec}$

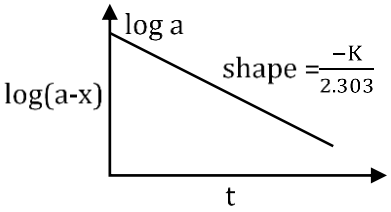
45. Ans. (2)

In 1st order

$$2.303\log\left(\frac{a-x}{a}\right)=-Kt$$

$$\log(a-x)-\log a=\frac{-K}{2.303}t$$

$$\log(a-x)=\frac{-k}{2.303}t+\log a$$



log (a-x) Vs time 't' plot is straight line for 1st order.

46. Ans. (1)

Given 1st order, K = 4×10^{-3} sec^{-1}, [A] = 0.02M

For 1st order reaction

r = K[A]

r = $4\times10^{-3}\times0.02=8\times10^{-5}\frac{M}{S}$

47. Ans. (4)

Given 1st Order

$[A]_0$ = 1M, $[A]_t$ = 0.25 M, t = 20 Min, K ?

$$K=\frac{2.303}{t}\log\frac{[A]_0}{[A]_t}$$

$$=\frac{2.303}{20}\log\frac{1}{0.25}$$

$$=\frac{2.303}{20}\log 4$$

$$K=\frac{2.303\times2\log2}{20}=\frac{0.6931}{10}=0.06931\,Min^{-1}$$

48. Ans. (1)

Given 1st order, $\frac{a}{a-x}=8$, t = 10 Min K ?

$$K=\frac{2.303}{t}\log\left(\frac{a}{a-x}\right)$$

$$K=\frac{2.303}{10}\log8=\frac{2.303}{10}\times\log2^3$$

$$K=\frac{(2.303\times3\log2)}{10}$$

49. Ans. (2)

Given 1st order

$t_{75\%}$ = 32 Min, $t_{50\%}$?

we know that $t_{75\%} = 2t_{50\%}$ = 32 Min

$t_{50\%}$ = 16 Min

50. Ans. (2)

Given 1st order, $t_{1/2}$ = 69.3 sec, [A] = 0.1 $\frac{Mol}{L}$

r ?

$$K=\frac{0.693}{t_{\frac{1}{2}}}=\frac{0.693}{69.3}=10^{-2}\,sec^{-1}$$

$r = K[A] = 10^{-2} \times 0.1 = 10^{-3}\ M\ S^{-1}$

51. Ans. (4)

1st order, [A] = 0.2M, $r = 1\times10^{-2}\frac{mol}{L\ Min}$, $t_{1/2}$?

r = K[A]

$$K=\frac{r}{(A)}=\frac{10^{-2}}{0.2}=5\times10^{-2}\,Min^{-1}$$

$$t_{\frac{1}{2}}=\frac{0.693}{K}=\frac{0.693}{5\times10^{-2}}=13.86\,min\simeq14Min$$

52. Ans. (4)

Given 1st order, $t_{99\%}$ = 32 Min, $t_{99.9\%}$?

$$Kt = 2.303\log\left(\frac{100}{100-P}\right)$$

P=% of reactant reacted

$$K\times32 = 2.303\log\frac{100}{100-99}$$

$= 2.303 \log 100 = 2.303\times2\log10$

$$Kt_{99\%} = 2.303\log\left(\frac{100}{100-99.9}\right)$$

$= 2.303 \log 1000 = 2.303\times3\log10$

$$\frac{(2)}{(1)}\qquad\frac{t_{99.9\%}}{32}=\frac{3}{2}$$

$t_{99.9\%}$ = 48 Min

53. Ans. (3)

Given radioactive reaction, $t_{87.5\%} = 40$ Min, $t_{50\%}$?

All radioactive reaction are 1st order.

$$Kt = 2.303\log\left(\frac{100}{100-P}\right)$$

$$Kt = 2.303\log\left(\frac{100}{100-87.5}\right)$$

P = % of reactant reacted

$$K \times 40 = 2.303 \log\left(\frac{100}{100-87.5}\right) = 2.303$$

$\log 8 = 2.303 \times 3\log 2$ ...(1)

$$K \times t_{50\%} = 2.303\log\left(\frac{100}{100-50}\right) = 2.303 \log 2$$ ...(2)

$$\frac{(2)}{(1)} \quad \frac{t_{50\%}}{40} = \frac{1}{3}$$

$$t_{50\%} = \frac{40}{3} \text{ Min} = 13 \text{ Min } 20 \text{ sec}$$

54. Ans. (3)

1st Order $(t_2\text{-}t_1) = 20$min,

$(A)_{t_1} = 1M$, $(A)_{t_2} = 0.6M$

$(t_2-t_1)' = ?$ $(A)'_{t_1} = 0.6M$, $(A)'_{t_2} = 0.36M$

$$K(t_2-t_1) = 2.303 \log \frac{[A]_{t_1}}{[A]_{t_2}}$$

$$K \times 20 = 2.303\log\frac{1}{0.6}$$

$$K(t_2-t_1)' = 2.303\log\frac{0.6}{0.36} = 2.303\log\frac{1}{0.6}$$

$$\frac{(2)}{(1)} \quad \frac{(t_2-t_1)'}{20} = 1$$

$(t_2-t_1)^1 = 20$ Min

55. Ans. (2)

In first order reaction in same time interval equal percentage of reactant convert into product and it is independent of the initial concentration of reactant.

P = 80

Alternate method

$$Kt = 2.303 \log \frac{100}{100-P}$$

P = % of reactant converted into product

Case II P=80

56. Ans. (3)

Given $t = 0$ $a = 10g/L$

$t_1 = 30min$ $a_{t_1} = 5g/L$

$t_2 = 90min$ $a_{t_2} = 1.25\ g/L$

$$10 \xrightarrow[t_{1/2}=30min]{} 5g/L \xrightarrow[t_{1/2}=30min]{} 2.5g/L \xrightarrow[t_{1/2}=30min]{} 1.25g/L$$

90 min

1st order

57. Ans. (4)

$t_{1/2} \propto (\text{Initial concentration})^{1-n}$

$t_{1/2} \propto (\text{Initial concentration})^{1-n}$

$t_{\frac{1}{2}} \propto a^{1-n}$ ...(1)

$2t_{1/2} \propto (2a)^{1-n}$ ...(2)

$\frac{(2)}{(1)}$ $2^1 = 2 = 2^{1-n}$

$1-n = 1$

(order) $n = 0$

58. Ans. (3)

$t_{\frac{1}{2}} \propto (\text{Initial Pressure})^{1-n}$

$4 \propto (50)^{1-n}$ ...(1)

$2 \propto (100)^{1-n}$ ...(2)

$$\frac{(2)}{(1)} \quad \frac{2}{4} = \left(\frac{100}{50}\right)^{1-n}$$

$2^{-1} = 2^{1-n}$

$-1 = 1-n \Rightarrow n = 2$

59. Ans. (2)

Given: $K = 10.8\times10^{-5}$ mol $L^{-1}S^{-1}$, order (n) = ?

General unit of K = $mol^{1-n}\ L^{n-1}\ time^{-1}$

On comparing power of mole

$1-n = 1$

(order) $n = 0$

60. Ans. (3)

Given : A decompose by 1st order

Flask I (1L)	Flask II(100 ml)
$[A]_0 = 1M$	$[A]_0 = 0.6M$
$T = 8hr$, $[A]_t = 0.25M$	t ? $[A]_t = 0.3M$

$$1M \xrightarrow[t_{1/2}]{} 0.5M \xrightarrow[t_{1/2}]{} 0.25M \quad 0.6M \xrightarrow[t_{1/2}]{} 0.3M$$

8 hr

$2t_{1/2} = 8hr$ $t = t_{1/2} = 4hr$

$T_{1/2} = 4hr$

61. Ans. (1)

$2A + B \longrightarrow$ product

t	k
15 sec	2.57×10^{-5} L $mol^{-1}s^{-1}$
30 sec	2.60×10^{-5} L $mol^{-1}s^{-1}$
50 sec	2.55×10^{-5} L $mol^{-1}s^{-1}$

General unit of $K = L^{n-1} mol^{1-n} time^{-1}$

On comparing power of mol : $1-n = -1$

$n = 2$

62. Ans. (3)

Given, 2nd order, $K = 8 \times 10^{-5}\ M^{-1}min^{-1}$

$[A]_0 = 1M$, $[A]_t = 0.5M$, t ?

$$Kt = \frac{1}{[A]_t} - \frac{1}{[A]_0}$$

$$8\times10^{-5} \times t = \frac{1}{[A]_t} - \frac{1}{[A]_0} = 1$$

$$t = \frac{1}{8\times10^{-5}} = 1.25 \times 10^4 \text{ min}$$

63. Ans. (2)

Given $t_{50\%} = t_{1/2} = 10$ min, $[A]_0 = 2$ mol L^{-1}

$t'_{50\%} = t'_{1/2} = t$ min, $[A]_0 = 2$ mol L^{-1}

From the graph it is clear that half life is independent of concentration, So it is 1st order and half-life also remains same, it doesn't depend on initial concentration.

$t'_{1/2} = t_{1/2} = 10\,\text{min}$

n = 1, $t'_{1/2} = 10\,\text{min}$

64. Ans. (3)

Given 1st order, $\frac{t_{99.9\%}}{t_{50\%}} = ?$

$$Kt = 2.303 \log \frac{100}{100-P}$$

P = % of reactant reacted

$$Kt_{99.9\%} = 2.303 \times \log \frac{100}{100-99.9}$$

$= 2.303 \log 1000 = 2.303 \times 3$ ...(1)

$$Kt_{50\%} = 2.303 \times \log \frac{100}{100-50} = 2.303 \log 2$$

...(2)

$$\frac{(2)}{(1)} \quad \frac{t_{99.9\%}}{t_{50\%}} = \frac{3}{\log 2} = \frac{3}{0.3} = 10$$

65. Ans. (3)

Given $t_{1/2} \times a$ = constant

$t_{1/2} \propto a^{-1}$ ...(1), n ?

In General $t_{1/2} \propto a^{1-n}$ ...(2)

Compare power of 'a'

$1-n = -1$

$n = 2$

66. Ans. (2)

Given $r = K$

$r = K[A]^0$

It is zero order reaction

* It is independent of concentration of reactant.

* $t_{1/2} = \frac{[A]_0}{2K}$, $t_{1/2}$ depends on initial concentration of reactant.

67. Ans. (4)

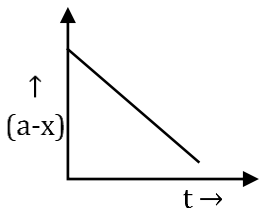
Given : 1st order

$$\log(a-x) = \frac{-K}{2.303}t + \log a$$

$y = mx + C$

$$m(\text{slope}) = \frac{-K}{2.303}$$

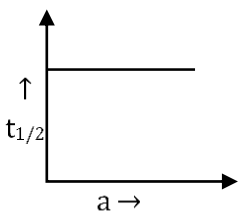
C (Intercept) = log a



$$t_{1/2} = \frac{0.693}{K}$$

$y = C$

$$\text{constant (C)} = \frac{0.693}{K}$$



68. Ans. (3)

$t_{1/2} \propto (\text{Initial pressure})^{1-n}$

$3.64 \propto (50)^{1-n}$ ...(1)

$1.82 \propto (100)^{1-n}$ ...(2)

$\frac{(2)}{(1)} \quad \frac{1.82}{3.64} = \left(\frac{100}{50}\right)^{1-n}$

$\frac{1}{2} = 2^{-1} = 2^{1-n}$

$1-n = -1$

$n = 2$

69. Ans. (1)

Given $K = xs^{-1}$

Rate constant, s^{-1} unit is the unit of 1st order reaction

(Old rate) $r = K[A]^1$ ...(1)

(New rate) $r' = K\{3[A]\}^1$ ...(2)

$\frac{(2)}{(1)} \quad \frac{\text{New rate}}{\text{Old rate}} = \frac{3}{1}$

New rate = 3×old rate

70. Ans. (2)

Zero order : $t_{1/2} = \frac{[A]_0}{2K}$

1st order $t_{1/2} = \frac{0.693}{K}$, $r = K[A]$

General unit of K = $mol^{1-n}\ L^{n-1}\ s^{-1}$

For 2nd order, unit of K = $mol^{-1}\ L^1 s^{-1}$

71. Ans. (2)

Hydrolysis of ester in alkaline medium

$RCOOR' + NaOH \longrightarrow RCOONa + R'OH$

Rate = K[RCOOR'] [NaOH]

Order = 2, Molecularity = 2

72. Ans. (4)

Given : 1st order, $t_{1/4} = t_{25\%}$?

$Kt = 2.303 \log \frac{100}{100-P}$

P = % of reactant reacted

$Kt_{1/4} = 2.303 \log \frac{100}{100-25}$

$t_{1/4} = \frac{2.303}{K} \log \frac{4}{3}$

73. Ans. (2)

Given : 0 order, K = 0.2 mol $dm^{-3}hr^{-1}$

t = 30 min = 0.5 hr, $[A]_t$ = 0.05 mol dm^{-3}, $[A]_0$?

$[A]_t = [A]_0 - Kt$

$0.05 = [A]_0 - 0.2 \times 0.5$

$[A]_0 = 0.05 + 0.1 = 0.15$ mol dm^{-3}

74. Ans. (1)

Given : 1st order,

Case-I $[A]_0$ = 100g,

x = 75g

t = 100 min

% of reactant decompose = $\frac{75}{100} \times 100 = 75\%$

Case II $[A]_0$ = 200g, x = 150g, t' ?

% of reactant decompose in case II

= $\frac{150}{200} \times 100 = 75\%$

In first order equal % of reactant decompose in same time period and it is independent of initial amount.

So in above both cases as 75% reactant is decomposing so time taken will be same, t' = 100min

75. Ans. (2)

For zero order r = K

Unit of rate = unit of rate constant

76. Ans. (4)

Given : 1st order, $t_{1/2}$ = 20 min,

$[A]_0$ = 0.2 mol^{-1}, K ?

$K = \frac{0.693}{t_{1/2}} = \frac{0.693}{20} = 0.034\ min^{-1}$

77. Ans. (4)

Given : 1st order

$t_{100\%} = t_{completion}$ = infinite

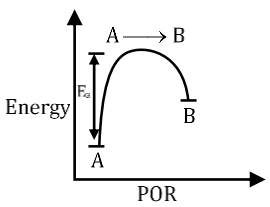
$t_{1/2} = \frac{0.693}{K}$

Unit of K = $time^{-1}$

$t_{1/2} \times K = 0.693$ (constant)

78. Ans. (3)

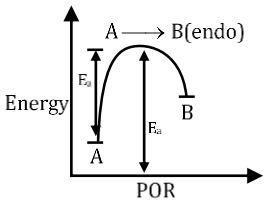
Activation energy is minimum extra energy required to overcome the potential barrier of reaction.



79. Ans. (3)

Activation energy is extra energy required by reactant molecules to enter into chemical reaction.

Threshold energy is minimum total energy required by reactant molecule to enter into chemical reaction.



80. Ans. (2)

$K_1 = 2K_2$ $\quad\downarrow K = Ae^{-\frac{E}{RT}}$

$K_1 > K_2$ $\quad$ if E↑ K↓

$E_1 < E_2$

81. Ans. (3)

$2NO + O_2 \xrightarrow{K_1, E_1} 2NO_2$ (Fast) ...(1)

$2CO + O_2 \xrightarrow{K_2, E_2} 2CO_2$ (Slow) ...(2)

$\downarrow K = Ae^{-\frac{E}{RT}}$

As E↑ K↓ reaction become slow.

Reaction (1)	Fast	$K_1\uparrow$ $E_1\downarrow$
Reaction (2)	Slow	$K_2\downarrow$ $E_2\uparrow$
	$K_1 > K_2$,	$E_1 < E_2$

82. Ans. (3)

Effective collisions are those collision which have energy enough to overcome energy barrier and have proper collision orientation. As Effective collisions increases, rate of reaction increases.

83. Ans. (3)

Given $A \longrightarrow B$, Endothermic,

$\Delta H = +5$ Kcal/ mol

$E_{af} = 15$ Kcal/mol

$B \longrightarrow A$, E_{af} ?

$E_{af}(B \longrightarrow A) = E_{ab}(A \longrightarrow B)$

For the reaction $A \longrightarrow B$

$\Delta H = E_{af} - E_{ab}$

$5 = 15 - E_{ab}$

$E_{ab} = 10$ Kcal/mol

84. Ans. (2)

As temperature increases, number of collisions increases, number of activated molecular doing effective collisions increases and rate of reaction increases.

85. Ans. (1)

As temperature increases, number of collisions increases for all reaction whether it is endothermic or exothermic, number of activated molecules doing effective collisions increases and rate of reaction increases.

86. Ans. (3)

Given : $\mu = 2$, $\Delta T = 60°C$

$r_{T_2} = r_{T_1} M^n$, $n = \frac{\Delta T}{10} = \frac{60}{10} = 6$

$r_{T_2} = r_{T_1} \times 2^6$

$r_{T_2} = 64 r_{T_1}$

So if temperature is increased by 60°C then rate of reaction increases by 64 times.

87. Ans. (3)

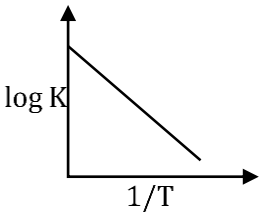
Arrhenius equation logarithm form is

$$\log K = \frac{-Ea}{2.303R}\left(\frac{1}{T}\right) + \log A$$

Y = mx + c

Slope (m) = $\frac{-E_a}{2.303R}$

Intercept (c) = log A



log K vs $\frac{1}{T}$ graph is straight line

88. Ans. (4)

When lnK vs $\frac{1}{T}$ = graph is plotted, it is straight line with slope $\frac{-E_a}{R}$. By measuring this slope we can get value of activation energy.

89. Ans. (4)

$$\log K = \log A - \frac{E_a}{2.303RT}$$

When $T \to \infty$, $\frac{1}{T} \to 0$, $\log K \to \log A$

90. Ans. (2)

Given T = 25°C, K = $3\times10^{-4}s^{-1}$, E_a=104.4Kj, A=$6\times10^{14}s^{-1}$, $T\to\infty$, K ?

Arrhenius $K = K = Ae^{-E_a/RT}$

as $T\to\infty$, $\frac{E_a}{RT}\to0$, $e^0\to1$

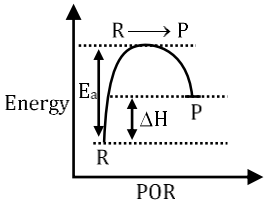
$K \to A$

When $T\to\infty$, $K\to A$

$K = 6\times10^{14}s^{-1}$

91. Ans. (3)

Given : Endothermic reaction



In endothermic reaction activation energy (E_a) is always greater than ΔH

92. Ans. (3)

Given : μ = 2.3, T_1 = 300K, T_2 = 310K

$K_1 = K$, $K_2 = ?$

$K_2 = K_1\mu^n$, $n = \frac{\Delta T}{10} = \frac{310-300}{10} = 1$

$K_2 = K(2.3)^1 = 2.3K$

93. Ans. (3)

Given : 1st order

$K = Ae^{-E_a/RT}$

Rate constant depends on activation energy and temperature. It don't depend on concentration. Hence if concentration is reduced by 'n' times then value of rate constant of 1st order or any order remains same of constant.

94. Ans. (4)

In general $R \longrightarrow P$

$Rate(r) = K(R)^n$

(R) is the concentration of reactant

n = order of reaction

$K = Ae^{-E_a/RT}$

K = specific rate constant

A → Arrhenius constant

E_a → activation energy

T = Temperature

Reaction rate depends upon reaction temperature, reaction concentration, specific rate constant, activation energy, catalyst.

95. Ans. (4)

Given : 1st order

$K = Ae^{-E_a/RT}$

rate constant depends upon activation energy and temperature. It don't depends on concentration of any reactant or product and volume of container.

96. Ans. (4)

Given : $2N_2O_5(g) \to 4NO_2(g) + O_2(g)$, E_a

$N_2O_5(g) \to 2NO_2(g) + \frac{1}{2}O_2(g)$, E_a^1

Activation energy depends on catalyst. It don't depends on stoichiometry coefficient of reaction. So $E_a = E_a^1$

97. Ans. (1)

$\Delta H = E_{af} - E_{ab}$

Of ΔH = 0

$E_{af} - E_{ab} = 0$

$E_{af} = E_{ab}$

98. Ans. (4)

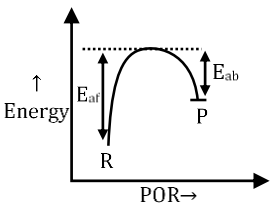
Given : E_{af} = 50 Kcal

There are two types of reaction

(1) Endothermic

(2) Exothermic

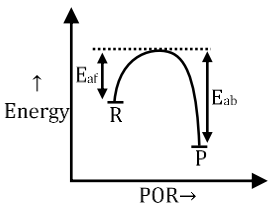
Endothermic



$E_{ab} < E_{af}$

$E_{ab} < 50$kcal

Exothermic



$E_{ab} > E_{af}$

$E_{ab} > 50$ kcal

Hence energy of activation of its backward reaction may be greater than or less than 50 kcal.

99. Ans. (3)

$x \longrightarrow y$, Exothermic, $E_{af} = 30Kj/mol$, $\Delta H = -20KJ$ E_{ab} ?

$\Delta H = E_{af} - E_{ab}$

$-20 = 30 - E_{ab}$

$E_{ab} = 50KJ$

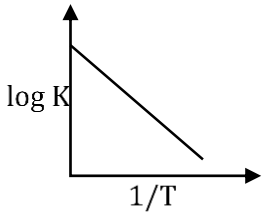
100. Ans. (3)

Arrhenius equation

$$\log K = \log A - \frac{E_a}{2.303R} \times \frac{1}{T}$$

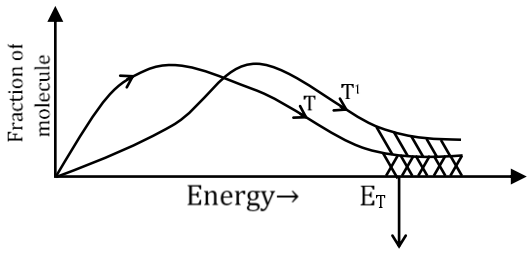
$y = C + mx$

slope (m) $= -\frac{E_a}{2.303R}$, Intercept(C) = logA



101. Ans. (3)

In Maxwell distribution carve



This area represent fraction of molecules possessing energy $\geq E_T$

As T↑, fraction of molecules' possessing energy $\geq E_T$ increases

102. Ans. (4)

Given : $T_1 = 25°C = 298K$, $r_{T_1} = r$

$T_2 = 35°C = 308K$, $r_{T_2} = 2r$

According to Arrhenius equation

$$\log \frac{K_{T_2}}{K_{T_1}} = \log \frac{r_{T_2}}{r_{T_1}} = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

$$\log \frac{2r}{r} = \frac{E_a}{2.303R}\left(\frac{308 - 298}{298 \times 308}\right)$$

$$0.3010 = \frac{E_a}{2.303R}\left(\frac{10}{298 \times 308}\right)$$

$$E_a = \frac{0.3010 \times 2.303R(298 \times 308)}{10}$$

$$E_a = \frac{0.693R \times 298 \times 308}{10}$$

103. Ans. (4)

According to collision theory,

For effective collisions, the colliding molecules most possess energy equal to or greater than threshold energy as well as proper orientation of collision.

104. Ans. (3)

As temperature increases, No of effective collisions increases, rate of reaction increases, so in lesser time half of the reactant covered into product so half-life decreases.

105. Ans. (2)

Activation energy depends on nature of reacting species, if activation energy is low than rate of reaction is test.

106. Ans. (2)

Activated complex has highest energy in the mechanism, then activated complex decompose, some energy release and stability increases.

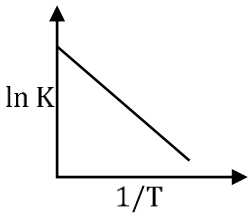
Exercise-II (Previous Year Questions)

1. Ans. (2)

Arrhenius equation :

$$\ln K = \ln A - \frac{E_a}{K}\left(\frac{1}{T}\right) \qquad \text{slope(m)} = \frac{-E_a}{K}$$

$y = C + mx$ Intercept (C) = ln A



ln K Vs $\frac{1}{T}$ graph is straight line by measuring the slope of line we can get the value of activation energy.

2. Ans. (1)

In General $t_{1/2} \propto (\text{Initial concentration})^{1-n}$

When n = 1

$T_{1/2}$ = constant

Hence for 1st order reaction half-life period is not affected by initial concentration.

3. Ans. (2)

Given : $A \longrightarrow B$

$K = 0.6\times10^{-3}$ M sec^{-1}, $[A]_0 = 5M$, $[B]_t = ?$, $t = 20min$

General unit of rate constant (K) = M^{1-n} sec^{-1}

$1-n = 1 \Rightarrow n = 0$

So the given reaction is zero order

$[B]_t = x = Kt$

$[B]_t = x = 0.6\times10^{-3}\times20\times60 = 0.72M$

4. Ans. (1)

Given : 1st order reaction

$t_1 = 10$ s, $r_{t_1} = 0.04\,mol\,L^{-1}s^{-1}$

$t_2 = 20$ s, $r_{t_2} = 0.03\ mol\,L^{-1}s^{-1}$, $t_{1/2} = ?$

$$\frac{r_{t_1}}{r_{t_2}} = \frac{K[A_{t_1}]}{K[A_{t_2}]} \Rightarrow \frac{r_{t_1}}{r_{t_2}} = \frac{[A_{t_1}]}{[A_{t_2}]}$$

Interval formula; $K\left(t_2 - t_1\right) = 2.303\log\frac{r_{t_1}}{r_{t_2}}$

$$K(20-10) = 2.303\log\frac{0.04}{0.03}$$

$$K = \frac{2.303}{10}\left[\log4 - \log3\right]$$

$$\frac{0.693}{t_{1/2}} = \frac{2.303}{10}\left[0.6020 - 0.4771\right]$$

$$t_{1/2} = \frac{0.693\times10}{2.303\times0.1249} \simeq 24.1\ s$$

5. Ans. (4)

When catalyst added to a chemical reaction, its activation energy decreases.

6. Ans. (3)

During decomposition phosphine (PH_3) on tangsten, first PH_3 adsorb on tangsten then it decompose.

Rate $\propto$ surface area of tangsten covered with PH_3 $\propto$ Partial pressure of $PH_3(g)$

at low pressure as $P_{PH_3(g)}$ $\uparrow$ surface area of tangsten $\uparrow$ rate $\uparrow$ covered with PH_3

at high pressure as $P_{PH_3(g)}$ $\uparrow$ surface area of rate is tangsten covered with constant PH_3 remains constant

rate $\propto$ (Pressure of PH_3)^0

7. Ans. (3)

Rate = K'(X) $[y_2]$ ...(1)

$\uparrow$

Intermediate

$$K_c = \frac{[x]^2}{[x_2]} \Rightarrow [x] = K_c^{1/2}[x_2]^{1/2}$$

Place [x] value in equation (1)

$$\text{Rate} = K^1K_c^{\frac{1}{2}}[x_2]^{\frac{1}{2}}[y_2] = K[x_2]^{0.5}[y_2]^1$$

Overall order = 0.5 + 1 = 1.5

8. Ans. (1)

Given : 1st order, $K = 10^{-2}s^{-1}$, $[A]_0 = 20g$, $[A]_t = 5g$, t ?

$$Kt = 2.303\log\frac{[A]_0}{[A]_t}$$

$$10^{-2}t = 2.303\log\frac{20}{5} = 2.303\log4$$

$$t = \frac{2.303\times0.6020}{10^{-2}} = 138.6\ s$$

9. Ans. (2)

First order reaction : $r = K(A)^1$, $t_{1/2} = \frac{0.693}{K}$

Second order reaction : $r = K(A)^2$, $t_{1/2} = \frac{1}{K(A)_0}$

10. Ans. (2)

Given : zero order

$$t_{1/2} = \frac{[A]_0}{2K}$$

$T_{1/2} \propto$ Initial concentration reactant.

As initial concentration of the reactant is doubled, half-life period of a zero order also doubled.

11. Ans. (3)

Given : first order, rate constant = K,

Time required for 99% completion of reaction ?

$$Kt = 2.303\log\frac{100}{100-P}$$ P = % of reactant converted into product

$$Kt' = 2.303\log\frac{100}{100-99}$$

$$t' = \frac{2.303\times2}{K} = \frac{4.606}{K}$$

12. Ans. (3)

Given : $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

$$\frac{\text{rod of } N_2}{1} = \frac{\text{rod of } H_2}{3} = \frac{\text{roa of } NH_3}{2}$$

$$-\frac{d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = +\frac{1}{2}\frac{d[NH_3]}{dt}$$

13. Ans. (4)

Given : 1st order, $K = 2.303\times10^{-3}s^{-1}$, $[A]_0=40g$, $[A]_t = 10\ g$, t ?

$$Kt = 2.303\log\frac{[A]_0}{[A]_t}$$

$$2.303\times10^{-3}\,t = 2.303\log\frac{40}{10}$$

$$t = \frac{2\log 2}{10^{-3}} = \frac{2\times0.3010}{10^{-3}}$$

t = 602 sec.

14. Ans. (2)

$E_a = 0$,

T	K
200K	$1.6\times10^6 s^{-1}$
400K	?

$$\log\frac{K_2}{K_1} = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$\log\frac{K_2}{K_1} = 0 \Rightarrow K_2 = K_1$$

$$K = Ae^{-E_a/RT}$$

If $E_a = 0$

$K = Ae^0$

$K = A$

When $E_{a.} = 0$ then rate constant don't depend on temperature. At all temperature, its value will be same. So at 200K of rate constant is $1.6\times10^6 s^{-1}$ then at 400K also its value will be $1.6\times10^6 s^{-1}$.

15. Ans. (4)

Given : 1st order, $K = 4.606\times10^{-3}s^{-1}$, $[A]_0=2g$, $[A]_t=0.2g$, t ?

$$Kt = 2.303\log\frac{[A]_0}{[A]_t}$$

$$4.606\times10^{-3}\,t = 2.303\log\frac{2}{0.2}$$

$$t = \frac{1}{2\times10^{-3}} = 500s$$

16. Ans. (1)

When Concentration of the reactant of a reaction increase then number of reactant molecule's per unit volume increases and collision frequency increases.

17. Ans. (1)

Given : zero order, $t_{1/2} = 100$ sec, $[A]_0 = 0.02M$, K ?

$$t_{1/2} = \frac{[A]_0}{2K}$$

$$K = \frac{[A]_0}{2t_{1/2}} = \frac{0.02}{2\times100} = 1\times10^{-4}.\text{mol}\,L^{-1}s^{-1}$$

18. Ans. (2)

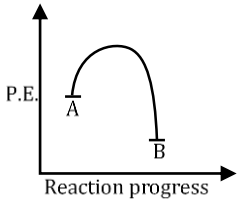
In collision theory Z_{AB} symbol means collision frequency of reactant A and B.

19. Ans. (2)

Given : $A \longrightarrow B$, $\Delta H = -4.2$ KJ/mol

$E_a = 9.6$ KJ/mol

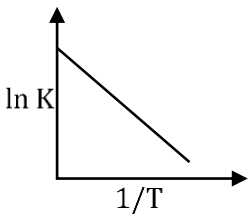
Given reaction is exothermic so potential energy of product will be less than potential energy of reactant.



20. Ans. (1)

$$\ln K = \ln A - \frac{E_a}{A}\times\frac{1}{T} \qquad \text{slope}(m) = \frac{-Ea}{R}$$

y = C + mx Intercept (C) = ln A



Given : $\text{slope} = \frac{-E_a}{R} = -5\times10^3$

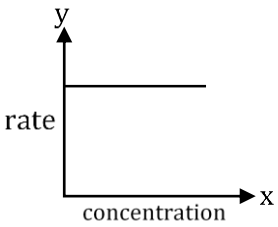
$Ea = 5\times10^3R = 5\times10^3\times8.314$

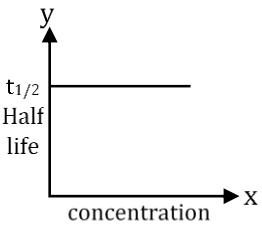
$Ea = 41.5\times10^3$ J ml^{-1}

$Ea = 41.5$ KJ mol^{-1}

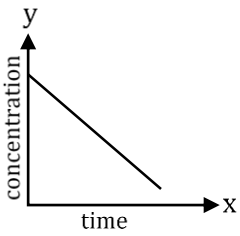
21. Ans. (2)

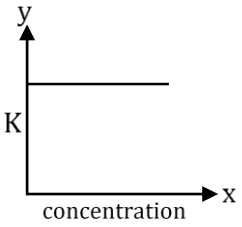
Zero order	First order
r = K	$t_{1/2} = \frac{0.693}{K}$
T = constant	$t_{1/2}$ = constant, T = constant





$[A]_t = [A]_0 - Kt$	$K = \frac{0.693}{t_{1/2}}$, K = constant



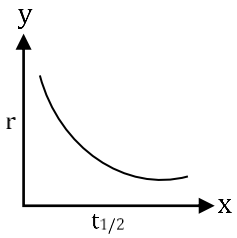


First order

$r = K(A) = \frac{0.693}{t_{1/2}}(A)$

T = constant

(A) = constant



22. Ans. (1)

Given : 1st order

$[A]_0$ = 0.1 M, $[A]_t$ = 0.001 M, t = 5min, K ?

$$Kt = 2.303 \log \frac{[A]_0}{[A]_t}$$

$$K \times 5 = 2.303 \log \frac{0.1}{0.001} = 2.303 \times \log 100$$

$= 2.303 \times 2 \log 10 \Rightarrow K = 0.9212 \text{ min}^{-1}$

23. Ans. (2)

Given : 1st order,

$t_{1/2}$ = 2000 year, t = 8000 years,

$[A]_0$? $[A]_t$ = 0.02 M

$[A_t] = \frac{[A]_0}{2^n}$ n = No. of half life = $\frac{\text{total time}}{t_{1/2}}$

$0.02 = \frac{[A]_0}{2^4}$ $n = \frac{8000}{2000} = 4$

$[A]_0 = 0.02 \times 16 = 0.32$ M

24. Ans. (1)

Given : rof of C = $6 \times 10^{-2} \frac{\text{mol}}{\text{Ls}}$,

rod of A = $4 \times 10^{-2} \frac{\text{mol}}{\text{Ls}}$

ror ?, $\Delta[B]$ = ?, ΔT = 10sec

$$\text{ror} = \frac{\text{rod of A}}{4} = \frac{\text{rod of B}}{3} = \frac{\text{rof of C}}{6} = \frac{\text{rof of D}}{9}$$

$$\text{ror} = \frac{\text{rod of A}}{4} = \frac{4 \times 10^{-2}}{4} = 1 \times 10^{-2} \text{ml L}^{-1}\text{s}^{-1}$$

$$\text{ror} = \frac{\text{rod of B}}{3}$$

$$\text{rod of B} = -\frac{\Delta[B]}{\Delta t} = 3 \times \text{ror}$$

$$-\frac{\Delta[B]}{10} = 3 \times 10^{-2}$$

$\Delta(B) = -30 \times 10^{-2}$ mol L^{-1}

Amount of B consumed = 30×10^{-2} mol L^{-1}

25. Ans. (3)

A reaction cannot have zero activation energy.

E_a is minimum extra amount of energy absorbed by reactant molecules so that their energy becomes equal to threshold value.

26. Ans. (4)

$$\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

Here k_1 is rate constant at T_1.

k_2 is rate constant at T_2.

27. Ans. (4)

$k = A \cdot e^{-E_a/RT}$

$\ell n\, k = \ell n\, A - \frac{E_a}{RT}$

$\ell nk = \frac{-E_a}{R}\left(\frac{1}{T}\right) + \ell nA$

$y = mx + c$

On comparing $\Rightarrow \ell n\, k \to y\,;\ \frac{1}{T} \to x$

This is a equation of straight line with negative slope.

28. Ans. (1)

$\frac{r_2}{r_1} = 4 = \frac{k_2}{k_1}$

$\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

$\log_{10}(4) = \frac{E_a \times 1000}{2.303 \times 8.314}\left(\frac{1}{300} - \frac{1}{330}\right)$

$E_a = 38.04\ kJ/mol$

29. Ans. (3)

$r = k[A]^x[B]^y$ ...(1)

from experimental data of I & II; order with respect to A is 1

from experimental data of II & III; order with respect to B is 2

30. Ans. (3)

$\ell nk = \ell nA - \frac{E_a}{R} \times \frac{1}{T}$ ....(1)

$y = c + mx$

slope = $-E_a/R$ = (–) Ve

intercept = ℓnA = (+)Ve

So ans is (3)

31. Ans. (3)

$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

$\log\left(\frac{0.14}{0.04}\right) = \frac{E_a}{2.303 \times 8.31} \times \left(\frac{200}{500 \times 700}\right)$

E_a = 18219 J

Exercise-III (Analytical Questions)

1. Ans. (4)

For zero order ; r = K

K depends on temperature, so rate (r) also depends on temperature. So statement I is false.

For zero order $t_{1/2} = \frac{[A]_0}{2K}$

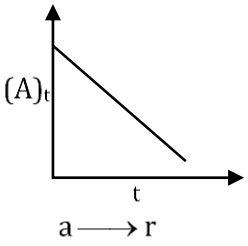
Half-life depends on initial concentration so statement II is also false.

2. Ans. (1)

(a) 'o' order

$[A]t = [A]_0 - kt$

$y = C + mx$

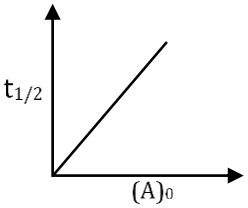


a ⟶ r

(b) 0 order

$t_{1/2} = \frac{1}{2K}(A)_0$

$y = mx$

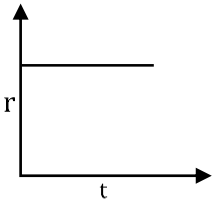


b ⟶ s

(c) '0' order

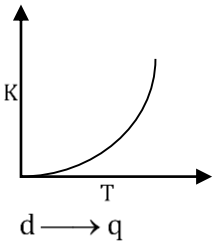
r = K

y = constant



c ⟶ p

(d) $K = A e^{-\frac{E_a}{RT}}$



3. Ans. (2)

$A \longrightarrow P$

rate = $K(A)^n$

when (A) = 1 M

rate (r) = K (rate constant)

Assertion is correct.

$A \longrightarrow P$

For zero order r = K

Reason as a independent statement is correct but it is not a correct explanation of assertion.

4. Ans. (2)

$$K_{eq} = \frac{K_f}{K_b}$$

Catalyst increases K_f and K_b equally and there is no effect on the value of K_{eq} statement I is correct.

Catalyst increases rate of reaction in forward as well as backward direction equally and equilibrium establish rapidly. Statement II is correct but it is not the proper explanation of statement I.

5. Ans. (3)

Order of a reaction is experimental quantity. Order is applicable for elementary as well as complex reaction unit of rate constant for 1st order is s^{-1}.

As temperature increases, rate of reaction increases, half-life period decrease.

Statement (a) and (d) are correct.

6. Ans. (1)

(a) When temperature increase, effective collision increases, rate of reaction increase.

(b) When activation energy decreases, rate constant value increases, more reactant molecule convert into product, rate of reaction increases.

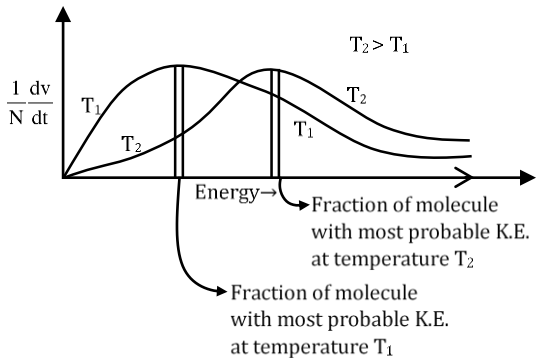
(c) $K = \frac{A}{e^{\frac{E_a}{RT}}}$ as $T \uparrow \frac{E_a}{RT} \downarrow e^{\frac{E_a}{RT}} \downarrow e^{\frac{A}{\frac{E_a}{RT}}} \uparrow K \uparrow$

Rate constant increases exponentially with increase in temperature.

(d) Statement (d) is wrong

Only statement (a) and (b) are correct

7. Ans. (2)



Fraction of molecule with most probable K.E. decreases at higher temperature.

$$V_{mp} = \sqrt{\frac{2RT}{M}}, \qquad (K.E.)_{m.p.} = \frac{1}{2}mv^2$$

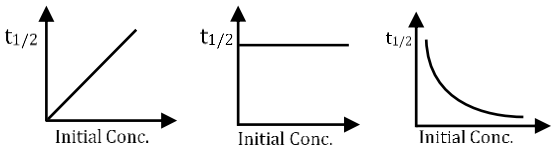
As $T \uparrow$ $V_{mp} \uparrow$ $(K.E.)_{mp} \uparrow$

Statement a and c are correct

8. Ans. (1)

0 order	1st order	2nd order
$t_{1/2} = \frac{[A]_0}{2K}$	$t_{1/2} = \frac{0.693}{K}$	$t_{1/2} = \frac{1}{K[A]_0}$

$$y = \frac{1}{2K}x$$



(ii) → (b) (i) → (a) (iii) → (c)